

GENETICS AND THE LOGIC
of
EVOLUTION

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KENNETH M. WEISS AND ANNE V. BUCHANAN

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*We wish to dedicate this book to our children, Ellen and Amie, for their forbearance,
and for the inspiration they have continually given to us.*

Apologia for advice not followed

And, freed from intricacies, taught to live
The easiest way; nor with perplexing thoughts
To interrupt the sweet of life, from which
God hath bid dwell far off all anxious cares,
And not molest us; unless we ourselves
Seek them with wandering thoughts, and notions vain.
But apt the mind or fancy is to rove
Unchecked, and of her roving is no end;
Till warned, or by experience taught, she learn,
That, not to know at large of things remote
From use, obscure and subtle; but, to know
That which before us lies in daily life,
Is the prime wisdom: What is more, is fume,
Or emptiness, or fond impertinence:
And renders us, in things that most concern,
Unpractised, unprepared, and still to seek.

J. Milton, Paradise Lost VIII: 182–197, 1667.

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Preface

WHAT THIS BOOK IS ABOUT

A PHILOSOPHY OF BIOLOGY

Our aim in this book is to develop some general principles to help describe the patterns to be found in the seemingly disparate facts about the diversity of life on Earth. It is an effort to assemble and digest observations made by naturalists and biologists from Aristotle to scientists publishing today—of organisms that live at temperatures above the boiling point of water and others that live in ice, organisms that fly and others that swim, those that inhale oxygen and those that expel it; how they are different and what they share. How can evolution, the single phenomenon that we invoke to explain how this endless diversity arose from one beginning, have produced it all?

Modern biological theory is thoroughly gene-centered, and this book is no exception. Genes are considered the essential storehouses of biological information and the mechanism through which evolution works. Thus, our specific interest in this book is in explaining the role of nucleic acids—DNA and RNA—in the evolution of complex organisms. At the same time, there is a danger in attributing too much to one cause, genetic or otherwise, or to one evolutionary process, or in considering the issues in such a detailed and itemized way that the broader picture gets lost. In this book we explore the ways in which an overly gene-based approach to biology can constrain our understanding of evolution.

Much of what we write about necessarily assumes evolution as its basic framework, that is, that organisms today have descended from ancestral organisms. But much of what we present considers alternative or supplemental general principles that we think are about as fundamental and ubiquitous in life as the core principles originally articulated by Darwin and Wallace. The theory of evolution was formalized as population genetics almost a century ago, but population genetics has little to say about the actual traits in organisms, how they are made, and how they evolve.

Natural selection is at the heart of the classical theory, but there is more going on than that, and we try to show what it is and where it might apply. Biology is forced to guess at the particulars of the evolution of traits and organisms because of millions of years of unobserved history that lies behind them. Natural selection is a rather generic explanation, which does not provide a very satisfying account of the particulars of the high degree of complexity found in organisms or even in cells themselves. To look at these, we consider aspects of life such as development, sensory systems, reproduction, and even perception. They illustrate some general

principles that provide a remarkably consistent picture of processes involved in very disparate traits across the spectrum of life.

Organisms confront their world in a multitude of successful ways involving a comparable diversity of pathways to, or consequences of, complexity. Complexity requires that an organism have, for example, extensive mechanisms for communicating internally among its cells, and many such mechanisms have evolved in all branches of life, which we will discuss. Externally, organisms are surrounded by a wide diversity of information with possible relevance to their safety, reproduction, and food acquisition, and organisms have evolved many ways to use (or dismiss or get along without) that information, and we will discuss these. Indeed, the external environment contains “information” only when or if it is needed or used. Most of our attention in this book is on complex multicellular species, but we consider simpler organisms as well.

For reasons that probably go back to the way life first began, an elegant few strategies have been employed to confront the challenges of life (we do not imply conscious intent here). The word “logic” in our title refers to the way that the diversity of complex organisms has come about through a few general mechanisms that, along with shared history, enable a trait or developmental pathway or gene, once it has arisen, to be used, reused, and modified. Very similar characteristics and relationships are found among entirely different and/or unrelated genes across the living world. These facts make it much easier to understand complex nature than did earlier and simpler views of genes as each individually coding for a specific protein with a unitary function. Many of these attributes of life have been long known, though not always to all persons working in diverse areas of biology, or well integrated into their work.

We discuss specific genes throughout, but it is the relationship or process, not the detail, that counts. We also cover aspects of life not yet explained very well in genetic terms, but a major point is that one can predict the *nature* of those genes and processes, based on generalizations derived from what we already know. Such predictions are possible because the logic of life can be reduced to a small number of basic, ubiquitous principles. Nevertheless, a main point will be that this is not a prespecified system that follows necessary rules, or “laws of nature,” the way formal mathematical logic does. Only in the broadest terms can there be a single theory of the contingent, largely chance-driven process that is the evolution of life. We can’t prove that some mechanism other than the one we try to reconstruct might not have yielded the same diversity of life we see on Earth; that is the nature of retrospective analysis that we are stuck with in trying to understand the unobserved past.

We try to develop a broad and unifying sense of life and the ways that organisms live it. Our attempt is intended for any reader wishing to understand some of the most important generalizations to emerge from recent biological research. This is not only edifying—it is to us—but can also provide a guide for future work.

We hope especially to stimulate students learning about biology and evolution to see that there are broad principles at work in life that go beyond the one darwinian view so often taught. A theory helps us construct a consistent worldview but is always at risk of becoming a constraining ideology. Although we are involved in molecular biology ourselves, we seek to understand the unity of life in broader terms, compatible with the effort to reduce an understanding of biology to an understanding of genes and their action, but that always keeps its eye on the organism as

a whole or on the phenomena that essentially rest on the interactions of molecules, cells, or even organisms.

OUR APOLOGIES

In this connection, it is certain that what follows will contain errors. Biology is a rapidly changing field, and facts are continually being amended or their importance reinterpreted, sometimes because of error and sometimes because of increasing knowledge. We are just two people confronting an enormous literature, and our own understanding will sometimes be flawed or we will have missed important papers; we will post errors and issues that we learn of on the <http://www.wiley.com> world-wide web page for this book. However, we think our general picture will be of some durability, and we hope that our attempt to go beyond the usually accepted principles of evolution, and to call some of those into question, will be useful and, if nothing else, thought-provoking.

We present some detail and technical material here but have tried to provide self-contained explanations; our intent is invariably conceptual rather than technical. Readers should be able to “read around” technical aspects that, because so much of the relevant genetics is of very recent vintage, are likely to be incomplete at best at this stage. We try to give a sense of what is known, with leads into the literature, without providing extensive lists of genes or pathways (we cite many excellent books, reviews, and scientific papers that do that). A reader interested in following up any particular points can easily find more about them through the literature and the internet—which would also help limit the damage that might be done by errors that we have made. If only because of the necessary lag time between writing and publication, no book can safely be regarded as definitive in detail, in a rapidly changing world.

A major risk in an era of exploding research and the sense of major discovery that now pervades genetics is that the firmness or importance of new results is probably overstated. However, we have tried to cite what seem to be reasonable interpretations of recent work that illustrate the generalities. We hope we have not been too restricted or parochial in doing so.

We have been unable even to approximate a thorough bibliography. As in the Technical Notes (below), internet web sources are so extensive and accessible that we think exhaustive citation is not as important for knowledge as it has been. We have cited primary literature to document our interpretation of various specific points, but as a rule we have preferentially cited recent reviews, texts, or convenient summary sources where we felt they would be useful, and/or that provide bibliographic entrée to the broader literature. Unfortunately, this nearly unavoidable way to handle information overload does undermine the proper assignment of credit for work and ideas because the authors of reviews are not always the sources of the material itself. We offer sincere apologies to many, many authors whose work we are aware of, staring us in the face from big piles on our floor, but that for practical reasons could not be cited.

Writing this book was a joint, interactive, and often grueling effort over several years as we tried to develop a credible understanding of fields entirely new to us, and to find the common threads among them. Although the illustrations new to the book were primarily done by one of us (AB), the writing itself was a joint, integrated effort in every respect regarding the ideas and the content.

If the ideas we present are interesting or stimulating to those who read this book, we will have succeeded, no matter how well our own particular views on life stand the test of time.

TECHNICAL NOTES

GENE NOMENCLATURE

Genes are being discovered by the hundreds, often by automated means. The nomenclature system is somewhat undisciplined and not entirely consistent. In this book we have discussed results from work in most areas of biology, many of which have their own conventions for gene nomenclature, not always even internally consistent. Designation conventions change and seem likely to do so even more as an ever-larger set of species and their genes are identified, and genes are grouped ever more accurately and into more extensive phylogenies. Thus we have tried at least to be clear and consistent within the bounds of this book, to minimize distracting readers with confusing gene designations. We generally use italics for gene names (*Bmp4*), and corresponding standard font for their respective coded protein (Bmp4). This may be the single most consistent general aspect of nomenclature in the field. Our own consistency with these guidelines varies from strict to yielding to well-established conventions in various fields where a strict adherence would strike the informed reader as strange.

We have hopefully been clear about whether we are referring to a gene or to its product, although the distinction can usually be inferred in context. We try to identify relevant homologous genes among species, when they have very different names. Above all, while we undoubtedly have missed things and not been perfectly consistent, nomenclature should provide only a minimal distraction.

BIBLIOGRAPHIC SUPPORT

Internet resources

We have not cited many internet URLs (worldwide web sites) in this book although the internet is a valuable resource for genetics. Readers who want to know more can usually use keywords to go right to major and minor resources for anything in the book and can follow up various issues by finding diagrams, DNA sequences, protein structures, technical descriptions, and even animations of many kinds and all levels. Unfortunately, the internet is a moving target, so that many URLs we would list here would be gone by the time a user wished to find them. The URLs we have cited seem to us to be likely to be relatively stable.

Reference citations

For the same reason, we have been especially sparing in our use of generic references. We usually give one or a few for broad topics. Similarly, it is utterly impossible to include all relevant technical references. The US National Center for Biotechnology Information (www.ncbi.nlm.nih.gov) provides many references and links, including PubMed (Medline) in which keyword searching can easily lead to the most recent literature. Readers should not rely on the accuracy of a conceptual survey such as ours, especially in an age in which so much is being learned so rapidly, and can be checked so easily.

Acknowledgments

We are pleased and grateful to acknowledge the help and guidance of a number of people as we wrote this book. Kris Aldridge, for an introduction into the literature on the brain, Mary Silcox for the tip on Linnaeus' classification of bacteria, Frances Hayashida for invaluable assistance with Adobe Illustrator, her provocative questions on the subject of every chapter and her delight in the answers (even though she's an archeologist), Ela and Janusz Sikora for cheering us on (even though they would prefer to talk about change by corrosion than by evolution), Nancy Buchanan for the photo of the flight feather from "the recent killing in the backyard," reminding us that nature can indeed be "red in tooth and claw," Bill Buchanan for his continual enthusiasm and support for this and all such endeavors, and Ellen Weiss for her photograph of Darwin's entangled bank (and her extra votes on the title).

We thank everyone, too numerous to mention by name, who gave us permission to use their figures or photographs, either as we redrew them, or the originals, and those we contacted for further clarification or expansion of their findings or ideas. People were invariably generous and helpful in guiding us to a better understanding of their ideas, whether or not we ultimately got it right or to their satisfaction.

And, we appreciate the tolerance of everyone who had to listen to us say for so long that we were "almost finished" with this book—all but our children were too polite to ask if it was really true this time. We thank Danielle Lacourciere and Rasa Hamilton at Wiley, for dealing with drafts rougher and more complicated than they probably expected. And last but certainly not least, we thank our editor at Wiley, Luna Han, for her patience and her belief in this project for more years than we would like admit.

PART I

Understanding Biological Complexity

Basic Concepts and Principles

In this section, we consider some of the general principles that characterize the nature and evolution of organized, functionally adaptive life on Earth. The mechanisms that determine the nature of organisms and the origin of the traits they possess can be approached at various levels of complexity. First, we will look at general principles. Inheritance is a vital component of diversified, specialized life, and we will consider just what it is that is inherited. We will then consider how that changes over time and relates to the processes we know as “evolution.”

Chapter 1

Prospect: The Basic Postulates of Life

Natural history is the *descriptive* study of the natural world. The ultimate objective of science is to go beyond natural history to find generalizations, or explanatory *theories*, to account for our observations of nature. Theory enables us to explain a set of observations with fewer “bits” (a “bit” being equivalent to the answer to a single yes/no question) of information than are contained in the observations themselves.

The more dramatic the reduction in the amount of such information needed to account for observations and the more accurate the predictions we can make, the more *explanatory power* we credit to the theory. Predictive power is the gold standard for confidence in a scientific theory. The more sweeping and accurate the better, so long as the predictions are not vacuously vague. Newton’s laws of motion, for example, apply broadly in the universe and are sufficiently accurate for many applications; Einstein’s modifications are even more accurate and comprehensive.

Scientific theory involves many assumptions that may not always be stated. We assume that the facts of nature are objective and can be explained in *natural* terms, that is, without intervention of nonmaterial (“supernatural”) factors. We also assume the universal validity of logical reasoning and mathematics. One of the most important assumptions that we make in building theory is that the fabric of causation in the cosmos is continuous and well-behaved, that facts are replicable—if we had the same conditions twice, we would have the same outcome. This may not be true in the ultimate sense (for example, if there is true randomness in the motion of atoms). More importantly for biology, our theory may assume replicability to a degree beyond what really applies, or, replicability may be the true state of Nature but our measurements too inaccurate. In fact, predictions and extrapolations can be almost completely inaccurate except in the short run, even for totally deterministic processes whose states or characteristics are not perfectly estimated (this phenomenon is sometimes characterized as “chaos” in the complexity literature).

The general belief among scientists is that we may not know the ultimate truth but that an ultimate truth does exist and that scientific methodology continually gets us closer to that truth. Philosophers of science debate whether this is actually so, noting that science is like other belief systems in resting on axioms—basic principles taken as givens and not to really be questioned. Indeed, science can be a kind of fundamentalism not unlike religion in its intolerance of challenges to its axioms. When, episodically, we become dissatisfied with the accuracy of this theoretical

edifice and an alternative explanatory framework is suggested, we experience what Thomas Kuhn called a scientific “revolution” (Kuhn 1962).

One rather curious basic assumption, the principle of parsimony (sometimes called “Occam’s Razor”), states that nature is no more complex than it has to be. In scientific practice, this means that we assume that the simplest explanation for an observation is the *best* one. We implicitly accept that this also means the *truest*. But of course we don’t know how complex nature really is or, in information terms, the degree to which any new theory could explain our current observations with fewer bits of information. This is a special challenge in biology because the biosphere is continually recreated through birth, death, and mutation in ever-changing environments. Unlike chemistry, we cannot replicate observations precisely at our will. Each new organism is unique, and life, unlike theory, does not always behave in the most parsimonious way. In the extreme, if life really were just as complex as our observations, then biology could not go much beyond descriptive “natural history.”

Evolutionary biology both describes and predicts. The history of life is generally assumed to have been a one-time affair, whose specific events are unique, contingent (that is, depend on unique circumstances), and hence not replicable. Yet, each individual is a new test of the challenges of survival, and in that and other ways the living world continually replays the general principles of evolution. We find regularities, and these have led to a formal theory of evolution. Nonetheless, this has limited power because *specific* events in the future cannot be predicted the way one can predict the nature of a chemical reaction, for example. What can be “predicted” (or if we look back in time, “retrodicted”) are patterns we might expect to see among descendants, based on postulated processes that affected their ancestors. A central problem is that in inferring how evolution produced what we see today we already know the outcome, so that much of what we do is to fit observations to theory rather than make truly deductive predictions.

One example of a very general prediction is that if different species share a recent common ancestor they will share more characteristics with each other than with species of more remote shared ancestry. If we could specify the extent of the similarity—say, in percent of difference between them on some scale—that specification could reduce the need to enumerate all the traits of each species. Linnaeus developed his systematic classification of life using morphological traits that he believed were important. The same idea can be extended to genes: related species will share genetic (DNA sequence) similarities to an extent that corresponds in some way to their phylogenetic history. This kind of divergence from a common ancestor was the basic idea underlying Charles Darwin’s metaphoric tree of life (Darwin 1859) (Figure 1-1), an image that Alfred Russel Wallace also used to express the diverging nature of life, and one similarly employed by evolution’s advocate in Germany, Ernst Haeckel, to show the nature of life diverging from “some one primordial form.” (In this book for their symbolic utility we will frequently mention specific prominent individuals, but historians of biology have shown clearly that most advances have come from the work of many, famous and less famous).

Relationships previously characterized by Linnaeus have generally held up to studies of genetic data; morphology is not a bad guide to taxonomy. There are exceptions, but they usually involve subtleties, very ancient splits, or traits that can change easily or rapidly with relationships that can only be resolved with extensive amounts of DNA data. Although Linnaeus knew about bacteria (they were first seen micro-

scopically in 1680 or so by Leeuwenhoek), he didn't understand them or their relationship to other living things and thus lumped them all into a category of miscellany that he called *Vermes*, in a class called *Chaos* (Magner 1994) (unrelated to the modern technical use of "chaos" referred to above). Sorting them out was left to future systematists. The complications are similar in nature to the complexity of nongenetic traits that have traditionally enabled debate among taxonomists.

In fact, genetic data are strikingly consistent with, and their characteristics were predicted by, darwinian principles, and it is significant that these findings were entirely independent of, and after, Darwin's formulation of his theory (in this book, we will use uncapitalized references, such as "darwinian," when discussing modified descendants of the original idea and capitalized references, such as "Darwinian," when discussing the specific notions of the person introducing them). Independent confirmation of theoretical ideas with new data is very important to the deductive aspects of science, and genetic taxonomy is an independent confirmation of Darwin. Of course, we know that morphological traits are affected by genes, so genetic data are not entirely independent; however, in a nonevolutionary world, for example, one made by a fixed creation event, there would not have to be any relationship between DNA sequence and morphological similarities.

If genes provide a kind of blueprint for life, genetic data should enable us to describe traits in different species or individuals with less information than is needed to describe each trait or individual separately. This is exactly the kind of reconstruction that Richard Owen and Georges Cuvier made famous in the early 1800s, when they used single bones to reconstruct whole animals, and why Thomas Huxley once exclaimed "A tooth! A tooth! My kingdom for a tooth!" (see Desmond 1994). Their theories were functional (not evolutionary): complex traits like a bone or

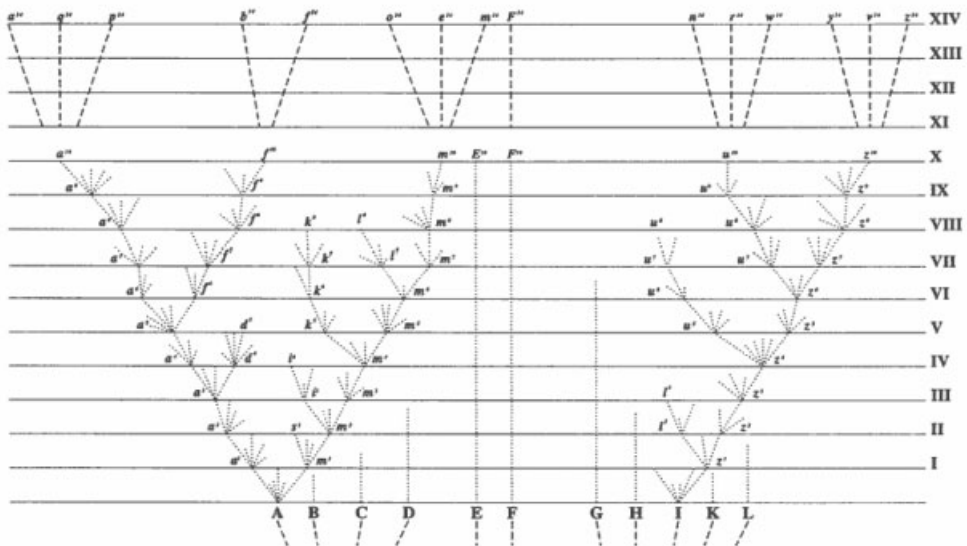


Figure 1-1. Trees of Life. (A) Darwin's from *Origin of Species*; (B) Haeckel's version from Haeckel (Haeckel 1906).

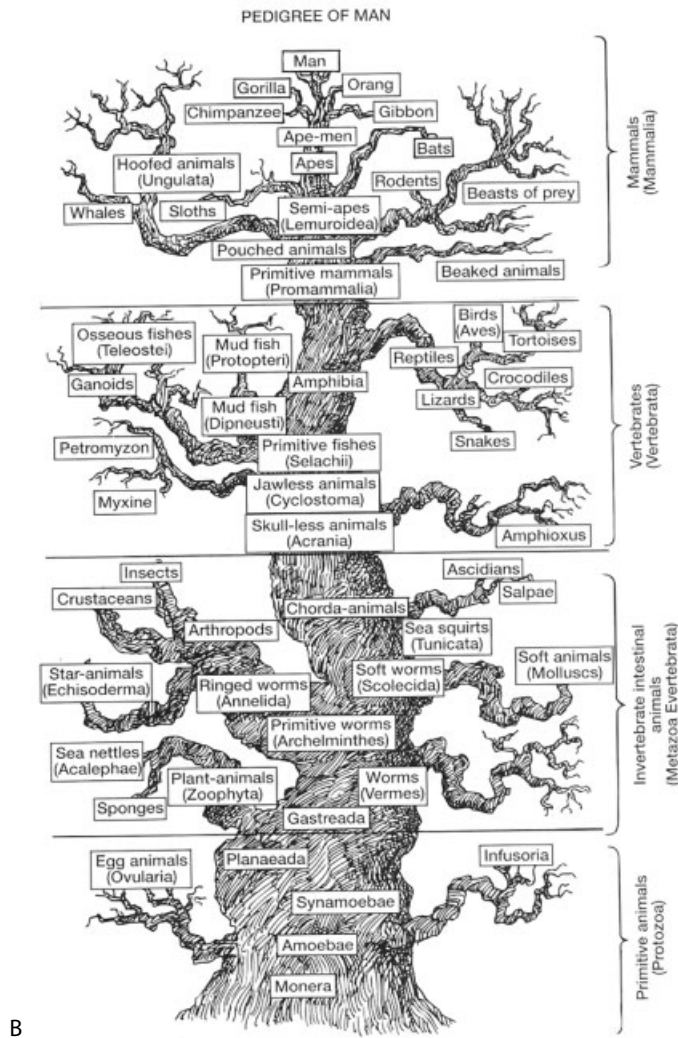


Figure 1-1. *Continued*

tooth reflect the function performed by the organism. A carnivore needs claws and teeth and speed, so to speak, and different carnivores share this general suite of characteristics. Genes provide similar kinds of relational information. One major purpose of this book is to ask how true are the simplifications that can be made from genes.

WONDROUS NATURE TO BE EXPLAINED

Naturalists, theologians, philosophers, and poets have written of their wonderment at the panoply of natural forms. Many have been struck by the adaptation of organisms to what they do in life; perhaps this insight alone is responsible for our modern

view of biology. Different explanations for the origins of adaptation have been offered, but it is worth quoting one of the first advocates of an evolutionary view, the naturalist Henry Walter Bates, who described the following observations on the butterflies in Ega, on the Upper Amazon (Solimoens), hundreds of miles upriver from Manaus (Bates 1863):

[They] vary in accordance with the slightest change in the conditions to which the species are exposed. It may be said, therefore, that on these expanded membranes Nature writes, as on a tablet, the story of the modifications of species, so truly do all changes of the organization register themselves thereon. Moreover, the same colour-patterns of the wings generally show, with great regularity, the degrees of blood-relationship of the species. As the laws of Nature must be the same for all beings, the conclusions furnished by this group of insects must be applicable to the whole organic world; therefore, the study of butterflies—creatures selected as the types of airiness and frivolity—instead of being despised, will some day be valued as one of the most important branches of Biological science.

This observation was made after Darwin and Wallace (a friend and co-Amazonian explorer with Bates) first publicized their views in 1858. In 1862, Bates laid out his view more formally in evolutionary terms, and it became known as *Batesian mimicry* (Bates 1862), the idea that tasty butterflies evolved to look like bitter ones that birds learn to leave alone. This is to this day one of the clearest cases of natural selection and is a textbook example cited to support the modern theory.

So what is this phenomenon called “evolution,” and how is it that this one phenomenon can explain the diversity of life?

THE PHENOMENON OF EVOLUTION

The theory of evolution developed out of earlier ideas, some clearly anticipating what Charles Darwin and Alfred Russel Wallace would introduce to the world in 1858. The important concept was that the diversity of life, present and past, was not static and produced by externally derived creation events but was the product of historical *processes*, operating since some origin time on Earth—and *still* operating. One can view these as the biological version of the prevalent idea of a universally applicable natural law. Darwin himself left open how the whole process may have started, but biologists almost uniformly assume it was a terrestrial, strictly chemical phenomenon (this, too, is an *assumption* that, while not necessitated by specific knowledge, reflects the purely materialistic working world view of most scientists). The theory of evolution is elegantly simple and requires only a few basic elements. Darwin and Wallace introduced a few, to which several additional broad generalizations about the phenomenon of life can be added.

DARWINIAN FUNDAMENTALS

The basic postulates of evolution are simple and well known, but it is worth listing them: (1) organisms vary, (2) some of that variation is heritable from parent to offspring, and (3) there is population pressure on resources related to survival and reproduction; that is, organisms produce far more offspring than the environment can support. From any system with these general properties, the *fact* of evolution could be predicted, so that once they were clearly stated, darwinian phenomena are

neither surprising nor really open to doubt. But this doesn't mean we could deduce any *particular* life form, even simple ones.

These basic principles can be summarized in Darwin's own phrase: "descent with modification." He deduced the consequence of persistent population pressure on resources: the variation that is able to reproduce more prolifically will be more commonly represented in future generations. He extrapolated this over long time periods, assuming it worked more or less consistently and gradually, to *hypothesize* that this *natural selection* explained the *adaptation* of organisms to their environment and accounted for the origin of new species from previous species over long time periods.

We should be aware of what these evolutionary premises do *not* say. They do not specify the resources under stress nor what it is that is heritable (or *how* it is inherited). Darwin did not specify correctly where new variation comes from, and there are widespread ideas in biology that are based on tacit additions to the basic principles (for example, some aspects of strong genetic determinism). We will see the implications of these assumptions.

We also don't need to argue about *whether* darwinian processes happen, as they are rather obvious and make very little in the way of specific assumptions. They not only apply to life but to any system built up of multiple, changeable units with competitive inheritance. But at the heart of Darwin's contribution to science was the theory that this is the process responsible for the transformation of species. Surprisingly, nothing in Darwin's premises necessitates species formation, even if his process adequately explained the form and structure of life. The separation into distinct species—which is what he was trying to explain—does not follow. At least one additional postulate is needed. This is (4) *sequestration*.

SEQUESTRATION AND DIVERSIFICATION

Sequestration is implicit in Darwin's postulates. Life forms are isolated from each other, so that differences can accumulate between individuals. Darwinian variation does not immediately blend away (ironically, Darwin's mistaken idea of blending heredity was a problem for his theory, a fact that bothered him greatly). We use mating barriers between organisms to define "species." Even assuming they are genetically based, such barriers can be established by genetic changes having nothing to do with response to environments. In fact, whether adaptive or even random processes per se lead to new species has never been adequately proved as a generalization, and partly depends on our definition of "species."

Darwin was trying to explain the diversification of life into many species. In fact, he felt, and it is often argued, that *phylogeny*, or branching (divergent) speciation, is predicted by his evolutionary postulates. He developed his theory with the species question in mind, but adaptation does not by itself imply speciation. A global primeval soup could in principle evolve by changes in its chemical composition, energy, or some other cyclical processes diffusing through it over time. This does not constitute divergence among the states of life, except in the sense that there would be variation, as there is among readers of this book.

Evolutionary thinking predicts branching because descent with modification produces variation, and if that variation does not freely mix, then eventually reproductive exchange between the different branches becomes no longer possible. This is the essence of "speciation." Variation is sequestered within lineages, which

accumulate increasing divergence over time. Because this process never ends, each lineage in turn diversifies. The result is a nested phylogeny.

It would seem from a superficial consideration of the similar nature of all cells that the basic machinery of life had developed before cells began to diverge. This assumes there was once only one cell population. Cells effectively isolate *very* localized packets of living matter from each other and from the surrounding “soup.” Internally, the cell maintains the special conditions for using DNA to code for protein, a system almost certainly already present when organized cells evolved. Higher-level organization of life into multicellular organisms depended on this so that even within an organism there is local isolation of material.

Sequestration of material into cells, however, can never be complete. Even the first cells had to evaluate their environment and interact selectively with it (bring in nutrients, release waste, control ion concentrations and pH, and so on). Multicellular organisms require interaction and hence exchange of “information” among cells. Elaborate mechanisms have evolved for this, including partially permeable cell membranes, with mechanisms for transporting material across them, signaling mechanisms that work across cell membranes, and mechanisms for direct contact or transfer between adjacent cells.

DNA sequences, which will be described specifically in Chapter 4, are inherited across generations and thus, by nature, retain a trace of the past. Indeed, the sequestration of DNA from direct modification by the cell is one of the cornerstones of modern evolutionary theory, as we will see. However, DNA replication is not perfect or evolution could not have occurred, and if we have some external means of calibrating species history, such as known points in the fossil record, we can compare sequences of fundamental genes in representatives of the major branches of organisms to make educated guesses about what the ancestral cell type and its

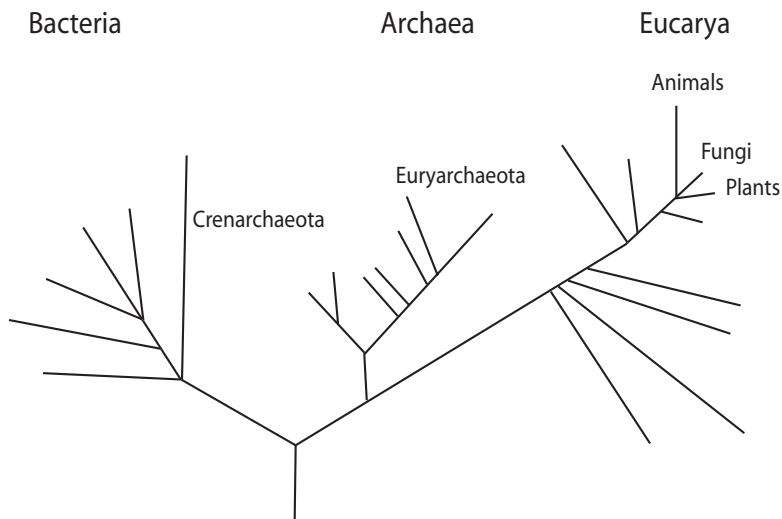


Figure 1-2. Tree of the major branches of life, based on ribosomal RNA. Redrawn from (Woese 2000) with permission. Original figure copyright 2000 National Academy of Sciences, U.S.A.

mechanisms may have been like (e.g., see Doolittle 1998; Doolittle 1998; Woese 2002). The process of accumulation of errors in DNA copying is highly stochastic (probabilistic); therefore, not all genes give precisely the same picture, so we have to aggregate data from many genes simultaneously. By grouping sequences that are most similar and roughly equating the amount of difference with time since common ancestry, we can reconstruct a hierarchical, treelike, representation of the history of life (e.g., Banfield and Marshall 2000).

The idea is based on the assumption that life had a single ancestry, here on Earth, represented by the trunk of our metaphoric tree. The tree of life reconstructed by genes presumably really is the tree of *cellular* life because basic biochemical mechanisms had to precede cells. Given a single origin of life, the principle of sequestration then leads naturally to diversification. Again, sequestration cannot be complete or we would never have aggregates of essentially similar cells that we call organisms or of essentially similar organisms that we call species. We will see, however, that this, like so many things in life, has important exceptions.

In addition to sequestration, three other aspects of life are so ubiquitous and fundamental that they should be added as generalizations about life as it happens to have happened on Earth. These are (5) *modularity*, (6) *duplication*, and (7) *chance*. Biological evolution could occur without them, but they have nearly comparable ubiquity and predictive power to the other postulates.

MODULARITY AND DUPLICATION

New structures from molecular to morphological are built by evolution from pre-existing foundations. One of the most important and fundamental aspects of this is *modularity*. From molecules to morphology, we see variations on similar themes. These comprise separate modules or units from which more complex structures have been constructed. And one of the most important ways this has taken place is by the *duplication* of structures, with subsequent differentiation. The pervasiveness of duplication of structure has been known since systematic biology began and has only been reinforced by the history of discovery in physiology and molecular biology.

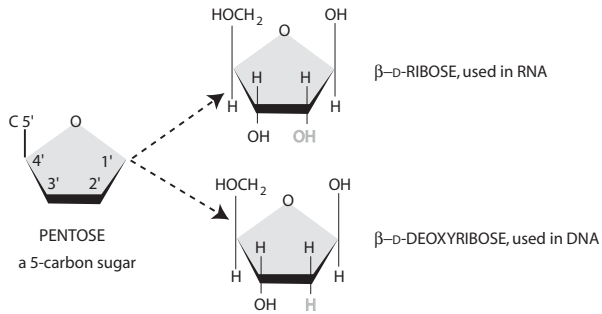
Modularity and Duplication Below the Level of the Cell

The modular nature of most of the basic biological molecules can be seen in Figure 1-3, which shows the chemical structure of nucleic acids, amino acids, and steroids. Variation on core structures as found in nucleic and amino acids was probably to a great extent a natural given, whereas variation in other molecules like steroids is at least to some extent manufactured by organisms. This is certainly true of protein families, as will be seen throughout this book.

The system of life today has been built on the modular nature of a corresponding concatenation of nucleic and amino acids into DNA/RNA and proteins. The nature of its ultimate origins is debated, but at some point biological information came to be stored in the form of the specific sequences, not the chemical nature, of these components. In particular, genetic coding is based on the *order* of concatenation of nucleotides in DNA and RNA, which has no chemical bearing on the *nature* of the protein being coded. The code for a given amino acid (see Chapter 4) is essentially universally used and has no bearing on the chemical nature of that amino acid nor on what that amino acid will do in a final protein. So it is in that sense a true *code*.

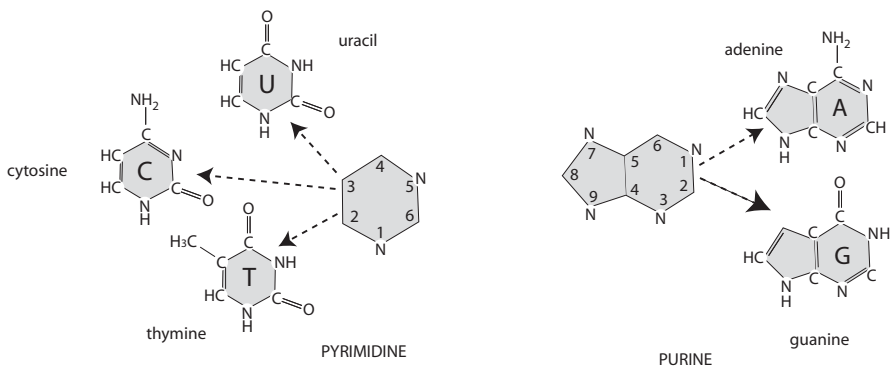
Nucleic Acids

SUGARS



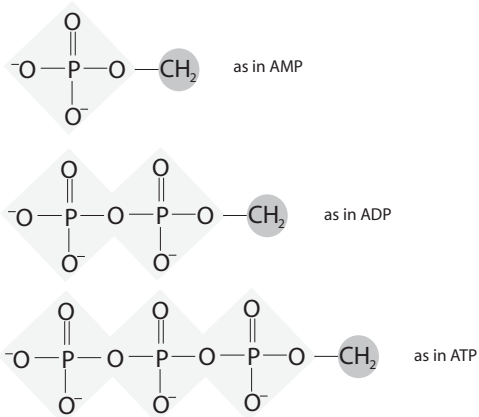
BASES

nitrogen-containing ring compounds



PHOSPHATES

join to the C5 hydroxyl of the ribose or deoxyribose sugar



A

NUCLEOTIDES

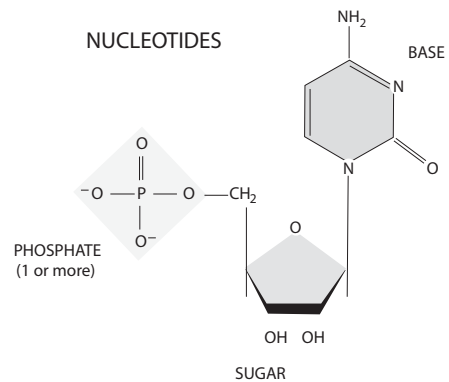
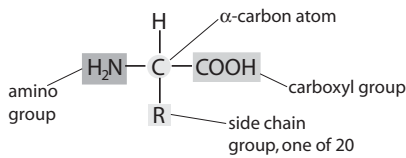


Figure 1-3. Modularity on basic chemical structure. (A) Nucleic acids; (B) amino acids, (C) Steroid molecules. (A) and (B) redrawn after (Alberts 1994).

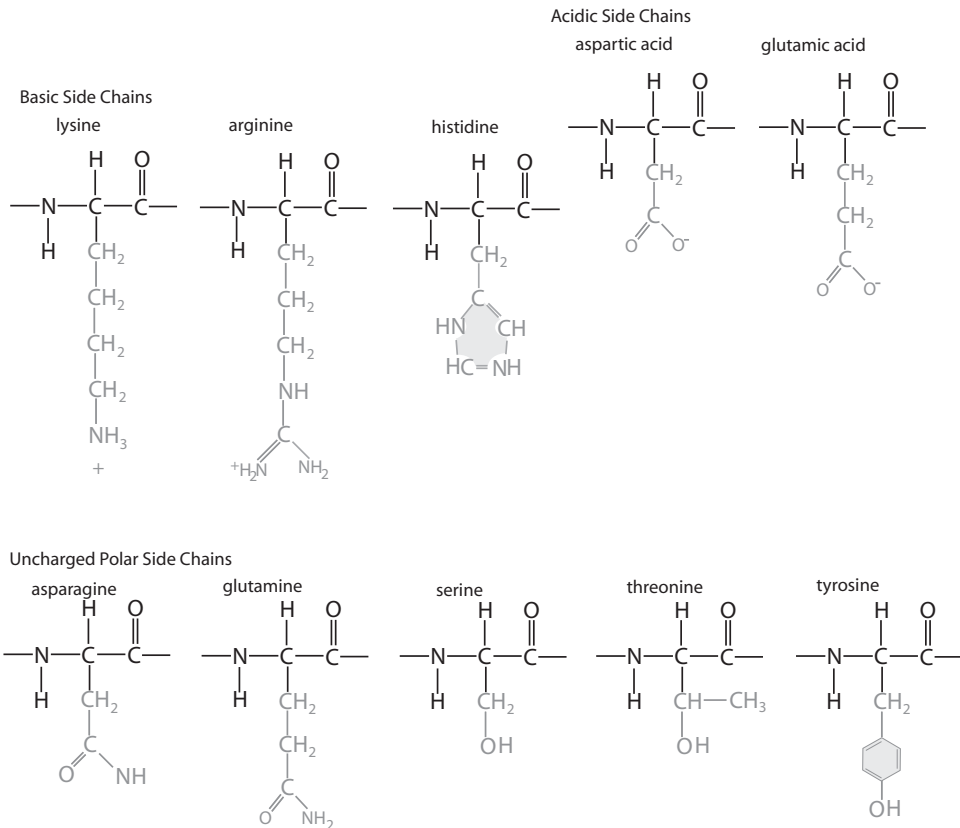
Amino Acids

general formula:



general families:

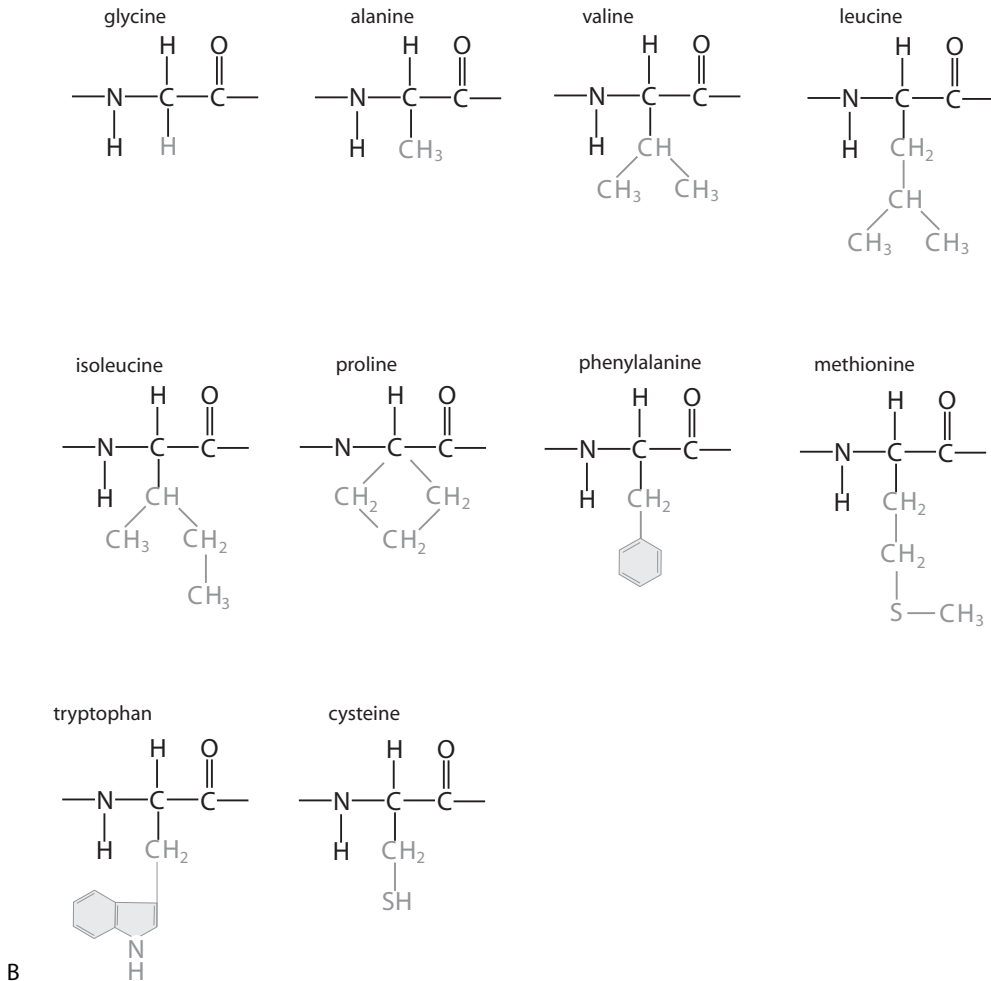
basic
acidic
uncharged polar
nonpolar

Figure 1-3. *Continued*

Much of what will be discussed in this book, and indeed in much of biology, is based on the elaboration of modular characteristics. Whether life had to evolve via modularity or whether some other form of aggressive energy-capturing self-replicating chemical system could have arisen from the conditions that existed when life began is difficult to say. But modular organization is certainly what happened and is so fundamental that one can surmise it would be inevitable.

Proteins and DNA/RNA are modular in that duplicate copies of the individual “beads” on these strings are used in the synthesis of new molecules. Indeed, new

Nonpolar Side Chains

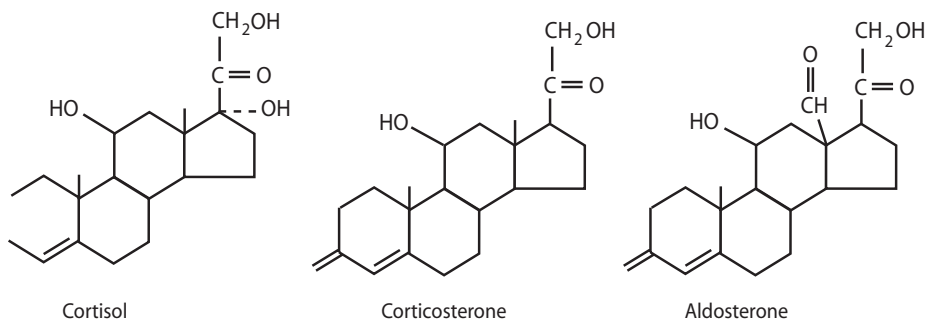
Figure 1-3. *Continued*

function also arises in a modular way and not simply by accretion. The molecules are occasionally modified when copied, and the copies can subsequently accumulate variation and modified or new function. This process can be termed *duplication with variation* and, as we will see, is a particular but important aspect of Darwin's major principle of descent with modification.

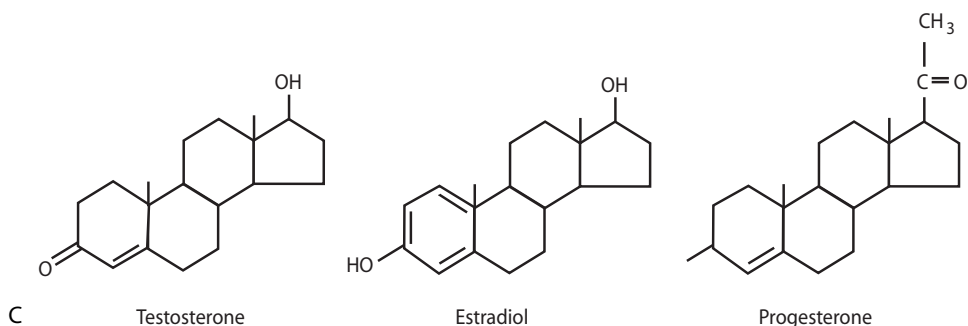
Duplication Above the Level of the Cell

Modular organization is related to sequestration as is seen by the important fundamental step of the evolution of cells, the modular units of which organisms are

Adrenal Steroid Hormones



Gonadal Steroid Hormones

Figure 1-3. *Continued*

built. Cell division provided the mechanism for reproduction of the cell as an organism. Multicellular organisms are aggregates of differentiated cells that ultimately descend from a single cell (e.g., the fertilized egg). Thus, large complex organisms are built on a process of duplication with variation.

Organisms are modular in many ways beyond being aggregates of differentiated cells. Many if not most higher-level structures, like organ systems, are also modular (each themselves built up of cells, of course). A limited number of basic processes seem to be responsible for this hierarchical modularity, which we will review in later chapters. These processes are responsible for initiating very local cellular division and differentiation to produce individual organ subunits like leaves, flowers, intestinal villi, feathers, teeth, nephrons in kidneys, ommatidia in insect eyes, or vertebrae. Somewhat similar interactions may be responsible for the *branching*, a related but somewhat different process, that produces repetitive pattern in plants, lungs, blood vessels, and other structures.

A duplication strategy applies to physiological as well as morphological systems. The lipid (fat molecule) transport, endocrine (hormone), and immune systems, for example, are characterized by the interaction of slightly different products of related

genes (the modules). How these various types of modularity arose and work will be discussed throughout this book.

Duplication and modularity were of course long known to be important in life but were not widely considered to be basic properties of evolution, because the phenomenon per se can occur without them. But early thinkers in the history of modern biology, for whom evolution meant the development of an organism, considered it; Goethe even likened the repetition of bones (as in vertebrae) to that in plants (leaves) (Richards 1992). The founders of biology were groping in the 19th century for such generalizations, or laws, of biological traits (at that time, essentially meaning morphology, because that was mainly what could be studied) and its variation, and for example how to explain the morphological changes through which embryos go, or the morphological or embryological similarities or differences (divergence) among species. Darwin used many of the same facts, but he accounted for them in terms of historical rather than developmental time. His idea provided a sweeping generalization about external processes that produce organismal diversity. After his and Wallace's exciting new idea, the central role of development, and the search for comparable internal generalizations, were somewhat shunted aside (e.g., Arthur 2002). They have returned.

CHANCE

Life as we know it depends on both precision and chance. The elegant precision of DNA replication gives life its predictability, generation after generation, but without chance mutations in genes, as well as the stochastic and error-prone nature of cellular processes in general, evolution would have no variation with which to work. We would all still be swimming in the primordial soup.

Darwin and Wallace had flashes of insight when they thought of the struggle for existence occasioned by overpopulation and the idea that the fittest organisms would prevail. This is the origin of the notion of natural selection, and both Darwin and Wallace saw selection as an immensely powerful tool for modifying organisms. To Wallace, and to some extent Darwin, the assumption was that the organized aspect of life is the product of selection, but in many ways this is as much a belief system for biologists as is that of fundamentalists who view the world of organisms as having been directly created by God. There are important reasons why we need to be careful about this view. One is our own human- and culture-dependent view of what it means to be "organized."

The second is a form of the *anthropic* principle, in essence, that things we see are not as improbable as they may seem, because had they not existed neither they, nor we, would be here to see them. Even in purely darwinian terms, whatever is here *had* to be "adaptive" in the sense of having been reproductively successful. Adaptations may seem highly refined, but among all the essentially infinite number of ways organisms might have evolved (even if by chance), *something* evolved. Thus, whatever is here is not as unlikely as we might otherwise think.

This of course cannot be used as an argument *against* natural selection; it should merely temper our after-the-fact reconstructions. However, the other side of causation is chance, and chance is pervasive and unavoidable—and often, as we will see, almost indistinguishable from causation. Darwin and Wallace were impressed by the universal ability of organisms to produce more offspring than could in the long run survive. Often, this means massive die-off. No matter how many acorns an oak tree

produces or how many eggs a fly lays, most of the time populations are relatively stable in total size. Even a very small growth rate leads to major population size changes in short numbers of generations (a property of exponential growth). Thus, an oak tree on average produces one descendant tree.

This massive die-off means that differential survival based on the characteristics of the particular lucky acorn may over time favor that variant on the oak, but it doesn't imply that it will do so. In fact, it seems clear that life on Earth is much more awash in the disorder of chance than most biologists tend to acknowledge. At least, if the world is as remorselessly ruthless as the usual darwinian view holds, it is wanton rapine, directed to no particular end.

Actually, evolutionary geneticists recognize that much—perhaps most—of the genetic variation in the world not only arose by chance mutation but probably is barely if at all affected by natural selection: the amount and nature of variation changes over time by chance aspects of survival and reproduction alone. The same seems as likely of traits as it is of genes, as we will discuss in Chapter 3.

But What Is “Chance”?

Formal discussion of chance is a profound subject in cosmology about which there is no consensus; in this book we are not really concerned with whether true chance events occur in nature or whether they follow regular textbook probability distributions (e.g., binomial, normal, etc.). Instead, by “chance” we mean literal, or at least practical, unpredictability.

A probability distribution essentially provides a formula by which we can compute the relative likelihood of a given outcome in an experiment that can be repeated, like coin tossing. Sometimes observations can't be repeated in practice, but we still assume that underlying what we can actually observe is such a distribution. The Brownian motion of molecules is an example. But what about the “chance” that a given enzyme molecule meets a molecule of its substrate in a given cell? This may be a case in which we can at least assume that this probability could be specified in principle. In practice, this is not so clear: we can't really come close to specifying the probability because a cell is not a uniform fluid space, and there is no such thing as an archetypal cell in the practical sense, except perhaps in the somewhat artificial conditions of an experimental lab. Genes are expressed in cells based on regulatory mechanisms that, while specific, entail a substantial component of chance, but one that would be very difficult to predict. Each cell of a tissue of seemingly identical cells is somewhat different. What about the chance that a given wildebeest will be eaten by a lion or that a given human will have a given number of children during her lifetime? These questions seem somewhat simpler and more meaningful, and we can imagine replication or sampling distributions that could help us answer them, in principle. Many aspects of life and evolution, however, involve chance in a more profound sense. Biologists might try to estimate the probability that frogs could have evolved “by chance,” for example, a question that is analogous to asking whether, if we could start life on Earth over again we would find the same outcome.

This is really a colloquial but scientifically misleading misuse of the term “chance” because there is no seriously meaningful sense in which the outcome of life represents a probability distribution in the sense of repeatable experiments. At best, the number of possible outcomes is so complex that it does not make real probabilistic sense to speak of the “chance” that frogs will evolve in the most stringent

use of the criteria of science. At worst, our understanding is so rudimentary that the question itself makes too many assumptions that cannot be verified. In the end, we cannot operationalize this statement by specifying any practical way to make it testable. It is a metaphor.

Much of life is this kind of metaphor. It is important to realize that, although chance plays a role throughout life and its evolution, much of the time what that means essentially is unpredictability for all practical purposes. Some of the time—an unknown amount of the time—it means literal, ultimate true unpredictability as far as what we understand of the world today can tell us.

IMPLICATIONS OF SEQUESTRATION AND MODULARITY

Communication, differentiation, nesting and repetition are seen throughout life. They depend on controlled degrees of sequestration. In multicellular organisms, cells are organized into tissues of different types, then into organs or organ systems, such as the vertebrate epithelial, nerve, muscle, blood, lymphoid, and connective tissues, or, in plants the dermal, vascular, and ground tissues. Individual cells are bound together in layers in various ways and need modes of both communication and adhesion (and, for some cell types, such as sperm or egg or circulating blood cells, means of nonadhesion). Intercellular contact and communication allow the “blending” of isolated material to a limited or controlled extent.

The sequestration among cells in an organism is less than that between individuals in a species or between species. One of the most important of the mechanisms for reducing isolation of individuals within a species is sexual reproduction. Recombination among homologous chromosomes during meiosis breaks down some of the isolation of individual genes, and recombination of *genomes* (the entire set of genes inherited by an organism) in the formation of diploid zygotes allows diversity that would otherwise accumulate separately to be mixed, presenting greater variation and ability to respond to environmental circumstances among other things. Sexual reproduction allows individuals to be different and, for example, subject to screening by natural selection but also allows a limited and highly controlled amount of exchange that means that species can exist.

SOME BASIC ASPECTS OF THE CHEMICAL “LOGIC” OF LIFE

Here it is worth mentioning two facts that are fundamental to the way life works from a chemical point of view. Many other generalities apply, such as the properties of carbon-based life with the other major molecules (hydrogen, nitrogen, and oxygen). But here we are not referring to basic biochemistry but to the *logic* of evolution.

The details will be seen throughout this book. The first basic principle of the logic of life is that the four nucleotide bases, commonly denoted **A**, **C**, **G**, and **T**, chemically pair: **A** with **T** and **C** with **G**. As explained in Chapter 4, this is the essential fact in the “information” storage property of life.

The second major fact is that proteins, the other basic functional constituents of life and evolution, function by combining with other chemicals in the cell. From an evolutionary point of view, this in particular includes interactions of proteins with each other. This is fundamental to communication of all sorts, to the basic biochemical aspects of life, and to the way that evolutionary history and information

are stored and used. An important example is the general phenomenon of one protein (known as a *ligand*) binding with another (known as a *receptor*) that has been evolved specifically to chemically recognize the ligand. We will see this phenomenon throughout the book.

CONTINGENCY

In the context of adaptation, sequestration, and the various levels of hierarchy seen across life from intracellular reactions to species evolution, it is important to keep in mind that these processes are *contingent*. That is, what happens now depends on the current state and not on previous or future states. The myriad biochemical reactions taking place within a cell are often localized within the cell, and each reaction depends on the current state, even if the raw materials of earlier states are still around or those prior-stage reactions are still occurring.

Two major points about this are worth raising in this context. First, life today depends on its history of evolution, and in that sense the nature of a cell or organism is dependent on its history as “written” in the genome (and in the nature of the environment around it, also the product of evolution). But what happens today is contingent on today’s state, rather than some previous one. And any apparent longterm trajectory (e.g., teleology, or changes aimed at a certain distant adaptive end) is illusory except to the extent that the nature of things that are interacting today make it possible to predict what will happen tomorrow.

Beyond “tomorrow,” it is generally agreed that life and its evolution are not specifically predictable, and the reason is contingency: what happens in the future depends on events that we view as purely chance, such as mutation, climate change, and the proverbial “acts of God” such as being struck by lightning. Similarly, it may be that many systems in nature are “chaotic,” meaning that, even if they are totally deterministic, only perfect knowledge would allow *us* to predict future states with specifiable accuracy (a hypothetical example is the well-known one of a butterfly flapping its wings, unseen and unmeasured, that eventually leads to major weather changes). Whether such systems exist is itself essentially unprovable, but if it were to be shown that there really is nothing in nature that is purely chance—that is, that *all* the “laws” of nature are perfectly deterministic—would we have to back away from the view of nature as contingent? If so, everything really would be predictable from the beginning, a form of omniscience that would be truly God-like.

There are traits in some species that do not seem to vary, yet the species can be shown to harbor unexpressed variation (e.g., Dun and Fraser 1959; Gottlieb et al. 2002; Lauter and Doebley 2002) that can be revealed by changes in circumstances or other means—a kind of potential that is already there in the organism. Some aspects of life do not seem as chemically unlikely as previously thought. Functional proteins have been shown to arise rather easily even in random mixes of amino acid strings (Keefe and Szostak 2001), that is, they can be a kind of natural state.

Except for the stochastic element affecting whether two molecules come in contact, it seems generally true that the nature of molecular interactions is built into their atomic structure, that is, the principles of basic chemistry. Chemistry determines how DNA does its complementary base-pairing, the specific interactions of proteins or how enzymes interact with substrate molecules to affect chemical reactions. If there were no stochastic factor affecting whether molecules come into contact, the genome and other molecules within a cell would have essentialistic

TABLE 1-1.
Principles of Evolution.

Descent with modification:

1. Organisms vary
2. Some of that variation is heritable
3. There is population pressure on resources so that not all organisms survive to reproduce equally well (when differential reproduction is related to heritable variation we call that natural selection)

Duplication with variation:

4. Sequestration
5. Duplication
6. Modularity
7. Chance
8. Contingency

Basic chemical logic:

9. Complementary base-pairing in RNA/DNA
 10. Protein-protein and receptor-ligand binding
-

properties such that life really would be the unfolding—the original, embryological development sense, of the term “evolution”—of the interactions latent in the molecular structure. In addition, life would be perfectly predictable from the day of the Big Bang onward.

If these kinds of molecular inevitabilities or chaos theory premises are literally true, it will have profound implications for cosmology, religion, and philosophical views of the nature of existence. However, based on what we know today, it seems just as true that the degree of any such predictability is so small, relative to the phenomena that we ourselves can observe, that evolution behaves for all practical purposes as if it were contingent and in that sense unpredictable.

CONCLUSION

In this book, we will be searching for generalities. We can think of these as the *logic* of evolution or of life, by which we mean the basic premises, properties, or processes, and their relationships to each other and to living forms. An important distinction is between how life *does* work, and how it might have *had* to work. Probably no one is really able to answer the latter question. That biology is not too far wrong in its assessment of the logic of life is shown by the fact that we routinely *use* the principles of evolution successfully to seek genes and understand processes across the entire span of biology. Using these principles allows us to account for a huge diversity of facts with only a modest number of basic units, processes, and principles. That is the goal of any science.

Chapter 2

Conceptual and Analytic Approaches to Evolution

The purpose of this book is to consider the genetic aspects of how various capabilities of complex life forms have come about, in the context of the basic postulates of evolution. There are many issues to keep in mind in the search for generalities. In this chapter, we consider various perspectives that may bear on this, given what we currently understand about both evolution and genetics.

SOME GENERAL CONCEPTS

UNIQUENESS AND GENERALIZATION IN A HISTORICAL SCIENCE

Evolution is an opportunistic process that builds solely on its present state. There is a large chance component in the environmental and biological variation that exists at any time and place. Evolution has been a one-time history. Each individual organism inherits, and must work with, the products of the events that happened in its unique *prior* history. Selection may mold organisms in a given direction *now* but does not—and cannot—aim toward anything in the future. Evolution does not require (and our theory does not even *tolerate*) any form of external force, as invoked by some religions, nor does it require internal animistic drive, invoked by Jean-Baptiste Lamarck and Henri Bergson (Bergson 1907), that directs organisms to evolve toward a particular *future* objective. Lamarck is today famously ridiculed for explaining evolution in terms of traits acquired by the striving of organisms toward some objective during their lives being inherited by their offspring.

But if evolution is contingent and without such directedness, can we expect to find generalizations or just local description? Might there be forms of biological necessity? Can we identify properties of life on Earth that should apply to life that might be discovered elsewhere, or are we simply describing history and calling it theory? In fact, general principles can be identified. Molecular and physical constraints and characteristics can be viewed as universal if we define life as varying,

self-replicating chemical interactions. Perhaps more specific to life on Earth are the intrinsic commitments of basing life on carbon and oxygen or on RNA and DNA and proteins.

To identify general principles, we make the core assumption of a single terrestrial origin, look comparatively at various forms of life to identify characteristics they share, and interpret them as mediated by shared historical processes. At a fundamental level, we compare gene coding mechanisms or biochemical usage among diverse species. To obtain ideas about higher-level phenomena, we can compare similar traits in multiple species—for example, forelimbs in mammals and birds. In such instances, we seek partial generalizations that apply to specifiable subsets of life. Many higher-order aspects of complex life are widely shared, even if the details vary, and a manageable number of broad generalizations are possible. Even so, there are usually exceptions because of the contingent, chance element involved in the historical phenomenon we call “life.”

IS EVOLUTION A BRANCHING PROCESS?

One of the most profound aspects of our notion of evolution is that it is a divergent or *branching* process. As shown in Chapter 1, the metaphor of a tree has been used as the fundamental image of evolution, at least since Darwin and Wallace. But is it correct?

It is true that mutation, natural selection, and other processes lead to *divergence*, or the accumulation of differences among different lineages descending from a common ancestor. If we look *forward* in time, the process produces a branching relationship as descendants of an individual, species, and so forth are isolated from each other, each acquiring unique new variation. Because of its ad hoc and complex nature and because of sequestration of the players, the process is, essentially, irreversible. If this is how life works, over time it would generate ever-proliferating diversity, with hierarchical, *cladistic*, or *nested* variation within branches sharing common ancestors.

In practice, we cannot really look forward in time, but must use the patterns seen among current individuals and species (calibrated by our understanding of mutational processes and the limited amount of geological and morphological information provided by the fossil record) to look *backward* in time. By grouping organisms with shared traits and assuming that to be the result of common ancestry, what we see involves the “coalescing” of today’s forms into ever-fewer ancestral forms—ultimately going back to the common origin of life. This does not generally imply that there were fewer forms around at that time, except in a very general sense in relation to the very earliest times, because many past life forms will have left no descendants living today.

However, and especially if we attempt to infer the nature of the first life, the picture is less nested and treelike than the usual conception of evolutionary relationships. For example, reconstructing this history from DNA sequence data reveals inconsistent phylogenies among genes, suggesting that some systems have been transferred *horizontally* among long-separated branches. The recipient branch’s original system is thus replaced in its descendants by the system transferred in from a different branch (Doolittle 1998; Doolittle and Logsdon 1998; Jain et al. 1999; Koonin et al. 2000; Ochman 2001; Ochman et al. 2000). As a result, today two groups *A* and *B* e.g. may appear to share a common ancestry based on studies of one par-

ticular gene, but groups *B* and *C* seem to have a common ancestor relative to another gene. The tree of ancestry is not the same for each gene. The extent of this phenomenon is unclear (Fitz-Gibbon and House 1999).

Essentially, horizontal transfer from one individual to a peer of a different species is a violation of the general darwinian postulate of variation transmitted *from parent to offspring*. Subsequently the immigrant gene, acquired horizontally, is passed by the recipient to its offspring in the normal way. Despite the occasional transfer of genes or sets of genes, one might nonetheless surmise that all of this reticulated ancestry at least goes back to a single ancestral *cell*. But even this may not be so.

An alternative to a single tree of all cells is that life evolved from a kind of communal pot of primordial soup, dished initially into rather imperfectly sequestered troughs, of incompletely different cell prototypes, and finally to more completely separated bowls. In the first stage of such a scenario, biochemical reactions took place communally, diffusing rather freely. As membranes or other barriers developed, some local, isolated specialization evolved tolerant to horizontal transfer (Woese 2000; 2002). Over time, local environments became more highly structured and organized and sequestration more important and impermeable—in the form of cells. Only by the latter stage were cell types among the major branches of life essentially isolated permanently from each other.

Woese (Woese 2002) questions when the effective isolation occurred, noting that basic gene replication mechanisms share little homology between archaea and other cell types. One possible example is DNA transcription and protein translation systems (the nature of these things will be described in Chapter 4), but mechanisms even more fundamental than that may also have transferred horizontally (Jain, Rivera et al. 1999). An incoming gene that cannot be used quickly becomes mutated into extinction, and it must be in the germ line of the recipient species to be transmitted (a major barrier to horizontal transfer in multicellular organisms). However, even among single-celled organisms, horizontal transfer more likely involves mechanisms that depend on the host having cell machinery compatible with that of the donor's.

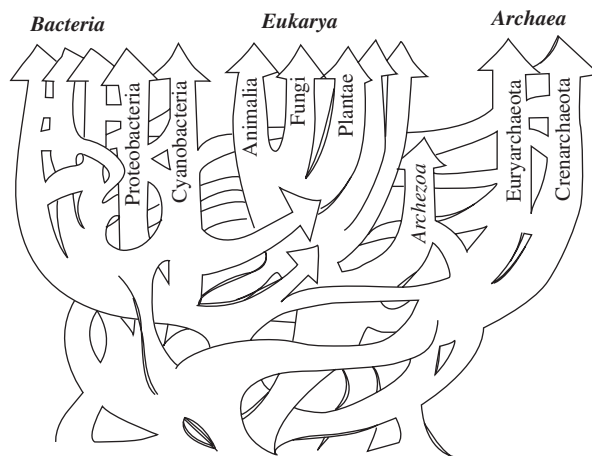


Figure 2-1. Tree of life showing horizontal transfer among major branches. Reprinted from (Doolittle 1999) with permission.

Eventually, according to this scenario, cells became too metabolically and genetically integrated across their diversity of functions to be tolerant to the importation of genetic mechanisms other than things basically self-contained. A “Darwinian threshold” is reached after which little horizontal transfer can occur, and cells and their descendants generate the standard kind of diverging, diversifying “tree” of relationships. There is, however, still debate about how much actual horizontal transfer has occurred, centering on, among other things, the analytic methods used to identify possible transfer products in present-day cells and species.

Eukaryotic cells (those with nuclei) are thought to have evolved by cell fusion (e.g., Hartman and Fedorov 2002) that introduced the subcellular organelles mitochondria and chloroplasts into a cell. This was initially a symbiotic relationship (a mutually beneficial arrangement between otherwise self-standing entities) but eventually the descendant cells and organelles were unable to survive without each other. Horizontal transfer may have been important in the evolution of algae, enabling them to engulf each other (or algal-bacterial transfer) and thereby acquire new function (Archibald et al. 2003). We also know that horizontal transfer at least occasionally still occurs, for example, by the insertion of viral genes into recipient genomes.

If there were complete sequestration, every cell lineage would be an entirely separate species. In fact, a number of normal mechanisms modulate sequestration and connect branches of life. Sexual reproduction within a species keeps the germ lines connected. This kind of transfer is easy and generally complete because donor (e.g., sperm or pollen) and recipient (egg or ovum) are so similar. Individual genes within species also are kept connected by recombination between an individual’s incoming paternal and maternal genomes.

Life of course is *generally* cladistic, especially with regard to complex organisms. As the barrier to horizontal transfer became basically impermeable, species and their diversification in the usual darwinian sense became possible. This shows the importance of sequestration in biology.

All metaphors limp to some extent, but because various mechanisms of horizontal transfer exist, there is no single tree of life. Instead, what we see is reticulated, or interconnected, *networks* of historical relationships. This is true of deep aspects of species relationships, and also of gene relationships in shallower time.

DIRECTEDNESS IN EVOLUTION: DO ORGANISMS SOLVE PROBLEMS?

It is difficult to write about evolution without using some kinds of convenient verbal shorthand, such as referring to existing adaptations as if they evolved *for* what they do today or to solve a problem confronted by the organism’s ancestors. That is, we often tend to equate today’s function with the selective forces of the past. Examples might be the ability to think, or fly, or digest cellulose. A cow’s ancestors developed the ability to digest grass. However, no ancestral insectivorous mammal faced a field of grass and pondered how to digest it, the way we face a field of grass and ponder how to mow it. Organisms have no known way to develop heritable means to solve problems identified prospectively.

It is easy to think of environments as presenting problems for organisms to solve. But this can mislead us into Lamarckian thinking. The presence of the atmosphere makes flying possible, and flight has evolved many times. But birds’ reptilian ancestors did not have to fly, as many contemporary land-bound reptiles demonstrate.

Nor has flight always taken the same form. An opportunity is not the same as a necessity. We can easily imagine opportunities that have not been taken.

Perhaps, genuine lamarckian mechanisms for directly producing heritable change in response to environmental circumstances will be discovered; some possible instances have been offered. There are examples to suggest that “evolvability” may exist, in that some organisms under stress respond by producing mutations, perhaps even in a context-specific set of genes (Caporale 1999; Fontana, 2002; West-Eberdard, 2003). The idea is that at some point organisms (and here we are speaking of single-celled organisms) had regions of DNA that were subject to mutations under, say, nutritional stress, and those mutations by chance led to tolerance of that stress and hence proliferation. However, the specific mutations themselves in this case are random, not directed to the specific need, and involve simple DNA changes rather than complex adaptations. In fact, organisms with a sequestered germ line are less likely to evolve such mechanisms because the mutations generated under stress would have to be inserted in the germ line and not just be in the body itself, yet it is the body that must survive the stress.

Our current understanding is that evolution is not *teleological*, that is, it does not work with future objectives in mind. We may some day discover lamarckian means of genetic evolution, but until then we have to hold to our view that mutational change is random relative to need. Neither directed, future-anticipating change nor the inheritance of acquired traits provides necessary explanations for the major functional characteristics of organisms. (Life may, however, someday evolve in a teleological way, if we develop genetic engineering methods to produce ends we envision in advance, such as sheep whose milk contains antibiotics useful for treating human disease.) The modern DNA theory of life, often called the Central Dogma, that a specific gene codes for a specific protein but that the gene’s structure is not directly affected by how that protein fares in life (see Chapter 4), is the theoretical guarantee of this lack of lamarckian inheritance. However, this has to do in part with how we *define* heredity, and there are numerous examples of parent-offspring transmission of acquired traits—one being your ability to read this book.

WHAT IS A TRAIT? WHAT EVOLVES?

It might seem strange to ask what a trait is, if the whole point of understanding evolution is to explain the diversity of traits in organisms. However, there have been and continue to be debates about what exactly it is that we refer to in this context. If traits are selected for or evolve, what are “traits”? Another way this has been put is this: What is the unit of selection?

This sounds simple but is not a trivial question. There is so much diversity in nature. Anything can be a trait. But if we want to relate our discussion to genes and adaptation, we need to know what we are considering relative to genes. Life cycle is a good example. This would seem to be directly related to the notions of darwinian fitness in the face of natural selection, since those who live longest or reproduce first might be declared the evolutionary winner. Does selection work directly on that or on the processes underlying the result? For example, much has been said of the notion that maximum lifespan is a characteristic of a species. This seems sensible, but does it imply there are genes for the timing of death? Does age at death evolve as a trait? Or is it just that causes of disease, that is, problems in cellular physiology, are screened and the net result is a statistical pattern of ages at death? This

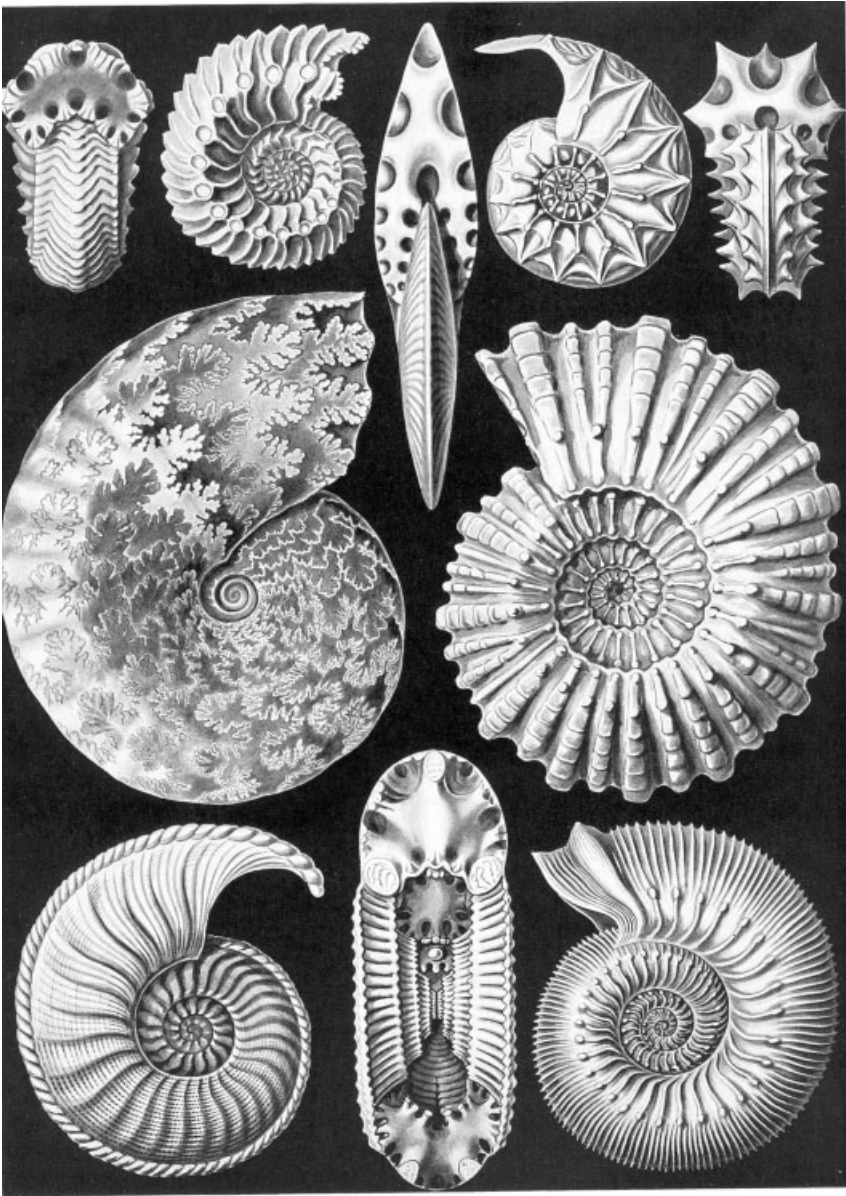


Figure 2-2. Ammonite shells drawn by Ernst Haeckel (a fine and avid artist) to represent diversity in nature. But what aspects are “traits” in the darwinian adaptive sense? From *Art Forms in Nature* (available in reprint as Haeckel 1899).

seems most likely (e.g., Finch and Kirkwood 2000); indeed, something we have already stressed is that there appears to be a huge component of chance even in life history events. This can easily seem to fly in the face of the universal appearance of adaptation in nature; but does it?

IS ADAPTATION A PROFOUND OR AN ILLUSORY CONCEPT?

The lack of foresight in evolution leaves *us* as scientists (not organisms) with a problem because so many traits *appear* to have evolved to “solve a problem.” Bats certainly fly! And it is not much of a misstatement that they evolved *to* fly or that they evolved because their ancestors *strove* to fly or that selection *avored* flight. Our theory holds that evolution is opportunistic. Selection screens variation that exists (by chance), favoring some functional variation that may or may not relate to flight but is useful at least in some way.

If an aspect of the environment remains relatively constant over long time periods, traits suitable for increasingly effective use of that aspect can be favored. Over time, the trait’s evolution can continue in a generally consistent direction because variation that arises is screened by the same factors. This can effectively *canalize* (channel) evolution in a persistent way (Wagner et al. 1997). Biochemical constraints (see below) can limit what selection can achieve, and can contribute to what the prominent biologist William Bateson (Bateson 1913) referred to a century ago as “positions of organic stability.” There is never any foresight involved, but a steady environment can lead to what *appears* to be directed evolution.

The resulting *teleological illusion* is what drove Lamarck, the Argument from Design, and numerous other responses to darwinian explanation. However, it can never be stressed enough that, if suitable variation had not arisen under particular circumstances, we would not have observed, for example, flying organisms today. Nothing we know about the mechanisms in biology suggests that flight arose through foresight or internal drive in any animal in the past or suggests that foresight or internal drive was necessary.

We are also somewhat trapped by the anthropic principle referred to in Chapter 1. *Every* organism we see today is the descendant of four or so billion years of *uninterrupted* success. Each has inherited genes that history blessed since the primeval soup. It cannot be otherwise. Critics note that, because of this fact, adaptive explanations verge on tautology because one can always invent an adaptive story that leads from past to present, and such explanations are sometimes applied so unconditionally by biologists as to be scientifically not much more meaningful than a collection of Just So stories because they cannot be verified. The issue might be ameliorated if we tempered these adaptive scenarios by keeping in mind that organisms clearly are not as finely tuned to their environment as is often casually assumed.

Similarly, our adaptive scenarios for complex traits might be tempered if we were obliged to specify *how* it happened. The human brain is considerably larger for our body size than the brains of other primates. We assume this is an adaptation for mental function. However, some humans have much smaller brains than others with no obvious defect (in behavior or, more importantly, reproduction), and genetic variation can seriously affect brain function without affecting brain size. It is easy to “explain” brain evolution by saying that a change of some very small amount in average size per generation (e.g., 1 mm^3) would be sufficient over many thousands of generations to increase brain size of the amount observed comparatively and from the fossil record. But how does 1 mm^3 of extra brain volume lead to increased reproductive fitness? If brain size did not increase in a gradual way, how did it increase? And why?

These important questions can be asked about most complex traits. For a scientific principle to have much meaning, or to be persuasive in a given circumstance,

there should be some constraint on when and how it can be invoked. Adaptive explanations raise fundamental issues about concepts of causation in biology. This can be seen by the fact that, from Darwin to today, many biologists effectively assume that any complex trait is mainly the result of steady, gradual selection. (Rapid change by sudden mutation leading to a new level of complex organization has seemed impossible except for certain special cases, such as segmented, serially homologous systems like hands and feet or legs and wings in insects, in which the number of elements might under restricted circumstances change quickly and in simple ways we now understand at the gene level.)

There is no satisfactorily provable way out of the teleological illusion, but this has not shaken biologists into eschewing the making of adaptive scenarios, mainly because a good enough alternative material explanation for directed change does not exist. Religious creationists scurrilously misrepresent what biologists mean when they say that evolution is due to chance. But ironically, in insisting on adaptive scenarios, biologists share with religious creationists the belief that complex traits cannot arise just “by chance.” However, we will suggest below that chance may be a more important factor in adaptive evolution than has been thought.

NATURAL SELECTION AND THE SPECIES QUESTION

One of the central concerns in the early stages of systematic professional biology in the late 18th and early 19th century was the “species question,” that is, explaining the existence of the diversity of species, each suited to its way of life. Everyone knew that plants and animals were variable, and breeders could modify that variation up to a point. Domesticated species could be bred to change, but when the breeder’s attention lapsed they seemed to “revert to type,” and breeding never extended to the production of new species. Creationist explanations were weakening, as evidence from fossils, biogeography, and systematic, comparative, anatomic, and taxonomic studies accumulated. These studies showed that life was some type of historic phenomenon, and evidence showing that species *did* change and that new species arose by diversification from earlier species increased. But *how*?

Darwin and Wallace provided a general, codified, plausible, and in a sense *observable* mechanism—natural selection—by which species could in principle arise and change (Patrick Matthew also expressed the same argument clearly in 1831, but it went unnoticed in most subsequent priority credits because it appeared as only a brief comment in an appendix to a paper in a specialized book on naval arboriculture). But this lengthened the prevailing sense of time and made it a critical factor concerning relationships among species. Previously, when evolution referred to development, time was on the embryological scale. Charles Lyell, James Hutton, and other geologists had discovered slow processes by which the Earth’s shape is changed, and that was an important factor lending plausibility to the ideas forming in Darwin’s mind. His theory required a lot of time, and he was concerned that there might not have been enough for natural selection to mold the wonderful and complex diversity seen on Earth. He was convinced that the biblical estimates of the age of the Earth, roughly 6,000 years, were incorrect, but he thought “We have almost unlimited time . . . there must have been . . . millions on millions of generations” (C. Darwin, 1858, paper announcing evolution read before Linnean society; available on the public domain and web).

At the time, it was impossible to know just how many millions of generations the age of the Earth might truly have supported, and Darwin struggled with the problem. In fact, he thought that hundreds of millions (perhaps 300) of years would be required, and was highly discomfited by the British Royal Astronomer Lord Kelvin's estimate that the Earth was only about a tenth that old.

We now know that the Earth is much, much older, and that life has been here for several billion years (perhaps even 4 billion). But is that "long enough," not just for the evolution of cells or butterfly wings but for the evolution of *all* traits in *all* species, from leaves to language, without exception? Does 4 billion years make complex evolution, or adaptive explanations more or less plausible than some younger age? This is really a moot point, which is why Darwin could persist in his views despite unclear and sometimes quite contrary arguments about the age of the Earth. There is no real way to know how long is long enough for selection to have done its job. So long as we explain adaptation *conditional* on the *assumption* that life has evolved by natural selection on Earth, it *must* be old enough—it *was* old enough! Debates about this today are usually waged over contending adaptive scenarios. But so long as we accept the theory, there is no issue of the adequacy of time; instead, it is our job as biologists to use our theory to understand the details of *how* a given being evolved during its respective historic interval, which we document, for example, by molecular "clocks," calibrating time by the number of mutations that occur in DNA sequences compared among several species, and from fossil dates and biogeography. As we will see in Chapter 3, general mathematical and statistical theories have been developed for the rate and rapidity of selection in populations under various specified characteristics, and, while oversimplified, this yields an understanding of the way the process works in principle.

However, we will see in the next section that in some sense the ancient age of the Earth may actually make our reliance on natural selection *less* necessary than we have generally thought, which may change the kinds of reconstructions we should make or, at least, the need to invoke selection as much or as determinatively as we do.

SOME CONCEPTS OF EVOLUTION ARE USEFUL, BUT SHOULD NOT BE OVERUSED

Biological theory, like all theory in science, is an attempt to order the diversity of facts of the world. But the history of science consistently shows that the same "facts" can be interpreted in various ways and that what is chosen or accepted as fact is often culturally rather than, or as well as, objectively determined. An excellent and in some senses founding treatment of this now commonplace idea was presented by the Polish physician Ludwik Fleck (Fleck 1979), in the context of the way that the history of Western culture affected the drive to understand the causal nature of syphilis (Weiss 2003). That science is in part a product of its history and cultural context often means that alternative interpretations that we have not thought of or have minimized or rejected might be of comparable utility. Better explanations might be rejected or not even considered because of such factors. It is interesting to consider some of the prevailing notions in evolutionary biology and how they may reflect the culture in which that theory developed.

MACHINE AND INFORMATION ANALOGIES ABOUT ORGANISMS

In the 17th century, René Descartes promulgated the view that organisms were machines and could be understood in mechanical terms. Professional biology developed during the flowering of the industrial or machine age of the 19th century. Many important biological advances, particularly in genetics, occurred during the computer or information age of the 20th century; today, the prevailing view is that an organism is not just a machine but is the computable product of the information stored in its inherited DNA.

The machine analogy is in a curious way related to the 19th century's Lamarckian view of evolution. A machine *is* teleologically designed. It works because its parts are individually and independently manufactured in advance to serve a particular function. It is assembled from the outside and can be repaired part by part. Modifications can be introduced in various ways, some by chance perhaps, by purposive testing and experimenting *with some objective in mind*. Overall, the important characteristics of a machine are that it is *prospectively and purposively designed and manufactured*.

However, the same biology that views organisms in this way unambiguously denies that organisms have come about through teleological, lamarckian processes. Instead, an organism is the product of *contingent* processes, not *de novo* design. An organism develops from the inside, rather than being assembled by an external factory, and it evolved by modifications of that process, which works on the whole organism and not part by part and without any end objective in mind. We know that organisms are a patchwork of messy construction, yet we persist in analyzing them as if they can be decomposed part by part.

In a similar way, the computer age has led us to view genes as the sole blueprint for an organism, as if it were a simple storehouse of digital information for the assembly of an organism, a computer program for an organism. This metaphor has many problems (Kay 2000; Lewontin 2000). A program is modified not by overall natural selection but by debuggers that look for syntax errors and logical errors *relative to the preconceived function and built-in syntax rules*. Programs might be very different if all that we require of them is that they do *something*, as opposed to *this thing*. Selection works on organisms, not DNA, and we know that much of the DNA in this world is affected at most only weakly by selection (selectively *neutral* DNA; see Chapter 3). Nonetheless, much of modern biology is dedicated to the treatment of organisms as if they can be decomposed—and, for genetic engineers, repaired—gene by gene, just as we can execute a computer program step by step.

Of course, at some levels of approximation and for carefully chosen purposes, machine and information analogies work very well. Recently, however, we are learning that in important ways these analogies are frustrating at best and can be seriously misleading, and examples of these will be presented. Descartes did say that the body was a machine driven by the spirit.

NATURAL SELECTION: COOPERATION AS WELL AS COMPETITION

Another element of the cultural context of the development of modern biology relates to ideas about the role of competition in evolution. We are not the first to observe that the formalized justification of competition is a core aspect of the industrial age in which evolutionary biology developed. It is often argued that were

Darwin not a wealthy industrialist in the world's Imperial power, he would not have produced his theory and it would not have been embraced by others. Nonetheless, for whatever reasons, although they differed in details, competition for scarce resources was central to the formulation of evolutionary ideas by both Darwin and Wallace. To both, the requisite force was the culling back of Malthusian overreproduction by an environment that favors those more "fit" for their circumstances. In the 20th century, it came to be widely believed this was to be explained not in terms of competition among individuals but by reduction all the way to a gene's eye view, in which the fundamental molecules of life—genes—are seen as "selfish", self-perpetuating units relentlessly trying to outdo each other and exploit each other for their own ends (e.g., see Hurst et al. 1996), most widely known through the uncompromising popularizations of this view by Richard Dawkins (Dawkins 1981).

We do not have to be stopped by the obvious fact that selfishness is a very human concept with many culture-specific nuances to agree that there is competition in nature and that those not fit to survive that competition are unlikely to do so. This was a natural way to view the world in the heady and dominant days of the British Empire and the competition and class hierarchies associated with industrialization and the growing organizational hierarchy of large, urbanizing societies. Selection does not screen on genes directly; however, it is true that over time some genetic variants become more common than others, thus winning a competition. But the historiographic explanation of evolutionary theory is weakened by the fact that Wallace was not a wealthy or socially conservative industrialist. Nonetheless it is fair to ask whether cultural circumstances affect how nature's observers interpret what they see; there are other ways to view the processes by which species might evolve. Competition is not all that occurs in nature.

Predation and Competition Between Species

As far as we know, life originated in some chemical mix in which the original organized molecules—whatever they were—increased by incorporating nearby molecules. Although initially life forms could have lived strictly by incorporating materials that were not previously involved in life or the decayed detritus of former life, the supply of such material would eventually have been exhausted. An important change occurred, in which living material depended on incorporating materials from other living organisms: the dawn of predation.

Obtaining nutrients from other living forms obviously became a successful way to do business and thus proliferated. It is easy to see how such a source of nutrition could be favorable, as the prey has been made to do the work of preparing the molecules needed by the predator (e.g., by making protein, carbohydrates, sugars, etc.). Consequently, a corresponding path to success emerged in the form of the ability to *avoid* being incorporated by another organism. Thomas Hobbes' war of all against all was on! Much of the structure of individual organisms and of ecosystems throughout the biosphere reflects billions of years of this arms race.

Recycling of dead matter by plants and many animals and, even better, capturing it in the form of living prey are major results of this history. Anyone reflecting on a meadow, forest, or pond (or on his own dinner plate) can see the importance and pervasiveness of this kind of interspecific competition. Systematic pressures of predatory or other limiting aspects of the environment can lead to increased organizational structure among living forms over time, and that is what Darwin and Wallace realized. It is easy to see how predation and other forms of competi-

tion that eliminate the sluggish could systematically produce organized traits in organisms *if* and *when* the traits of winner or loser are heritable. But this is not the whole story.

Cooperation and the Not-So-Selfish Gene

If we look at descendants over time, we can view those that increased in relative number as having acted in a way that served selfishly competitive ends. The notion of competition transformed scientific thinking by adding a dynamic process to a static worldview driven by religious dogma. However, much can also be learned by viewing things in terms of cooperation. Competition and cooperation are often viewed as antithetical, but biological processes almost by definition involve the interaction among molecules or other components of life. Success in life often depends on successful interactions. A purely selfish gene cannot even replicate itself. One can translate many of the phenomena now viewed in competitive terms into terms of cooperation (of course, as with competition, the entities need not be aware of what they are doing). A gene that gets along with others in its community—of molecules, cells, organisms, or even species—may have a better chance of succeeding. “Better chance” may be viewed as competition but “gets along” is just as necessary.

Within an organism, cells of different types are produced by a tree of descent from a common ancestral cell (the fertilized egg). These different cells perform vital complementary functions in very intricate ways. Without these functions, the organism perishes. Cooperation has been a basic strategy for building larger, complex organisms, as many chapters to follow will illustrate. Cells communicate with each other so that each differentiates in a way that is good for the organism. There is cooperative housekeeping, as nutrients and oxygen are delivered to all cells. Some cells protect other cells from damage by pathogens. Some communicate behavioral instructions from a central nervous system. At many stages in animal development, in the process known as *apoptosis*, cells self-destruct to make way, so to speak, for other cells. Animals require energy fixed by plants to survive. In plants, cells in roots deliver water to leaves.

Sexual reproduction has been viewed as fundamentally competitive at least since Charles Darwin wrote *Descent of Man or Selection and Selection in Relation to Sex* in 1871. Darwin quickly realized that competition to get or to avoid becoming food was not all there was to nature. Obtaining mates could also provide a strong selective force. Individuals of one sex choosing the “best” mate they can get or of the same sex competing with each other for access to the opposite sex, can in principle generate major phenotypic changes. This kind of competition can become exaggerated, a positive-feedback display race (Figure 2-3). Of course it is always difficult to infer after the fact whether this, or some more material type of competition, might actually have been responsible.

Beyond these kinds of organismal behavioral aspects of sexual selection, many modern darwinists peer into the intricacies of sexual reproduction to see all sorts of elements of competition—among individuals, among sperm, among fetus and mother, and so on (Hurst 1995; Hurst, Atlan et al. 1996; Parker and Partridge 1998; Partridge and Hurst 1998). Sexual reproduction has evolved many times, and the consensus reason is that this enabled species to be more diverse and responsive to changing environments, but it also has fundamental elements of cooperation. In addition, organisms from bacteria to humans manifest all sorts of cooperation,

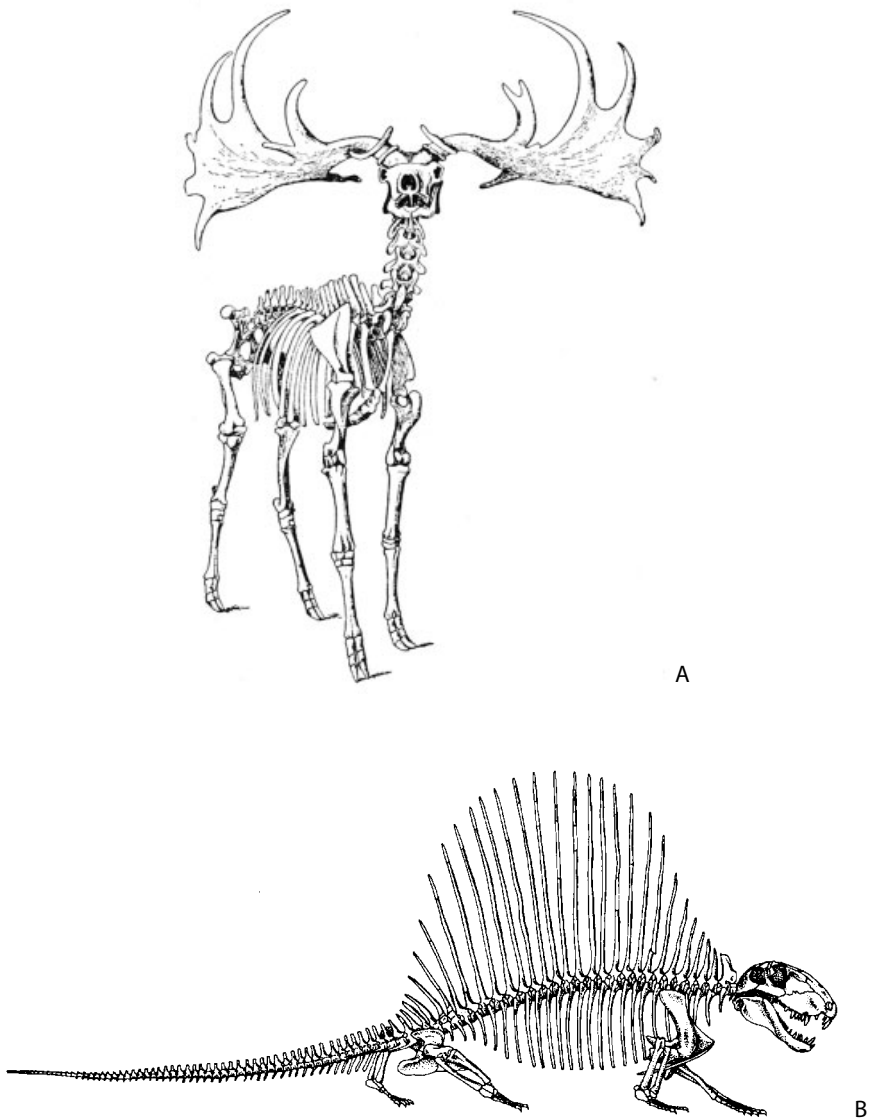


Figure 2-3. Were these exaggerated traits due to sexual selection, competing for mates, threatening predators, thermoregulation, or something not yet thought of? (A) Irish elk; (B) large-sailed *Dimetrodon*. (A) from (Romer and Price 1940); (B) from (Reynolds 1913).

including altruism and special dependency relationships such as that between plants and their pollinators, bees and their queen, or even bacterial cells and each other.

What is real here? A historiographic analysis can suggest that today, in the intensely competitive world of industrialized science, we view life through competitive lenses. If life must be viewed as competition for success in circumstances of limited resources, one would feel compelled to explain these phenomena as favor-

ing genes leading to this behavior at the expense of less competitive genes. We see such interpretations at every turn.

However, from another point of view, this sounds quite forced. The best way to understand life in many cases is to view these phenomena in terms of their *driving* force—the cooperation that they are. Like Janus, the Roman god of change from past to future who has two faces, we should not ignore one view at the expense of the other, because that may blind us to understanding. How far we can take this is probably a matter of judgment and the viability of our interpretations. As we'll see in Chapter 8, prey can be viewed as cooperating with predators (by providing them food), as the essential nature of ecosystems. That cooperation is vital is easy to see: when the body dies, its perfectly normal cells die, but not the bacteria that were in the organism at the time.

Organismal Selection

Besides organized cooperation, there is at least one other way in which adaptive evolution can occur. Nature, like Hell in the play *No Exit* by Jean-Paul Sartre (Sartre and Gilbert 1947), can be portrayed as trapping organisms with each other, like people in a small room, for all eternity—or like throwing one piece of meat into an Roman arena and watching the lions kill each other to get it.

This is an accurate way to think of many circumstances in which species or local populations sometimes find themselves. But it is not the only way that life is lived, and there are ways to succeed other than by direct competition to the death. In many if not most situations, environments are complex. Organisms—even plants—actively seek environments in which they can survive, where their particular skills are suitable. A major characteristic of organisms is this kind of “plasticity” or adaptability and it is thoroughly integrated into their nature (West-Eberhard 2003). This is manifest by the great diversity of species of closely related animals (plants, fish, insects, mammals, etc.) found within the same lake or forest. The organisms have found many subenvironments in which to live.

As John McPhee (McPhee 2002) said offhandedly when observing the shape of the fins of American shad fish, “One look at that tail and you know that the fish is active in the middle of the water column and not sitting around on the bottom like a bullhead catfish, whose tail is so rounded it looks like a coin.” In their preferred environment, they will meet and mate with like individuals. If migration is rare, or some mating or physical barrier arises, the individuals can become sequestered enough that they can diverge and eventually produce new species.

This is *organismal selection*, that is, active selection of the environment by organisms rather than passive selection of organisms by the environment, a reversal of the usual darwinian notion of natural selection. Over time, variation can accumulate, leading organisms to seek ever more specialized environments or to be more reluctant to leave the current one.

This self-selection may, but need not, initially involve genetically determined aspects of the organism. Divergence among populations that leads to subsequent speciation can then be aided by genetic changes that are unrelated to environment but provide mating barriers between individuals in the new and old environments. It may only take a very few such changes to create a reproductive barrier, and that is the definition of new species (Navarro and Barton 2003; Via 2001). Somewhat similar ideas have been expressed under rubrics like sympatric speciation or the Simpson-Baldwin effect, or “organic” selection, referring to the evolutionary effects

of nongenetic transmission, that can eventually be canalized genetically (e.g., Hall 1999; Schlichting and Pigliucci 1998, and see Chapter 3). Though some of these terms have connotations from earlier disputes about evolution, sometimes offered in opposition to the notion of darwinian evolution, that need not be the case.

Of course, whether organismal selection, cooperation, classical competition, or just chance provides the best account for any given case is a separate question and sometimes difficult to answer.

PHENOTYPIC DRIFT: COULD COMPLEX TRAITS HAVE EVOLVED BY CHANCE?

As noted earlier, Darwin hungered for an old age to the Earth because that would provide enough time for natural selection to do its work. However, in an ironic way, an old age might relieve the necessity of invoking specific and highly discriminating natural selection to hurry along the process of adaptation. If the Earth is old, slower mechanisms can lead to organismal changes.

Defenders of creationist views of life accuse biologists of saying that creatures arose by chance. According to the alternative Argument by Design, finely tuned biological structures like the eye could not plausibly arise by chance. Biologists agree with this, and they never argued that adaptation occurs by chance in the sense that their opponents suggest. Biological organization is viewed as having arisen *systematically* by natural selection, but without any teleology because selection feeds on genetic change arising by chance *relative to the needs of organisms*. Selection is systematic, but opportunistic, working only on variation that happens—by chance—to be present.

Actually, classical darwinian explanations do not have to assume that selection is persistent and systematic enough to enable a complex trait to arise gradually from nothing. Instead, the evolution of such traits is assumed to be by a stepwise rather than via a continual teleological process. A complex state like bat flight or eyes with focusing lenses evolves through a series of earlier states. Each state itself became adapted by selection relevant to conditions *at each time and place*, unrelated to future states or needs. The variation that selection worked on at each stage had itself arisen by chance, without regard to states or needs. Because it only works with whatever is at hand and under the local circumstances at the time, evolution is known as a *contingent* process.

What the local selective reasons may have been at any stage is open to debate (or, perhaps, inherently speculative). Whatever their adaptive reasons for being, the earlier states, sometimes called *exaptations* (Gould and Vrba 1982) became available for selection that led to the next step, ending up in their final—that is, their present—use. For example, rudimentary wings useless for flight may have been selected for their value in thermal regulation, mating display, surface-gliding on streams (e.g., Thomas et al. 2000), and so on. Darwin called these traits “incipient stages,” but that implies that they were functional rudiments of the present state, which is probably not the case for the evolution of novel function.

This view is essentially deterministic in that each stage is thought to be adaptive in the usual darwinian sense. But chance *not* aided by selection may have played a much larger role than usually assumed. Most organisms die for no reason related to their particular genetic makeup compared with that of their peers. Think of the millions of acorns whose unlucky fate it is to fall into shade or be eaten by squirrels.

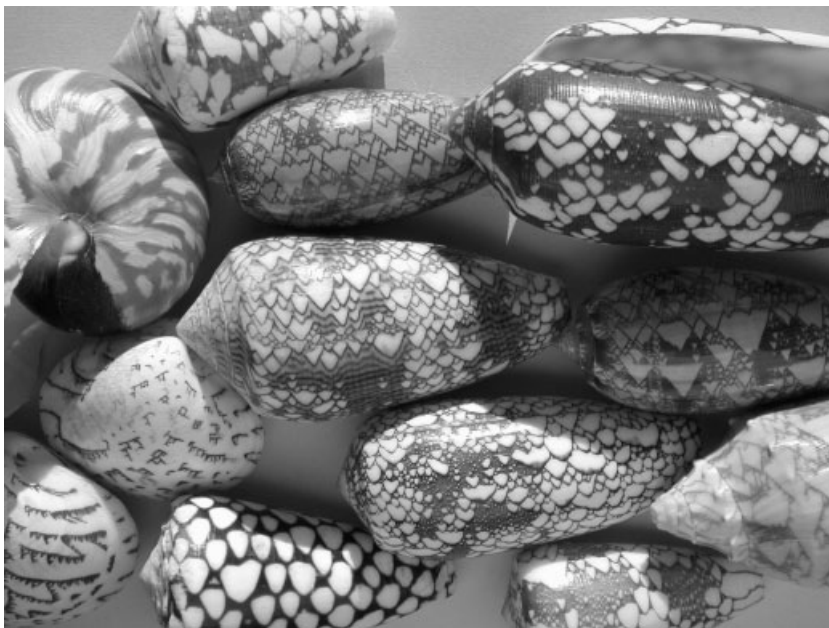


Figure 2-4. Variation in sea shells coloration patterns, showing that patterns are highly plastic. Is this due to chance or adaptation? Courtesy H. Meinhardt.

Chance effects on reproductive success will affect heritable phenotypic variation, and the traits involved will thus change randomly over time. For example, their means, variances, or other characteristics will change in ways unrelated to their environment. Organisms might become larger, greener, rounder, and the like, by chance. Such *phenotypic drift* has nothing to do with natural selection except to the very general extent that the phenotypic changes must be compatible with successful reproduction.

Biologists are generally comfortable with the notion that a substantial fraction of *genetic* variation is due to chance aspects of reproduction from generation to generation (Chapter 3). In fact, random change of selectively neutral variation in DNA sequence has replaced selection as a theoretical baseline by which evolutionary dynamics are evaluated, and it can be shown theoretically that the effects of chance can even prevail over weak selection. But evolutionary biologists have been reluctant to apply the same view to phenotypes because of our long-standing adaptationist bias.

The working title of Darwin's *Origin of Species* was *Natural Selection* (1859), and he said his purpose was to show "that there is such an unerring power at work in natural selection." Wallace felt the same. Darwin later mused in *Descent of Man* (1872) that if he had overstressed the role of selection it was to show that creationist explanations were not needed to account for biological diversity. Selection provided a way out of creationist explanations.

Based on what could actually be observed in nature and domesticated species, adaptation could in principle occur—as Darwin repeatedly stressed—through the accumulation of *slight* variations, that is, slowly, gradually, and essentially determin-

istically. Even to Darwin, weak selection was the most prevalent kind, which is why he wanted the Earth to be old enough. These are just the circumstances in which phenotypic drift will occur and perhaps predominate. The general notion has been around for some time (e.g., Gulick 1872).

The difference between phenotypic drift and slow, local stepwise selection can be so little as to be philosophical rather than testable. In fact, darwinian selection is not even so parsimonious an explanation. It requires environmental factors that are steady, persistent, durable, and sensitive enough to mold organisms systematically in some way. Stepwise models help in some senses, but the meandering course of a series of exaptations is another way in which evolution is essentially “random.” A stepwise scenario becomes nearly vacuous as an explanation if we take to heart the anthropic principle. It is at least as parsimonious to view chance as the ground state of phenotypic change and to invoke selection only when we have cause.

ENERGETICS AND EVOLUTIONARY EXPLANATIONS: THE NOT-SO-THRIFTY GENOTYPE

Characteristics of our age and culture include concepts of energetics, productivity, and efficiency. The evolution of biological traits is commonly explained the way we account for successful industry, as if natural selection can detect and favor subtle differences in form or physiology if they are more energy efficient. The idea is that efficiency would be favored because it costs less metabolic energy and hence less food and less struggle to acquire it. This essentially asserts that nature, like science, favors parsimony.

Energetic efficiency may be favored in many instances, especially in selection against very inefficient mutants, but this is not the only plausible view of life. Purging inefficiency in the face of mutations, organismal imperfections, and the sloppiness of the environment might levy a higher energetic cost on organisms than simply tolerating a degree of inefficiency. Such “noise” is pervasive in nature, and it is by no means clear whether natural selection is stringent enough to detect it. What reproduces, reproduces. Contrary to widespread notions of the genotype as a “thrifty” product of a prescriptive natural selection, *tolerance* of inefficiency may actually be a baseline—and more important—characteristic of evolution.

It is worth bearing in mind that it is we human interpreters who determine what is “efficient.” One type of locomotion may be more efficient than another in terms of ground covered per calorie, but animals cover ground for particular purposes. Does a bird or bat expend fewer calories to catch a bug than a spider? Is it a mark of reptilian inefficiency that they cannot prey on big ungulates as a lion can? Clearly, energetics alone cannot be viewed as a very useful predictive or interpretive criterion. We should invoke energetic arguments only when there is a good, specific reason for them.

THREE C’S OF EVOLUTION

Evolutionary change involves the three C’s of chance, competition, and cooperation in a way that is sometimes inextricable or even philosophically nondiscriminable. Different explanations can be offered for the same facts. For example, chance in small populations can have the same effect on variation as selection in larger ones. Afterward or even at the time, the empirical fact of change in heritable variation

over generations can be translated as the result of chance, competition, or cooperation. A genetic variant that advances an organism's prospects has to be compatible with the cooperation among molecules and cells, and sometimes organisms, to succeed. If it succeeds relative to other variants at the same gene by leaving more copies in the next generation, this can be viewed as having out-competed the other variation and as "selfish." But the same result could be due to good luck.

In Chapter 3 we will see ways in which we may sometimes be able to distinguish between chance and selection, but this is usually a statistical criterion and not a direct observation of cause. Only in limited circumstances can we make convincing inferences of selection, and these usually require us to have a clear mechanism. But even classic cases of selection purportedly directly observed in action, such as industrial melanism in the peppered moth and the evolution of beak size and shape in the Galapagos finches, are not entirely clear-cut (Weiss 2002).

These points do not just relate to the degree to which our explanations are satisfying and consistent but can affect how we design experiments, draw generalizations, assess the role of genes in biological traits, and view Darwin's entangled bank of nature, the image with which he concluded his famous book (see Chapter 17).

GENES ARE INVOLVED IN EVERYTHING, BUT NOT EVERYTHING IS "GENETIC"

THE IMPORTANCE OF GENES IN INHERITANCE AND PHENOTYPE DETERMINATION

A century of unprecedented work has led to an understanding of the importance of genes as inherited material, as the molecule that stores the information from the history of life, and as determinants of the traits of organisms. Before genes were discovered and understood, it was difficult to explain inheritance and the evolution of organized traits. Genetics has become the central, theoretical organizing principle of biology.

However, recent work, and ideas that will be considered in this book, raise tempering questions about several aspects of the present view of genes. First, the connection between genes and traits is in many ways more indirect and subtle than most biologists have thought (or than many still seem to think). Second, although DNA is one of the most important and widespread constituents of new organisms, most of which begin life as single cells, some aspects of inheritance are not strictly based on DNA sequence. These include parental RNA and proteins, DNA packaging and modification, and other chemical characteristics of the cell. Many aspects of behavior and the construction of environments are also inherited (Chapter 3), and additional nongenetic aspects of inheritance may be identified in the future. Despite the undeniable and continued power of genetics to organize biological thinking and research and to account for the evolution of life, the degree to which variation, inheritance, or evolution should be described strictly in genetic terms is a more open question.

A MENDELIAN ILLUSION

Both Darwin and Gregor Mendel, the Moravian monk who first characterized the segregation of heritable traits with his experiments on pea plants, worked with very deterministic notions of "genes" (the term itself was not yet in use, of course), that

is, with the heritable determinants of phenotypes. Darwin's theory of pangenesis turned out to be quite wrong, perhaps because he was thinking of a gradual change of continuous variation. Mendel, however, chose traits with simple transmission patterns that bred true in hybrids and that were essentially controlled by single "factors." That knowledge eventually led us to the genes themselves.

Mendel's discovery transformed biology, but in some ways we have become entrapped by the elegant simplicity of his choice of dichotomous traits manifesting "mendelian" inheritance. Many of the driving concepts of biology are built on this, and we still often pay little more than lip service to what goes on between the genotype an organism is born with and the phenotypes that develop in its lifetime. This is especially true of phenotypes not closely connected to the protein products of individual genes—the ones of most interest to Darwin and evolutionary biologists ever since.

In fact, the inner workings of life are far more complex than had been expected. In important ways, we attempt to force classical darwinian-mendelian theory in circumstances in which the fit is not so good. This does not mean that inherited traits do not involve genes, but genes are not always good predictors of traits, as will be seen in Chapters 3 through 5. Put another way, the mendelian inheritance of *genes* does not imply the mendelian inheritance of *phenotypes* (traits). Phenotypes are not inherited; organisms begin life as single cells with genes but not with arms, stomachs, or flowers.

DETERMINISM, REDUCTIONISM, AND GENETICS

One of the fundamental aspects of most Western science is *reductionism*. The idea, attributed generally to influential thinkers such as Descartes and Francis Bacon, derives from our notions of empirical experimental design that the phenomena of nature can—indeed, perhaps should—be studied and understood part by part, ultimately all the way down to the most fundamental parts. This does not mean that each part acts independently, nor that we necessarily ever will understand all aspects of a trait. But it does assume that in principle we can come to a fundamental understanding of a phenomenon by isolating and analyzing its component effects. Just the way we disassemble a machine into its parts (discussed earlier).

The ultimate belief of reductionism is that the universe is (only) a space filled with matter and energy. If this view is true, then everything can in principle be "reduced" to, that is, ultimately explained in terms of (only), molecular and energetic phenomena. Biological phenomena, too, will ultimately be understood best in terms of the molecular biology of genes, the "atoms" of biological information (in some ways, biochemists would extend this even further down, of course). In this spirit, geneticists seek to study each trait in terms of genes, as separable causal elements. The objective is to explain a phenotype in terms of the effects of the individual genes that affect it—just the way we reassemble a machine from its parts.

Reductionism in biology works at various levels. Functional anatomists attempt to reduce traits like locomotion to the contribution of individual bones and muscles. To a psychologist, learning may be comprised of recognition, memory storage, memory recall, units of meaning, and ultimately to neurons and neurotransmitters. To an ecologist, an ecosystem consists of predators, prey, a food chain, etc.

Each biologist chooses his/her level of reduction. Some may choose to look only at the role of frogs in the biodiversity of a swamp without feeling compelled to try

to explain the croaking of the frog in terms of its genes. Some wish to go farther and to “reduce” the croaking to hormone receptors, neuronal pathways, and the like. A biochemist may see this all as a problem in ligand-receptor binding, signal transduction, and gene regulation by action potentials of auditory hair cells.

Some biologists, although acknowledging that one can account for a frog in terms of chemicals, believe that reductionism cannot adequately *explain* the croaking. From this point of view, the phenomenon is an *emergent* one, that must be understood at its own level of organization. Field biologists may not even care to try to understand a bullfrog’s croak and his mate’s response in terms of DNA sequence data or hormone kinetics, a level of accounting in which they have no interest—any more than you might think the words you are reading could be understood by analyzing the chemistry of their ink and paper.

To some reductionists, higher-level studies barely count as important science, as they are too superficial. A common reason given is that we are not as good at making “operational” the study of complexity as we are at reductionist, experimental methods. The latter have a long history, and the triumphs of modern science and technology are the fruit. This view holds that higher-order phenomena are not *fundamental* and that eventually we *will* be able to predict “emergent” biological—or even cultural—traits by analyzing their components (e.g., Wilson 1998). Science does not allow nonmaterial causation, so how can an understanding of any phenomenon of nature not follow from an *adequate* understanding of its parts?

A reductionist perspective does not assert that causation is always one-to-one, but only that if we know all the actors, we will explain the play. An illustration of the issues involves gene action. Geneticists have long recognized that genes are often *pleiotropic*, that is, have many functions. Similarly, different genotypes can be found in individuals with the same phenotype. The genotype-to-phenotype relationship is often many-to-many in nature. Even if each component can be characterized in molecular terms, the overall effect of a gene on an organism, or of natural selection on a gene, may depend on the *set* of interacting constraints. We may not be able to predict the trait from any one of its components, but we should be able to do so from the set. An important question is when or how well we can ascertain or even define what that set is.

Arguments about reductionism are not new, and they are probably not resolvable, but the points are important in this book, whose aim is to understand the role of genes in how organisms manage their lives. But what it means to “understand” a phenomenon depends to a great extent on the question being asked.

CONSERVATION, VARIATION, AND HOMOLOGY

With caveats raised earlier about horizontal transfer, evolution is generally diversifying over time. Based on this, a central organizing fact in evolutionary biology, indeed a key to what Darwin and Wallace contributed, is *homology*, the conservation of traits in descendants of a common ancestor. It is historical connectedness that differentiates evolution from creation. In the 20th century, as genes took the throne of the biosphere, the key element of homology tended to shift from shared traits to shared genes or DNA sequence elements. However, the more we learn about genes and how they are used, the more rather than the less elusive the concept of homology has become (e.g., Hall 1999; Wagner 1999; Wagner and Gauthier 1999).

For example, the limbs of tetrapods are considered to be descendants of fins in the common fishlike ancestor of these four-legged vertebrates, so limbs and fins are

homologous in the classical sense. As expected, homologous genes are involved in limb development in different tetrapods. By contrast, eyes appear to have evolved independently in insects and vertebrates and have long been used as an example of *analogous* traits. However, developmental geneticists have recently discovered unexpected similarities in the mechanisms used to generate eyes in both groups of species—but only in some of the mechanisms and not all the same ones in all species. We will discuss this specific question in detail (e.g., Chapter 14). Here, the point is that recent discoveries of the genetic connectedness of diverse life forms show the need for a reformation of the very important homology-analogy question.

Ever since Mendel, and reinforced by the Central Dogma that one gene codes for one protein, the view that genes evolve “for” traits has been predominant. But this can be misleading in at least two important ways. On the one hand, unlike the eye situation, the genetic basis of similar, seemingly *homologous* traits sometimes turns out to be different (e.g., Hall 1999; Raff 1999; Wagner and Misof 1993; Weiss 2002; Weiss and Fullerton 2000). When this happens, the trait itself may be homologous in the traditional sense but not its underlying genes. On the other hand, it may be that a trait is produced by a developmental *process* that is completely conserved (homologous among species under comparison) but that the details of the process vary. For some traits with multiple elements, like body segments, the process and the nature of the trait may be conserved but the individual elements may not be homologous.

Another common view is that traits that have been shared since lineages diverged from a common ancestor have been conserved by natural selection. However, it is possible that the trait is conserved simply by the genealogical relationship. Even if selection is not acting, a trait will only change slowly between related lineages. Genes of humans and mice are around 80 percent identical on average, even in regions of the genome unlikely to have been seriously affected by natural selection. Perhaps of more potential importance, there are reasons to believe that in some instances, involving either particular chemical interactions or the interactions of many components, natural stable states or “attractors” may exist (such as “positions of organic stability” mentioned above). These may conserve a trait over long time periods or even provide an element of inevitability (e.g., Kirschner et al. 2000; Laughlin et al. 2000; Monod 1971; Morowitz et al. 2000; Schuster 2000)—in fact, as noted earlier, many aspects of life that seem complex may not be all that improbable in the first place (Keefe and Szostak 2001; Schuster 2000). Furthermore, a phenotype may be conserved but not its underlying genetic basis; the molecular structure of transfer RNA may be one example (Fontana and Schuster 1998). Similarly, as referred to earlier, natural chemical constraints would not need competitive natural selection in the usual sense, to be maintained. Indeed, once reached, it may be very difficult to get out of such canalizing constraints to try another way.

ON BEING A “BEING”

For many centuries, the array of natural organisms was viewed in Western thought as a natural hierarchy, often referred to as the Great Chain of Being. The long-standing view was that a creation event had produced this natural order of life. In this book, we will see how the same sense of relationships was transformed by the idea of evolution. However, we will suggest ways in which, even from a genetic and evolutionary perspective, the notion of “being” has been too restrictive.

Biology is about beings, plural, not just the more philosophical phenomenon of

“being”—the existence of life forms. The term is generally applied to organisms in the colloquial as well as scientific sense: a bird, ant, person, or birch tree. However, in many ways this is arbitrary: species, cells, and even genes can also be viewed as “beings.” At these various levels of observation, the interactions within and among these entities are logically similar and, as we will see in various chapters, involve similar or identical genetic mechanisms. A broader, more flexible sense of the term “being” helps unify biology and make the origin of its complexity more readily understood.

WHY THESE CONCEPTUAL ISSUES ARE IMPORTANT

This chapter raised a number of issues, caveats, and perspectives that may impact our ability to understand the fundamental generalities of complex life. It is important to examine even the most basic of what are often tacit assumptions. Exceptions and alternative viewpoints can have considerable if sometimes unexpected merit, as we will try to show in various ways throughout the book.

The purpose of searching for generalizations about life is to be able to extrapolate from observations necessarily limited to only a subset of individuals, species, or model systems to as broad a scope as possible. Without observing everything, how far can we extrapolate? The answer is important for animal model work, agriculture, and biomedicine, as well as for understanding basic biology.

We would also like to reconstruct the unique history of life by taking specific account of the trace of past events left in DNA sequences or in their indirect manifestation in biological traits. For more than a century, this has been based on a strong branching model, but suppose this is not the right model? Suppose homology loses its clear-cut meaning before we get too far in the past, or life turns out not to have a single-trunked genealogy, or that genes turn out to be less determinative than has been thought.

In a historical field like evolutionary biology, the challenge of making retrospective evolutionary reconstructions is daunting. Unlike chemistry or physics, almost any general statement will have nontrivial exceptions, and for many biological observations there are multiple comparably plausible explanations (for example, chance, competition, and cooperation), and it is all the more important to put our notions to the test.

Classic darwinian theory has been exceedingly powerful at providing coherent explanations and has transformed thinking in biology essentially by equating similarity with genealogy. We caution against the overuse of the old Cartesian notion that an organism is a machine, but with linguistic irony our own evolutionary explanations are *fabricated*, in the literal sense. We make them up, hoping they are accurate re-constructions of a species’ or trait’s history. But we can rarely be too sure.

The historical connectedness of organisms and the consequent storage of historical information in genes provide unifying tools for understanding. But we can still ask important questions about how apparent order can emerge from the disorder of an undirected universe.

Chapter 3

Evolution By Phenotype: How Change Happens in Life

For the first part of the 20th century, the study of genetics was considered by many biologists to be separate from, or even irrelevant to, the processes by which darwinian evolution occurred. It did not seem possible that natural selection—if it was the mechanism of speciation—could work gradually, as Darwin had suggested, through the discrete particles of mendelian inheritance, whose known changes caused discretely different, or worse, grossly disruptive changes to the organism. However, as genetic understanding grew, it became possible to see how a unified theory of biology might work. By the 1930s, a group of leading biologists proposed what they referred to as the modern, or evolutionary, synthesis (Mayr 1982) that united the study of taxonomic relationships among species, the fossil record, and the theory of genetic inheritance into a single formal theory of evolutionary biology.

Before this time, it was difficult to have a rigorous, quantitative theory about the pace or nature (sometimes called the “tempo and mode”) of evolution, and the theory was largely conceptual. But an assumption made by the modern synthesis, with widespread implications, was that genes are the fundamental elements of life—much as atoms are the units of chemistry and physics—and that evolution is to be explained in principle in terms of the processes of genetic change. The subsequent discovery of the nature and inheritance of DNA and its function as a protein and regulatory coding system greatly strengthened the gene-based view of life and provided a general research approach that predominates biology today.

Whatever the inherited material, if it is variable and particulate, so that each variant can be identified and not blend quantitatively with other variants, then the behavior of such variation over time and place can be quantified. If genes are the root units of biological causation, then the behavior of genetic variation over time will illustrate—and will *be*—evolution. The formal mathematical theory of evolu-

tion which describes this is called *population genetics*. Of course, organisms are more than genes, but to the extent that phenotypes (traits) can be ascribable to specific genotypes, it should be possible to subsume phenotypic evolution under the same theory. Here, we will present some of the basic principles of population genetics (thorough treatments are given by Gillespie 1998; Hartl and Clark 1997; Hedrick 2000; Lynch and Walsh 1998) and then discuss some of the subtleties that arise when considering that evolution works through phenotypes rather than directly through genotypes. We will refer to aspects of the nature of genes that will be explained specifically in Chapter 4 for readers not familiar with them.

EVOLUTION BY GENOTYPE: THE DARWINIAN POSTULATES FROM A GENE'S POINT OF VIEW

Population genetics is a kind of rigorous formalism about the evolution of genes. It can be done with no consideration of phenotypes and indeed many population geneticists specifically avoid dealing with the latter or with the connections between the evolution of genes and that of phenotypes. Population genetics is essentially a mathematical theory and hence is as rigorous as mathematics, which means that the theory and its usefulness depend on the values of various parameters, the degree to which they can be accurately known, and the degree to which the assumptions and formulas realistically reflect what goes on in life.

BASIC FREQUENCY CONCEPTS

The basic dynamics of variation over time can be described without being too specific about what is meant by “gene” beyond an identifiable, discrete, heritable unit of inheritance. We can assign a *relative* frequency to each of the alternative states, or *alleles*, that are found in a gene, *in some specified population of inference*; the latter can be a local deme or an entire species, as long as we realize that the analysis depends on this choice. Variation requires at least two alleles in our specified population.

Conceptually, an allele frequency can be viewed alternatively as the fraction of copies of the genes in the reference population that are of the specified type or as the probability that a randomly sampled gene from that population will be of that allele. Because the discussion is usually framed in terms of *relative* variation for a specific genetic unit in a specified population unit, by definition the frequencies of the alternative alleles in our population must sum to 1.0.

To present the logic of genetic theory, a simple situation is usually envisioned, in which the gene in the population has only two alleles that, ever since Mendel, have been conventionally labeled as *A* and *a*. We denote the frequency of *A* as p_A ; in addition, because there are only two alleles, *a* must comprise the rest of the population, so its frequency must be $1.0 - p_A$. That is, $p_A + p_a = 1.0$. This can be seen schematically in Figure 3-1. It is important to keep in mind that this refers to the proportion rather than the number of copies of each allele *in the population of inference*. When working only with *samples* from that population, all of this refers to the characteristics of the sample, which if one is careful can be used as estimates of the true situation in the entire population.

Another important concept is the *genotype frequency*, again with regard to a reference population or a sample from it. Many species, including most animals, are

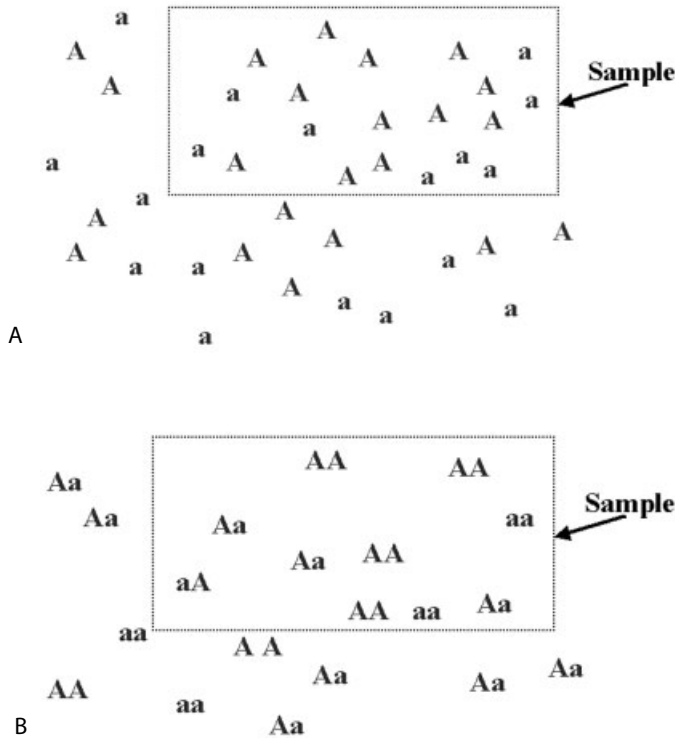


Figure 3-1. Frequency concepts. (A) Frequencies of two alleles, A and a , in a sample of 20 alleles from a population; $p_A = 12/20 = 0.6$, $p_a = 8/20 = 0.4$; (B) frequencies of genotypes $p_{AA} = 0.4$, $p_{Aa} = 0.4$, and $p_{aa} = 0.2$ in a sample of 10 individuals.

diploid, meaning that (like humans) they inherit a copy of each gene from each of two parents. This means that what happens to alleles in the population happens to them as they are carried around in pairs. The variants in a given individual are referred to as its *genotype*. (Some species like bacteria are *haploid* and have only a single copy of each gene; for them their genotype is the same as their allele. There are other ploidies in nature, but these need not be considered here to understand the basic principles.) In a diploid species in the simple case (Figure 3-1), the genotype must be a *homozygote*, AA or aa , with two copies of the same allele, or a *heterozygote*, Aa , with two different alleles. The relative frequencies of these genotypes can be denoted as p_{AA} , p_{Aa} , and p_{aa} . Again, because these are the only possible genotypes for this gene, $p_{AA} + p_{Aa} + p_{aa} = 1.0$.

Because a diploid genotype represents the contributions to the individual from each of two parents, the genotype distribution in a given generation is the product of the mate choice and reproductive pattern in the previous generation. Of specific interest is a baseline condition called *panmixia*, in which individuals choose their mates randomly relative to the genotype of mates choosing each other. When this occurs, the frequencies of the genotypes are just the product of the frequencies of each allele. Thus, the probability that a random individual has one A allele is p_A ,

and the probability that this individual has a second A is also p_A , so that we have $p_{AA} = p_A p_A = p_A^2$ and by similar reasoning $p_{Aa} = 2p_A p_a$ and $p_{aa} = p_a^2$. These are known as the *Hardy-Weinberg* genotype frequencies, and when they characterize a population it is said to be in Hardy-Weinberg Equilibrium (HWE) because, as has been shown mathematically, under idealized conditions the genotype frequencies will not change over time.

Mating is in fact typically strikingly close to random with respect to most genes because mate choice is unaffected by a mate's specific variants at most genes, and HWE serves as a baseline from which to judge observed genotype frequencies. If these differ in a statistically significant way from what is expected under random mating, there may be reason to investigate why this is so, and there are many possible reasons, including mating that is affected by the gene in question, or by natural selection.

These are the basic frequency measures needed in order to describe the essential concepts of evolutionary change from a gene's eye view.

Mutation: Change of State in the Genome

All genetic change ultimately comes about through mutation. Mutation can do many things that will be discussed in Chapter 4 and beyond. These changes can alter a gene's function, expression pattern, or structure. In some instances, new genes can be inserted from outside the individual, as for example when viral particles integrate into the genome. Such a change, if it occurs in the germ line and is transmitted to the next generation, constitutes horizontal transmission described earlier.

Frequency Change Over Time

Each new variant arises with allele frequency $1/N$, where N is the number of copies of the gene in the population of reference. (The frequency is $1/2N$ in a diploid species, since each of the N individuals in the population has two copies of the gene.)

Over time, alleles experience changes in their frequency. From one generation to the next, the proportion of a given allele may differ for several reasons, the primary one being chance. In a diploid species, for example, one of the two alleles that came together to form the parent randomly segregates into the germ cell to be transmitted to any given offspring. This is mendelian *segregation*. Which of these two *possible* outcomes occurs in any given case is inherently *probabilistic*.

Mendelian segregation introduces a fundamental element of chance in allele frequency change when individuals reproduce, and many other aspects of chance in survival, mate acquisition, or fertility affect whether that reproduction will occur in the first place. Under various assumptions, there are ways to quantify the relative probabilities of various outcomes of mendelian segregation in any family or in an entire population, and hence the distribution of possible allele frequencies in an offspring generation, if the frequencies in the parental generation are known.

The phenomenon of allele frequency change due strictly to chance is known as *random genetic drift*. We encountered the metaphor of "drift" earlier in discussing random changes in the distribution of phenotypes in a population. Frequency "drifts" randomly up or down over time, until a variant is eventually either fixed in or lost from the population, as illustrated in Figure 3-2.

Genetic drift is *inevitable*. Based on assumptions about the population, the expected (average, over many replicates could they occur) change in allele frequency from one generation to the next and the probabilities of any particular

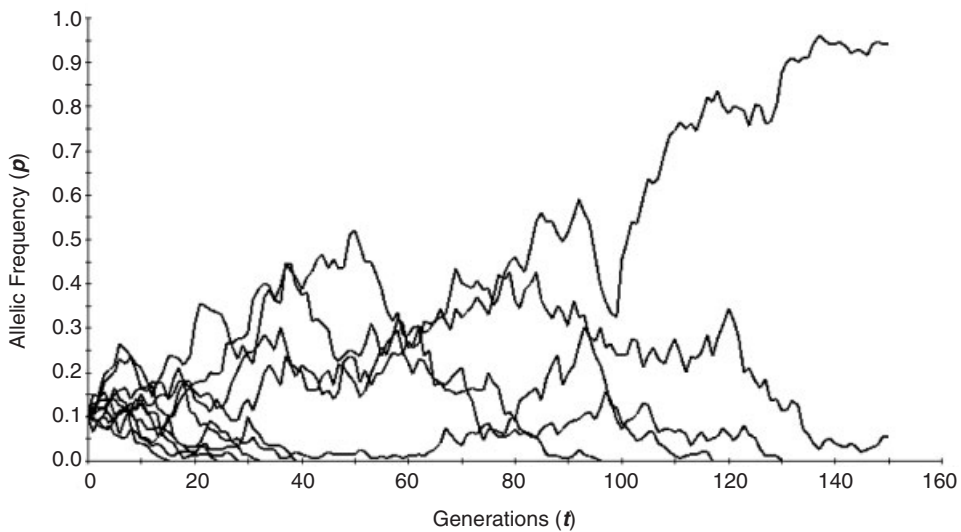


Figure 3-2. Schematic of genetic drift. Each line represents a separate “experiment” of an allele starting with a frequency of 0.10 followed over time in a population of size 100. In these computer-simulated data, frequency change is due to chance alone. Any frequency is achievable, but most new alleles quickly disappear while a few are lucky and rise to high frequency. In any population multiple alleles may be segregating, their relative frequencies drifting up or down over time, some being lost, new alleles generated from time to time by mutation. The fact that an allele has high frequency does not imply a history of adaptive natural selection having favored it. Usually, only statistical arguments can determine when that is most likely. Data simulated with Populus (Alstad 2003).

outcome can be specified. It is, however, not possible to specify what *will* happen in any given situation, Drift calculations are almost always approximations, but data often seem at least broadly consistent with the drift expectations—that is, as if drift were the *only* factor affecting allele frequencies.

Whether due just to chance or to other factors, allele frequencies can change substantially over time. If chance alone is operating, descendant copies of any selected allele will *eventually* either drift to a frequency of zero (be lost) or to 1.0 (at which point it is *fixed*, that is, replaces all other alleles at the locus in the population). The reason is simple: there is always *some* probability of fixation or loss, and an allele can never return from the dead or change in frequency once fixed.

The *rate* of drift, or probability that fixation or loss will happen in any given time period, depend on population size and structure (mating patterns, subdivisions, and so on). This is because genetic drift is a kind of sampling from one generation to the next, and the variance of a sampling process depends on the sample size. In a large population the sample is large, and the sampling variance small, so that allele frequency changes little over a generation. However, actual species may be divided into small local populations in which drift can be important and rapid. Thus, *population structure as well as gene function are important determinants of evolutionary change*. This may seem to be a trivial consideration because species typically seem to have very large numbers of individuals. But they may not always have that, and

in any case the Earth is old enough relative to population size that there has been time for drift to have been a major, if not *the* major, factor in genetic change during the history of life.

There is a continual flow of *substitutions* (one allele replacing another in the population) as new variation arises in each generation, and some fixation or loss of variants takes place. If the population size doesn't change, there is no in or out migration, and mutation rates stay the same, a steady state will arise between fixation, loss, and mutation. An ideal steady-state population will come to have a standing level of variation that depends on the relative strengths of these factors. However, and again and most importantly, the *specific* variants that are present, and their frequencies, will continually change—and will vary among local populations. Different genes and different populations accumulate differences over time in a statistically regular way.

Because this is a random process, it is impossible to say which of the alleles present today will be around at some point in the distant future. However, under idealized conditions, we can estimate various aspects of the process such as how long it would take for a given number of genetic differences to accumulate in copies descended from a single individual. One can thus look backward in time to estimate how long it has been since copies of a gene we see today shared a common ancestor. With some important technical caveats, genetic variation can serve as a *molecular clock* of evolution (e.g., Li 1997; Nei 1987). These are purely probabilistic phenomena, and population genetic models are at best crudely approximate; therefore, time estimates have a large statistical error. This can be reduced by looking at enough data. For example, the clock can be calibrated by external information such as experimental estimates of mutation rates and the age of relevant fossils representing the common ancestor of species being considered.

If the calibrating parameter values like population size change, then the timing and/or form of the numerical results will change, but the basic idea of a dynamic flux of variation holds. Effects can be estimated by making some simplifying assumptions such as that the population is in demographic equilibrium. Looking *forward*, a new mutation has very little chance of surviving very long and only about a $1/N$ chance that its descendant copies will become fixed in the population. But looking *backward*, from *any* gene, all copies extant today must be descendants of a single ancestral copy at some point in the past. This common ancestral copy is known as the *coalescent*. The existence of a coalescent is implied by the assumption that all life derives from a common ancestor and that mutation is a divergent, random process that generates new branches on the tree of descent from that common origin. Even if there is no single tree of life, as discussed earlier in regard to horizontal transfer of genetic material among species, copies of the different genes present today can in principle individually be traced back to a common ancestor, and molecular clocks are used to estimate when that was. This is true even if the individual genes in an individual each go back to a different ancestral copy, in a different individual, in a different place, and at a different time.

Common ancestry is perhaps the most important organizing factor in biology and was Darwin and Wallace's major contribution. Expressed in modern genetic terms, Darwin and Wallace's theory was based on branching divergence in genes produced by mutation aided by natural selection screening on allelic variation. Genetic drift will also bring about divergence, if not so systematically as Darwin's idea of continual natural selection. Even just by genetic drift, lineages (especially if sequestered

by geographic isolation) will accumulate enough genetic differences over time that their members can no longer (or do no longer) mate with members of other lineages. This mating criterion is generally used to define “species.” Darwin and Wallace provided a *process-based* explanation for the patterns of similarity among species that were already known. Population genetics tries to emulate that process, at least in regard to the genes that are involved.

SYSTEMATIC ASPECTS OF EVOLUTION AND BIODIVERSITY

To say that mutation, mendelian transmission, and genetic drift all involve fundamentally random elements does not mean that they appear random because of inadequate data or insufficient knowledge on science’s part about the underlying truth. Chance is an *inherent* aspect of life. Despite this, the prevailing view in biology is basically deterministic and is concerned with the molding of biological traits by natural selection. We discussed this phenomenon conceptually in Chapter 2, but here we can see how population genetic theory deals with it.

Differential Success: What Is “Adaptation”?

Everyone has an informal idea of natural selection. Usually it centers around the idea that the fittest organisms are those that are the most successful, that is, that reproduce the most. Over time, the population will comprise an ever higher frequency of these fittest organisms as they disproportionately leave descendants. From a genetic point of view, the *reason* they reproduce the most is not relevant so long as the outcome is systematic over time and, because genes carry information about organisms, that information persists and proliferates. We say that as a result the successful organisms have *adapted* to whatever was imposing the selective screen. Of course, the environment can change and new variation is always being introduced, so that the process may never reach an end. That is one aspect of the stepwise adaptation model described in Chapter 2.

Population genetics formalizes these notions by assigning *relative* selective or “fitness” coefficients to each genotype being considered. For example, if Aa and aa genotypes are lethal because of the effects of the a allele, these genotypes can be assigned a *fitness* coefficient (often denoted by w) of 0, and 1.0 is assigned to the AA genotype because in relative terms it is the most fit genotype. The corresponding *selective* coefficients, denoted s , are the complements of the fitnesses, ranging from 1.0 for traits with no chance to reproduce and 0.0 for genotypes experiencing no deficit in fitness (in some treatments, s is used for the relative beneficial or harmful effect on fitness of a specific allele rather than genotype, but this is a technical detail about the way population genetic models are used).

If in such a situation the a allele has frequency p_a at some time, we can compute how fast that frequency will change in the face of selection. As shown by a simple example in Figure 3-3, a favored allele increases steadily in frequency and will eventually become fixed in the population (that is, will be at a frequency of 1.0). There are many subtleties and various ways that selection may act, but this is the essence of the genetic theory of darwinian adaptive evolution.

Because selective coefficients are usually treated in relative terms, the fitness assigned to a genotype is measured not in terms of the inherent value of the organism but by comparison to the most fit genotype *at a given time and in the population of inference*. In the simple illustration of Figure 3-3, only one allele is ultimately

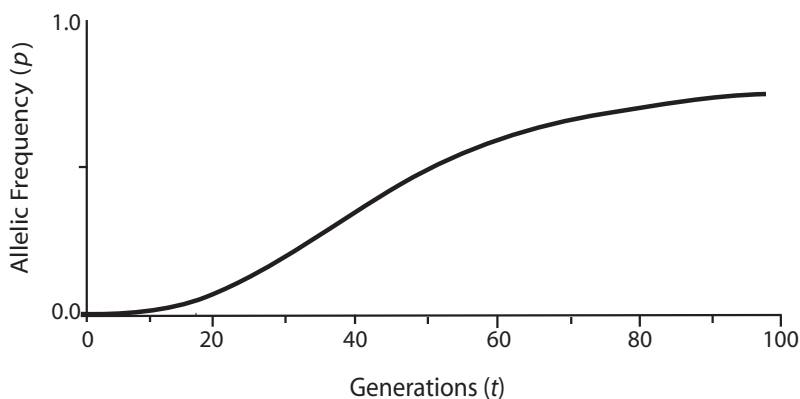


Figure 3-3. Effect of natural selection on allele frequencies. The increase over time of the frequency of an allele, A , in a population in which the aa genotype had a 10 percent disadvantage. These data are simulated by the method used in Figure 3-2, but here deterministic selection is simulated, with no factor of genetic drift. The allele's rise to high frequency in this case is due entirely to selection.

successful, but the theory doesn't imply "survival of the fittest" in the literal sense that in real populations only one genotype is successful while all others fail. There are typically many genotypes with comparable fitness, and even were there a single most-fit genotype, it can become disfavored the moment a new allele arrives by mutation or gene flow (incoming migrants). Something different may occur in other populations of the same species.

This is a formally competitive model: if an allele increases in frequency we can define it as having outcompeted other alleles in its given time and place. However, this says nothing about how that is brought about in real organisms, or what "competition" means in any particular case. And of course, having done better in this sense does not mean having been good or perfect.

Selection can act in many ways, including *balancing selection* (sometimes called *heterosis*, for heterozygote advantage), that maintain rather than exhaust variation by favoring or disfavoring multiple alleles in different genotypes. In the classic example, sickle cell anemia, the AA and SS genotypes are both harmful (one causes anemia, the other susceptibility to malaria), whereas the heterozygote AS genotype is better in both regards. A stable *balanced polymorphism* results over time, with unchanging allele frequencies that depend on the relative fitness values of these three genotypes. Other evolutionary strategies than allelic competition can generate heterogeneity in systems we will see later on, such as in immune resistance and olfaction, and we will see why that is important.

It is not unusual for individuals with extreme values for a trait to do less well in life than those near the average, presumably because the average is in some sense the result of an adaptive history. But this is not always the case, nor does it imply that variation in any individual gene associated with the trait is maintained in this way. Each trait and each gene are different.

Empirically, there are typically many alleles at a locus all the time, continually changing in frequency. Assigning fitness in relative terms, with 1.0 given to the best available genotype, means that the theory in this form does not specify whether the

population grows or shrinks in absolute size as a result of selection. Population growth can be taken into account, and although it complicates the theory somewhat it typically does not change the way that relative fitness competition is at the core of the theoretical model.

What's In That "s"?

Genetic drift is omnipresent. Nonetheless, in classroom and some classical versions of population genetics (and essentially in Darwin's notions), selection coefficients are treated basically as inherent, deterministic properties of genotypes (this is what was simulated in Figure 3-3 as well). In truth, selection or fitness are probabilistic phenomena. An individual with the fittest genotype in our scheme might be struck down by lightning before reproducing. Does that mean that that individual's fitness was 0? The answer is a rather curious kind of "no" because we conceive of fitness or selection coefficients as applying to a genotype in a probabilistic sense, on average. That is, a genotype's average fitness does not mean that every individual with that genotype will have identical reproduction. In computer simulations, even if chance is built into models that include selection, we know the truth (a value of s is assigned to a genotype by the programmer). But it is much more ambiguous in reality.

What appears to be clearly selective may not be in the causal sense. Did that moth really get eaten because it did not have good camouflage and the bird saw it sitting on the tree trunk? Or because the bird noticed a leg moving, or smelled it—or chanced upon it? How can we know? These are not simple questions to answer, even in what seem like classically easy cases (Weiss 2002).

At best, the fitness of a genotype has to be estimated by observing many individuals who bear it and seeing how well they did. But if it were necessary to observe every individual event, and then determine which instances of survival, death, reproduction, or infertility "counted" specifically as being causally due to the effects of the specific genotype, rather than to other characteristics of the individuals, or to chance, we would be delving into epistemological quicksand. Usually, only when selection is consistent and rather strong can its effects be convincingly detected. However, it can be shown that the less that selection coefficients vary among existing genotypes in a population, or the more they appear to vary from one generation to the next, the more the frequency changes of the alleles will behave as if they were selectively neutral (affected by chance alone). With massive die-offs, it might seem that selection would be quite strong, and we use the evolution of antibiotic or pesticide resistances as examples. But massive die-offs in nature need not involve such selection, because they can affect individuals regardless of their genotypes. But massive die-offs, by their consequent reductions in population size, reduce the amount of variation available for selection to work with. The reason is that in a massive die-off much variation will be lost by chance.

Quantitatively, if we treat selection as a fixed property of genotypes, and apply it probabilistically, and if the absolute value of the product of population size and selective coefficient (Ns) is small, say much less than 1.0 or 2.0, then drift will predominate over selection in determining the future course of allele frequencies (e.g., Hedrick 2000). In fact, most genotypes most of the time in nature seem to be only weakly affected by selection, as discussed in Chapter 2, which essentially means that natural selection is not as discriminating or prescriptive as would be expected in the deterministic gradual adaptation described by a strictly darwinian view of life.

In general, it is difficult to appreciate the degree to which the fitness of various genotypes is empirical and contextual rather than inherent properties of the genotypes (Lewontin 2000; Schlichting and Pigliucci 1998). When genotypes are nearly neutral, their small selective coefficients are unlikely to be stable over time or across environments. Because they are formally treated as relative values of competition among peers, the coefficients depend not only on the constraints of the environment but also on the other genotypes in the population (and the genetic variation in *all* other genes in each individual).

After the fact, we might be able to estimate an *average* or net fitness for a given genotype over some time period, which could account mathematically for the observed net allele frequency change. However, this is not necessarily a good way to explain what happened to individual organisms in their individual lives over long periods of evolutionary time, much less *why* it happened. Yet this is important if we are to give biological meaning to a probabilistic *s*. For example, selection might work only through occasional episodes of intense screening or may only trim away genotypes associated with extreme phenotypes, otherwise leaving the field to drift. Except under unusually favorable circumstances, population genetic models of a given situation are basically schematic—sometimes even when the causal process seems quite clear (Weiss 2002).

With so much uncertainty in this system, or weak and perhaps changeable effects, it could require studies that were themselves on the evolutionary or whole-species scale to generate a satisfactory understanding (e.g., see Tautz 2000). This is consistent with the general contingent, step-by-step view of the evolution of complex traits. Although rarely put this way, it means that even when a trait is adaptively evolving, most of the time in most individuals most vital events are due mostly to chance (although, like every trait, every gene can be viewed as “adaptive” by definition, because it is here to be observed). And this also means tolerance of variation by the screen of selection, and that selection is correspondingly a less precise or prescriptive mold of traits.

The average or net fitness of a genotype does not say much about daily experience, but fitness does not just suddenly occur. The determinants of fitness act over the lifetime of an individual, and depend inherently on the age-specific aspects of survival and reproduction of organisms. These can be expressed in terms of the vital rate schedules, $l(a)$ and $f(a)$, the rates (or more properly, probabilities) of survival and reproduction, respectively, of organisms age a . Fitness is related to the product of these two, $l(a)f(a)$, over the lifespan, and these schedules have to be estimated specifically for each genotype. This must be done in the same population at the same time, and for a long enough time and large enough sample size to obtain useful estimates. In fact, these are highly probabilistic and not necessarily stable over time, even when the gene is having some effect. Does the genotype (or a trait with which it is associated) only affect reproduction or survival early in life? Or at certain life history stages such as larval or embryonic periods? Does it affect longevity?

Fitness also depends on successful mating in sexually reproducing species, which means that each individual's fitness depends on the genotype (and hence fitness) of its mate(s)—and this is especially complicated with overlapping generations, when mates are not chosen strictly from birth cohort peers.

The formal genetic theory of evolution by natural selection was famously articulated by one of the founders of population genetics, R. A. Fisher (Fisher 1930), and has been augmented by many others (e.g., Cavalli-Sforza and Bodmer 1971;

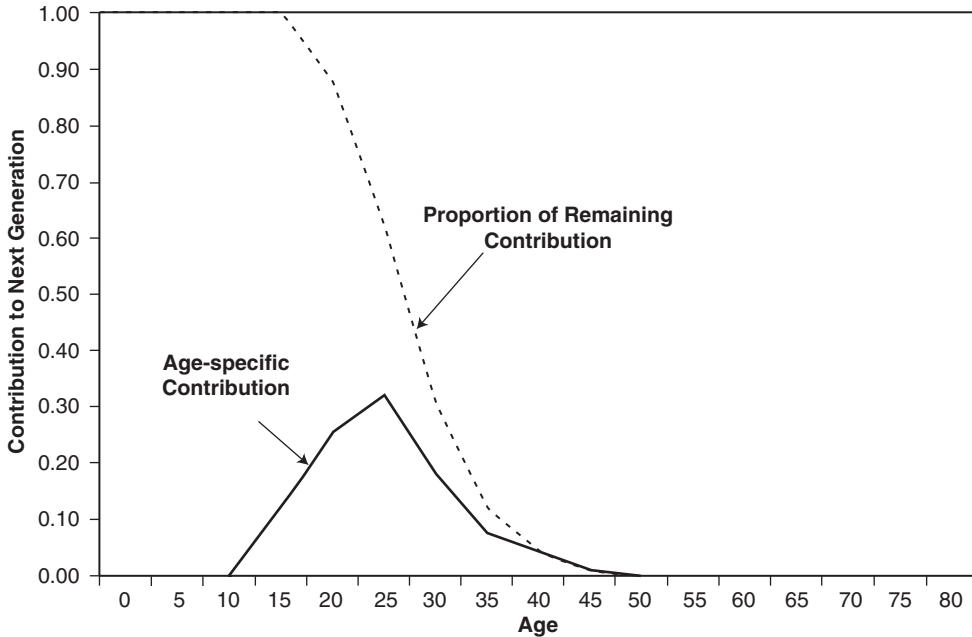


Figure 3-4. Schematic of human age-specific survivorship and fertility $l(a)f(a)$ schedules, showing age-related contribution to the next generation, and that the relative future contribution is greatest at puberty. The dashed line shows the proportion of remaining lifetime contribution as a function of age. Conceptually, each genotype would have its own such schedule, and that determines fitness or selection coefficients. In practice, these are usually weak inferences at best. The important point is that this is how fitness actually works on the ground and what must be estimated to truly understand its quantitative effects. It is worth keeping this in mind as we go through the diverse traits in nature and attempt to provide evolutionary explanations.

Charlesworth 1994; Crow and Kimura 1971). While we stress the problems with the concept to draw attention to the temptation to apply it uncritically or too universally, we can use it conceptually to see aspects of how selection might be expected to act. For example, using a term introduced by Fisher, an individual at a given age has a *reproductive value* that is determined by its genotype's vital rate schedules. This has been used to demonstrate the differential potential for selection to have an effect at various ages in the organism's lifespan (e.g., Crow 1958). This concept can help us understand how selection might work.

Figure 3-4 provides a schematic of human $l(a)f(a)$ schedules (Figure 3-4). Fertility is zero until puberty, then peaks for a while before gradually tapering off. Meanwhile, mortality rates increase from puberty onward. Overall, as shown by the dashed line, the amount of remaining reproductive potential is greatest at the beginning of adult life and declines thereafter. Events affecting young individuals can have a disproportionately great impact on reproductive success than events affecting them at later ages. So selection impacting young individuals will affect allele frequencies more rapidly than effects with later impact. Genotypes aiding in early reproduction more rapidly produce offspring than other genotypes, and their alleles

outgrow their competitors over time. But late in life, differential mortality or fertility have little if any impact on the individual's total reproductive output. Even if a genotype leads somehow to an awful disease at such an age, it is not selected against in the evolutionary sense. To that extent, the alleles are selectively neutral. Thus, for example, one might in vain search for evolutionary explanations for the frequency of alleles that led to high late-onset cancer risk in humans. But an allele with high frequency that causes a serious childhood disease should have been eliminated by selection; if it isn't, one suspects some form of balancing selection has been at work.

The Illusion of Adaptation Revisited

It may now be easier to understand why in Chapter 2 we said that the notion of adaptation was in some senses a seductive illusion, if not a tautology. To be here today, an organism had to have been “adaptive”—sufficiently suited to its ancestral environments for an uninterrupted 3 billion year ancestry. What criterion can we use to infer anything beyond that? With genes, unlike phenotypes, we can at least define more directly specific units of evolutionary change, and this allows for somewhat more believable quantitation of effects.

Typically, we *define* selection as taking place when an observed pattern is unlikely to have occurred by chance alone. “Unlikely” is defined by choosing a statistical significance cutoff level, like 5 percent; we can use this convention to attribute to selection a pattern less likely than that cutoff value to have occurred by genetic drift alone. If the cutoff is 5 percent, that means we accept a 1 in 20 chance of an inference being wrong. Because drift is a sampling process, the probability that we estimate is only as accurate as our understanding of relevant evolutionary parameters such as sample size, population size, and the stochastic factors in birth, death, competition, and cooperation. These are rarely very precise, and the choice of a significance level is also, of course, entirely arbitrary. That this can be problematic can be seen in Figure 3-2. Consider the allele that became fixed. How can we tell whether it arrived at that frequency by selection or by drift?

In practice, indirect tactics are used. If we can make plausible estimates of the parameters, such as population size and mutation rates, then the amount of standing variation that would be produced by genetic drift alone can be estimated, and compared to that in a sample. If the time since divergence can be estimated (e.g., from the fossil record), population genetics theory can be used to estimate the amount of difference that would have accumulated by drift.

The amount of variation can be compared in different parts of the genome, searching for regions of reduced variability. There are biological reasons to think that some parts of the genome are more affected by selection than others. Figure 3-5 shows that this is reflected in the relative rates of accumulation of genetic divergence in different parts of genes observed between species. The relative amounts of standing genetic variation within species shows similar patterns. The reasons are the same: we expect and typically find less variation in regions with known function on the assumption that the effects of natural selection will generally prune variation that arises in such regions, relative to selectively neutral regions.

Particular attention is paid to the pattern of variation in regions that can most safely be assumed to have no function that could affect fitness. This variation then becomes the baseline. Regions showing less variation are assumed to have been constrained in some way. Likewise, parts of genes that are highly conserved (similar)

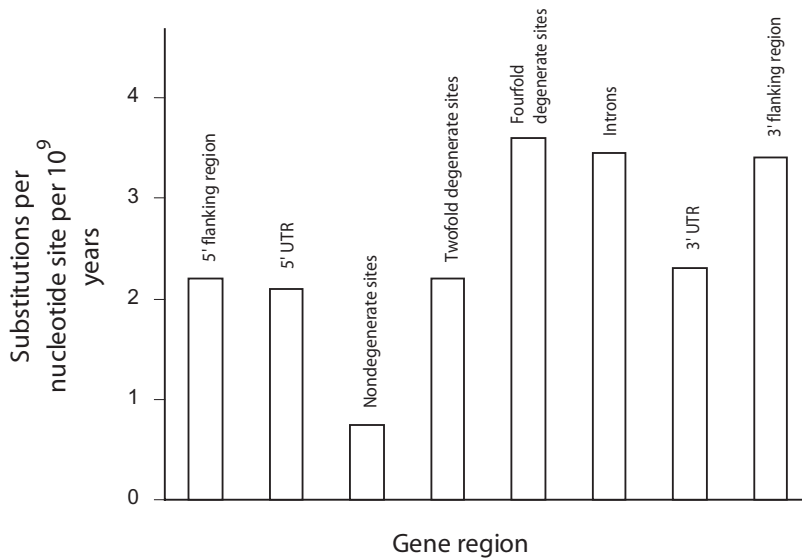


Figure 3-5. Relative rate of accumulation of variability in different parts of a stereotyped gene over evolutionary time. The units are the rate of nucleotide (basic DNA unit) substitution per 10^9 years. 5' flanking, 5'UTR (untranslated region), and introns are not used to code for protein. 3'UTR is part of the protein coding mechanism. Degenerate sites are parts of DNA for which nucleotide variation has no difference on a coded protein. Fourfold degenerate sites are essentially freely mutable in that sense. Twofold degenerate sites, and nondegenerate sites will, if mutated, generally affect the coded protein. These regions are less variable. For details, see Chapter 4. Redrawn from data in (Graur and Li 2000).

between distantly related species are assumed to have been maintained by the effects of selection.

These indicators of the existence of past selection work because there are thousands of units that can be studied in a single analysis, so that the sample size at least approximates a replicable process (on the assumption that similar types of regions in different genes behave similarly). But the analysis is usually only statistical and approximate. It may indicate *that* selection has occurred, but nothing about the *particular* cause and may not support any particular adaptive story. There may be alternatives that do not involve selection, or involve it in some other way. A specific hypothesized scenario has to be supported with knowledge of specific function or by showing experimentally that a given DNA variant has an effect on fitness. But even this does not tell us directly whether the function that we observe today is the same as that favored by selection in the past.

The idea of “adaptation” is in some sense fundamentally philosophical. It is by a kind of idealism that fitness or selective coefficients are viewed as being permanent, inherent, or deterministic properties of the genes themselves rather than changeable and contextual phenomena. In experimental or agricultural selection, the theoretical principles can be observed and applied because we can control the situation to be, essentially, what is specified in the theory. Longterm adaptation, however, is too slow to observe directly. Most real selective coefficients are quite small, so that the evolutionary process is usually meandering, having a changeable opportunistic

direction, and highly stochastic. The overall statistical analysis of DNA variation in identified functional elements generally suggests only rather weak average selection. Bursts of local intense selection do occur; but as a general rule, drift is the baseline characteristic of evolutionary genetic change.

AT THE SEQUENCE LEVEL, A HIERARCHY OF STRUCTURE

Population genetics is designed to explain and quantify change in genetic variation over time. It was initially developed from a consideration of genes simply as variable “things.” Discoveries during the 20th century about what those things are like have revealed additional complexities and characteristics of variation.

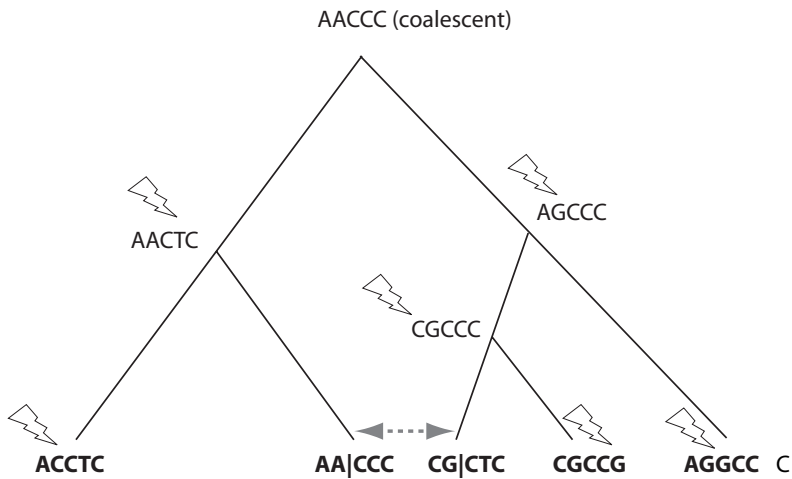
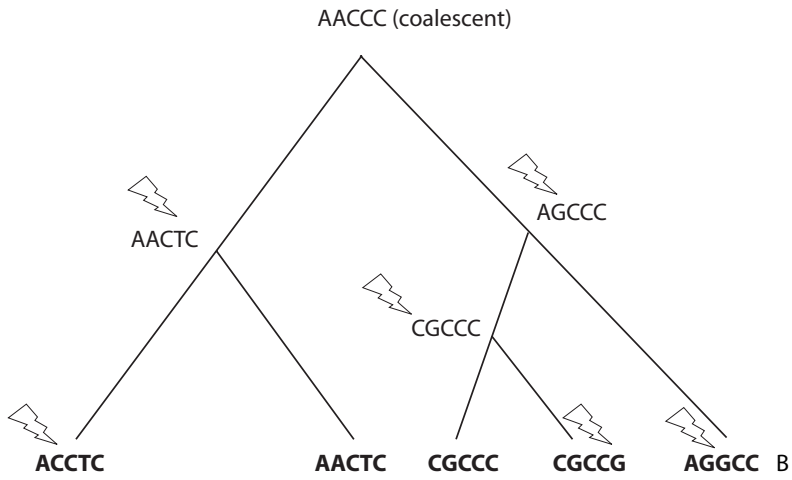
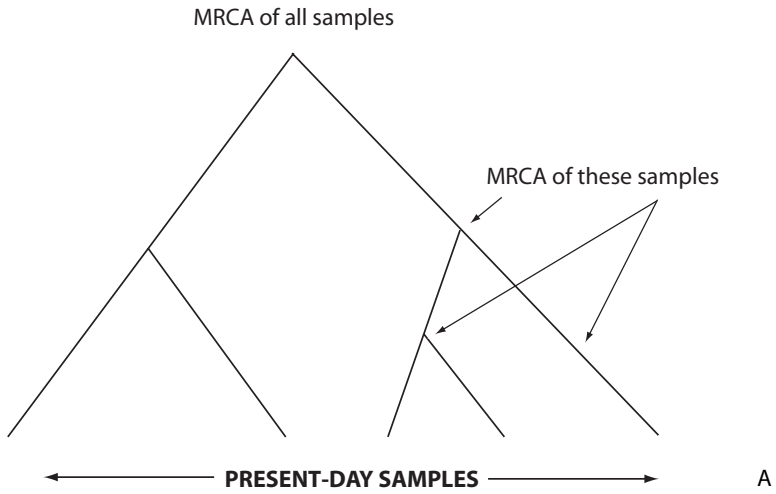
Hierarchy of Change Due to the Modular Nature of DNA Sequence

As we will see in Chapter 4, DNA is organized as a string of modular units. This leads to the phenomenon of *molecular evolution* on which so much of life is based. To a great extent, new mutation is random with regard to its specific location in a gene, and, although some mutation occurs every time the DNA is copied during cell division, mutation in any *particular* location in any particular replication is rare. As a consequence, over time, mutations generate a nested or hierarchical pattern of variation. A central fact in life and its evolution is that this hierarchical accumulation of genetic divergence is what allows organisms to accumulate functional divergence, and it is in this sense also that traits as well as genes bear the history whose common origin it was Darwin and Wallace’s insight to realize.

We can illustrate molecular evolution by a simple contrived example, in which a starting sequence of units, say **AACCC**, experiences a mutation that modifies the sequence to **AGCCC** in a descendant copy, and a subsequent change may change it further to **AGGCC**. Meanwhile, a different descendant copy of the original **AACCC** may mutate to **AACTC**, and so on. These sequences bear the trace of their history: the molecule evolves. That cannot be said of sugar or lipid molecules.

Figure 3-6 shows this kind of descent tree. Panel A shows the *demographic* history of the sequences we observe today. Regardless of its actual sequence details, each copy of a gene today has an ancestor, and all copies coalesce to a common ancestor in the past. Note that this is based on the *assumption* that evolution has proceeded as our understanding of life would have it. Panel B shows how mutational variation accumulates in descendant copies of the original sequence (the coalescent). If we assume this kind of process, the figure shows why we view the set of today’s copies as having a nested, hierarchical pattern of relationships and population genetics theory allows us to make an educated guess as to the history of

Figure 3-6. The nature of gene sequence evolution. (A) Demographic (parent-offspring) ancestry of samples collected today, going back to the MRCA (most recent common ancestor) of the samples; (B) with no recombination, sequence relationships go back to a common ancestor (the coalescent) in a simple nested way; (C) with recombination (double arrow), segments of the sequence (in middle two samples) have separate paths back to the MRCA shared by all copies. Present-day samples in **boldface**, ancestral copies normal font. Lightning represents mutations.



mutation (Li 1997; Nei 1987; Nei and Kumar 2000; Nordborg and Tavare 2002; Rosenberg and Nordborg 2002) and sometimes to reconstruct probable ancestral sequences. Viewed top down, Figure 3-6 shows an actual history, but looking bottom up it would have to be viewed as an inferred hypothetical history, because the ancestral sequences are not directly observed.

Panel C introduces the concept of *recombination*. Each generation, before reproduction in diploid organisms, the two copies of a chromosome align and occasionally exchange pieces. The figure shows that as a result different parts of the gene have different ancestral histories, though eventually going back to a common ancestor.

This type of analysis of DNA sequence data can be done to reconstruct the evolution of individual genes, or for multiple genes, sampled from various populations, to reconstruct species phylogenies (evolutionary histories) (Nei and Kumar 2000). Whole genes occasionally duplicate, and each copy then accumulates variation over time. The result is a set of divergent genes known as a *gene family*. Duplication events are a kind of mutation and the history of the set of genes can be reconstructed using coalescent concepts. We will see how this is used in many examples in subsequent chapters.

Within a species, we can examine the pattern of variation observed in different populations or different parts of its range. A specific mutation (e.g., the change from **A** to **G** on the right side of the sequence tree) is rarely recurrent, each change is likely to occur uniquely on a different sequence background, and to arise in a different part of a species' geographic range. As a result, DNA sequence variation is variegated over time and space. The elements of such a sequence are said to be *linked*, that is, colocated on the same chromosome, and the spatiotemporal nature of mutation produces statistical association between linked variants. For example, a given variant will only be found on related sequences (e.g., on the same branch of the sequence phylogeny in Figure 3-6B). The variation represented by the sites in a given branch is not independent because the sites have a shared history; we refer to this statistical association as *linkage disequilibrium* (LD).

When there is LD between sites in the genome, it means that knowing what is found in one site provides information about what is statistically more likely to be found in the other(s). Recombination breaks LD down over time because, as shown in Figure 3-6C, segments from different branches of the tree are exchanged, reducing the association between their respective variants and the consequent trace of history. Selection and population mixing or size instability can generate LD, and this and other subtleties mean that molecular evolution is often less straightforward to reconstruct than presented here.

Reticulated Variation

In nature, sexual reproduction is widespread and allows for the scrambling of the hierarchically nested variation that a history of mutation produces. In many animal and plant species, this means that individuals bear more than one copy of their species' genomes (humans have two—one copy from each parent). Scrambling of variation occurs because of recombination, or crossing-over, between homologous chromosomes during meiosis: prior to reproduction, the homologous (corresponding) chromosomes align and exchange pieces at one or more essentially random locations. This exchange reduces the LD among mutational variants along the chromosome, dissociating potentially important functional variation and allowing for new combinations of variation to occur together and to affect a trait or be presented

to selection. The same phenomenon reduces the “hitchhiking” that can occur if an allele in one place on a chromosome is favored by selection or drift, raising the frequency of nearby alleles, so recombination enables each gene to be subject to its own history over time.

Sometimes the sequence on one chromosome is used as a template for correcting differences found on the other chromosome (e.g., Lewin 2000), a process called *gene conversion*. The effects on the resulting sequence relationships are similar to those of recombination.

This scrambling of variation produces reticulated rather than simple treelike branching relationships among gene sequences that constitute a kind of horizontal transfer between peer (contemporary) copies of a chromosome. This is different from horizontal transfer between species, although both create reticulated sequence relationships. Among other things, individuals in a species are more closely related (have more similar sequence, share a more recent common ancestor) in some parts of their genome than they are in others.

Recurrent mutation is another phenomenon that causes reticulation of sequence similarity relationships. In a sample, we are likely to identify multiple instances of the same allele. We say that these are *identical by state* (ibs). If mutation were strictly unique, all copies would also be *identical by descent* (ibd) from a single ancestral mutational event. Most mutations may be unique, but recurrent mutations generate ibs variants that are not ibd. Recurrent mutations are notably common in sequence repeat elements (Chapter 4). Descendant copies of each event will share ancestry with that event; however because we can only observe their present state, it is not always possible to sort out which are which. Nonetheless, we can accommodate most of these complications, and it is the generally unique nature of the mutational process that allows us to make plausible reconstruction of gene history and hence of gene evolution.

EVOLUTION BY PHENOTYPE: THE DARWINIAN POSTULATES FROM A TRAIT'S POINT OF VIEW

What we have described so far is a general quantitative theory of the inheritance and evolution of genes. The assumption is widely made that population genetics is the rigorous theory of evolution and that the study of phenotypes and their development is rather descriptive and ambiguous by comparison. If genes are the ultimate units of evolution, the theory of evolution really only need consider what happens to genes. This needs a bit of clarification.

Population genetics gives a plausible sense of constraints on values of various parameters like times of separation among species, mutation rates, selection coefficients, and typical population sizes. The theory is highly valuable when experiments can be done, as with bacteria or fruit flies, to study genetic change (and even relate that to the effects of experimental selection on traits). However, this may not provide much information regarding phenotypic evolution in nature, or about what genes do, especially over the long term, or how gene function comes about. Yet, evolution works through phenotypes (traits in whole organisms), not directly on genes, so there is much to learn by looking at evolution from a trait's rather than from a gene's eye view.

Darwin and Wallace gave us a general overarching theory of evolution by phenotype, but we still do not have a good formal theory of the relationships between

genotypes and phenotypes beyond the simplest level. We can see some of the issues by reference to phenomena discussed in Chapter 2.

SOME SLIPPAGE BETWEEN GENE AND TRAIT

When organisms select local environments or in some other way associate with similar individuals, phenotypic variation can become partitioned, and this can in principle lead to speciation. We referred to various ways this might happen in Chapter 2 as *organismal* selection. One of the most important characteristics of organisms, especially complex organisms, is that they actively explore their environment and occupy favorable subenvironments that they find. Fish able to swim in strong currents may tend to stay in midstream, and weaker fish towards the margins, for example. What was a single initial population will become divided. Fish congregating (and mating) in a given part of the stream will be more alike (e.g., stronger fins), and there will be differences between groups inhabiting different parts of the stream.

Any kind of isolation of populations eventually leads to the accumulation by drift, mutation, and selection of genetic differences between them. Behavioral isolation may ultimately lead to the accumulation of variation that prevents mating should members of the two subdivisions encounter each other. Such changes may involve genes related to mating ability unrelated to the sorting trait itself (e.g., Rieseberg and Livingstone 2003). Variation in genes associated with the traits on which self-assortment was based will also be partitioned, in a way that is similar to the effects of competitive selection: reduce variation within populations because like individuals aggregate, increase it between them. This is a speciation scenario that is not about classical darwinian competition but can be difficult to discern from genetic data alone.

By a process known as *genetic assimilation* (Hall 1999; Waddington 1942; 1953; 1956; 1957; Wilkins 2002), something that is nongenetic can become genetic. If an originally acquired (environmentally or behaviorally produced) trait is favored by natural selection, mutations that increase the chance the bearer will have the trait can rise in frequency. It would be difficult to know from genetic data that such a process has occurred; yet it is not one initiated by genes. Canalization can lead a trait to become less variable in its environment, once the genes have been “focused” in this way. The extent of canalization depends on circumstances, that have recently been explored with a variety of theoretical approaches (Ancel 1999; Ancel and Fontana 2000; Fontana 2002; Siegel and Bergman 2002).

When there is phenotypic drift, a trait changes over time without the aid of selection. Genes associated with the trait will also change, but a rigorous or general kind of relationship between the two rates of change and the individual genes involved is not specifiable in practice. Although if we knew what the genes were we would find that their net pattern of variation today is consistent with selective neutrality, this tells us little if anything about the nature and timing of the phenotypic drift history.

What is needed is a good *phenogenetic* theory (if one is indeed possible), that is, for the relationship between genotypes and phenotypes: the number of genes involved, how those genes and their variation work, or the nature or extent of selection that may be operating over time.

When there is phenotypic drift, one consequence over time can be that a new “version” of the trait can replace a previous one. This is logically comparable to

allele substitution by genetic drift. Although it is always chancy to cite examples retrospectively, an interesting possible instance of phenotypic drift might be the stridulation mechanisms observed in grasshoppers. A curiosity noted in observations of the Amazon by the prominent 19th century naturalist, W. H. Bates (Bates 1863) was that some species of grasshopper make their noises by rubbing asymmetric structures on their two wings, but others achieve the same kinds of objectives using legs or legs and wings, etc. Whether these chirping mechanisms could have evolved totally independently is debatable. More likely is that the behavior—chirping—has been shared by these related organisms since their common ancestor, and that one mechanism has gradually replaced another in different lineages. We might eventually identify the genes involved in each type of chirping behavior, but we would have to know a lot about the phenotypic history if we were to make proper sense of how the genetic differences actually evolved.

One thing we know about phenogenetic relationships is that for most traits a variety of genotypes can be found in individuals with essentially the same phenotype. Similarly, among individuals with the same genotype, we commonly observe a distribution of different phenotypes, presumably because of the effects of chance and the many-to-many relationship between traits and their associated genotypes. This is *phenogenetic equivalence* and can occur for various reasons, including redundant genes, alternative biological pathways, and additive (dose-like) contributions of many different genes to the trait.

Phenogenetic equivalence is very widespread and weakens our ability to predict underlying genotypes from knowledge of phenotypes and vice versa. Natural selection cannot see “underneath” the phenotype any better than we can and will be correspondingly weak at affecting genotypes by screening phenotypes. Even if selection were precisely prescriptive at screening phenotypes, genetic variants underlying equivalent phenotypes will evolve neutrally relative to each other (Clark 1998; Hartl and Campbell 1982). That is, there can be *phenogenetic drift* (Wagner and Misof 1993; Weiss and Fullerton 2000): some alleles may disappear, and new ones may appear. Some genes may lose their effect on the trait, while other genes may become involved. But the trait stays the same. Many examples of phenogenetic drift are being identified, in experimental species (Rutherford and Henikoff 2003), in developmental processes that will be of interest in this book (Gellon and McGinnis 1998; Kissinger and Raff 1998; Robert 2001; True and Haag 2001), and in many others.

Because phenogenetic drift is not restricted to selectively neutral traits, over time, classical darwinian selection can preserve a trait unchanged while its associated genes change, sometimes dramatically. Phenogenetic drift can have an important impact on our understanding of evolution and on routine work in biology. Conservation of genetic mechanisms is often tacitly assumed in research with animal models, for example, in extrapolating from one model species (e.g., mouse, *Arabidopsis*, zebrafish, *C. elegans*, fruit flies, yeast, *E. coli*) as well as to other unstudied species. Biomedical research depends heavily on this kind of extrapolation because we cannot experiment on humans. To the extent that phenogenetic drift occurs, these assumptions will be inaccurate. The common finding that genetic interventions have different effects in different strains of laboratory mice is but one disturbing example that is well known but conveniently ignored in the effort to lay out tidy phenogenetic scenarios.

The actual extent or impact of phenogenetic drift in nature is unclear, but given the prevalence of phenogenetic equivalence it must occur. Nonetheless, as we will see in subsequent chapters, there can also be extensive and strikingly deep phylo-

genetic conservation of phenogenetic mechanisms. Similar traits even among distant species often seem to share at least some aspects of their underlying mechanisms. Nobody now would suggest that complex traits like eyes or limbs or leaves, even in distantly related species, are controlled by entirely different mechanisms. But the conservation is typically not complete, and sometimes it does turn out to be fundamentally different. Explaining how such strong conservation has been maintained by selection in the face of phenogenetic drift and equivalence is not yet possible, though some ideas have been advanced and will be discussed below.

The unpredictability of evolution in terms of phenotypes can be seen in a different way. Many examples of alternative pathways are being discovered in experimental species and traits. Computer simulations of evolutionary processes (Yedid and Bell 2002) and replicate experiments in flies (Fry et al. 1995; Mackay 1995; 1996; 2001; 2001) and bacteria (Cooper et al. 2003; Lenski et al. 2003; Papadopoulos et al. 1999; Travisano et al. 1995) show that, from the same beginning, even with the same starting variation (as far as can be determined), different genetic bases may result even if selection generates the same resulting trait.

These various phenomena are logical extensions of the fact that variation in quantitative traits is contributed to by allelic variation at many genes simultaneously. To what extent we can formulate a general theory that goes beyond description of individual instances is not clear. But the absence of a good phenogenetic theory contributes to the difficulty in inferring from genetic data how evolution of organisms and their traits actually happened.

WHAT EXACTLY IS INHERITANCE?

Perhaps because of the predominance of the physical sciences through the 19th century, the prevailing worldview of science today is that the ultimate units of causation in biology must be chemical or molecular. Genes are molecules that provide the codes for protein sequences, and proteins form the basic chemistry of life. Genes also retain the sequence structure that has evolved over billions of years, and genes are universally important elements transmitted from organisms to their descendants. For these reasons, many in biology have come to view genes as *the* units of inheritance and, conversely, inheritance as the study of genes. The study of genes, however, is not the only way to view inheritance.

A new organism starts life as much more than a set of genes. A fertilized egg contains membranes, sugars, carbohydrates, messenger RNA (Chapter 4), salts, and so on. These components are not generally regarded as an important aspect of inheritance because their production is viewed as dependent ultimately on instructions from genes acting in the parent that produced the egg or sperm. This is only partly accurate. Genes themselves are highly modified by chemical packaging and the like, and their modification can vary and hence is an important component of inheritance. Genes are not really even self-replicating, because self-replication only occurs in the context of the complex machinery of a cell.

There are nonmaterial aspects of inheritance as well. There are many ways in which the environment of organisms is not just what they must adapt *to* but what they themselves *create* (Lewontin 2000; Turner 2000). These aspects of the parents' environment will also be inherited as part of the environment of their offspring. Humans are not the only organisms that directly and systematically transmit environments or behavior to their offspring. Language and culture are easy human

examples, but similar characteristics apply to birds' nests, the local environment into which coral offspring are shed, the training to hunt that lion cubs receive, or the parental sounds that chicks respond to. Physiological effects can be directly transmitted transplacentally in mammals or indirectly via the nutritional or immunological environment an offspring is born into.

These traits are not directly genetic, but they are surely "biological" phenomena of animals and plants; they are meaningfully inherited, and they are widespread or even to some extent universal. They are as much the product of selection (and chance) as other traits; survival depends on nest-building abilities that birds have been passing to their offspring for countless generations as much as it does on their metabolism. The transmission patterns are different from those of genes; sometimes they are transmitted horizontally or depend on characteristics of the population as well as the parent. Nongenetic traits change, although not with the relatively specifiable characteristics of genetic mutation.

To some extent, it is rather arbitrarily that biologists have chosen to exclude nongenetic elements from the formal theory of inheritance and evolution. There are good reasons for making the distinction between genes and other aspects of life, but the boundaries blur and both should be considered (e.g., for perspectives on this, see Gottlieb 1997; Oyama 2000; Oyama et al. 2001). Population genetics generally treats nongenetic inheritance as an aggregate of "environmental" correlation in trait values observed among relatives, making very generic assumptions about these non-mendelian effects (e.g., that they have a normal distribution in the population) and treating them as temporary and of no interest to genetic evolution. Nonetheless, nongenetic inheritance affects the properties of organisms and populations, which can have direct, longterm effects on genetic evolution itself.

ELEMENTS OF EVOLUTIONARY PHENOGENETICS

Darwin gave us a conceptual theory for the evolution of *phenotypes*, not genes, and we have developed a formal theory that applies to the evolution of genes. Population genetics and darwinian concepts would be subsets of a rigorous theory for the evolution of phenotypes and would be united by incorporating aspects of phenogenetics, such as the complex regulation of gene expression, the interaction among genes, the relationship to environmental factors, a more inclusive definition of inheritance, and an adequate accommodation for the role of chance.

Population genetics largely puts genes into a black box by being nonspecific about how they work, which in turn weakens our ability to understand the actual nature of evolutionary change even if we can examine its net results. Here we can mention a few things that might be considered, for example, constraints that might explain some of those aspects of phenogenetic relationships that seem to have been conserved over very long time periods.

Genes involved in fundamental biological processes, like basic cellular energy metabolism, are generally expected to be less variable and more slowly evolving than genes related to developmentally later and/or more specialized functions. The reason is that, once installed early in the history of life, too much has come to depend on them. However, there are exceptions, and even basic housekeeping genes do evolve.

Pleiotropic genes, genes with multiple functions, are expected to experience less change than genes with only a single function (like coding for a protein used only

in one tissue). Genes related to embryological development can be highly pleiotropic, for example. Variation in such genes may be constrained because a mutation favorable to one of the gene's functions might be unfavorable to its other functions. Variation related to such genes is more likely to be found in aspects of its tissue-specific *expression* (usage) rather than in its coded protein product. But as we will see, the mutational and evolutionary dynamics of the elements that control gene expression are very different from those of protein-coding elements. We have theory for the behavior of pleiotropic genes under selection, but not the evolution of pleiotropy in the first place.

Along with some phenogenetic mechanisms, some other traits seem hardly to have changed from what is seen in fossils hundreds of millions of years old. Cockroaches and horseshoe crabs are famous examples. Some aspects of DNA sequences are also conserved among very distantly related species. What maintains the invariance? The usual explanation is that morphological conservation reflects the constraining force of adaptive natural selection. Yet this seems somewhat perplexing, because it is not as if there cannot be variation among crabs or cockroach relatives, or that their genomes have also remained frozen in time (they haven't). One contributing possibility is that, as mentioned in Chapter 2, longterm stability can occur when traits are canalized by developmental constraints, genetic commitments made in the past that are hard to modify. The organ systems of present-day species involve a complex network of genetic effects. These systems are the product of hundreds of millions of years of evolution, and their interactions seem to provide at least some canalization, or restraint on variation (Siegal and Bergman 2002). In a sense, they are so entrenched that they no longer need to be maintained by natural selection, or perhaps a better way to put it is that survival of the organism can tolerate little variation in the trait. But this is a weak general explanation, and there are exceptions.

Also mentioned in Chapter 2 was that some basic features of molecular interactions may simply, because of their chemical nature, remain in Bateson's "positions of organic stability" (Fontana and Schuster 1998; Kirschner et al. 2000; Laughlin et al. 2000; Monod 1971; Morowitz et al. 2000; Schuster 2000). It may be difficult or impossible for selection to alter these.

Genes that quantitatively affect patterning processes responsible, for example, for periodically repetitive traits like digits, leaves, hair, and the like may experience small mutational change with large phenotypic effect, very different from the Darwinian notion of gradual evolution. As will be seen, a substantial fraction of traits are patterned repetitively or in other similar ways. Darwin accepted the prevalent notion that nature makes no big jumps; to suggest otherwise is a kind of biological heresy. Population genetic models easily accommodate traits with a few states that can be altered by single mutations (like Mendel's peas) but generally treat complex trait evolution in a gradualistic way that does not provide a useful explanation of pattern traits.

The degree of change in phenogenetic mechanism over time is constrained by the availability of relevant genetic variation; this is hard to model without a better sense of the workings of the mechanism. Phenogenetic drift can only occur if alternative genetic mechanisms are available. Most traits probably have at least some phenogenetic equivalence, but it is not clear how to model the amount or its generation. Different alternative mechanisms may be subject to the effects of selection, drift, and mutation in different ways.

However, one important implication is that while selection may favor the same trait in different populations, it only works indirectly on genetic variation, so that in different populations selection can favor different genotypes that lead to similar phenotypes. That is, *phenotypic convergence* can lead to *genotypic divergence* between populations. An example is the diversity of genetic mechanisms by which humans in the Old World tropics have adapted genetically to malaria. Similar phenotypes do not imply similar genotypes.

CONCLUSION: DETERMINISM AND INDETERMINISM IN GENETIC INFERENCE

A trait can in principle be 100 percent genetically determined, in the sense that *if* we could replicate an entire genome *and* if we could ignore subsequent environmental and random effects, organisms would turn out to be identical. Each individual, however, is different in practice, making it impossible to view the inherited genome as “the blueprint” for life. We have experience with clones, twins, inbred animals, and other kinds of experimental evidence to show the degree to which a given genotype can determine traits (especially under similar environments). There is predictability, but it is not perfect and usually we only have observed the genes under a restricted range of environments, which can be deceiving (Lewontin 2000; Schlichting and Pigliucci 1998).

The issue of how deterministically genes drive evolution is more problematic, because evolutionary events cannot be replicated, so that the inference we can make from genetic data is usually indirect and statistical in nature. Common observations such as the strain-dependence of gene-inactivation experiments or the complex control of quantitative traits provide evidence that phenogenetic drift and phenogenetic equivalence are important aspects of evolution.

Phenogenetic drift illustrates this in a philosophically interesting way. Because new offspring must develop their final form from incomplete beginnings, often only a single cell, traits must be regenerated each generation. This justifies our search for the underlying “information” that makes this possible, and we attribute that information essentially to genes. From this perspective, the genes are real and a trait is their ephemeral product. But when a trait is preserved by selection while its underlying genetic basis changes, we can view the trait as real and its underlying genetic basis as an ephemeral servant.

We have raised and probably belabored some problems and incompleteness in current evolutionary genetic theory. The strengths of population genetics as a way to quantify the dynamics of genetic change and to reconstruct its history are well known. But it does not really address or predict the role of genes in what organisms actually do in life and how they came about. One purpose of this book is to try to identify some of the generalities that might be part of a phenogenetic theory of life.

PART II

Building Blocks of Life

A Genetic Repertoire for Evolving Complexity

Life is a complex, diversified phenomenon comprising species of many different kinds, some of them only very distantly related. Nonetheless, the basic system on which this diversity is built is common to all of life. A few principles and characteristics are shared, including the way biological “memory” evolves and is preserved in DNA, the cell, and a small set of basic genetic mechanisms that make it and all of life possible.

Chapter 4

The Storage and Flow of Biological Information

As discussed in Chapter 3, genes are generally given the place of honor at the core of biological and evolutionary theory. Genes are viewed as the carriers of biological “information.” This metaphor may be somewhat overstated, perhaps fitting in too easily with the worldview of the computer age (Kay 2000), but given that caveat, in Chapter 4, we will describe the basic ways information is stored and used by DNA and related molecules.

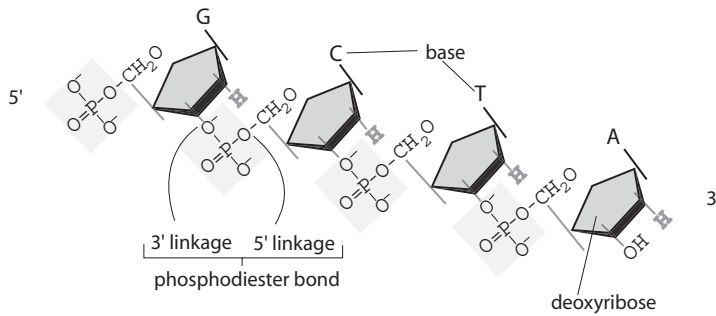
GENETIC ASPECTS OF INFORMATION

BASIC GENETICS: A BRIEF PARTLY GUIDED TOUR OF DNA

The large macromolecules, the *nucleic acids* DNA and RNA, serve many functions. The vital structural property of both molecules is that they are *modular*: each is a string of concatenated nucleotide *bases*, of which there are four basic types: adenine (**A**), cytosine (**C**), guanine (**G**), and thymine (**T**) (or in a modified form as uracil (**U**) in RNA). These are linked by a phosphate-sugar backbone. Each of the two purines (**G** and **A**) easily forms a chemical bond with one of the two pyrimidines (**C** and **T/U**): **C** pairs with **G** and **A** with **T**. This *complementary base pairing* allows the molecule to take on higher-order three-dimensional structure, and these basic properties are used both by nature and by experimenters to manipulate DNA and RNA, including the important function of replication in which a DNA molecule is copied and passed down from one cell generation to the next. Indeed, much of life and of our research methods to *understand* life is based on complementary base pairing.

The most important characteristics of both molecules are that (1) the length of the concatenated string of nucleotides is chemically arbitrary and (2) the sequence of successive nucleotides can be arranged in any order. Clearly, both molecules are modular in nature and the subunits are variations on a basic chemical theme—and this is how and why it works.

Sugar Phosphate Backbone of DNA



DNA Double Helix

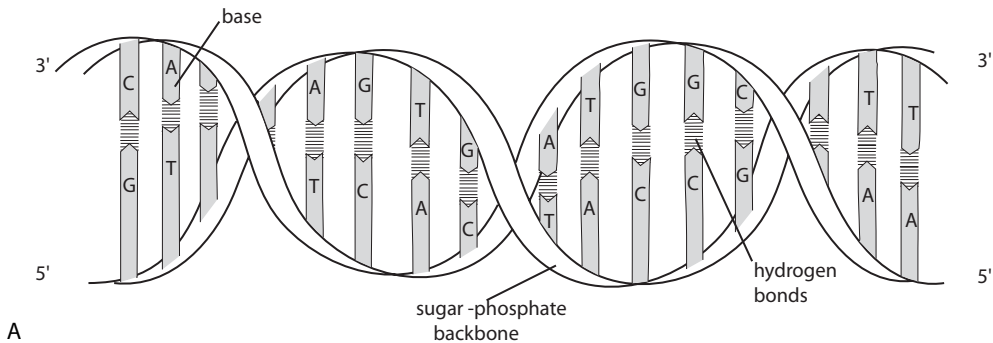
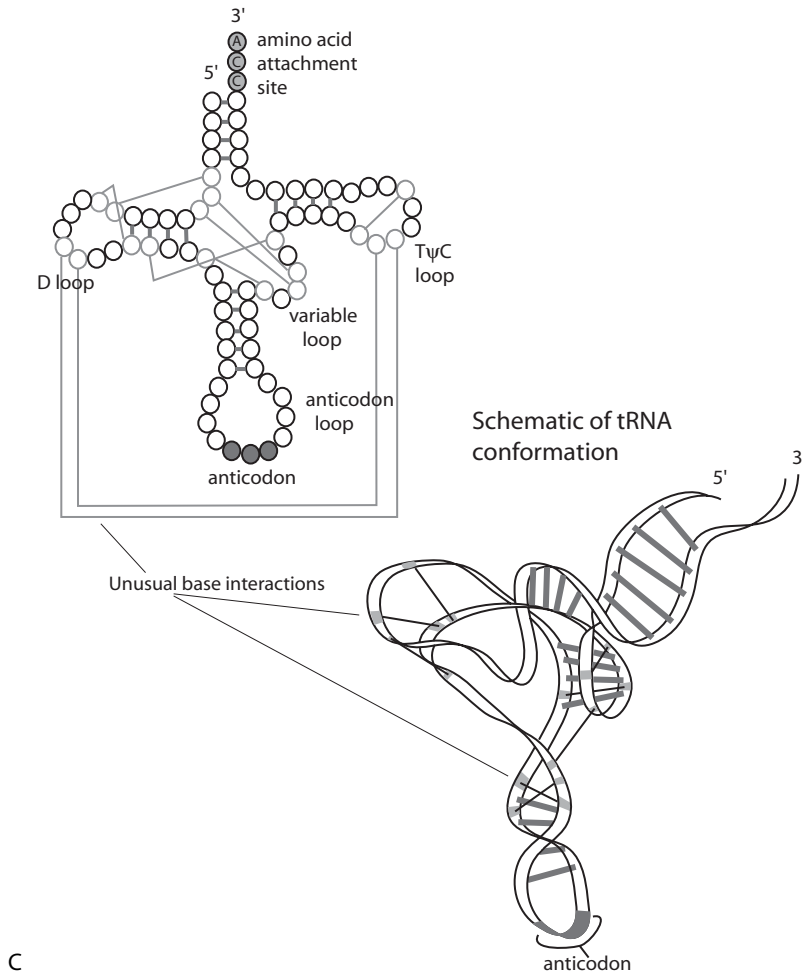


Figure 4-1. (A) Basic chemical structure of DNA; (B) basic chemical structure of RNA; (C) a tRNA molecule in natural conformation. Redrawn from (Alberts 1994).

DNA and RNA are illustrated schematically in Figure 4-1. DNA is double-stranded, with one strand having the complementary nucleotides to the other (e.g., **AAACGTA** would be paired with **TTTGCAT**). For physicochemical reasons, this arrangement leads to the well-known basic double-helix shape of DNA. However, the DNA molecule in a cell would, if stretched out linearly, greatly exceed the space available in the cell (e.g., in a human cell, the DNA would be more than one meter long); therefore, the molecule is wrapped and packaged into much tighter, more structured form.

The double-stranded nature of DNA has to do with its stability and its copying process; its information, however, is mainly contained in the sequence. RNA is usually viewed as a single-stranded molecule (Figure 4-1B), but autoannealing can occur by base pairing, as it folds upon itself, forming higher-order partly double-stranded structures, such as seen in a *transfer RNA* (tRNA) molecule (see below; Figure 4-1C). As we will see later in this chapter, although it has coding functions



C

Figure 4-1. *Continued*

related to sequence as does DNA, RNA has many other direct functions having to do with the conformation it takes when autoannealing. RNA is considered a transitory source of information storage because, except for some groups of viruses, it is no longer the primary inherited coding material, for reasons to be explained below.

A segment of a DNA sequence (one strand only) is shown in Figure 4-2. DNA is more passive than RNA but is of much greater interest because of its role in the storage of inherited protein-coding information. However, like RNA, DNA is involved in other functions; these involve the contextual *use* of protein coding, as well as DNA's own maintenance, packaging, and replication during cell division.

Along with many other, simpler types of molecules such as carbohydrates, the diversity of life is characterized by a diversity of proteins. Proteins are modular concatenations of amino acids whose function depends on their specific sequence. There

Figure 4-2. DNA sequence of the coding strand of the human apolipoprotein E (*ApoE*) gene, in standard 5' to 3' orientation. GenBank accession number M10065. The core coding functions of DNA and RNA are determined by their *sequence*, that is, the identity of the successive nucleotides along the molecular chain. The specific sequence has specific implications, and the sequence information is used in many different ways in both DNA and RNA.

AGCCGGACCCGCGACCGCCTGGACGAGGTGAAGGAGCAGGTGGCGGAGGTGCGCGCCAAGCTGGAGGA
 GCAGGCCAGCAGATACGCCTGCAGGCCGAGGCCTTCCAGGCCCGCCTCAAGAGCTGGTTCGAGCCCT
 GGTGGAAGACATGCAGCGCCAGTGGGCCGGGCTGGTGGAGAAGGTGCAGGCTGCCGTGGGCACACGCG
 CCGCCCTGTGCCACGCGACAATCACTGAACGCCGAAGCCTGCAGCCATGCGACCCACGCCACCCCGT
 CCTCCTGCCTCCGCGCAGCCTGCAGCGGGAGACCCTGTCCCGCCCCAGCCGTCTCTGGGGTGGACCC
 TAGTTTAATAAAGATTACCAAGTTTCACGCATCTGCTGGCCTCCCCCTGTGATTTCTCTAAGCCCCAGC
 CTCAGTTTCTCTTTCTGCCACATACTGCCACACAATTCTCAGCCCCCTCTCCATCTGTGTCTGTGT
 ATCTTTCTCTCTGCCCTTTTTTTTTTTTTTAGACGGAGTCTGGCTCTGTCACCAGGCTAGAGTGCAGTGGCA
 CGATCTTGGCTCACTGCAACCTCTGCCTCTTGGGTTC AAGCGATTCTGCTGCCTCAGTAGCTGGGATTACA
 GGCTCACACCACACACCCGGCTAATTTTTGTATTTTAGTAGAGACGAGCTTTCACCATGTTGGCCAGGC
 AGGTCTCAAACTCCTGACCAAGTGATCCACCCGGCCTCCCAAAGTGCTGAGATTACAGGCCTGAGCC
 ACCATGCCCGGCCTCTGCCCTCTTTCTTTTTAGGGGGCAGGAAAGGTCTCACCTGTCACCCGCCATC
 ACAGCTACTGCAGCTCCACCTCCTGGACTCAAGTGATAAGTGATCTCCCGCCTCAGCCTTTCCAGTA
 GCTGAGACTACAGGCGCATACCACTAGGATTAATTTGGGGGGGGTGGTGTGTGTGGAGATGGGGTCTG
 GCTTTGTGGCCAGGCTGATGTGGAATTCCTGGGCTCAAGCGATACTCCACCTTGGCCTCCTGAGTAGCT
 GAGACTACTGGCTAGCACCACACCCAGCTTTTATTATTATTGTAGAGACAAGGTCTCAATATGTTG
 CCCAGGCTAGTCTCAAACCCCTGGCTCAAGAGATCTCCGCCATCGGCCTCCCAAAGTGCTGGGATTCCA
 GGCATGGGCTCCGAGCGCCTGCCAACTTAATAATATTGTTCTTAGAGTTGCACTC

Figure 4-2. *Continued*

are 20 amino acids, each with its own molecular characteristics, and this diversity allows proteins to be highly diverse in their own functional characteristics. Proteins can be hundreds of amino acids long. Because each amino acid has its own particular chemical characteristics, the characteristics of the concatenation that makes a particular protein can be very diverse indeed. It is for this reason that proteins are considered to be the basic biochemical difference between animate and inanimate matter.

Proteins carry the biological information, so to speak, for the many chemical reactions that take place in life, but the term “information” is usually not applied to them because they are generally ephemeral molecules that are destroyed or recycled but not inherited. Instead, for every protein, there is a DNA sequence that directly corresponds to its amino acid sequence—and the former sequence *is* inherited. Because of its modular structure, we can see how mutation and the processes that subsequently determine their frequency, reviewed in Chapter 3, generate sequences that are a cumulative reflection of their history; that transmitted legacy accounts for our viewing DNA as bearing the essential information in biology.

This same system is used throughout the entire biosphere: in present-day trees, bacteria, mushrooms, trout, and humans, as well as 3.5-billion-year-old stromatolites and 100-million-year-old dinosaurs. Indeed, we would not define as “life” anything earthly that did not in some way involve this system. This single modular system of building blocks (nucleotides) evolved from a very rudimentary system to be able to incorporate open-ended functions and to preserve the information required to carry the diversity of functions of all life, across all generations.

Over billions of years, by growth in size and change in sequence, and 3 to 4 billion years of *uninterrupted* copying and descent, “endless forms most beautiful and wonderful have been, and are being, evolved,” as Darwin (Darwin 1859) stated, describing the history of life on Earth (see Chapter 17). He did not know about DNA, but he sensed the unity and common ancestry of all life which we now know is due in large measure to the incredibly flexible sequence-variation storage properties of these modular molecules.

DIGITAL STORAGE OF BIOLOGICAL INFORMATION

The Information Structure of DNA

DNA can be a very long molecule, and we often refer to it in terms of kilobases (kb, or thousands of sequential bases) or megabases (Mb, millions of bases). If we were to look along a molecule of DNA from end to end and observe the nucleotide sequence, it can superficially appear to be random; for example, knowing one nucleotide might provide essentially no power to predict the next nucleotide in line. But in a strange and interesting way, there is rich and complexly structured order in the sequence, only some of which can be discerned from the sequence alone. To identify, much less understand the patterning, chemical or biological function studies are required. For example, proportions of **CG** vs. **AT** pairs can vary considerably from one region to the next; one would have to scan a lot of sequence, like that shown in Figure 4-2, to identify pattern reliably and to infer that regions rich in **CGs** have particular meaning—in vertebrates protein-coding regions tend to be characterized by high **CG** content.

Figure 4-3 shows schematically the kinds of patterns that can be found in DNA sequence; each identified element varies both within and between individuals and species. In a virus, the DNA (or RNA) of a few thousand base pairs is a single molecule packaged within a protein coat. In bacteria, and in organelles (mitochondria and chloroplasts) within cells, the DNA is in the form of a closed circle or ring.

In eukaryote cells between cell divisions, DNA is found in the nucleus, wrapped periodically around a complex of proteins called *histones* and coiled into even more complex, compact form. At some points, some sequence motifs related to the incorporation sites of the histones are present. Several separate DNA molecules—the *chromosomes*—are in each cell. This set of chromosomes is referred to as the *genome*. Each species has its own number of chromosomes; a stereotyped display of the lengths and structures of the chromosomes in a species is called a *karyotype*.

The total length of each chromosome is tens or hundreds of millions of base pairs. Each end of each chromosome comprises a variable number of roughly similar (but also somewhat variable) sequence repeats, for example, **(TTAGGG)_n** in vertebrates, where *n* refers to the number of adjacent copies. These are *telomeres* and function to protect the chromosome ends from enzymatic degradation. Somewhere along the sequence another motif (also variable in sequence and location among and within

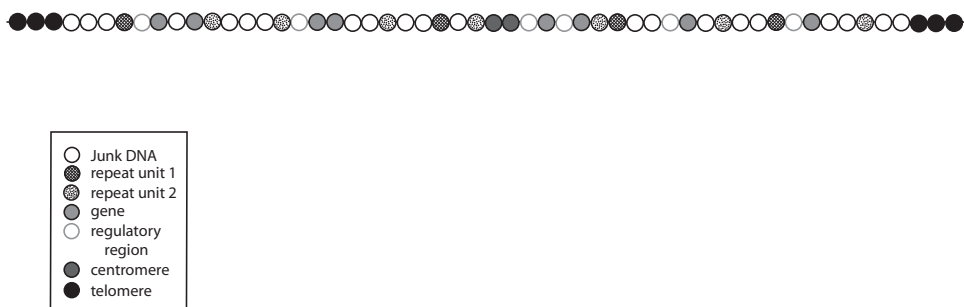


Figure 4-3. Stereotypical chromosome structure. Bead representation of various elements labeled as in key. “Junk DNA” is a colloquialism for DNA currently of unknown function.

species and characterized by complex patterns of sequence repeats) is found; this is the *centromere*, which is used to separate chromosomes into daughter cells during cell division (Sun et al. 2003; Tyler-Smith and Floridia 2000). Some of this DNA has been copied and inserted into this region from elsewhere in the genome, perhaps facilitated by repeat sequences such as **CAAAAAGCGGG**, and in flies, at least, there are also many short **AATAT** and **TTCTC** elements in centromeres. This structure may allow the DNA to loop out from its chromosome packaging and fold up upon itself, to provide an attachment of the spindle fibers that separate the chromosomes in cell division.

Along the sequence, numerous runs of the same nucleotide (e.g., **TTTTTTTTTT** . . .) of variable lengths or sets of tandem (adjacent) short nucleotide motifs, like **(ATTT)_n** or **(CA)_m**, are present. Some of the repeat copies are exactly or nearly exactly the same in different (but more or less randomly spaced) locations along a chromosome. Many of these repeated motifs have been catalogued (Vossilenko 2003). (See Table 4-1 for a sampling of these elements.) Depending on the motif or species, the repeat motif may occur tens or hundreds of thousands of times in the genome. When scanning different individuals of the same species in their corresponding chromosomal regions, one finds variation in the number of times a motif is repeated. In fact, the two sets of chromosomes in a diploid individual vary considerably in this respect. Thus, an individual may have **CACACA** on one chromosome and **CACACACACA** on the other *homolog* (copy of the corresponding chromosome), in a given location. These highly variable length repeats are called *microsatellites* (depending on the details, they are also known as short tandem repeats, or minisatellites). More than 50 percent of the human genome, in fact, consists of repeats (Lander et al. 2001). The proportion of the genome that consists of repeat regions varies considerably between species, however, from around 1 percent in some species to over 50 percent in others. The high variability in repeat numbers, within species as well as individuals, raises an important but perhaps little-appreciated point that there is no single length of the genome for a species or even in the two copies within an individual.

Some of these repeated elements have complex structures, such as a particular sequence at each end of a short (few hundred or fewer base pair) stretch of DNA, sometimes the sequence is in inverted order at the two ends, and sometimes such elements occur multiple times in tandem. Variation in short elements appears to be due to error in DNA replication during cell division. Some elements, however, are *transposable*, that is, have mechanisms known to make pieces of DNA move around among chromosomes from time to time, or move from parasites like viruses into a host's chromosomes. Repeat element sequences often resemble some other functional element that was captured in some way and subsequently distributed in copies around the genome. The *Alu* elements in primates are an example (Mighell et al. 1997); they are about 300 base pairs long and are distributed in hundreds of thousands of places in human (and other primate) genomes. The core sequence suggests that the *Alu* is a transposable element, including what once was a small RNA gene (Mighell, Markham et al. 1997). Sequence comparison shows a hierarchy of variation as if these elements episodically insert copies of themselves around the genome, accumulating mutations in the interim.

Clearly, selection has tolerated the presence of this repetitive DNA. It may be the harmless detritus of imperfect DNA repair or replication or of random insertions due to viruses or other “unintentional” processes manipulating DNA in the

TABLE 4-1.

A Sampler of Some Sequence Repeat Elements.***Tandemly repeated DNA***

Minisatellites such as di-((**TA**)_n) tri- ((**CAA**)_n) or tetranucleotide ((**GAGA**)_n) repeats
 Microsatellites (10–40 bp) and
 Macrosatellites (3–20 kb)

Telomeres

Long arrays of **TTAGGG** repeats

Centromeres

A complex of highly variable duplicate repeats of variable length, nature, and origin,
 often **A–T** rich, elements often having inverted repeat ends

Interspersed repeats

Short Interspersed Nuclear Elements (*SINES*), generally evolved from small RNA
 species, usually tRNA, but also 7SL cytoplasmic RNA
 Alu repeat family (primate specific); ~1,500,000 copies in the human genome each about
 280 bp long, usually flanked by 6–18 bp direct repeats
 B1, B2 (mouse); ~150,000 copies in the mouse genome 140–190 bp long
 Mariner (Mariner-like) elements, about 80 bp long, two inverted 37 bp regions, flanked by
 TA dinucleotide
 Long Interspersed Nuclear Elements (*LINES*) (mammal specific); called
 “retroelements” because they are related to retroviruses and retrotransposons
 L1 element (Kpn repeat), generally 6–8 kb long, but as small as 500 bp

Transposable elements (TEs)

Transposable elements with Long Terminal Repeats 1.5–10 kb
 Retroposons derived from RNA and transposed to DNA via cDNA
 DNA transposons, transposed directly from DNA to DNA

Transposable repeat elements

Miniature Inverted-Repeat Transposable Elements (*MITE*) (eukaryotes) 80–500 bp,
 terminal inverted repeats (TIRs)
 Maize transposable elements; Ds/Ac; *Ac* is 4563 bp long, and contains 11-bp terminal
 inverted repeats; *Ds* are truncated versions of *Ac*. *Ds* requires *Ac* to move, and is
 called “nonautonomous;” *Ac* can move without *Ds*, and is called “autonomous”
P elements (*Drosophila*), 2907 bp, terminal 31 bp inverted repeat

Whole Gene or Cluster Duplication

Tandem gene duplications (e.g., *Hox*, globin, olfactory receptors, immunoglobulin, and
 R-genes)
 Gene cluster duplications (e.g., 4 clusters of *Hox* and associated genes)
 Whole genome duplications, for which there is some evidence in vertebrates

cell. These repeat sequences can have functions; for example, they may be used to help the packaging or replication of the genome (e.g., see papers in Caporale 1999). Regardless of their origin, some appear subsequently to have experienced mutation that, in their new context, affects protein coding and expression. This is interesting because the Japanese puffer fish (*Fugu rubripes*) has basically the same genes as other fish but a substantial scarcity of repeat (and other noncoding) elements. Like-

wise, despite the highly structured nature of centromeric DNA, cell division can apparently take place without it (Amor and Choo 2002).

Genes have arisen historically through duplication events. Functional pieces of genes, or whole genes, are occasionally duplicated or transposed during meiosis. Gene duplication produces a family of related genes. They may remain in tandem on the same chromosome or may be inserted in other chromosomes. Gene families constitute another type of repeat element in the genome; the number of members of a gene family can range from only a few to hundreds. As with other duplicates like *Alu*, the individual copies can accumulate subsequent mutational variation. Gene duplication is a fundamental characteristic of evolution (Ohno 1970).

Figure 4-4 exemplifies the origin and evolution of gene families with a famous example, the *Hox* genes, that are involved in many aspects of embryological pat-

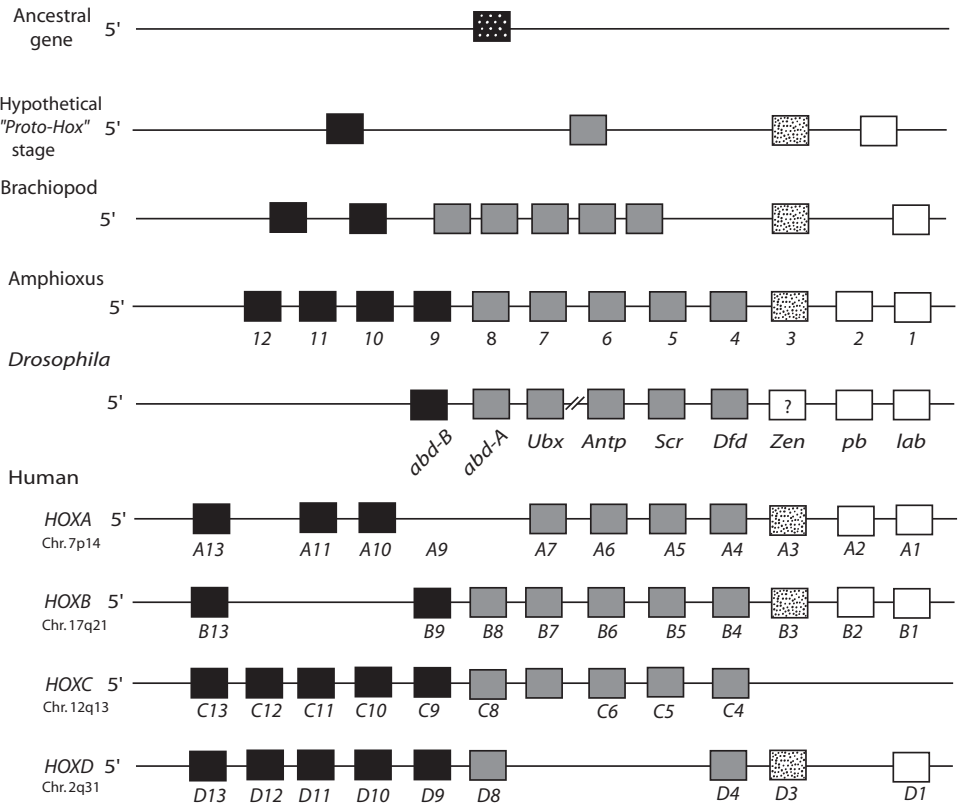


Figure 4-4. Evolution of the *Hox* subfamily of homeobox-containing genes involved in developmental patterning (see Table 7-5 for a description of this family). From an original gene, a small set of “proto-*Hox*” genes evolved by duplication. From these, subsequent duplication has created chromosomally linked clusters in invertebrates and early vertebrates. These clusters continued to gain and lose genes by tandem duplication and the clusters themselves were duplicated on at least two occasions (perhaps more in fish). Shading indicates likely homologies, that is, genes thought, based on sequence comparisons, to be descended from a specific common ancestor. Gene names are shown for the vertebrate human and *Drosophila* clusters and for the stem chordate *Amphioxus*.

turning. Figures of the evolution of the *Hox* genes are so often published that perhaps everyone is aware of them and they have become trite. But the example is important because the discovery of these genes had a transforming effect on biology. As will be seen in later chapters, not only are these genes used in corresponding structures in very diverse animals, but they are used in a way that was striking when first discovered. The *Hox* genes persuasively, and dramatically, showed the continuity of animal life and the much greater than expected homologies of structure and process across the animal world.

Hox genes and their action were first identified in patterning of the major fruit fly body axis. That was remarkable enough, as it was one of the first examples of complex patterning to be understood genetically. But then homologous genes were found in vertebrates. In addition, the gene arrangement structure was similar to that in flies, and indeed vertebrates have four separate clusters that resulted from major cluster-duplication events. Comparable sharing has now been extended to most groups of animals, and indeed following these discoveries many more instances of deep conservation of genes have been found, to the extent that it is now perhaps expected rather than surprising. We will see examples in several subsequent chapters.

With some understanding of what is generally found in a molecule of DNA, we can now look briefly at some of its major sequence-based elements that serve to *code* for various functions. The biology of DNA itself (its replication, packaging, and the like in cells) encompasses core biological traits but is not of primary interest in this book; therefore, we next consider the role of DNA in coding for the production of other substances.

Genes: Coding for RNA

Located at various places in the chromosomes of every species are the codes for the various types of RNA molecules. These include the tRNA molecules that “transfer” (carry) a specific amino acid for use in protein assembly, the *ribosomal RNA* (rRNA) molecules that are major constituents of *ribosomes*, where the tRNA-borne amino acids are concatenated, and the “small nuclear” snRNA molecules that participate in a variety of functions such as splicing mRNA and attending to telomeres (chromosome ends). These genes code for RNA that is itself directly functional, depending on its nucleotide sequence and autoannealing conformation.

RNA *transcription* is achieved by enzymes that move along a specific one of the two DNA strands at a given location (the two strands locally separate for this to happen); the enzyme “reads” the sequence incorporating (ribo)nucleotides one at a time into a chain by complementary base pairing to each nucleotide on the DNA template strand. The resulting RNA molecule is identical in sequence to that of the other “coding” DNA strand, except that ribonucleotides that are chained together to make an RNA molecule have a slightly different sugar than DNA and U replaces the T of DNA. Different polymerase (chain making) enzymes are used to transcribe different types of RNA; the control of which polymerase is used is encoded in the flanking sequence on the chromosome that represents a physical binding site (known as a *promoter*; see below) for the enzyme.

DNA is the “permanent” codebook. Once the RNA is transcribed, it leaves the DNA and proceeds to go about its business, while the DNA template—the “master” code—remains intact and can reanneal into a stable double-stranded state. When a cell divides, for example, the whole intact DNA molecule is copied and passed to

A. Lysine tRNAs, Various Species

Consensus	--GCCCGG-TA-GCTCAGT-CGGT--AGAGC-ATCAGACTTTT-AATCTGAGG--GTCGTGGGTCGA-GTCCACGTTGGCGCXXX
Archealglobus fulg	GGGCTCG-.....GCCA..C-.....GACGGG..TTT...CCCCTCG--GTCGCG...T.AA-AT..CGTCGAGCC.G
Bombyx mori	--GCCCGGC.....GT-C..T-.....ATGAGA..CTT...TCTCAGG--GTCGTG...T.GA-GC..CACGTTGGG.G
C elegans	--GCCCGGC.....GT-C..T-.....ATGAGA..CTT...TCTCAGG--GTCGTG...T.GA-GC..CACGTTGGG.G
Chicken	--GCCCGGC.....GT-T..T-.....ATGAGA..CTT...TCTCAGG--GTCGTG...T.GA-GC..CACGTTGGG.G
Chlorella virus k2	--GCCCGTC.....GT-C..T-.....GCCAGA..CTT...TCTGGTG--GTCGTG...T.GA-GC..CACGATGGG.A
Drosophila	--GCCCGGC.....GT-C..T-.....ATGAGA..CTT...TCTCAGG--GTCGTG...T.GA-GC..CACGTTGGG.G
E coli	--GCCCGGC.....GT-T..T-.....AGTTGA..TTT...TCAATTG--GTCGCA...T.GA-AT..TGCACGACC.ACCA
Haemophilus influ	--GCCCGTC.....GT-C..T-.....AGCGGA..TTT...TCCGTTG--GTCGAA...T.GA-AT..TTCACGACC.ACCA
Helicobacter pylori	--GCCCGTC.....GT-C..T-.....AATTCC..TTT...GGAATGG--GCCGTT...T.AA-AT..AACACGGGT.A
Human	--GCCCGGA.....GT-C..T-.....ATCAGA..TTT...TCTGAGG--GTCGAG...T.AA-GT..CTGTTCGGG.G
Loligo bleekeri	--GCCCGGC.....GT-C..T-.....ACGAGA..CTT...TCTC-GG--GTCGTG...T.AA-GC..CACGTTGGG.GCCA
Methanococcus jan	GGGCCCCG-.....GTCT..C-.....GCCTGG..TTT...CCAGGTG--GTCGAG...T.AA-AT..CTTCGGGCC.G
Methanotherm fer	GGGCCCCG-.....GTCT..T-.....GCCTGG..TTT...CCAAGTA--GTCGCG...T.AA-AT..CGTCGGGCC.G
Mouse	--GCCCTGA.....AT-T..T-.....ATCAGA..TTT...TCTGAGG--GTTTCAG...T.AA-GT..CTGTTCAGG.G
Mycoplasma capric	--GACTCGT.....GC-C..T-.....AACTGG..TTT...CAGTGG--GTCGCG...T.GA-AT..CCGACGAGT.ACCA
Rat	--GCCCGGC.....GT-C..T-.....ATGAGA..CTT...TCTCAGG--GTCGTG...T.GA-GC..CACGTTGGG.G
Treponema pallidum	--GGCCAT-.....GT-T..T-.....AACAGA..CTT...TCTGTGG--GTCGCA...T.GA-AG..TGCATGGCT.A
Wheat germ	--GCCCGTC..M.....GD-D..T-.....MGCAAGG..CTTT...CCTGTGMDMCGTG...Y.GMAGC..CACGGTYGG.GCCA
Xenopus laevis	--GCCCGCA.....GT-C..T-.....ATCAGAC..TTT...TCTGAGG--GTCGAG...T.AA-GT..CTGTTCGGG.G
Phage T5	--GGGTGCG-.....AC-T..TTT.....ACTGGT..TTT...ACCATAG--GTTACA...T.GA-GT..TGTGCAACC.ACCA

B. Human Serine tRNAs

Consensus	GTAGTCGTGGC--CGAGTG-GTTAAGCGCATGGACTTGAATCCATT-GGGGTT-TCCCCGCG--GCAGGTTCCGAA-TCCTGCCGACTACGXXX
Hum Serine1	.T..TCG...C--CG.GTG-GTT.AG..G.T....TGA..T..A.T-G.GGTT-.CC.CGC---.A...TCGA...T.CCGA..ACG
Hum Serine2	.T..TCG...C--CG.GTG-GTT.AG..G.T....AGA..T..A.T-G.GGTT-.CC.CAC---.A...TCGA...T.CCGA..ACG
Hum Serine2.2	.T..TCG...CACCG.GDGMGDD.AG..G.Y....TGA..Y..A.TMG.GGTMC.CC.CGMC---.A...TCGA...T.CCGA..ACG
Hum Serine3	.T..TCG...C--CG.GTG-GTT.AG..G.T....AGA..T..A.T-G.GGTC-.CC.CGC---.A...YCGM.A...T.CCGA..ACGCCA
Hum Serine4	.T..TCG...C--CG.GTG-GTT.AG..G.T....TGA..T..A.T-G.GGTT-.CC.CGC---.A...TCGA...T.TCGG..ACG
Hum Serine5	.G..ATA...C--AC.ATCTGGT.CT..A.C....CGA..T..G.CA.CCGCG.TT.GGGTGT..G...TCAA...C.CTAT..CCT

Figure 4-5. Conservation and variation in tRNA. (A) Alignments showing conserved (dots) and variable elements in the DNA sequences coding for lysine tRNA in various species; (B) alignment showing variation among the multiple copies of genes coding for serine tRNA within the human genome. Consensus (generally, the most common nucleotide at each position, even though no gene has this actual sequence) shown above each part. Bold face triplets are the mRNA-binding anticodon (this shows the redundancy of the coding system—see below).

descendant cells. In contrast, RNA is much less stable and is often enzymatically destroyed.

Multiple Functions via Modularity in RNA

A tRNA molecule folds upon itself via complementary base pairing into a general “cloverleaf” shape that is characteristic of tRNAs. Over evolutionary time, a considerable amount of variation on this structure has accumulated, although the basic shape is retained. The sequence at one end of a given type of tRNA determines which amino acid will attach there, and a sequence of three nucleotides at the other end of the (folded) tRNA molecule, called the *anticodon*, determines its specificity to a particular *codon* in the assembly of polypeptide (protein) chains.

The genomes of individual organisms contain multiple copies of codon-specific tRNA genes, some of them clustered on their respective chromosomes. Whatever other variation may have occurred, each codon-specific tRNA has a recognition triplet in the proper location (relative to the folded tRNA molecule) and the sequence structure that enables it to carry its specific amino acid. The “same” (amino-acid-specific) tRNA within or among organisms can vary in many details of its sequence, as can be seen by aligning them. Because of the redundancy of the coding system (see below), different recognition triplets that specify the same amino

acid can be found in the appropriate positions on their respective tRNA molecules. Thus, within the flexibility of the cloverleaf conformation, there is considerable variation that serves this variation-on-a-theme function (amino acid specificity).

From DNA to RNA to Protein: A Digital Coding System for the Diversity of Life

The most familiar role of DNA is as a code that specifies the structure of a protein, the role to which the word “gene” was first applied in regard to DNA. The diverse functions of life are brought about by the chemical properties of the 20 different amino acids (listed in Table 4-2); their number and order determine the interactions of a protein with other molecules. These properties depend on how it is folded, chemically modified, and combined with other molecules (including other proteins). Many proteins function as complexes of several polypeptides, each coded by region of DNA that may or may not be located close to each other on the same chromosome. This simple logic accounts for the diversity of function observed in the biosphere.

Amino acid specification is based on a three-nucleotide genetic code, as listed in Table 4-3. Because there are 20 amino acids but only four nucleotides, at least three nucleotides are required to specify 20 different amino acids. However, because there are $4^3 = 64$ possible different three-nucleotide codes and the code is *redundant*, most amino acids are specified by more than one triplet, in ways that probably reveal the evolutionary history of the coding system (see below). Within a protein-coding segment of DNA, each successive three-nucleotide stretch comprises a codon, whose sequence specifies a single amino acid in the coded protein. Most amino acids are specified by more than one codon (code redundancy), and three codons specify

TABLE 4-2.
Amino Acid Single Letter Designation Code.

A —Alanine (Ala)
C —Cysteine (Cys)
D —Aspartic Acid (Asp)
E —Glutamic Acid (Glu)
F —Phenylalanine (Phe)
G —Glycine (Gly)
H —Histidine (His)
I —Isoleucine (Ile)
K —Lysine (Lys)
L —Leucine (Leu)
M —Methionine (Met)
N —Asparagine (Asn)
P —Proline (Pro)
Q —Glutamine (Gln)
R —Arginine (Arg)
S —Serine (Ser)
T —Threonine (Thr)
V —Valine (Val)
W —Tryptophan (Try)
Y —Tyrosine (Tyr)

TABLE 4-3.

Genetic Code, Showing the Amino Acids Coded for By Three-letter Codons.

		Second Letter of Codon											
		T			C			A			G		
First Letter of Codon	T	TTT	Phe	(F)	TCT	Ser	(S)	TAT	Tyr	(Y)	TGT	Cys	(C)
		TTC	Phe	(F)	TCC	Ser	(S)	TAC	Tyr	(Y)	TGC	Cys	(C)
		TTA	Leu	(L)	TCA	Ser	(S)	TAA	STOP		TGA	STOP	
		TTG	Leu	(L)	TCG	Ser	(S)	TAG	STOP		TGG	Trp	(W)
	C	CTT	Leu	(L)	CCT	Pro	(P)	CAT	His	(H)	CGT	Arg	(R)
		CTC	Leu	(L)	CCC	Pro	(P)	CAC	His	(H)	CGC	Arg	(R)
		CTA	Leu	(L)	CCA	Pro	(P)	CAA	Gln	(Q)	CGA	Arg	(R)
		CTG	Leu	(L)	CCG	Pro	(P)	CAG	Gln	(Q)	CGG	Arg	(R)
	A	ATT	Ile	(I)	ACT	Thr	(T)	AAT	Asn	(N)	AGT	Ser	(S)
		ATC	Ile	(I)	ACC	Thr	(T)	AAC	Asn	(N)	AGC	Ser	(S)
		ATA	Ile	(I)	ACA	Thr	(T)	AAA	Lys	(K)	AGA	Arg	(R)
		ATG	Met	(M)	ACG	Thr	(T)	AAG	Lys	(K)	AGG	Arg	(R)
	G	GTT	Val	(V)	GCT	Ala	(A)	GAT	Asp	(D)	GGT	Gly	(G)
		GTC	Val	(V)	GCC	Ala	(A)	GAC	Asp	(D)	GGC	Gly	(G)
		GTA	Val	(V)	GCA	Ala	(A)	GAA	Glu	(E)	GGA	Gly	(G)
		GTG	Val	(V)	GCG	Ala	(A)	GAG	Glu	(E)	GGG	Gly	(G)

Note the redundancy of the code, that is, an amino acid can be coded for by more than one codon.

the instruction *stop*, that is, the end of the amino acid coding sequence. As can be seen from the table, the redundancy is mainly in the third position. The level of code degeneracy (e.g., whether two or four nucleotides in a given position code for the same amino acid) was what was shown in Figure 3-5 concerning the relative amounts of diversity in genes. Thus, the bar for variation found in two-fold degenerate sites refers to sites like the third position in the codons that specify the amino acid lysine, which can be an **A** or a **G**.

A functional gene is more than a protein-coding region. The structure of a typical gene is shown schematically in Figure 4-6A. By convention, relative position in DNA is denoted 5' and 3' (in a left-to-right diagrammatic layout), referring to the location number of the carbon atoms in the sugar ring that links the nucleotide “before” and “after” the position being considered, respectively. In the 5' DNA flanking a coding region, sequences are present that carry a different kind of information—that for the regulation of the gene in question (discussed below), not for protein structure. The major features in the immediate gene-flanking region include various generic sequences used in gene transcription (for example, the “**TATA**” box is a sequence element containing **TATA**), a site where the transcription starts, and an **ATG** codon (for methionine) that is the *start* signal for the translation of the code into protein (seen in the second exon of the *Apoe* gene in Figure 4-6B). This is followed by a series of consecutive codons, called an *open reading frame (ORF)*, which specifies a corresponding sequence of amino acids.

Similar to the production of rRNA and tRNA, transcription of a gene is brought about by an appropriate RNA polymerase that binds the promoter DNA itself (along with a complex of other regulatory proteins, themselves coded for by other genes) and then copies one strand of the locally unwound double helix of DNA into a single-stranded complementary molecule of *messenger RNA* (mRNA). Transcription ends when the polymerase encounters a *stop* signal (one of several particular codons, shown in Table 4-3) in the DNA.

Most eukaryotic genes are interrupted by sequences of variable lengths that do not code for amino acids and typically may not code for anything or even have a function. The coding regions are called *exons*, and the intervening sequences are called *introns*. An exon ends when an appropriately recognized **GT** . . . is reached, and this is followed by intronic sequence until an appropriately recognized **AG** . . . is reached. After that, another exon may be encountered. The details, especially of the DNA flanking these splice donor and acceptor sites, are somewhat variable. Different proteins have different numbers of exons, a number that is the result of history but generally seems not to be constrained by anything biochemical; some genes have no introns, whereas others can have tens of them.

The freshly transcribed RNA is a temporary structure whose introns must be excised and consecutive exons *spliced* together by chemical machinery that recognizes the **AG-GT** signals. The RNA is further altered by the addition of a methyl cap on one end and a poly(**A**) tail (long string of **As**) at the other to enable translation into protein. All of these modifications are functional, that is, affect chemical reactions at later stages, and/or the stability and processing of the RNA itself. The molecule is now mature mRNA. It is a messenger because it embodies the codon sequence that will be used to specify the amino acid sequence of a protein.

The mature mRNA is transported from the cell nucleus to the cytoplasm for *translation* into polypeptide (the corresponding system is somewhat different in prokaryotes, which have no nucleus, but the basic coding mechanism is similar). Protein synthesis is catalyzed by ribosomes, which are composed of multiple rRNA subunits (each transcribed from a locus in the chromosomal DNA), and other constituents. The mRNA serves as template; moving through the ribosome three nucleotides at a time, tRNAs whose anticodon is complementary to the mRNA

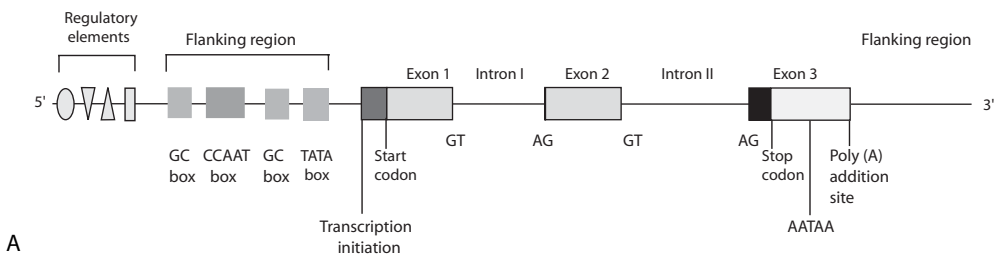


Figure 4-6. (A) Stereotype of a typical gene, including flanking regulatory regions, introns and exons, and so on. The number of exons varies among genes; (B) these components identified in the *Apoe* gene shown earlier in Figure 4-2, also indicating the location of repeat elements.

GGAAC TTGATGCTCAGAGAGGACAAGTCATTTGCCCAAGGTCACACAGCTGGCAACTGGCAGACGAGAT
 TCACGCCCTGGCAATTTGACTCCAGAATCCTAACCTTAACCCAGAAGCACGGCTTCAAGCCCTGGAAACC
 ACAATACCTGTGGCAGCCAGGGGAGGTGCTGGAATCTCATTTCACATGTGGGAGGGGCTCTGTGTCT
 CAAGGTACAACCAAAGAGGAAGCTGTGATTAACCCAGGTCCCATTGCAAAGCCTCGACTTTTAGCA
 GGTGCATCATACTGTTCCCAACCCTCCCATCCCACTTCTGTCCAGCCGCCTAGCCCCACTTCTTTTTC
 TTTTGTGAGACAGTCTCCCTCTTGCTGAGGCTGGAGTGCAGTGGCGAGATCTCGGCTCACTGTAACTCC
 GCCTCCCGGTTCAAGCAATTCTCCTGCCTCAGCTCCCAAGTAGCTAGGATTACAGGCGCCCGCCACCA
 CGCCTGGCTAACTTTTGTATTTTAGTAGAGATGGGGTTTACCATGTTGGCCAGGCTGGTCTCAAACCTCC
 TGACCTTAAGTGATTCGCCCACTGTGCGCTCCCAAAGTGTGGGATTACAGGCGTGAGCTACCGCCCCCA
 GCCTCCCACTCCCACTTCTGTCCAGCCCCCTAGCCCTACTTTCTTCTGGGATCCAGGAGTCCAGATCCC
 CAGCCCCCTCTCAGATTACATTATCCAGGCACAGGAAAGGACAGGGTCAGGAAAGGAGGACTCTGGG
 CGGCAGCTCCACATTCCTCTCCACGCTTGGCCCCAGAATGGAGGAGGGTGTCTGTATTACTGGGCGA
 GGTGTCTCCCTTCTCTGGGAGCTGTGGGGGGTGGTCAAAAGACCTCTATGCCCCACCTCTCTCTCTCT
 GCTCTGTCTGTGGCTGGGGCAGGGGAGAACAGCCACCTCGTGACTGGGCTGCCAGCCGCCCTATCCCT
 TGGGGAGGGGGCGGGACAGGGGAGCCCTATAATTGGACAAGTCTGGGATCCTTGAGTCTCTACTCAGC
 CCCAGCGAGGTGAAGGACGTCTCTCCCAAGGAGCCG GTGAGAAGCGCAGTCGGGGGCACGGGGATGA
 GCTCAGGGGCTCTAGAAAGAGCTGGGACCTTGGGAAGCCCTGGCCTCCAGGTAGTCTCAGGAGAGCTA
 CTCGGGGTCGGGCTTGGGGAGAGGAGGAGCGGGGTGAGGCAAGCAGCAGGGGACTGGACCTGGGAAG
 GGCTGGGCAGCAGAGACGACCCGACCCGCTAGAAGGTGGGTGGGGAGAGCAGCTGGACTGGGATGTA
 AGCCATAGCAGGACTCCACGAGTGTCTACTATCATTATCGAGCACCTACTGGGTGTCCCCAGTGTCTCA
 GATCTCCATAACTGGGAGCCAGGGGCAGCGACAGGTAGCTAGCCGTGATTGGAGAACTTTAAATG
 AGGACTGAATTAGCTCAATAATGGAACACCGCGCTTAAGTGTGAGGTTGAGCTTAGAATGTGAAGGA
 GAATGAGGAATGCGAGACTGGGACTGAGATGGAACCGCGGTGGGGAGGGGGTGGGGGGATGGAATTT
 GAACCCCGGAGAGGAAGATGGAATTTCTATGGAGGCCGACCTGGGGATGGGGAGATAAGAGAAGAC
 CAGGAGGGAGTTAAATAGGAATGGGTGGGGCGGCTTGGTAAATGTGCTGGGATAGGCTGTTCAG
 ATAATGCAACAAGGCTTGAAGGCTAACCTGGGGTGAAGCCGGGTGGGGGGCTGGGGGTGGGAGGA
 GTCTCACTGGCGGTGATTGACAGTTCTCTCTCCCAAG ACTGGCCAATCACAGGCAGGAAGATGAAGG
 TTCTGTGGGCTGCGTGTCTGGTCACATTCCTGGCAG GTATGGGGGGGGGGCTTGTCTCGTTCCCGCCGCT
 CTCCTCTCTCCTCTCCTCACTCAACTCTGGCCCATTCAGACAGACCTGGGCGCCCTCTCTGAGGCT
 TCTGTGCTGCTTCTGGCTCTGAACAGCGATTGACGCTCTCTGGGCTCGGTTTCCCCATCCTTGAGAT
 AGGAGTTAGAAGTGTGTTGT
 TCTGCGCCAGCTGGAGTGCAGTGGCGGATCTGGCTCACTGCAAGCTCCGCTCCCACTGACGCCAT
 TCTCTGCCTCAGCTCCCAAGTAGCTGGGACTACAGGCACATGCCACCACACCCGACTAACTTTTGTGA
 TTTTCACTAGAGACGGGGTTTACCATGTTGGCCAGGCTGGTCTGGAACCTCTGACCTCAGGTGATTCG
 CCGTTTCGATCTCCCAAAGTGTGGGATTACAGGCGTGAGCCACCGCACCTGGCTGGGAGTTAGAGGTTT
 CTAATGCAATGACGAGATAGTGAATACCAGACACGGGGCAGCTGTGATCTTATCTCCATCACCCG
 ACACAGCCCTGCCTGGGGCACACAAGGACACTCAATACATGCTTTTCCGCTGGGCGCGGTGGCTACCCCT
 GTAATCCAGCACTTTGGGAGGCCAAGGTGGGAGGATCACTTGAGCCAGGAGTTCACACACAGGCTGG
 GCAACATAGTGAGACCTGTCTCTACTAAAAATACAAAAATTAGCCAGGCATGGTGCCACACACTGTGC
 TCTCAGCTACTCAGGAGCTGAGGAGGAGGATCGCTTGAGCCCAAGAGTCAAGGTGCAAGTGAACCA
 TGTTACAGCGCGCTGCACTCCAGCCTGGGTGACAGAGCAAGACCCTGTTTATAAATACATAATGCTTTCCA
 AGTGATTAACCGACTCCCCCTCACCCTGCCACCATGGCTCCAAAGAGCATTGTGGAGCACTTCT
 GTGTGCCCTAGGTAGCTGATGCTTGACGGGGTCAAGAGACCTTGACCCGACCTTGAACCTTTGTTCCA
 CACAGGATGCCAGGCCAAGGTGGAGCAAGCGGTGGAGACAGAGCCGAGCCGAGCTGCGCCAGCAGA
 CCGAGTGGCAGAGCGGCCAGCGCTGGGAAGTGGCACTGGTTCGCTTTTGGGATTACCTGCGCTGGGTGCA
 GACACTGTCTGAGCAGGTGCAGGAGGAGTGTCTCAGCTCCCAAGGTACCCAGGAAGTGAAGTGAAGTGT
 CCGATCTGCGCCCTTGACCCCTCTGGTGGGCGGCTATACTCCCAAGTCCAGGTTTCACTTCTGCCCTGT
 CGCTAAGTCTTGGGGGGCTGGGTCTGTGCTGTTCTAGCTTCTCTTCCCATTTCTGACTCTGGCTTTAG
 CTCTCTGGAATCTCTCTCTCAGCTTTGTCT
 CTCTGTCTCTGTCT
 TGGTCTTGAACCTCTGGGCTCAAGCGATCTCCCGCTCGGCTCCCAAAGTGTGGGATTAGAGGATG
 AGCACCTTGCCCGGCTCTCTAGCT
 CTCTGTCT
 GCCCCCACTCCAGCCCTTCTCCCGGCTTCCCACTGTGCGACACCTCCCGGCTCTCTCTCTCTCTCTCT
 GCGCTGATGGACGAGACCATGAAGGAGTTGAAGGCCTACAAATCGGAAGTGGAGGAACAACTGACCCG
 GTGGCGGAGGAGACGCGGCGACGGCTGTCCAAGGAGCTGACAGCGGCGCAGGCGGGCTGGGCGCGGAC
 ATGGAGGACGTGCGCGGCGCGCTGGTGCAGTACCGCGGCGAGGTGACAGGCCATGCTCGGCCAGAGCAC
 GAGGAGTGTGGGTGCGCTCGCTCCACCTGCGCAAGCTGCGTAAGCGGCTCTCCGCGATGCGCGATG
 ACCTGCAGAAGCGCTGGCAGTGTACCAGGCCGGGGCCGCGAGGGCGCCGAGCGCGGCTCAGCGCCA
 TCCGCGAGCGCTGGGGCCCTGGTGAACAGGGCCGCGTGGGGCGGCCACTGTGGGTCCCTGGCCG
 GCCAGCGCTACAGGAGCGGGCCAGGCTGGGGCGAGCGGCTGCGCGCGCGGATGGAGGAGATGGG
 AGCCGGACCCGCGACCGCTGACAGGAGGTGAAGGAGCAGGTGGCGGAGGTGCGCGCCAAGCTGGAGGA

B

Figure 4-6. Continued

GCAGGCCAGCAGATACGCCTGCAGGCCGAGGCCCTCCAGGCCCGCCTCAAGAGCTGGTTTCGAGCCCT
 GGTGGAAGACATGCAGCGCCAGTGGGCCGGGCTGGTGGAGAAGGTGCAGGCTGCCGTGGGCACCCAGCG
 CCGCCCCCTGTGCCAGCGACAATCACTGAACGCCGAAGCCTGCAGCCATGCGACCCACGCCACCCCGTG
 CCTCTGCCTCCGCGCAGCCTGCAGCGGGAGACCCGTGTCGCCCGCCCCAGCCGTCTCTCTGGGGTGGACCC
 TAGTTTAATAAAGATTACCAAGTTTCACGCATCTGCTGGCCTCCCCCTGTGATTTCTCTAAGCCCCAGC
 CTCAGTTTCTCTTTCTGCCACATACTGCCACACAATTCTCAGCCCCCTCTCTCCATCTGTGTCTGTGT
 ATCTTTCTCTCTGCCCCTTTTTTTTTTTTAGACGGAGTCTGGCTCTGTACCCAGGCTAGAGTGCAGTGGCA
 CGATCTTGGCTCACTGCAACCTCTGCCTCTTGGGTTCAAGCGATTCTGCTGCCTCAGTAGCTGGGATTACA
 GGCTCACACCACCACCCGGCTAATTTTGTATTTTAGTAGAGACGAGCTTTCACCATGTTGGCCAGGC
 AGGTCTCAAACCTCCTGACCAAGTGATCCACCCGCCGCTCCCAAAGTGCTGAGATTACAGGCCTGAGCC
 ACCATGCCCGGCCTCTGCCCCTCTTTCTTTTTAGGGGGCAGGGAAAGGTCTACCCCTGTACCCGCCATC
 ACAGCTCACTGCAGCTCCACCTCCTGGACTCAAGTGATAAGTGATCCTCCCGCCTCAGCCTTTCCAGTA
 GCTGAGACTACAGGCGCATACCACTAGGATTAATTTGGGGGGGGTGGTGTGTGTGGAGATGGGGTCTG
 GCTTTGTGGCCAGGCTGATGTGGAATTCCTGGGCTCAAGCGATACTCCACCTTGGCCTCCTGAGTAGCT
 GAGACTACTGGCTAGCACCACCACCCAGCTTTTTATTATTATTGTAGAGACAAGGTCTCAATATGTTG
 CCCAGGCTAGTCTCAACCCCTGGCTCAAGAGATCTCCGCCATCGGCCTCCCAAAGTGCTGGGATTCCA
 GGCATGGGCTCCGAGCGCCTGCCCACTTAATAATATTGTTCTAGAGTTGCACTC

Legend:

SEQUENCE: Short repeat element

SEQUENCE: Alu repeat

SEQUENCE: Exon, untranslated regions

B SEQUENCE: Exon, coding region including signal peptide, mature peptide, stop.

Figure 4-6. *Continued*

triplet carry their amino acid to the site, where it is concatenated to the previously joined amino acid, thus producing a polypeptide, the precursor of a functional protein. This codon-anticodon matching is how the sequence of DNA specifies, through mRNA as a temporary information carrier, the sequence of amino acids, and this is why the DNA is referred to as a modular, digital code.

Correspondence Between Coding and Coded Units Enables the World of Unconstrained Biological Complexity

This system is built up of modular units at all levels and depends on that fact to make it open ended and flexible. A mutation that replaces one nucleotide by another in a codon may change the amino acid that is specified: for example, **GCA** specifies the amino acid alanine; an **A→C** mutation, resulting in **GCC**, will have no effect on the protein because the latter is also a codon for alanine; this is known as a *synonymous* mutation. However, if the **G** mutates to **A**, the resulting **ACA** codon specifies threonine, and hence is a *nonsynonymous* mutation because it changes the amino acid that is coded by this piece of DNA.

Modularity does not imply independence among the constituent parts, however. A change in one can have dramatic effect on the others. An insertion or deletion of a nucleotide will alter the *reading frame*, throwing all subsequent amino acids out of order. Thus, removing the second letter (h) of the following sentence, while preserving the number of letters in each word, destroys its meaning completely: “The fish swam well” becomes “Tef ishs wamw ell.” The following single-letter *substitu-*

tions make relatively little change: “The fish swim well” and “Thy fish swam well.” The following change also makes sense—but with quite a difference: “The dish swam well.”

With a modular amino acid coding mechanism, matched by the modularity in tRNA, a system evolved by which a sequence of amino acids could be assembled and “remembered” across replication, cell division, or organism reproduction. The number of ways a polypeptide can be put together via different amino acid sequences is essentially limitless, and life evolved to build structures and catalyze reactions around the kinds of interactions that depend on the chemical properties of folded, modified polypeptides. This constrains, but also enables, the huge diversity of life and form on Earth.

RNA Regulating Gene Expression

If it is true in physics that for every action there is an equal and opposite reaction, there is something like this in genomic information as well. If mRNA represents activation of a coding signal, the same signal can be dampened by *antisense* RNA, which can bind to mRNA by complementary base pairing. In Chapter 7, we will see that important means to do this do in fact exist. Of course, were this “reaction” truly equal and opposite, nothing in life would get accomplished. But then, biology is not the same as physics!

RNA is also used in an interesting and direct physical way in the control of chromosomal inactivation during sex determination in mammals, by affecting expression of genes related to functional differences between males and females. This will be described in Chapter 8.

DNA: A Protected Code

The beauty of the DNA/RNA system is that it solves many functions by the one phenomenon of base pairing. The double-stranded nature of DNA and its packaging make it a stable molecule resistant to damage. When a cell divides, the entire complement of DNA must be replicated so that each daughter cell receives a complete copy. To achieve this, the two strands, call them A and B, separate and are matched nucleotide-by-nucleotide by a new concatenation of nucleotides. This generates two new double helices, AB and BA (and these are chemically identical—the order we write them in is strictly arbitrary). Uncorrected copying errors (mutations) can occur in the replication process, which is vital to evolution, but basically the DNA is preserved for posterity.

Meanwhile, RNA is the local, contextual, temporary tool for making polypeptides. A given gene can be transcribed many times to make for abundant protein, but the mRNA itself is not replicated and is eventually degraded. RNA transcription is based on base complementarity and so is the tRNA anticodon system. Thus all these functions are based on the same base-complementarity system. While this is general and applies between strands, the linear sequence along strands is used to carry the *specific* coding information.

The “Central Dogma”

The discovery of the logic DNA → mRNA → protein as a fundamental characteristic of life led to the formulation of the “Central Dogma,” referred to in Chapter 2. This states that this information flow goes only one way: there is no flow back of information (that is, to DNA sequence) that reflects life experience directly in a her-

itable, lamarckian way. The causal arrows go only one way. Evolution unfolds but does not involute back upon itself.

The Central Dogma asserts an aspect of biological sequestration—of information—at the heart of life and evolution that, unlike other kinds of sequestration, is widely and almost uniquely considered to be complete and inviolate. Whether exceptions exist or will be found is uncertain; we know that DNA does many other things, some more codelike than others, and, as we will see, the nature and level of sequestration between those aspects of the genome and the organism are more variable and incomplete. They have to do, for example, with the regulation of gene transcription. The “Dogma” is fundamental to the genetically centered view of life that provided a powerful prescription for research that transformed biology. It is an elegant system and is the core of molecular biology now being done, and was one of the most important discoveries in biology in the 20th century.

INFORMATION CHANGE: GENETIC MUTATIONS

We introduced the general notion of mutation in the last chapter; functional aspects of genes can be altered in several important ways. As illustrated by the alterations in the fish sentence above, mutations can alter the amino acids for which a gene codes, can interrupt or even destroy the coding system and make the protein wholly functionless, or can alter the amino acids in subtle ways that make small changes in the action of the coded protein. But because of the redundancy of the codon system, a high fraction of single nucleotide substitution mutations do not even change the amino acids coded for; all they do is change the particular tRNA species that is used to supply that amino acid. Sequence-altering mutations can involve simple nucleotide substitutions, deletions, or insertions, rearrangements of chromosomal fragments, or the transfer of repeat elements into or out of a region.

As well as affecting coding sequences, mutations can also affect regulatory sequences, creating or destroying them, altering their location relative to the gene itself, or altering the binding efficiency of regulatory factors. The sequence itself can be affected, or change can occur in the way that the nucleotides are modified (e.g., methylation of stretches of adjacent **Cs** and **Gs** flanking mammalian genes). These various changes can have subtle or dramatic effects, depending on the particular gene or context.

Gene function can also be altered by the way DNA is packaged, which can make a gene more, or less, available for transcription in a given cell or context. The determinants of the location of histones or other packaging elements are currently not well understood and may depend on factors other than just sequence signals that could constitute a kind of nonsequence-based inheritance.

Another type of mutation occurs by horizontal transfer of genes among individuals or species. It appears to be rare, especially today, with a few major exceptions. One is the transfer of plasmids among bacteria, for example, to confer resistance to natural (or human made) antibiotics. Viral DNA survives mainly by horizontal transfer because it functions only in the context of a host cell and its genome; viral transfer can cause cancer or other diseases but is of longterm evolutionary interest relative to hosts only if it integrates into the germ line rather than just somatic cells.

As we have noted in several places, we know of no truly lamarckian mechanisms by which a mutation occurs in order to bring about a specific adaptive change in a protein. As we noted in Chapter 3, however, aspects of nongenetic inheritance

should perhaps be more seriously included in evolutionary biological theory. Some *gene-related* changes may be inherited, such as gene-expression levels that may affect traits like blood pressure, the stimulus for which can be transmitted across the placenta. But this is reversible in different environments, and does not permanently alter DNA, and hence is not lamarckian in the genetic sense.

CODING CELLULAR DIFFERENTIATION: INFORMATION REGULATING INFORMATION

Except for simple organisms, the number of genes in a given species is somewhat a matter of speculation, even when the complete DNA sequence is known. This is partly because of the multiple splicing of mRNA and overlapping transcription that are sometimes found. The definition of a “gene” also changes as we continue to learn of ways new functions are encoded in the genome. Even with the simplest classical definition of genes as protein codes, a complex organism typically has tens of thousands of genes.

This raises an important biological question. Why, if essentially all cells in an organism contain the entire genome, whatever the number of genes that includes, do individual cells differ in their structure and behavior? The answer is that this is mainly because, in any given cell or cellular context, only a subset of these genes are used. Genes are turned “on” or “off” (are expressed or repressed) in a highly controlled active way that is affected by circumstances. This control is mediated through DNA sequences, another form of information in the genome—information that controls the use of coding information. How is this done?

TRANSCRIPTION REGULATION

The key is *cis*-regulation, whose information signals are found in sequence on the same chromosome (hence, *cis*) as the protein-coding gene (Figure 4-7) (Davidson 2001; Davidson et al. 2002). Usually the regulatory signals are in the nearby 5′ flanking region. Gene expression is also regulated by the local state of the DNA in terms of its packaging or chemical modification. Typically, a gene is expressed when a complex of specific regulatory proteins including an appropriate polymerase is bound to promoter sequences just upstream (5′) of the transcription start site.

This is a generic signal found in corresponding places in most genes, but specific gene expression is affected by other sequence elements flanking (and sometimes within) a gene; these are known as *enhancers*, *repressors*, or other *response*, or *regulatory*, *elements* (REs), which are bound in the appropriate cellular context by a variety of regulatory or DNA-processing proteins to make the DNA accessible to the polymerase. “Binding” means that the chemical structure of DNA with a particular nucleotide sequence—the RE—fits the charge and shape of a binding domain of the regulatory protein.

The Logic of Gene Transcription

Gene transcription begins with RNA polymerase binding to the promoter site, a short sequence about 25 nucleotides upstream of a gene’s transcription start site. Sites like those labeled **TATA** and **CAAT** (Figure 4-6A) are found in about the same location flanking most eukaryotic genes. The rate of transcription is modified by the binding to enhancer sequence elements by DNA-binding proteins known as

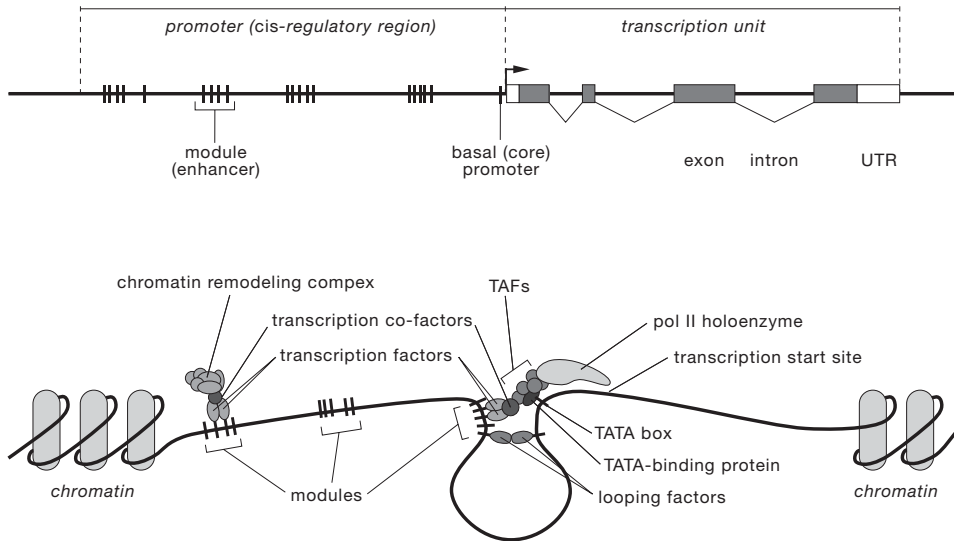


Figure 4-7. Basic *cis*-regulation of gene expression. Top line shows schematic organization of regulatory regions in relation to a gene. These can vary in copy number, number of elements, number of repeats and their arrangement relative to the gene. Bottom of figure shows the nature of protein complexes in a chromosomal region of a gene being expressed. Various components of the regulation process, including Transcription Activating Factors (TAF), are shown generically. Reprinted from (Wray et al. 2003) with permission.

transcription factors (TFs). These DNA-binding regulatory factors are encoded by genes elsewhere in the genome.

REs are typically very short (e.g., 5 to 10 base pairs) sequences, often assembled into regulatory modules or cassettes. This is shown schematically in Figure 4-7. A module typically contains binding sites for four to eight TFs, but there is no fixed number or arrangement or even location relative to the gene. A cluster of such elements is sometimes jointly called a *locus control region* (LCR), and can be quite distant from the gene itself. The TFs binding in such a region are often from different gene families. When in place, the configuration of TFs attract and enable the polymerases and other basal RNA transcription machinery.

Many regulatory proteins typically assemble in the regulatory region of a gene when it is to be expressed and interact in a variety of ways, including direct protein-protein as well as protein-DNA binding. Products of the *Sox* family of genes produce bends or loops in DNA near a gene, to enable other regulatory proteins to bind there. The number, location, or specific sequence details (e.g., binding affinities) of the REs can quantitatively affect expression levels. There are other sequence elements, for example, insulators that prevent multiple neighboring genes from being expressed at the same time.

Table 4-4 provides some examples of TF recognition sequences. This sequence information is not unambiguous, in that related TFs may recognize similar binding motifs, so there can be crossreaction; for example, many *Hox* genes (that is, their coded proteins) recognize motifs like **ATTAAATTA**. POU domains recognize sequences like **ATTGCAAT**. Some leucine zipper TFs bind **CCAAT**.

TABLE 4-4.
Known motifs in yeast.

Known motif	
Factor	Motif
<i>ABF1</i>	RTCRYnnnnnACG
<i>UME6</i>	TCGGCGGCTA
<i>CBF1</i>	RTCACRTG
<i>NDT80</i>	TCGGCGGCTDW
<i>REB1</i>	TTACCCCG
<i>MCM1a</i>	TTWCCCnWWWRGGAAA
<i>SWI6</i>	ACGCGT
<i>PHO4</i>	CACGTG
<i>MBP1</i>	ACGCGTnA
<i>SWI4</i>	TTTTTCGCG
<i>DAL81</i>	GATAAG
<i>RPN4</i>	TTTTGCCACC
<i>MSN2</i>	CCCCT
<i>MSN4</i>	CCCCT
<i>PDR1</i>	CCGCGG
<i>ESR2</i>	AAAAWTTTT
<i>MIG1</i>	CCCCRSWWWW
<i>MIG1b</i>	CCCCGC
<i>BAS1</i>	TGACTC
<i>GCN4</i>	ATGACTCAT
<i>GAL4</i>	CGGnnnnnnnnnnCCG
<i>HSF1b</i>	TTCTAGAA
<i>ESR1</i>	GATGAG
<i>MET31</i>	AAACTGTGGC
<i>AFT1</i>	YRCACCCR
<i>TEA1</i>	CGGnCGG
<i>PUT3</i>	CGGnnnnnnnnnnCCG
<i>HAP2</i>	TGATTGGC
<i>RAP1</i>	ACACCCATACATTT
<i>LEU3</i>	CCGGnnCCGG
<i>MCM1b</i>	YTTCTAATTWGnnCn
<i>INO4</i>	CATGTGAAAT
<i>INO2</i>	CATGTGAAAT
<i>GLN3</i>	GATAAK
<i>ADR1</i>	GGAGA
<i>FKH2</i>	TTGTTTACST
<i>FKH1</i>	TTGTTTACST
<i>RLM1</i>	CTAWWWWTAG
<i>SWI5</i>	KGCTGR
<i>HAP1</i>	CGGnnTAnCGG
<i>XPB1</i>	MCTCGARRRnR
<i>MAC1</i>	TTTGCTCA
<i>TBF1</i>	TTAGGG
<i>MSE</i>	TTTTGTG
<i>STE12</i>	RTGAAACA
<i>DIG1</i>	RTGAAACA
<i>MET4</i>	TGGCAAATG

TABLE 4-4.
Continued.

Known motif	
Factor	Motif
<i>HAP4</i>	TnRTTGGT
<i>SMP1</i>	ACTACTAWWWWTAG
<i>ACE2</i>	GCTGGT
<i>YAP1</i>	TTACTAA
<i>CIN5</i>	TTACTAA
<i>RME1</i>	GAACCTCAA
<i>HAC1</i>	CAGCGTG
<i>GCR1</i>	GGAAG

Source: (Lander, Linton et al. 2001).

Like the codon system for amino acid specification, the recognition sequences are *degenerate* or redundant; that is, the RE sequences for a given TF can vary. An example is given in Table 4-5 for the “paired” domain of *Pax6*, a gene involved in the development of eyes and other structures. These enhancer sequences may not have equal binding efficiency, but all are recognized by a given TF protein. The nominal or “consensus” binding sequence for the (vertebrate) *Pax6* double-bHTH paired domain is shown on the rightmost column (“N” means any nucleotide). *Pax6* also has a homeodomain (a separate bHTH domain) and binds **TAAT(T/C)(A/C/G/T)(A/G)ATTA** (simplified somewhat from Callaerts et al. 1997). See Table 7-5 for a general description of the different TF classes that bind these domains.

Unlike the codon system, however, in which redundancy is amino acid specific, variation in recognition sequences can lead to binding by different TFs. Variation in the copy number or location of REs can also contribute to variation in the strength of gene expression and hence have qualitative or quantitative effects. This kind of variation is shown in Figure 4-8 for REs for *Pax6* among various lens protein genes. Figure 4-9 shows variation in RE location among related genes used in rhombomere (hindbrain) segmentation.

The location of REs also varies among genes and among species for the corresponding gene and can be on either side of the gene, within the gene, near the gene, or tens of kilobases away.

This variation is clearly constrained by selection because there are so many similarities, but the pattern also shows the tolerance that evolves. In life, the functional meaning of REs is determined not just by their sequence but by their context-specific combinations. For example, a 30-base pair region called DC5 is needed to express a lens protein in the eye in chick embryos; this works if particular *Sox* and *Pax* class transcription factors jointly bind the 5' and 3' region of this enhancer (Kamachi et al. 2001).

Because RE sequences are short, they can be erased or generated relatively easily by chance in any DNA sequence. That is, any mutation has a substantial probability of generating a new RE, turning an existing RE into a different one, or altering the binding efficiency of an existing one. Mutations can also erase a binding site. Over reasonably short periods of evolutionary time, it appears that the blinking on

TABLE 4-5.

Variation in binding site sequences for the TF Pax6 ("paired" domain).

Pos	A	C	G	T	Consensus
01	15	7	6	10	N
02	21	9	3	10	N
03	10	9	10	18	N
04	8	14	9	16	N
05	3	2	4	38	T
06	2	0	1	44	T
07	3	29	1	14	C
08	40	5	1	1	A
09	3	39	0	5	C
10	1	0	44	2	G
11	1	36	7	2	C
12	23	2	1	21	W
13	1	4	0	42	T
14	2	13	26	3	S
15	40	1	6	0	A
16	14	11	15	7	N
17	2	4	3	37	T
18	1	0	20	25	K
19	13	17	9	4	N
20	14	8	4	6	N
21	4	12	3	9	N

The first column is the position along a *Pax6* binding site as experimentally determined; the next 4 positions give the frequency of each nucleotide in this position observed among 47 tested sequences bound by this TF. The last column gives the predominant nucleotide (if any). N indicates no predominant nucleotide, S is C or G predominates, W is A or T predominates.

Source: Transfac data base: <http://bioinformatics.weizmann.ac.il/transfac/>)

and off of REs in this way can alter the number and nature of REs, or their arrangement, near a given gene. Therefore, even if the coding part of a gene is not altered, its usage may be.

Sequence bears other sorts of expression-control information. CpG regions (CpG rather than CG to indicate successive nucleotides along a strand rather than C-G base pairs between strands) located just 5' to a gene can be enzymatically methylated to make the region unavailable for binding by regulatory factors. Histone modification by acetylation or methylation appears, respectively, to allow or suppress access by transcriptional machinery and hence to affect gene expression (e.g., Bird 2001; Struhl 2001).

Linkage relationships can present a kind of arrangement-based information that allows clusters of genes to be coordinately regulated via the information in shared REs. An LCR can be the initiation point for a process that moves down the chromosome, for example, by starting a competitive or sequential binding process that activates downstream genes. Or each gene can have its own copy (or copies) of the same RE(s). There are many examples, including coregulated, corelevant cellular housekeeping genes (Lercher et al. 2002). However, linked genes may be expressed together without being regulated by the same mechanism, and not all the genes

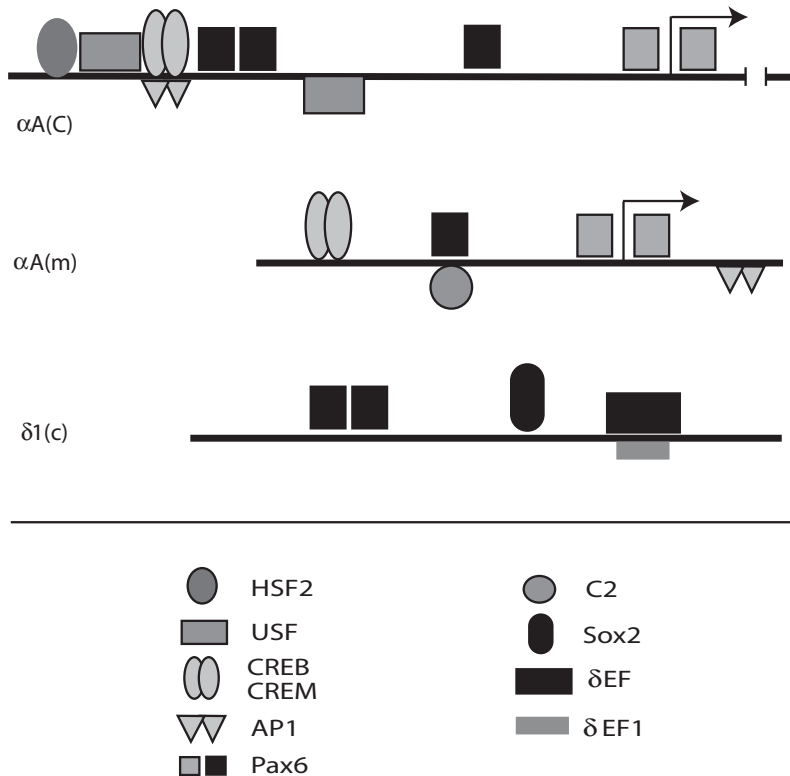


Figure 4-8. Variation in enhancer binding patterns in $\alpha A(C)$ chick, $\alpha A(m)$ mouse, and $\delta 1(C)$ chick lens crystalline genes (that make the lens protein in these species). Symbols show the location of REs for various TFs. In particular, note the variable location and number of the *Pax6* sites, in these genes that are each expressed in developing lens tissue. The broken arrows indicate RNA transcription start positions. Modified from (Cvekl and Piatigorsky 1996).

coexpressed in a given cell are phylogenetically or even functionally related (Caron et al. 2001; Spellman and Rubin 2002).

Similar REs may be, but are not necessarily, involved in coregulating genes in the same cells. For example, expression of *Hox*, *immunoglobulin*, and some color vision and olfactory genes is closely tied to their linkage relationships, but the alpha and beta hemoglobin genes are coregulated in erythrocytes by essentially unrelated mechanisms (Hardison 2000; 2001). Typically, only one alpha and one beta gene from each cluster of related genes is active in a given cell, and similar statements apply to opsin, immunoglobulin, olfactory, and other linked, related genes. Sometimes several, but not all, of the linked genes are expressed in a given context but others of the genes in other contexts; this is the case with the *Hox* developmental genes.

In addition to the mechanisms of *cis*-regulation by *trans*-acting TFs (that is, TFs whose own gene is on a chromosome other than that of genes it is regulating), there is also evidence for some true *trans*-regulation, that is, by sequence elements on one strand interacting with those on another. For example, there can be direct interac-

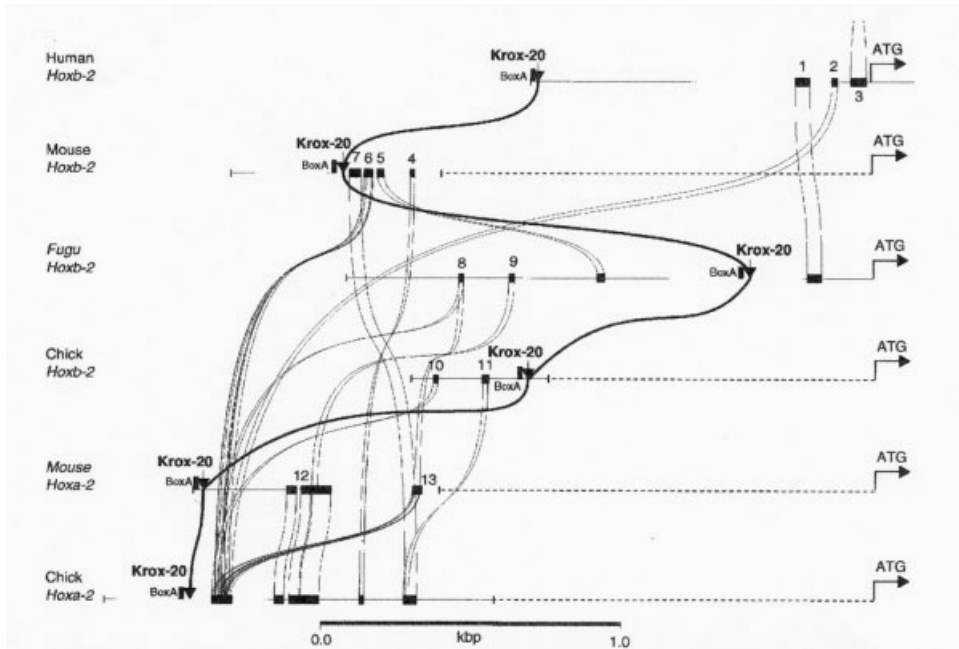


Figure 4-9. Schematic of variation in location of similar regulatory sequences. This is an alignment of the hindbrain r3/r5 segments enhancer elements of group 2 Hox genes from mouse, chicken, pufferfish, and human. Curves connect homologous blocks (solid squares) of sequence in the regulatory domains. The binding sites for the transcription factor *Krox20* (indicated by a solid triangle) and the BoxA REs (indicated by a solid rectangle) are situated at similar locations with respect to the start codon **ATGs** in all six fragments and joined with a thicker curve. The solid boxes (blocks 1–13) represent short stretches of at least 70 percent sequence identity dispersed along the enhancer elements of all four species. Regions of the mouse and chicken genomic DNAs between the **ATGs** and the enhancers that had not been sequenced are depicted with a broken line. Broken arrows are RNA transcription start positions. Reprinted from Nonchev et al., 1996, with permission. Copyright 2000 National Academy of Sciences, U.S.A. For details consult this source. See Chapter 16 on hindbrain segmentation genes.

tion of mRNAs from different homologs or genes. Direct contact-based influence of a sequence on one chromosome on a gene on another chromosome occurs regularly in at least some plants (called *paramutation*), and similar phenomena have been seen in at least some animals (e.g., Hollick et al. 1997). Paramutation often is associated with inhibiting gene expression. Although these phenomena are rare, we will see peculiar instances of gene regulation (e.g., in olfaction) where they may have an important application.

Evolving, But Not Treelike

One of the keystones of evolution, in the usual model, is that it generates divergence among descendants relative to their common ancestor. We have seen how mutation generates this kind of relationship among gene sequences and how gene duplication will generate a tree of divergent sequence relationships among the

members of a gene family. Interestingly, this is not quite how RE sequences evolve. A given TF recognizes a variety of RE sequences that are variations on a common theme (e.g., Table 4-4). They have relationships constrained perhaps by selection related to the binding efficiency, location near a regulated gene, and so on. But because they are at least largely generated by local mutation, not duplication and translocation, there need be no orderly tree of evolutionary relationships even among the enhancer sequences used by the same gene.

A Protein is More Than a Polypeptide

Genes are usually portrayed as coding for proteins, and a simple schematic of this relationship is given by Figure 4-10. This shows the modular nature of the coding logic, but it is not entirely accurate. Genes code for polypeptides—strings of amino acids. Polypeptides are not mature proteins. Functional proteins often comprise several polypeptides—from the same gene (homomultimers) or different genes (heteromultimers). Its shape gives a protein its ability to interact with some other specific compound, but a polypeptide does not fold into this shape without help, and other enzymes (that each must itself be coded by a gene) fold, modify, repair, protect, transport the polypeptide, and/or complex it with other polypeptides, to form the active protein. Many active proteins have *signal peptides* or other domains that must be enzymatically cleaved for the protein to be activated (or secreted from the cell, and so on, as needed in each case).

This processing machinery largely consists of other proteins, which means that, like the TFs that regulate the expression of a protein, these processing enzymes are also the product of genes. In general, although much of the information for a protein's function is in a sense contained in its amino acid code, a gene does not really code for a protein.

EVOLUTION OF THE DNA SYSTEM OF LIFE

DNA EVOLVED FROM AN RNA WORLD

Because of the diverse enzymatic roles now known for RNA, it is thought by many who speculate on the origin of life that some kind of “RNA world” preceded the DNA-based coding system (Gilbert 1986; Joyce 2002; Maynard Smith and Szathmari 1995). This was suggested by the discovery in the 1980s that RNA has enzymatic properties that include catalyzing some aspects of RNA replication (e.g., Cech 1986; 1990), and many additional active functions of RNA have since been discovered. Indeed, RNA can fulfill all the elemental functions of life as we know it (self-replication, catalysis of biochemical reactions) (Altman 1990; Cech 1990; Doudna and Cech 2002; 2003; Joyce 2002). Today, we see the remnants of this earlier history in the various fundamental uses that small, nontranslated RNA molecules still play. tRNA, rRNA, and snRNAs described earlier are examples. However, there is no clear consensus about the relative roles of protein, DNA, RNA, or other components, nor the stages through which this early evolution went. For our purposes, it is enough that the basic functions are essentially universal, so that how *organisms* manage has to do with how they use this system rather than how the system originated.

Similar questions are asked about the evolution of the DNA-protein coding system. This is a modular system, and it has been suggested that it evolved from a

START

```

M   K   V   L   W   A   A   L   L   V   T   F   L   A   G   C   Q   A
atg aag gtt ctg tgg gct gcg ttg ctg gtc aca ttc ctg gca gga tgc cag gcc

K   V   E   Q   A   V   E   T   E   P   E   P   E   L   R   Q   Q   T
aag gtg gag caa gcg gtg gag aca gag ccg gag ccc gag ctg cgc cag cag acc

E   W   Q   S   G   Q   R   W   E   L   A   L   G   R   F   W   D   Y
gag tgg cag agc ggc cag cgc tgg gaa ctg gca ctg ggt cgc ttt tgg gat tac

L   R   W   V   Q   T   L   S   E   Q   V   Q   E   E   L   L   S   S
ctg cgc tgg gtg cag aca ctg tct gag cag gtg cag gag gag ctg ctc agc tcc

Q   V   T   Q   E   L   R   A   L   M   D   E   T   M   K   E   L   K
cag gtc acc cag gaa ctg agg gcg ctg atg gac gag acc atg aag gag ttg aag

A   Y   K   S   E   L   E   E   Q   L   T   P   V   A   E   E   T   R
gcc tac aaa tcg gaa ctg gag gaa caa ctg acc ccg gtg gcg gag gag acg cgg

A   R   L   S   K   E   L   Q   A   A   Q   A   R   L   G   A   D   M
gca cgg ctg tcc aag gag ctg cag gcg gcg cag gcc cgg ctg ggc gcg gac atg

E   D   V   C   G   R   L   V   Q   Y   R   G   E   V   Q   A   M   L
gag gac gtg tgc ggc cgc ctg gtg cag tac cgc ggc gag gtg cag gcc atg ctc

G   Q   S   T   E   E   L   R   V   R   L   A   S   H   L   R   K   L
ggc cag agc acc gag gag ctg cgg gtg cgc ctc gcc tcc cac ctg cgc aag ctg

R   K   R   L   L   R   D   A   D   D   L   Q   K   R   L   A   V   Y
cgt aag cgg ctc ctc cgc gat gcc gat gac ctg cag aag cgc ctg gca gtg tac

Q   A   G   A   R   E   G   A   E   R   G   L   S   A   I   R   E   R
cag gcc ggg gcc cgc gag ggc gcc gag cgc ggc ctc agc gcc atc cgc gag cgc

L   G   P   L   V   E   Q   G   R   V   R   A   A   T   V   G   S   L
ctg ggg ccc ctg gtg gaa cag ggc cgc gtg cgg gcc gcc act gtg ggc tcc ctg

A   G   Q   P   L   Q   E   R   A   Q   A   W   G   E   R   L   R   A
gcc ggc cag ccg cta cag gag cgg gcc cag gcc tgg ggc gag cgg ctg cgc gcg

R   M   E   E   M   G   S   R   T   R   D   R   L   D   E   V   K   E
cgg atg gag gag atg ggc agc cgg acc cgc gac cgc ctg gac gag gtg aag gag

Q   V   A   E   V   R   A   K   L   E   E   Q   A   Q   Q   I   R   L
cag gtg gcg gag gtg cgc gcc aag ctg gag gag cag gcc cag cag ata cgc ctg

Q   A   E   A   F   Q   A   R   L   K   S   W   F   E   P   L   V   E
cag gcc gag gcc ttc cag gcc cgc ctc aag agc tgg ttc gag ccc ctg gtg gaa

D   M   Q   R   Q   W   A   G   L   V   E   K   V   Q   A   A   V   G
gac atg cag cgc cag tgg gcc ggg ctg gtg gag aag gtg cag gct gcc gtg ggc

T   S   A   A   P   V   P   S   D   N   H
acc agc gcc gcc cct gtg ccc agc gac aat cac tga

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STOP

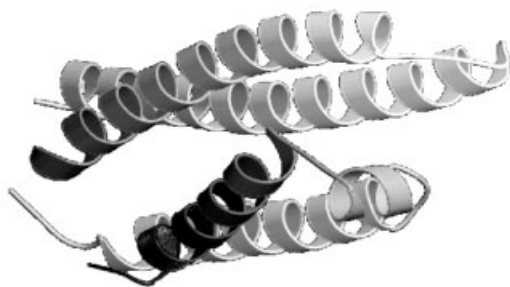


Figure 4-10. Modular nature of the information transfer system illustrated with human *Apoe*. Single letter amino acid sequence codes (translation of mRNA) sit above the three letter codons which specify them. Below is the protein structure, showing the *Apoe* lipid binding domain only (for the “e4” allele). Based on GenBank accession number M10065.

simple beginning (de Duve 1991; Maynard Smith and Szathmary 1995; Trifonov 1999). It seems very likely that a more direct relationship between RNA and protein existed before the conversion of mRNA to an intermediary message bearer. The most likely scenarios that have been suggested have this evolving from primitive RNA-amino acid interactions, perhaps related to amino acid synthesis, and there are relationships between the similarity of codon sequences, the genealogical relationships among the tRNA sequences, the chemical properties, and biosynthetic relationships of their corresponding amino acids (Maynard Smith and Szathmary 1995; Ronneberg et al. 2000). In this sense, the coding system can be viewed in adaptational terms as being selected to minimize the impact of mutational error on protein characteristics (Freeland and Hurst 1998; Freeland et al. 2000; Haig and Hurst 1991).

Today, the coding system is nearly universal. There are a few exceptions (for example, there are slight changes in the codons used by mitochondria, the organelles in cells that are used for energy metabolism), but its core elements seem to go back to a single beginning.

A DIGRESSION ON SELECTION: GENETIC LOAD

We have raised several points about the potential illusory aspects of adaptation and that selection is generally a probabilistic process (even if the common notion is more deterministic) whose characterization in nature can be elusive. We've suggested various reasons why darwinian selection by fitness competition may not need to be invoked as universally or strongly as is often done to account for evolution. But, as so often happens, there is a reverse problem as well. When we look at genetic variation across life, we may have a problem providing an explanation for *too much* evidence for selection. It is clear from the patterns of relative variation (shown in Figure 3-5) that functional regions of genomes vary much less within and between species than regions for which either there is no known function or strong evidence only for weak function.

Pseudogenes are the clearest case, and provide an approximate baseline for the rate of accumulation and amount of standing sequence variation to be expected when selection is not acting at all. A pseudogene is a gene, or duplicate copy of a gene, that no longer has the regulatory sequences needed for transcription; it is a sequence like a gene or a former gene that is clearly "dead" today and hence without function and whose variation is invisible to selection—a truly neutral sequence. By contrast, splice junctions, first and second positions in codons, regulatory sequences, and regulatory sequences are systematically less variable, and this constraint on variation is explained today as the result of selection. That is entirely plausible, indeed, the only good explanation until or unless some chemical error correction mechanism is found that applies specifically to such regions of DNA.

However, the order of magnitude of the amount of selection required to maintain this reduced variation is worth considering: for every 10^4 genes there are about 10^2 amino acids (and hence codons), in each and the variation in the first two nucleotides is constrained relative to the third, most redundant position—amounting to roughly 20,000,000 constrained sites per 10,000 genes, not counting enhancers, splice junctions, start signals, polyadenylation sites, **TATA** boxes, and so on. From today's estimates, the human genome has somewhere between 30,000 and 50,000 genes. This means something on the order of magnitude of 100,000,000 constrained

variation sites. Yet, in the long term, each human has on average only one surviving child (two per couple) each generation.

Not only are codons less variable than noncoding sequences, presumably because of selection related to function, but there is sometimes a further *codon bias* related to the availability of the alternative tRNA genes, based on codon redundancy, that use the possible codons specifying each amino acid. The genes for alternative tRNA molecules for a given amino acid vary in their number of copies in the genome. Codon bias is a slight statistical excess of codons for the most abundant of the alternative tRNAs. Codon bias has been found in screens of many species, although at least one search in mammals found no statistically meaningful evidence for codon bias (Urrutia and Hurst 2001) and the bias is not found uniformly among genes.

The subtle and basically statistical conservation of so many parts of genomes raises the problem known as *genetic load*. Genetic load became a critical issue in biology when protein electrophoresis methods made it possible to characterize variation in detail. How is such variation maintained? There would seem to be simply too much variation to be sustained by the amount of overreproduction needed to screen out mutational variation in the millions of conserved sites by mechanisms of darwinian selection. It was this problem that led to the development of the neutral theory of evolution (sometimes for this reason called “nondarwinian evolution”), because selectively neutral variation places no reproductive burden on the organism. The neutral theory holds that most variation most of the time is not affected, or barely affected, by selection and proposes neutrality and genetic drift in finite populations as the baseline for evolutionary population genetics.

The genetic load problem has been largely ignored recently but was never very satisfactorily solved. Even if we accept that introns and so on are neutral, the remaining subset of the genome that we take as constrained in variation involves a huge potential amount of selective screening. If selection is illusory in some ways and very difficult to document in nature, there is also strong evidence for pervasive selection. Several explanations have been suggested to reconcile this seeming paradox.

Species with high rates of reproduction, in which each individual produces thousands or even millions of eggs or pollen grains, can sustain a lot of selective loss, since on average an individual in a species needs to only just replace itself so that only one of these eggs or pollen grains will be successful (populations are not typically growing in size). Think again of the number of acorns shed by a single oak tree over its lifetime, only one of which on average will become a new tree. Bacteria, too, can reproduce rapidly and have smaller genomes and may be able to sustain extensive but subtle selection. But the same kinds of constraints on variation across the genome are also seen in vertebrates, who are comparatively slow reproducers. This suggests but does not prove that there may be common mechanisms not wholly related to high or low reproductive potential.

One possible way to account for ubiquitous selection is that gametes even in slow reproducers are generated in profusion and may be screened even before fertilization (prezygotic selection), at least for genes related to cell division and normal metabolism, which gametes must do successfully.

Another possibility is that selection is not even approximately gradual the way Darwin suggested; the idea of selection against a variant because of a single trait is probably also too simple. Because of pleiotropy, selection on several different functions of a gene could constrain its variation so that there may be many ways to select

against a given variant. Furthermore, because of *epistasis*, or the functional interaction among genes, any given instance of selection may simultaneously affect many genes. Individual variants might be harmless in themselves but become harmful in rare circumstances or in combinations with many other variants in the genome, so that a selective event removes many of them at a time. Theory for epistatic effects of this kind of process has been advanced (e.g., Eyre-Walker et al. 2002; Keightley and Hill 1983) but does not seem able to account adequately for the amount of constrained variation seen across genomes.

We assume that mutations occur more or less randomly with respect to function and that selection can only purge them after they occur and in comparison with contemporary alleles in the population. Unless understanding of evolution is missing something basic, there is no known way for mutations in specific regions of a gene to be prevented from happening. For that to be possible, parts of codons would have to be specifically protected from mutation; perhaps this could somehow be achieved by DNA packaging in areas to be transcribed, but it is not something generally known as yet.

No matter what, it seems clear that for most variation of this kind the average selection coefficient has to be very small, probably so small as to be immeasurable even in large samples at any given time. In practice, most of these variants are probably essentially neutral most of the time. How to test some of these notions about genetic load is unclear, and the issue this raises is whether or to what extent there is something else important and widespread going on in life besides selection that we may not yet have stumbled upon.

Part of the problem applies generally to evolution. It is difficult to understand things that happen so slowly that we can hardly perceive them. In a famous Amazonian travelogue, H. M. Tomlinson noted that a dense tropical rain forest seemed eerily silent and almost lifeless as one moved through the dark floor under the remotely high canopy. But if time were accelerated, we would have seen the movements in that “war of phantoms . . . we should have seen . . . the greater trees running upwards to starve the weak of light and food, and heard the continuous collapse of the failures, and have seen the lianas writhing and constricting, manifestly like serpents, throttling and eating their hosts” (Tomlinson 1912).

IF DNA IS A BOOK, HOW DOES IT GET READ?

The metaphors are overstated, sometimes misleadingly so, but DNA is widely characterized as the Book of Life—a repository of the information of biology and the program from which organisms are computed. But this is very different from a book a human would write for the same reason that an organism is not truly a machine. A book or machine is assembled with a purpose in mind, from parts derived from all sorts of other knowledge. An organism is built from within, and its DNA must function that way.

We have described many ways that DNA fills a number of physiological roles in the cell. Only some of these are related to protein coding itself. Looking at the genome from without, as we have done, shows its proliferating, diversifying, modular nature. Remarkably, one might say that there are too many modular units to make any sense (biological or otherwise): any stretch of DNA contains nothing but sequences identical to codons. Every three base pairs of every enhancer has a codon sequence. Similarly, because of the tolerance of TFs for variation in their RE

sequences, and the fact that they are short, the genome is also nearly saturated with possible REs. Even a given 10-mer (that is, stretch of 10 consecutive nucleotides) occurs every 4^{10} nucleotides on average, about 1 Mb or about 3,000 times in the human genome. In fact, the critical regions of REs are often shorter than 10 base pairs. This also means that any stretch of sequence is simultaneously saturated with wholly overlapping codon and enhancer sequences.

Even if we only consider exons, the same codons are repeated millions of times in the coding parts of the genome, in no detectable order from the point of view of the sequence itself. Likewise for RE sequences. This is not an illusion because, for example, it is the functionally open nature of polypeptides, and hence of codons, that makes the diversity of life possible. As a consequence, one could view DNA sequence naively and say that it would be laughable to think it actually contained any information at all! Yet, these short otherwise essentially random motifs carry the *most* important biological information in the genome.

Thus, if DNA is so densely packed with information, how can an organism open and read the book? We don't have a very good answer to this, other than to say that it happened incrementally over eons of time. Also, because of cellular continuity from one generation to the next, the apparatus for reading the genome properly may always be changing as cells do what they have to do, but is never entirely missing. How scientists learned to read genomes (to the extent that we can) is more easily explained.

SCIENTISTS LEARNING TO READ

By what authorization do we turn Figure 4-2 into Figure 4-6B? We do not learn about DNA just by examining DNA. For example, it is not possible even to identify all the genes in a given genomic DNA sequence. There is no known signature by which all genes can be identified, and indeed it is not even clear what should constitute a "gene." If our understanding of the sloppy, contingent nature of evolution is accurate, there undoubtedly is no single signature or definition.

Instead, we have let organisms (sometimes enslaved by us experimentally) teach us their workings step by step. But it is important to understand the degree to which what we ask of organisms affects what we see and how we interpret what we see. Thus, our knowledge is built on observation filtered through and directed by theory. The theoretical understanding that DNA codes for mRNA that codes for proteins was discovered through experiments, but the information now can be applied to entirely new sequences.

For example, it is possible to separate mRNA from DNA in cells. We can then use mRNA sequences to determine the sequence of the corresponding exonic DNA sequences in the genome. The exons are separated by introns, which are excised when a gene is transcribed and thus are not part of the messenger RNA, but computer-based sequence alignments can be done to find the sections of genomic DNA that code for the successive exons that assembled to form the RNA. This shows us where the gene is in the genome, and, since only exonic sequence is included in mRNA, we can identify exon-intron boundaries by finding how an mRNA sequence is interrupted in the genome. This in turn makes it possible to identify the sequence characteristics of exon-intron boundaries and transcription start/stop sequences in the genome. (We can also work backward to DNA by first identifying the amino acid sequence of a protein, which can be done chemically).

Of course, the Central Dogma of DNA $\rightarrow$ RNA $\rightarrow$ protein is only partly correct. DNA does many other things. Once we knew that genes were selectively expressed in ways affected by DNA itself, we could identify the sequence elements (REs) that are responsible. mRNAs are informative of gene regulation because genes used in the same tissue may share some regulatory mechanisms. For example, once we find their genomic location, we can search the chromosomal regions of coexpressed genes to see whether they share any sequence elements in their flanking regulatory regions. Genes and their flanking regions can also be inserted in expression systems (letting bacteria, flies, or even mice express our test region); systematic deletions can identify sequence elements necessary for particular expression patterns. Then, because our theory is that response elements are bound by transcription factor proteins, we can use these elements experimentally to “trap” proteins that are bound to them in cultured cells and use this in turn to identify the transcription factor genes themselves.

There are developing databases of known RE sequences (pdap1.trc.rwcp.or.jp/research/db/TFSEARCH.html). It is clear that simple catalogs of REs cannot suffice to identify or characterize regulation from bioinformatic (sequence data) approaches alone. Some success with such computerized searches has been reported, however, and candidate sequences have been experimentally confirmed (e.g., Bonifer 2000; Bussemaker et al. 2000; Chiu et al. 2002; Dermitzakis and Clark 2002; Dermitzakis et al. in press; DeSilva et al. 2002; Hardison 2000; 2001; Michelson 2002; Pennacchio and Rubin 2001; Spellman and Rubin 2002).

The theory of evolution instructs that related organisms share similar traits because they share common ancestry. The same is true of genes. Through our understanding of evolution and the nature of DNA and mutation, the amount of sequence variation in regions that are functionally important to selection will be constrained, relative to the amount of drift that occurs in less stringently functional elements. This allows us to identify homologous regions of genomes of one species, to find regions that resemble the sequence of known genes in other related species. Similarly, because of our theory that genes arise by duplication, we can find new genes by looking for different sequences within the same genome that are similar to those of genes we have identified, and we can guess—often rightly—about what types of cells the new genes will be expressed in from the expression pattern of the known gene and the notion that related genes have related function.

We can align sequences from different species, first anchoring the alignments by known regions such as homologous genes. Then, we can search the aligned regions for conserved sequence shared between them. Since theory suggests that highly conserved sequence is likely to have some function, an analysis of the candidate sequence can show us what that function is. This approach can be used to identify at least some of the control elements that are conserved between the tested species (Schwartz et al. 2000). Alignments can identify previously known elements and other good candidates. Phylogenetic methods may be useful (e.g., Blanchette et al. 2002; Blanchette and Tompa 2002; Chiu, Amemiya et al. 2002; Dermitzakis and Clark 2002; Shashikant et al. 1998) in the search for conserved REs in subsets of related species that share a particular trait (e.g., type of teeth or limb or leaf structure).

We read the book of life by applying external information—viewing life from outside, whereas organisms have to do it all from within. They were also “designed” from within, that is, through evolution by phenotype, bit by bit over billions of years.

It is easy to see why we have trouble stereotyping the function of sequence too tightly, or thinking too deterministically about how it all works, or that it must be as neatly organized and regular as something we might design from the outside. Nonetheless, the various methods described have allowed scientists to peer rather successfully into the private business of every species we choose to look at.

CHICKENS AND EGGS—LITERALLY

Because this all occurs in *biological* context, elements that on the surface do not appear patterned or modular or meaningfully repeated are in fact densely filled with absolutely vital information. The trick is not the sequence alone but in its context along the chromosome in living cells because that is what determines what the sequence means.

So then, what is primary? DNA or its context—the chicken or the egg? Life may not exist without DNA, but DNA by itself would just sit there, inert. Nor does DNA control the expression of proteins or their structure. That only happens in the natural cellular context of proteins and other elements already in a cell (which is why viruses are not “alive” by themselves).

TFs themselves are proteins, which means that, if they are actively regulating genes in a cell, their own regulating mechanisms must be active in the same cells. This shows the rolling circle of developmental regulation because these TFs must thus be activated by other TFs, and so on. The mechanism that activates each TF in a cell must either have been active in that cell’s mitotic ancestors or must be stimulated in the new cell by developmental signals coming into the cell from outside.

Biologists give primacy to DNA and say that the egg only makes a chicken because it starts out with DNA. But this reasoning, that the proteins that make all this possible are encoded in the DNA, is inextricably circular. This chain of events goes back to the organism’s first cell(s), and hence to its parent, and . . . as fleas on fleas on fleas on dogs . . . ad infinitum. To break the circle, and settle the ultimate chicken-or-egg argument, we would have to go back 4 billion years—to the hypothesized RNA world. If the original “soup” comprised chemical reactions between RNA and amino acids, then there was neither chicken nor egg: the code developed by the addition of function to elements already interacting. Neither came first in time or importance. Nor has one been able to function without the continued presence of the other for the consecutive billions of years of life on Earth.

It is basically for our own *practical* reasons, to provide a research and interpretive strategy, that today we assign evolutionary primacy to DNA: the circle of life continually rolls through a DNA component, and that codes for the proteins that do everything else (including regulate DNA). This means that DNA is the entry point for new heritable variation, and to that extent stream of evolution is then mediated through this process. But that perspective on life is a *decision* of the culture that is science, not the only way we might view life, if we chose to feature other of its aspects. As discussed in Chapter 3, the flow of genetic information it is not even the exclusive element of inheritance.

CONCLUSION

We have seen some of the many ways in which what is conventionally and conveniently called information is contained within the coding sequence of DNA and how

this is used to produce protein and serve other functions. There are probably many other ways in which information is stored in nucleic acid sequence, some perhaps quite vital but simply not yet known. Although the information metaphor is inexact or even misleading, understood in their contexts, the coding functions of DNA are certainly fundamental to life, and we will see the importance of this as well as the modular nature of genetic information throughout this book. In particular, we can now look at some of the basic elements of the way genotypes are related to phenotypes.

Chapter 5

Genotypes and Phenotypes

We have seen that the genome provides a highly elaborate code, largely based on its nucleotide sequence. We have also seen how the code works, how it evolves, and the indirect nature of the relationship between the code and phenotypes in an organism. In Chapter 3, we explained why having a genetic theory of evolution does not automatically provide a satisfactory phenogenetic theory, that is, a theory that explains the evolution of phenotypes. Here, we will try to give a sense of what is known about the internal connections between the genome and phenotypes. How are genes used, and how precise is the “code”? Does it actually code for phenotypes? In what senses can we read the organism from its code? How do we determine these relationships?

Phenogenetic relationships are the product of nature’s single *de facto* criterion, that of the screen of evolutionary survival. Science looks at these relationships through different filters, however. In addition to wanting to understand evolution, we may wish to understand the genetic basis of disease or develop better agricultural products. These are not necessarily “natural” measures of a trait relative to its evolutionary origins. We may or may not be able to use our chosen filters to infer how phenogenetic relationships evolved, but we have been able to further our understanding of the relationship of the genetic code to phenotype.

THE RELATIONSHIP BETWEEN GENOTYPES AND PHENOTYPES

A number of important aspects of the relationship between genotypes and phenotypes relate to both the biology of the relationship and the problems we face in trying to identify and understand the underlying genes (Weiss 2003c). Some of these relationships are shown in Figure 5-1. The terms and components shown in this figure are those commonly used in the study of human disease, where we try to infer genetic causation. However, the same issues apply to traits studied in nature or experimentally (after all, humans are studied in nature). The difference is that in experiments we can control some of the variables better or more explicitly. However, the overall idea is to identify genes associated in some way (mechanistically or in terms of variation) with a trait of interest to us.

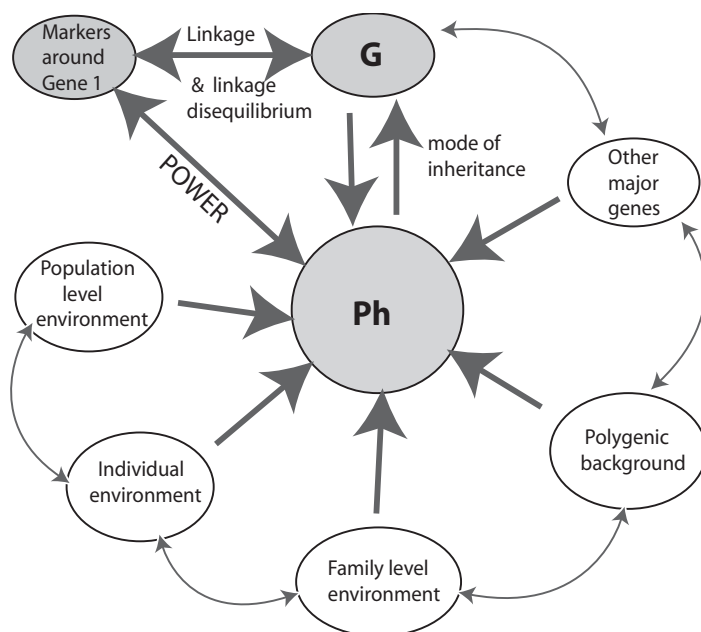


Figure 5-1. Gene mapping and Genotype-Phenotype relationships. **G** designates a gene that is assumed to have major effect on the phenotype, **Ph**. Other factors are also involved as indicated. To find the chromosomal location, or to “map,” **G**, we rely on a sufficient predictive power from **Ph** to **G**, as well as a statistical association between nearby genetic markers and **Ph** (see text). After (Weiss and Terwilliger 2000).

The primary relationship of interest is shown in the vertical arrow between the phenotype **Ph** that we actually identify or measure and some important genotype **G** that we wish to find. Associated with this are many other genes (the two ovals on the right) that contribute individually, in small interacting sets, or as a “polygenic” aggregate of unspecified numbers of individually small contributions.

Environmental factors—known or unknown—contribute in various ways. We often model them as an aggregate with some tractable behavior, which can be separated into various subaggregates: the effects of factors shared by siblings or other specified sets of relatives, factors unique to the individual, and cultural factors shared by the whole population.

We can refer to the number of genes, their interaction, and perhaps frequency relationships of their alleles in a population or species as the *genetic architecture* of a trait. An important point to make right away is that the genetic architecture is not inherent in the genome. Genes do a lot, but not everything, and the same allele or gene may do different things in different individuals, even within the same species; this is another way of expressing the many-to-many **G-Ph** relationship discussed in Chapter 3.

PENETRANCE CONCEPTS FOR ALLELES AND GENOTYPES

The effects of genes on an organism can be assessed in many ways, although typically we are interested in some particular trait, such as the level of expression of a

specific protein like insulin, or in more indirect genetic traits, such as wool quality, blood pressure, or flower morphology. We can ask the two genetic questions: (1) what is the mechanism of action of the genes? and (2) how is variation in the genes related to variation in the trait? Much of this book will concern the former question, but our interest at present is in questions of the second type. We can define the probability that a particular phenotype will be found in an individual with a particular genotype as the *penetrance* of that genotype.

For qualitative traits (ones with a discrete number of possible states, like presence/absence or the number of petals on a flower), the penetrance is the probability that a specified state will be found in an individual with the specified genotype. For quantitative traits, like blood pressure, which can take on an infinite number of possible values across some range, the penetrance of a genotype (sometimes called the expressivity) is the probability of a given value of the trait in bearers of the genotype. Actually, this is more properly referred to as a probability density, but since the probability of any *exact* trait value—like being 2.000001 meters tall—is zero, in practice we divide the possible values into intervals of a certain width, such as being 1.95 to 1.99 meters, 2.00 to 2.04 meters, etc.

Penetrance is inherently a *statistical* concept that is tied to a particular sample. Only in the limit (extremes) is a trait never or always observed (penetrance equal to 0 or 1, respectively). The extent to which this statistical nature is due to observational error and finite samples or to inherent chance factors in the actual **G-Ph** relationship is a major question with no single (or simple) answer, and clearly varies from situation to situation. The penetrance distributions provide the *spectrum* of genetic effects on the phenotype, at least relative to the alleles and genotypes being considered.

Gregor Mendel dealt with selected traits in crosses of hybrid pea plant strains. He deliberately chose traits and strains in which each parent strain was qualitatively different, meaning there were only two causal factors, one in each parent (one with strong and the other with weak effects). Only decades later was the nature of the factors, alleles at an individual gene, understood. Because of the simple situation he studied, Mendel got the general principles of genetic inheritance right. In fact, much of the 20th century, the century of genetics, was spent working out the details of the theory and designing experiments to understand genes in which there were basically only two alleles, or at least relatively simple relationships like Mendel's. In natural populations, or experimental organisms, the standard way of viewing things is to think of one state as more common and new states that arise as usually harmful; thus the alleles came to be called normal, or “wild type,” and “mutant.” These terms are still with us, although they somewhat misportray the situation outside the laboratory, where things are more variable and more complicated. Mendel might not have been able to reach the conclusions he did, if he had had to work with wild peas (Weiss 2002b).

For complex traits, like weight, stature, or blood pressure, the possible role of genes could until rather recently only be estimated crudely and indirectly. Relationships in trait values could be observed among individuals in specifically structured samples, such as sets of parents and offspring, controlled crosses among inbred strains, twins, or parents and offspring raised under conditions of artificial selection. This observed relationship was compared with the expected relationship that would arise under theoretical assumptions about how genes would affect the trait if they contributed quantitatively to it *in aggregate*. The genetic effects of many individu-

ally unidentified genes could be considered in aggregate, for example, as if the alleles contributed additively to the trait. In experimental situations, selection could be done by the investigator to follow the changes in the nature and amount of variation over many generations and then could be related to the model of underlying aggregate effects.

As long as one was satisfied to study effects and not specific genes—which was all that was possible for most of the 20th century anyway—this largely empirical approach to genetics was an effective way to study quantitative traits. Essentially, the underlying assumption, was that the many genes were individually inherited like Mendel’s pea genes, and that the alleles at each gene, although not identifiable, were related to the trait in the same *kind* of way. Artificial selection, either on the farm or in the laboratory, is more deliberate and systematic than selection in nature, but it does mirror the natural way that selection works on phenotypes in whole organisms.

Advances in genetic technology in the past quarter of the 20th century made it possible to identify genes directly and thus to identify their alleles and assess the association of their observable variation with traits of interest. It is now fairly straightforward to do this for simple traits, in which genotypes have high penetrance—that is, are directly associated with the measured phenotype. However, we have also learned how favorably artificial Mendel’s experiments were. In general, there are multiple alleles at a typical locus in natural populations (including Mendel’s traits in wild peas).

For a diploid species, multiple alleles means many possible genotypes. A locus with m alleles has $m(m + 1)/2$ possible genotypes. For example, 2 alleles can form 3 genotypes (**AA**, **Aa**, and **aa**), 3 alleles 6 (**AA**, **Aa**, **aa**, **Aα**, **aα**, and **αα**), and 100 alleles can produce 5,050 different genotypes. Unlike Mendel’s and many experimental situations, natural alleles are not equally frequent. Based on Hardy-Weinberg proportions, the genotypes will have comparable variation in frequency. If the three allele frequencies at a locus are 0.7, 0.2, and 0.1, the six genotypes will have frequencies 0.49, 0.28, 0.04, 0.14, 0.04, and 0.01.

Many possible genotypes are too rare to be found in a given population, and many may never be found at all (for example, if an allele has a frequency of 0.001—not unusual for a rare allele—then its homozygote will have a frequency of 0.000001, or one per 1,000,000 people). In addition, not all alleles that exist at a locus in a species will necessarily exist in any given population, and rare alleles usually will be found to be geographically localized near the area where the mutation that generated them originally occurred. Thus, not all possible genotypes, and hence phenotypes, will occur to be screened by evolution, certainly not in every population of the species. In addition, unlike Mendel’s situation, even at a single locus, there is often a quantitative relationship between genotype and phenotype rather than a simple mendelian penetrance probability. An allele that is potentially favorable only in some genotypes may never get the chance to shine.

As we discussed in Chapter 3, new alleles arise as mutations mainly unique at the DNA level, at least in terms of the variants in their nearby chromosomal background. Because of the effects of genetic drift, migration, and natural selection, the distribution and even the identity of many alleles will vary among populations, and the penetrance may be affected by variation at *other* loci (often referred to as the genetic “background” relative to the gene(s) in question).

This complex of variation is consistent with the ability of experimental breeding systems, as found in agriculture and model systems like *Drosophila*, to respond to

selection in the laboratory and also provides the fuel for natural selection to act on almost any trait. However, the amount of variation also generally means a relatively loose or nondeterministic relationship between genotypes and phenotypes. Over time, the variation can accumulate and change, especially between populations as they diverge into separate species or for other reasons cease to exchange mates.

One implication of quantitative genetic models, which is routinely confirmed by studies that look at the genes involved in complex traits, is that there is extensive phenogenetic equivalence, resulting in different genes in different individuals or populations being associated with the same phenotype (e.g., blood pressure). This provides fuel for phenogenetic drift. When we address the question as to how various biological traits are produced genetically, it may be that the answer differs among species even when they have shared the same trait since they shared a common ancestor.

HOW WE KNOW: MAPPING AND INFERENCE ISSUES

The elusive and statistical nature of penetrance relationships and the potential allelic and genotypic complexity even of individual loci in nature should be kept in mind when we think about genetic causation. This is especially so when a trait of interest is many steps removed from the direct action of a specific gene product. The safest, and most general way to conceive of the role of genes in life is to assume a priori that any gene or system of genes that we may wish to consider may be associated in some way with any phenotype. If natural selection can't effectively screen on genotypes via a given phenotype, we can't expect to predict the genotype from the phenotype (Weiss and Buchanan 2003).

When we attempt to or identify genes associated with a particular trait, for each genotype in the system, **G**, we can express penetrance as the probability, $\Pr(\mathbf{Ph}|\mathbf{G})$ that some phenotype, **Ph**, will be found. Of course, a system of genes may empirically be found *not* to be related to the phenotype; the evidence would be that the penetrance functions were identical (that is, to statistical accuracy) for all genotypes under consideration. The red and green visual pigment (*opsin*) genes, for example, have to do with color vision, but all genotypes in those genes probably are associated with identical distributions of body weight or insulin levels. However, we have many times been surprised at the degree of pleiotropy in natural organisms, a major problem in contemporary biology. Apolipoprotein E was originally studied for its lipid-transporting effects, but it turns out to have important neurological function; its alleles may even be related to the ability to recover from childhood head trauma, it is associated with Alzheimer's disease, and it seems to be involved in the embryological development of the brain.

These issues are important in both a "forward" causal and a "reverse" inferential sense. In the forward sense, they relate to the way in which individual genes affect traits. This reflects the biology of an individual and the products of evolution. In particular, we generally use a model that has genes "causing" traits, following the downward $\mathbf{G} \rightarrow \mathbf{Ph}$ path in Figure 5-1, based on the role of genes as protein codes. To evaluate the effects or associations of genotypes with phenotypes, we should be careful to consider the range of environments in which a genotype might find itself, including the background genotypes. This is impossible, actually, because we have no way to know that range except approximately. But it is an important point of caution (Schlichting and Pigliucci 1998).

The “reverse” approach is also important; this has to do with how we *infer* the role of genes, or identify genes, associated with a trait we wish to understand, using natural variation to map the chromosomal location of genes that affect the trait. What we would like to do is to go from **Ph** directly to **G**, but that requires that we first *map*, or find the gene in its chromosomal location.

To map a gene, we try to take an indirect route through the correlation arrow to *genetic markers* in Figure 5-1. These are variable sites all along the genome, each with known chromosomal location. We genotype these markers in all sampled individuals. If the model (the main causation we hypothesize) is reasonably close to being true (*in the sample we decided to collect*), then at markers chromosomally near to **G** there may be alleles in LD with causal alleles at **G**. If so, then when we find such a marker allele in an individual we are also likely to find a causal allele at **G** in the same individual. That is, we find association between the marker and the phenotype **Ph**; because we know the location of the marker, we now know chromosomally where to look for **G**.

There can be many a slip twixt **Ph** and **G** in this situation, ranging from statistical sampling variation, to our choice of sample, to our actual causal guess. Biologists often assume something about the forward sense, how genotypes predict phenotypes, but then use it in the reverse sense, assuming how phenotypes predict genotypes; too often, that “something” is highly oversimplified. The challenge of genetic inference is a subject beyond the scope of this book and dealt with in many places (Millikan 2002; Sham 1997; Weiss and Buchanan 2003; Weiss and Terwilliger 2000). However, it is appropriate to make a few comments here because this concept affects how we will interpret the evidence that exists for traits that we will consider.

Mapping Qualitative and Quantitative Traits

Mapping is an empirical approach to finding genes, that is, it is a technique that makes use of naturally existing or experimentally arranged phenotypic variation to find associated genetic variation, rather than the direct experimental manipulation of known genes. Traits can be treated as qualitative or quantitative, but they are approached in logically similar ways.

In mapping a qualitative trait, our statistical analysis searches for associations between the presence, or probability of presence, of a trait quality with specific underlying alleles or genotypes. The idea here is classically mendelian—directly related to Mendel’s experiments with peas. We use genome-spanning markers to find the gene involved. If more than one gene can generate a qualitative outcome, things become more complicated but are conceptually related.

In mapping a quantitative trait, however, the idea is to find mendelian segregation, but of alleles or genotypes that contribute to a quantitative trait measure. Usually, this means finding genotype-specific shifts in the mean or variance of the trait measure. This is a dose-like concept, in which one gene does not make or break the trait. A locus whose allelic variation in a given sample contributes to statistically meaningful fractions of the observed phenotypic variance is called a Quantitative Trait Locus (QTL). It does not “cause” the trait in the general way that we conceive of a particular allele causing wrinkled vs. smooth peas.

What Mapping Genes Tells Us and Doesn’t Tell Us

When we do mapping studies, the idea is to find genes that contribute quantitatively or qualitatively (if probabilistically) to a trait. This is an important objective

because, without identifying the genes, we can't understand what they do. However, at this stage of knowledge, much of what we do remains in the black-box category.

In many if not most instances, especially for complex or quantitative traits, variation in genes identifiable by mapping methods accounts for only a fraction, often a small fraction, of the apparently genetic variation in the population. Referring to Figure 5-1, we search for **G**, but we know that there may be other genes (again, the ovals on the right) with identifiable effects. That is, most of the phenotypic correlations among relatives remain unexplained after we statistically remove the effects of the already identified genes. As shown in Figure 5-1, this may be due to the effects of shared environmental factors of various kinds; however, even when genes are involved, the residual genetic correlation is treated perforce as a polygenic aggregate in the figure. When many genes vary but each contributes only a small amount to variation in a trait, there is almost inevitably a large amount of phenogenetic equivalence in the trait, and the concept of genetic causation becomes almost inherently elusive if not actually ephemeral (not to be exactly repeated in the next generation of organisms).

What mapping studies do is to pick out the genes whose alleles happen to be playing a relatively important role *in our particular sample*. We sometimes say that, although many factors are at work, we are identifying the "rate limiting" ones. In a metabolic pathway, one substrate is converted by a particular enzyme into the substrate for another enzyme and so on, until the original materials are converted into the phenotype being measured. If in our sample, alleles at one of these genes reduce the substrate concentration for subsequent steps, that allele may have a greater effect on the phenotype than alleles at other genes in the pathway. When this is replicated among samples, populations, or species, it may be that the gene in question really is more important in the normal range of variation than the others (or, perhaps, the least buffered by alternative pathways).

Such a rate-limiting step may appear as a QTL in a given sample; however, we may have a misplaced sense of inherency of genetic effect. If the same alleles are missing or alleles at other genes are newly present in subsequent samples, or environments have changed, the original gene may not have the same kind of effect. For these reasons, the generic term "genomic background" is invoked; that is, one strain of a model system responds differently to a particular genetic manipulation than do others (or families differ in the genes responsible for a trait, as detected by mapping studies).

Gene mapping can be done in various ways not described here, but the idea is generally the same—to find genes not already known to affect a trait. But mapping is an imperfect tool for identifying all such genes or characterizing the genetic architecture.

ORGANIZATIONAL ASPECTS OF THE GENOME: MODULES, MODULES

A major theme of this book is that the genomes of complex organisms essentially comprise a huge collection of scattered, sequence-nested, gene families and other repeat elements (e.g., Edgell et al. 1997). This modularity is closely connected to the relationship between phenotypes and genotypes.

PROTEINS FUNCTION IN MODULAR WAYS

Proteins are molecules whose shape, charge, miscibility, and so on determine their interactions with other molecules with which they come in contact. A protein typically interacts with a limited number of the thousands of other compounds in a cell, and these have been screened by evolution so that they are not just chance specificities, although a certain amount of chance determines which interactions happen in a given cell.

Different regions of proteins have specific functions. These regions are often due to distinct physical domains of the protein. Some physical parts of proteins are stable in the neutrally charged lipid membrane of a cell, where the active protein resides. Some regions bind specifically with other compounds to serve enzymatic function (to bring about or speed up a reaction involving those compounds). Still other regions of some proteins interact specifically with another copy of the protein itself to form homodimers or with other (often related) proteins to form heterodimers.

Proteins take a few basic shapes. Some form what are called beta sheets, a flat conformation, whereas others form helical shapes. These depend on many factors, which are shared generically among proteins. For example, sulfide bridges can connect cysteine amino acids in the polypeptide to hold it in a folded conformation. This is a complex subject, the point being that proteins are functionally modular beyond the simple fact that all proteins are basically chains of different combinations of the same set of amino acids.

THE ORIGIN OF PARTS OF GENES: MODULAR CODING FOR PROTEIN DOMAINS

We saw in Chapter 4 that much of the repeated structure of the genome is due to the duplication and rearrangement of whole genes or even groups of genes.

The evolutionary origin of genes is as interesting as the prior evolution of coding and codons. New genes provide opportunity for new function. Genes as whole units appear largely to evolve by duplication, but what is the source of their internal structure? This is important to gene function and its evolution. A few important facts help us here. First, genes in eukaryotes typically contain coding exons interrupted by noncoding introns; second, exons often correspond to the functional domains of the coded proteins. In addition, exons with very similar sequence can be found as parts of very different genes.

This led to the idea that the exons were functional or evolutionary units in their own right and that genes have arisen by the shuffling around of individual exons. There is considerable debate between this exon-shuffling “introns early” view and the contrary introns-inserted “introns late” view of the origin of genes. The latter is plausible because bacteria do not possess introns; if introns arose early, how did bacteria lose theirs? One way to examine the question is to see whether exons tend to be broken at natural codon boundaries (that is, are multiples of 3 base pairs in length) rather than having their terminal codons interrupted by introns. A statistical bias in favor of unbroken coding units has been found (Gilbert et al. 1999), favoring the exon-shuffling view. However, it may be that not all genes have arisen by exon shuffling and that only some, such as those associated with some early aspects of multicellularity, may owe their origins to this process (Patthy 1999).

The idea that exons can be shuffled around, and the ubiquitous evidence for duplication events in the genome, is consistent with important aspects of not just

modular but repetitive structures within genes. Many examples are known in which the phenotype directly reflects repetitive internal gene structure.

The important animal structural material, collagen, is built of long chains of inter-connected collagen subunit proteins. Collagens have repeat amino acid motifs that allow the molecule to form long fibers. Apolipoproteins have repeated cyclic domains that allow the protein to wrap around and hence provide an internal core that sequesters lipid molecules so that they can be transported in the aqueous environment of the bloodstream. Calcium-binding proteins provide another example of internal repeats (Kawasaki and Weiss 2003).

The source of the repeated coding sequences clearly seems to be duplication. Important evidence for this is that sometimes the copies have nested sequence structure, reflecting episodic duplication events and subsequent mutations. Thus a general initial core sequence **TCGGACGAGGC** can be duplicated as **TCGGACGAGGCTCGGACGAGGC** (using font to indicate different copies), then experience mutation to **TCGGACGAGGCTCGGACGAG_tC**, and later duplicate to **TCGGACGAGGCTCGGACGAG_tCTCGGACGAGGCTCGGACGAG_tC**. These various ancestral and descendant elements may be found in the same gene and/or in related genes, so that the history of the events can be reconstructed.

The duplication example here is hypothetical, but the core sequence used to illustrate it has been suggested as an ancestral core apolipoprotein gene sequence (Luo et al. 1986; Rajavashisth et al. 1985). The example is hypothetical because these genes are old, and the actual variation that has accumulated has been complex and considerable. The reason is that, for genes with internal repeat motifs, it is the protein properties of the repeat domains and not their codons or specific amino acids that count. Because there is codon redundancy, and some amino acids are chemically similar to others, there can be substitutions in the DNA sequence that are compatible with keeping a similar overall protein structure (that is, phenogenetic drift).

THE ORIGIN OF GENES: GENE FAMILIES AND WHAT THEY DO

Duplication events can produce copies of entire genes and their regulatory regions. Such events, although unlikely in any given meiosis, are a characteristic of life and have occurred regularly over evolutionary time (Ohno 1970). Through gene duplication, a single ancestral gene can spawn many descendant genes, creating a gene family. The related genes may remain in tandem array on the same chromosome or may be scattered around the genome or both. Examples are shown in Figure 5-2. Initially, if the required regulatory sequences and coding domains are copied intact, the organism has redundant genes with the same function. Subsequently, mutation can inactivate the gene (producing a pseudogene, detectable by its sequence relationship to the ancestral gene until further mutation eradicates the similarity). Or, mutation can modify the function of one or both genes, through the usual evolutionary processes of chance and selection.

Like the duplication of domains within a gene (or of dispersed repeat elements), nested sequence structure can arise as mutations occur in different members of a gene family, which themselves may then become the parents of new genes through subsequent duplication. The nested sequence relationships can be used to reconstruct the history of these events.

Human Globin Genes

Consensus	MXLSXXDRTXVXAXWXKXGXXAGXYGTEALERXFLSFPXTKTYFPHFDLSPGSAQVRAHGXXKVDALXXA
	10 20 30 40 50 60 70
Theta Globin	.A.SAEDRALVRAL.K.LGSNMGVYTT.A...T..AFPA....S.L..SP..S.VRA..Q..AD.LSL.
Zeta Globin	.S.TKTERTIIIVSM.A.ISTQADTIGT.T..L..SHPQ....P.F..HP..A.LRA..S..VA.VGD.
Alpha Globin	.V.SPADKTNVKAA.G.VGAHAGEYGA.A..M..SFPT....P.F..SH..A.VKG..K..AD.LTN.

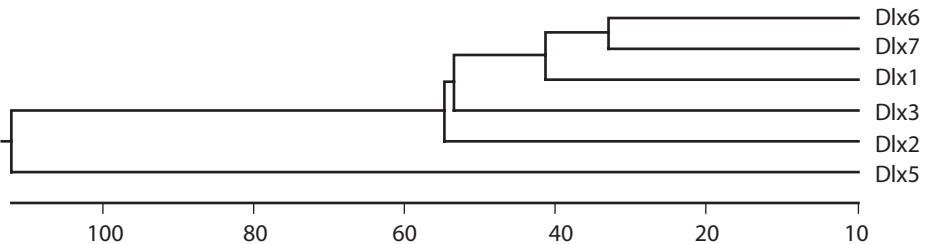
Consensus	VXXXDDXPXALSALSXLHAXLXLRVDPVNFKLLSHCLLVTLAAHXPADFTPAXHASLKDKFLSXSVSVLTSKYR
	80 90 100 110 120 130 140
Theta Globin	.ERL..LPH...A..H...CQ....AS.Q..G.....RHY.GD.SPALQ.SL...SH.ISA.VSE..
Zeta Globin	.KSI..IGG...K..E...YI.....VN.K..S.....ARF.AD.TAEAH.AW....SV.SSV.TEK..
A Alpha Globin	H.AV..MPN...A..D...HK.....VN.K..S.....AHL.AE.TPAVH.SL...AS.STV.TSK..

Consensus MTG-FDSLVLXSLX--ISSS-VFH---PP-S---ASP-G-----L--PGXXSAPD-SYS--SSYG---HPXG-----Y-H-GS---HAS-----YSPKSQY--G-
Dlx1 MT--MTMPESLNS-PVSGKAVFMEFGPPNQMSPPSMHGHYSMH--CLH-SAGHSQPDGAYSSASSFS---RPLG-----YPYVNSVSSHASSP---YISSVQSYGPS
Dlx2 MTGVFDSLVDAMHSTQITASSTYHQHQQPPSGAGAGPGGNSNSSSSSLHKKPQESPTLPVSTATDSSYYTNQHPAGGGGGGASPYAHMGSYQYHASGLNNVSYSAKSSYDLGY
Dlx3 MTGVFDRRVPSIRS-----GDFQAP-----PPTS-----
Dlx4 -----
Dlx5 MSGSFDRKLSSILT-DISSSLSCHAGSKDSPTLPESTVTD-----LGYYAPQHDYYSQQPYG---QTVN-----PYTYHHQPNLNLGLAGTG-AYSPKSEYTVGG
Dlx6 ---MTSLPCPLPD-RGASNVVFPDLAPALSVAAYPLG-----LS-PGTAASPDLSYS--QSYG---HPRS-----YSHPGPATPGDS-----YLPQQOQLVAP

Consensus AAAAQSGRA---ESP---DEPESEVLEXGEVVRNGK--GKKVR**KPRTIYSSLQLAALNRRFPQQTQYLALPERAELAAASGLTQTQVKIWFQNKRSKFKLLKQGSXPLEQSP**
Dlx1 ASLAQSQL--EDPG--ADSEKSTVVEGGEVRFNGK--GKKIR**KPRTIYSSLQLAALNRRFPQQTQYLALPERAELAAASGLTQTQVKIWFQNKRSKFKLLKQGSXPLEQSP**
Dlx2 TAAITSYAPYGTSSSPV---NNEPDKEDLEPEIRIVNGK--PKKVR**KPRTIYSSFLAALQRRFPQTQYLALPERAELAAASGLTQTQVKIWFQNKRSKFKLLKQGSXPLEQSP**
Dlx3 AAMHHPSQ--ESP---TLPESSATDSDYSPAG-----AAPHG-YCSPTSASYKALNPYQYTHGVNGSAAG-----YPAKAYADYGYHYPHYQYG
Dlx4 -AAQTRG--DDT---DQKTTVIENGELRFNGK--GKKIR**KPRTIYSSLQLAALNRRFPQQTQYLALPERAELAAASGLTQTQVKIWFQNKRSKFKLLKQGSXPLEQSP**
Dlx5 SYRQYGYR---EQPLAQDPVSVKEPEAEVRMVNGK--FKKVR**KPRTIYSSYQLAALQRRFPQQAQYLALPERAELAAQGLTQTQVKIWFQNKRSKFKLLKXNGEVPLEHSP**
Dlx6 SQPFHRA---EHPQ---ELEAESEKLALSVPSSQQSLTRKLR**KPRTIYSSLQLAALNRRFPQQTQYLALPERAELAAQGLTQTQVKIWFQNKRSKFKLLKQGSXPLEQSP**

Consensus GASXSALSAS SPA-LPASWD-----SASGKSSXPSXSXYAS--FL-----SWYHSAS-----QDAMQQPQLM-----
Dlx1 LANGRALSAGSP-VPFGWNP-----NSSSGKSGSGSAGSYVPS--YT---SWYPSAH-----QDAMQQPQLM
Dlx2 GASASPPCASPPVSAPASWDPGAPQRMAGGGPGSGGGAGSGSSPSSAASA-FLGN--YPYHQASGSASHLQATAPLHLPSQTPQAHHHHHHHHAGGAPVSAGTIF
Dlx3 GAYNRVPSATSQP--AFSWP-----LYREG--FR-----RLSTSPC-----QNASRWPL
Dlx4 LPGSAAALSPPSPA-LPPVWD-----VSASAGVSMPPNSYMPG--YS---HWYSSPH-----QDTMORPQM
Dlx5 NNSDSMACNSPPS--PALWDTSSHS-----TPAPARNPLPPPLPYASPNYLDPTNSWYHTQNLGSG-----PHLQQQPPQP---ATLHHASPGPPPPNGAVY
Dlx6 SGRPPSLSPHSPA-LPFIWG-----LPKADTLSSGYDNSH-FG---AWYQHRSS-----PDVLALPQM

B (Homeobox in bold)



C (Nucleotide substitutions X 100)

Figure 5-2. Amino acid sequence comparison of gene family members within a single species' genome. (A) Alignment of human globin proteins (components of hemoglobin); (B) alignment of the mouse *Dlx* gene family, involved in embryological development (head, limbs, etc.); (C) phylogenetic tree of the *Dlx* genes based on degrees of sequence similarity, relationships that reflect the genes' history of descent by gene duplication from a common ancestral gene. Here, the topology of the tree also reflects duplication of two-gene clusters (see text).

In some instances, a cluster of genes duplicates (perhaps as part of a large chromosomal or even genomic duplication event), as in the case of the *Hox* genes. The genes within the original cluster will have accumulated some sequence divergence because they arose from an initial gene duplication event. Subsequently, the copies of these genes in the duplicated clusters will accumulate their own mutations. However, the *orthologous* genes between clusters (those corresponding to the gene in a given position in the original cluster) will be closer in sequence than the *paralogous* genes (those within each cluster, relative to each other). This is because the cluster duplication was subsequent to the individual gene duplication. We infer that this has happened when the orthologous genes are found together in a tree of comparative gene sequences (Figure 5-2C shows this for the six *Dlx* transcription factor genes that arose as a single pair followed by cluster duplication). The pairs of genes *Dlx1/2*, *3/7*, and *5/6* are each located on a separate chromosome, these having arisen by cluster duplication. This is shown by the fact that *Dlx1/7/6* and *Dlx2/3/5* form separate branches on the tree of sequence relationships. These relationships occur because the origin of the first pair was earlier than the cluster duplications. At any stage, additional individual gene duplications can occur in each cluster.

Transcription and other regulatory factor genes evolve just as other classes of genes do. There are numerous TF gene families, whose members are sometimes found in linkage groups, but these may also be distributed as isolated single genes around the genome. The genes diverge in their sequence details and hence in their specific and protein recognition properties. This allows members of a TF class to attain different function after they are produced by gene duplication, again just as other genes do. The result today is a regulatory tool kit with sets of similar tools; for example, related TFs may recognize related but not identical enhancer sequences (REs). We can make analogy with the different socket wrenches, screwdrivers, and so on found in a mechanical tool kit. As discussed in Chapter 4, there is a corresponding littering of the genome with potential response elements (REs) that these regulatory proteins recognize.

Another common way in which modularity works to achieve complex results is by the interaction of gene products. An important class of such interactions allows cells in one part of an organism to affect others. This allows an organism to be more “whole” or to respond to environmental situations in an organized way. Thermal homeostasis, locomotion, and diurnal or seasonal reproductive or growth cycles all involve such interactions. Genes code for molecules that travel to other cells where they complex with molecules in those cells to affect gene expression and hence cell behavior.

Overall, a combinatorial model of gene regulation explains complex cellular behavior, differentiation, and development. That is, in a given situation a specific set of regulatory factors activate or repress an appropriate set of genes. There are tens to hundreds of factors available in complex organisms. The number, location, and/or binding efficiency of the specific sequences that occur in and around a gene provide a kind of quantitative regulatory power, a “tuning knob” for gene expression (Trifonov 1999). Similar quantitative effects might occur by other kinds of copy number variation, such as in redundancy due to multiple crossreacting members of gene families.

This regulatory system also allows for variability and flexibility in evolution. We have mentioned that the genome is saturated with potential RE sequences. Regulatory sequences are short—often only four or five base pairs—and they can arise

in multiple copies in one location and multiple locations in the genome by mutational chance more easily than by the highly coordinated duplication and translocation events that might otherwise be needed to change gene expression.

The members of a gene family usually retain some common aspects of function, or at least the general nature of their function (Marshall et al. 1994). The combination of constraint and variation allows for the evolution of complex structures. Members of a TF family may retain the property of being a TF—activating other genes—but do so in unrelated contexts (e.g., forming vertebrae, limbs, or gut; cell adhesion or immune reaction; regulating ion transport in and out of cells in muscle; or neurotransmission). This allows related genes to contribute to related but not identical structures, such as fore- and hind wings or mouth parts, antenna and legs on insects, anterior and posterior vertebrae, and so on. Whole new mechanisms need not be invented for each such modification.

OTHER ASPECTS OF INFORMATION AND CONTROL

Much more is in an egg than DNA. There are thousands of chemical constituents without which the egg could not live or function, much less duplicate to begin the development process. We tend to view the chemicals in the egg as the products of the parents' biochemistry, driven by *their* genes, and hence genetic at its root. The reason is that the biochemicals (in particular, the proteins that synthesize or use them) are exhausted and replaced (coded for by the DNA, which remains intact). In cell division, DNA but not protein or RNA is *directly* replicated. Still, DNA itself cannot be used or copied without those elements.

However, a number of elements are inherited that are not directly coded in DNA sequence. These include the way in which DNA is packaged and chemically modified. Sequence-specific modification, such as methylation of some CpG sequences, occurs in the parents and differently between males and females. This is known as *imprinting* and affects the expression of genes early in embryonic life (and, in some instances, throughout life: there are human diseases that depend on whether a mutant allele was inherited from the father or the mother). Other aspects of chromatin modification may affect gene expression and, in turn, the range of phenotypes presented to selection (Sollars et al. 2003). These traits do not depend on the DNA sequence of the genes themselves, but on modifications that affect their expression; the sequence itself is transmitted faithfully.

Every time a cell divides it is subject to mutation. This applies to somatic (body) cells as well as to the germ line. An organism is a mosaic of slightly variable genotypes in its different cells.

CONCLUDING REMARKS

Genes are not the only elements of inheritance but broadly defined seem to be the single most important information-carrying system. The genome, however, is not just a digitally arranged set of beads on a string coding for amino acids determining the traits that make up an organism.

During the previous century, our conceptual approach to genetics was heavily influenced by Mendel, who provided a fundamental understanding of how *genes* were inherited. However, he used traits very closely tied to individual genes to show that inheritance was particulate in nature. Whatever our definition of a gene finally

ends up being, this discovery led to a century of unprecedented progress in understanding the behavior of *genes*. However, we may have followed Mendel's conceptual world too closely, by overinterpreting the inheritance of genes to be equivalent to inheritance of *traits*. There is conceptual trouble as a result because this has led to thinking of a gene for or per trait and hence of traits as simple products of unique genes.

Mendelian thinking does work very well for carefully chosen traits and for those aspects of natural variation that are at the extremes of the penetrance functions. Much if not most phenotypic inheritance does not seem to be of that type. The greater complexity provides a greater challenge to understand but also probably provides greater natural stability for traits.

The modular nature of the genome is of tremendous help in explaining the origin, nature, and arrangement of genes and for reconstructing the history of duplication and rearrangement events between species. Gene families with related and partially overlapping function, and similar characteristics or regulatory elements, provide understandable ways to account for the "evolvability" and developmental nature of complex organs and systems.

But accounting for something like a wing or a peach does not really explain what it "is" because there is so much complexity in the myriad pathways between a fertilized egg cell and a wing or a peach. We can use the machine analogy to identify parts and connections; at least up to a significant point, we are already learning tremendous amounts about the inner workings. Overall, the most important generalization may be the combinatorial use of related pathways and genetic "parts" that are also used in other structures as will be discussed in subsequent chapters.

The power of complementary base pairing in DNA and RNA leads to an essentially open-ended diversity of possible function. Floating all around a cell are thousands of DNA or RNA sequences, providing many possibilities of complementary hybridization or protein-DNA/RNA binding; this means that opportunistic evolution can in a sense "create" new function if some sequence turns out to be recognized in one of these ways (which can affect transcription or translation). The complex mix of molecules looking for partners raises an additional issue—how to *avoid* uncontrolled binding and/or inhibition of these many elements. This is a largely unsolved problem for science (but not for organisms, who have figured it out).

Modularity is so fundamental a principle of life that new manifestations are likely to be found, and these may be entirely different from sequence-specific modularity. Such modularity may involve the conformation of DNA near expressed genes, the shape of chromosomes as a key or code for particular differentiation, or any number of other characteristics and conformations of genes. A whole different set of simple processes can explain the nature and evolution of segmented traits with variable numbers of repeated similar structures, like petals, vertebrae, or toes. We will see many examples.

Modularity, gene and gene family evolution, and the like provide a beautiful satisfaction—these few general characteristics of genomes and their "strategies" for producing complex traits apply widely across both the animal and plant worlds, providing elements of a unified view of the way life uses the information stored in genes.

Chapter 6

A Cell is Born

“Omnis cellula e cellula”—all cells come from other cells. This is an insight we now take for granted, but when it was made by one of the founders of modern pathology, Rudolf Virchow, there was still uncertainty over the origin of new life. The uncertainty was a continuation of the age-old debate over spontaneous generation. Virchow’s observation was one of the most important contributions to modern biology in the 19th century and is the cornerstone of the Cell Theory, the unifying idea in biology that all living things are composed of cells, that cells are the basic unit of life, and that cells come only from preexisting cells. Indeed, it is only in this way that the deeper point, that all life comes from other life, settled the long-standing question of spontaneous generation. Before the invention of microscopes and key experiments that followed, the appearance of organisms arising from dead matter (maggots in dead meat or widespread fermentation phenomena) suggested to many scholars that life could arise totally on its own. Today, we assume that that happened only at the beginning, *before* cells had evolved.

In Chapter 6, we will describe some of the basic characteristics and functions of cells with the objective of showing how, within this fundamental unit of life, the same kinds of primary phenomena of controlled sequestration and modularity, as well as many functional elements for “information” handling presented earlier, are used in the systems to be discussed in the remainder of the book. These components of the cell are largely familiar; the genetic concomitants are rapidly being identified. The organization of biochemistry into the compartments we call cells is an important, perhaps *the most* important, aspect of life on earth, since everything subsequently is essentially an elaboration of cellular life. If primary biological molecules are frequently modular as we have seen, and perhaps even come from different points on a modular cycle of basic energy metabolism, then the cell can be viewed as a very much higher level of intricate, modular physically and chemically hierarchical organization, internally, as well as the fundamental sequestration phenomenon that enabled more complex life to evolve its diversity of function and adaptation.

Cells show that complex organization has required highly orchestrated cooperation at every level—within as well as among cells, tissues, and organs. Even single-celled bacteria, completely self-sufficient as isolates, can enhance their survivability by massing together into sheets of cells that, among other things, communicate, pass genetic material, increase their antibacterial resistance, and better survive periods

of starvation (Hoyle and Costerton 1991; Miller and Bassler 2001; Pratt and Kolter 1999).

THE BASIC CHARACTERISTICS OF A CELL

A cell is a pool of controlled chemical composition bounded by an outer membrane. It is the main structure of undifferentiated single-celled organisms and the essential building block of highly complex multicellular animals composed of many different cell types. A cell (1) sequesters biological resources relative to the outside world, and also internally; (2) maintains the necessary concentrations of chemical components, pH, and so forth; (3) localizes, transports, exports, and imports select molecules; (4) uses selected, controlled, context-dependent subsets of its genes and controls whether gene products are kept local, for use by this cell, or are sent outside the cell; (5) allows differentiation from the surrounding medium, and (6) provides a building-block mechanism by which life can evolve more complex traits.

Although there is extensive variation in cells found in the biosphere, generalizations can be made about them and hence about life, a fact basic to our understanding of how life works and evolved. These generalizations have placed at least some constraints on what has evolved and on what can or will evolve in the future. All known organisms (except very primitive “life” forms like viruses and prions, which, although not cells themselves, depend on cells to replicate and to continue to exist) are composed of one of two basic cell types, *prokaryote* and *eukaryote*, shown schematically in Figure 6-1. Figure 6-2 provides a detail of the cell membrane. See Table 6-1 for details of the structure of prokaryotes and eukaryotes.

All cells use the DNA-RNA coding system for replication and for coding proteins. All cells are bounded by lipid membranes, although the composition of the membrane varies across cell types. All cells use ATP (adenine 5'-triphosphate) as their source of energy for carrying out metabolic processes. They do this by glycolysis, the anaerobic (in the absence of oxygen) breakdown of glucose into lactic acid with a net gain of ATP molecules, or by photosynthesis, the conversion of sunlight into energy either nonoxidatively or oxidatively with a net gain of many more ATP molecules than glycolysis yields. Actually, as we currently understand early life, the evolution of these processes both depended on and drastically altered the Earth's atmosphere, and knowledge about how particular cells make ATP and carry out metabolic processes such as movement, synthesis of various cellular constituents, and the like tells us something about the evolution of life itself. Because much of cellular physiology ultimately rests on basic energy metabolism and that is highly constrained at the molecular level, some of the basic nature or constraints may be predictable solely on chemical grounds (Morowitz et al. 2000).

Beyond these generalities, prokaryotes and eukaryotes are quite different. Prokaryotes, the simpler of the two, are generally much smaller. Two distinct groups of prokaryotes are known: bacteria and archaea. Bacteria are single spherical or rod-shaped cells, found living in almost every known niche—in the water and on, above, and deep below the surface of the earth. Often found protruding from the cell wall are *flagella*, which allow the bacterium to move, and *pili*, which are flagella-like, but are involved in the transfer of genetic information between cells or in anchoring the cell to other cells. Archaea have some features in common with Eukarya and some with bacteria. They are often found in extreme environments—intense heat, cold, salinity, alkalinity, acidity, and the like.

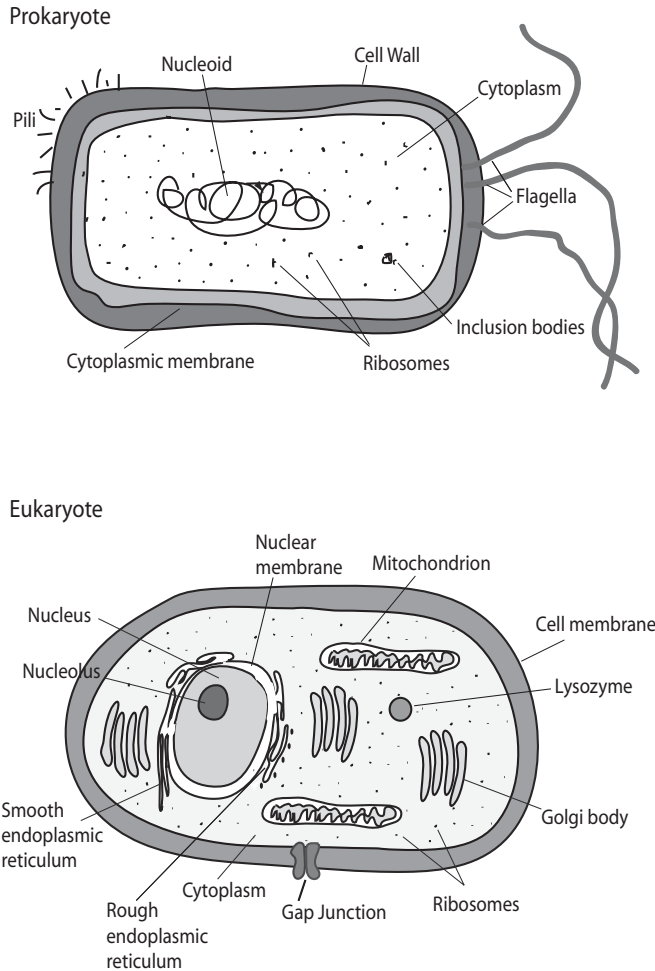


Figure 6-1. Schematic of prokaryote and eukaryote general structure.

Prokaryotes typically are bounded by a tough *cell wall*. However, among eukaryotes, plants, fungi, and algae do have cell walls, whereas animal and protozoan cells lack them. Bacterial cell walls are complex, but generally are composed of a type of petidoglycan called *murein*, a complex polysaccharide, whereas archaeal cell walls are made of protein, glycoprotein, and carbohydrate or *pseudomurein*, but never petidoglycan. Plant cell walls are made of cellulose and other polymers, and fungal cell walls are different still. The common characteristic of these cell walls, even if they differ in composition, is that they provide rigid structural support for the organism.

Within the cell wall is a plasma membrane, composed of lipids, mainly phospholipids and cholesterol, and proteins (see Figure 6-2). In bacteria, the lipids are saturated or monounsaturated. The lipids in archaeal membranes are quite different from those of either bacteria or eukaryotes. Among other things, they are polyunsaturated. The specific components of these membranes are probably what allow archaea to live in the extreme conditions in which they are found. In all cells,

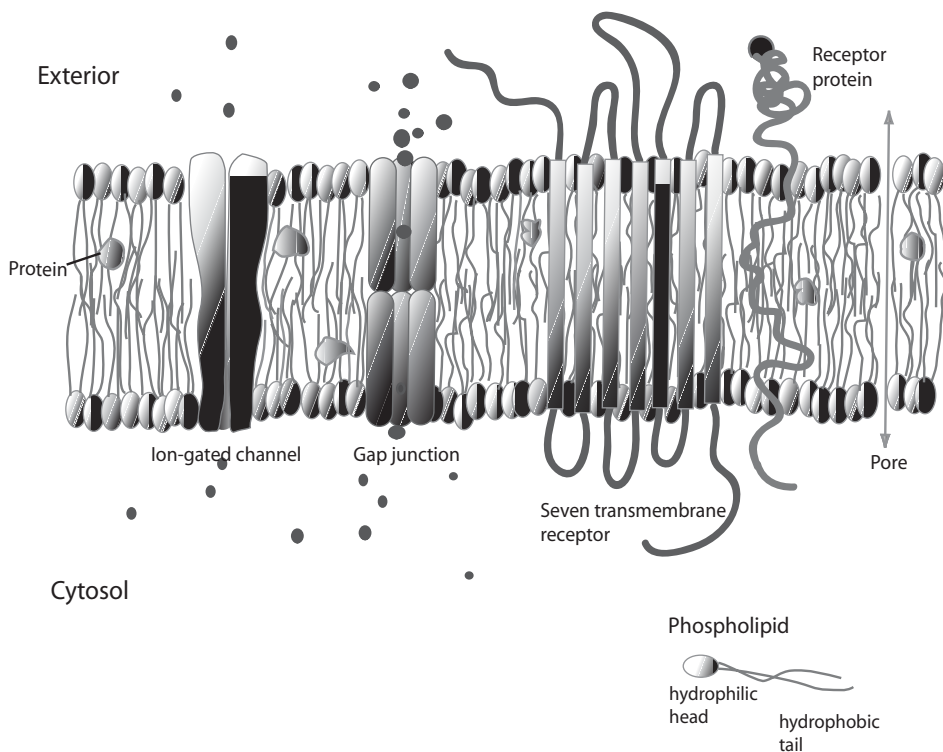


Figure 6-2. Schematic of constituents of prokaryote and eukaryote cell membrane/wall. The elaboration of the original bilipid membrane now enables cells to undertake a complex variety of functions, and provides sophisticated levels of sequestration within the cell itself.

no matter what their composition, the plasma membrane encloses the genomic material, proteins, and small molecules, all of which float free in the cytoplasm of prokaryotic cells, but are encased in a variety of organelles in eukaryotes.

In prokaryotes, the genome consists of one circular molecule wound tightly to form the *nucleoid*. Also in the cytoplasm are *ribosomes*, which are composed of ribosomal RNA and associated proteins and which translate RNA into proteins; *inclusion bodies*, which are globules of starch, glycogen or lipid, and stored nutrients; and *metachromatic granules*, which are storage sites for phosphate. Some bacteria also have *magnetosomes* in their cytoplasm, which help them orient in their environment.

Many of the metabolic processes of the prokaryotic cell take place on the cell membrane, including energy metabolism and waste removal. Other processes, such as protein synthesis and glycolysis, occur in the cytoplasm.

An important class of substructures in prokaryotic cells, and occasionally in lower eukaryotes, is *plasmids*, small circular extra-chromosomal DNA molecules found in many bacterial and archaeal cells. They contain genes for proteins that a bacterium can generally function and replicate without, although plasmids are also replicated when the cell divides. They provide functions like virulence factors, which are important for pathogenesis in some bacteria, and they can code for antibiotic resistance,

TABLE 6-1.

Components of Prokaryotic and Eukaryotic Cells.

Cell wall is a rigid structure that surrounds a plant, bacteria, fungus or alga cell. It maintains the structural integrity of the cell, and resists osmotic pressure. Although its structure depends on the species of bacteria or plant, it is generally multilayered. It does not regulate transport into and out of the cell, as the cell membrane of an animal cell does, but instead allows the passage of anything that fits through its pores.

Plasma membrane is the outer boundary of a cell. It is a continuous sheet of phospholipid molecules, with proteins studded throughout that serve to regulate transport of ions and chemicals into and out of the cell and receive and coordinate hormonal and other signals from adjacent cells or the animal's bloodstream.

Flagella and pili are often found protruding from the cell wall of prokaryotes. Flagella are involved in cell mobility, and pili are involved in the transfer of genetic information from one cell to another, or in anchoring the cell to other cells. Protists have flagella as well and their own form of anchoring device.

Cytoplasm is the aqueous contents of the cell, excluding the contents of the nucleus of eukaryotic cells, and is the site of most of the chemical activities of the cell. In prokaryotes, the genomic material floats freely in the cytoplasm. In eukaryotes, other membrane-bounded organelles are found here.

Nucleus is the largest organelle within the eukaryotic cell.

Nuclear envelope bounds the nucleus, is composed of two lipid bilayers, and is punctuated by nuclear pores through which transcribed DNA (mRNA) travels to the cytoplasm for translation into proteins by ribosomes.

Inner membrane gives the nucleus its shape.

Outer membrane is joined to the inner membrane at the nuclear pores, and is in contact with the rough endoplasmic reticulum, also joined at the nuclear pores.

Nucleolus is the site of assembly of ribosomes. Nucleoli form on specific sites on specific chromosomes, called "nucleolar organizer regions," which contain multiple genes for ribosomal RNA. The two ribosomal subunits are produced here and transported into the cytoplasm through the nuclear pores. The assembled ribosome is too large to reenter the nucleus, so stays in the cytoplasm, often attaching to the rough endoplasmic reticulum, to carry out synthesis of proteins.

Chromatin, complex of DNA, histones, and nonhistone proteins, of which chromosomes are made.

Chromosome, a long molecule composed of DNA and associated proteins. In eukaryotes, the chromosomes are packaged in the nucleus; in prokaryotes, they float freely in the cytoplasm. The number of chromosomes in a genome is species-specific. Bacteria have one circular chromosome, whereas humans have 23 pairs of linear chromosomes.

Endomembrane system of eukaryotic cells is composed of organelles with a single membrane or membranes connected via tiny vesicles, with perhaps a common origin. This system greatly increases the membrane area of the cell. The organelles of this system are as follows.

TABLE 6-1.

Continued.

Endoplasmic reticulum (ER) which maintains structural continuity with the cell membrane, is continuous with the nuclear envelope, and synthesizes and transports lipids and membrane proteins. There are two kinds of **ER** in the cell; the **rough ER** is a flattened sheet, studded with ribosomes carrying out protein synthesis, and the **smooth ER** is more tubular with no attached ribosomes, but with major lipid synthesis function and for the storage and release of calcium ions into the cytosol, where calcium catalyzes many processes. However, the function of **smooth ER** varies from tissue to tissue;

Golgi apparati (GA) play an active role in the synthesis, modification, storage, sorting, and secretion of the cell's chemical products. Lipids and proteins produced by the ER pass through the **GA** to the other organelles in the endomembrane system.

Lysosomes are made by the Golgi apparati, and are responsible for intracellular digestion of particles like other organelles that are no longer functioning properly, food molecules, foreign particles like bacteria, or antigens.

Endosomes are one of the bodies in the endocytic pathway. They carry newly endocytosed materials to the lysosomes for degradation.

Microbodies

Peroxisomes destroy hydrogen peroxide, a potentially dangerous byproduct of cell metabolism.

Glyoxysomes, in fat-storing tissues of germinating plant seeds, contain enzymes that start the conversion of fats into sugars.

Vacuoles (plants) store enzymes and waste products, as well as provide the turgor pressure that allows plants to stay upright. These organelles can comprise up to 90 percent of the volume of a plant cell.

Mitochondria, about the size of bacteria, are bilipid membrane bounded, and are the energy producing organelles found in all eukaryotic cells. The inner membrane usually contains many folds, or *cristae*, which give the organelle its characteristic appearance in cross-section. Communication between the mitochondrial matrix and the cytoplasm takes place through *porins*, junctions through the outer membrane. Mitochondria convert energy from the oxidation of foodstuffs into ATP, or adenosine 5'-triphosphate, a nucleotide present in every cell and the principal carrier of chemical energy. Mitochondria have their own genomes, which are circular chromosomes floating in the mitochondrial matrix, and they replicate by binary fission.

Ribosomes float free in the cytoplasm, or, in eukaryotes, some are attached to the endoplasmic reticulum and are responsible for the translation of mRNA into proteins. Mitochondria and chloroplasts have ribosomes for cDNA translation, but they resemble bacterial ribosomes more than they do those of eukaryotes.

Cytoskeleton, in eukaryotes is a highly flexible set of protein filaments that extends throughout the cytoplasm and is responsible for the shape of the cell, cell movement, cytokinesis (the division of the cytoplasm after the nucleus has split during cell division), and the organization of the organelles within the cell.

Chloroplasts (plants), like mitochondria, are energy producing organelles in plant cells. Unlike mitochondria, they convert sunlight, rather than food, into ATP. Again, as mitochondria, they are thought to have been introduced into cells early in the evolution of eukaryotes by endocytosis.

which is essential to bacteria that live in environments awash with natural antibiotics (that is, this predated by aeons the human development of antibiotics, in the battle by other organisms to protect themselves against bacterial attack).

The more complex types of cell, eukaryotes, are also bounded by a lipid bilayer membrane. The bulk of the genomic structures in eukaryotic cells are further sequestered within a *nucleus* encased in its own membrane that controls molecular transport in and out. Eukaryotic metabolic processes are carried out within and between membrane-bounded organelles floating in the *cytosol* (the contents of the cell minus the nucleus and organelles). Eukaryotic chromosomes are as a rule linear and vary greatly in size and number; as in prokaryotes, they are tightly wound but, like archaea but not bacteria, in association with periodically spaced protein complexes called *histones*, which package the DNA so that it will fit into the cell (see Figure 6-3).

Additional organelles in eukaryotic cells include the ribosomes, where protein synthesis takes place, as in prokaryotes, the *Golgi apparatus*, involved in transport of macromolecules between organelles, the *endoplasmic reticulum*, which carries out the synthesis and transport of lipids and membrane proteins to the cell membrane for transport out of the cell, *endosomes*, *lysosomes*, and *peroxisomes*, which are involved in intracellular transport, digestion, and maintenance of the proper concentration of specific chemicals, the *cytoskeleton*, which consists of protein filaments and most importantly *actin filaments*, involved in movement of the cell, *microtubules*, which are most likely the primary organizers of the cytoskeleton, *intermediate filaments*, ropelike structures that are thought to be responsible for the

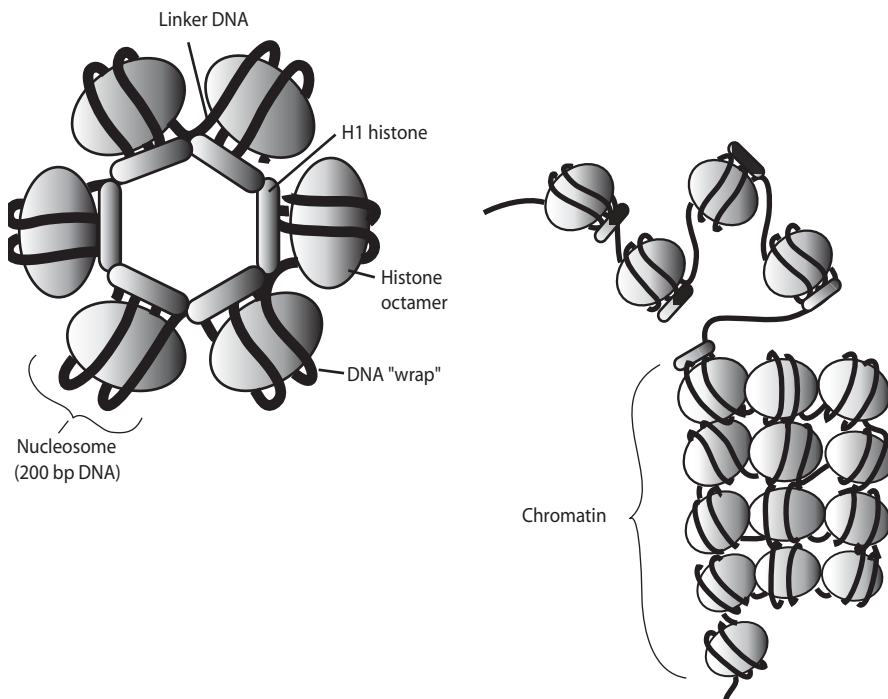


Figure 6-3. General characteristics of DNA packaging.

TABLE 6-2.
Genome Size and Number of Chromosomes in Selected Eukaryotes.

Organism	Genome size (megabases)	No. Chromosomes (2n)
Fruit fly	100	4
Mosquito	30	7
Housefly	900	6
Rice	400	12
Frog	3100	13
House mouse	3454	20
Wheat	1650	21
Human	3400	23
Tobacco	3800	24
Carp	1700	50–52

Genome size data from: <http://www.cbs.dtu.dk/databases/DOGS/>, <http://mcb.berkeley.edu/labs/cande/pages.dir/genomesizes.html>; chromosome numbers from: <http://www.kean.edu/~breid/chrom2.htm>, <http://www.wikipedia.com/wiki/Chromosome>.

mechanical strength of the cell, and *mitochondria*, where cellular respiration takes place—the production of energy by the combining of oxygen with food molecules. Mitochondria actually contain their own DNA, a circular molecule of about 16,000bp. Listed in Table 6-2 are the genome sizes and chromosome numbers for a selection of eukaryotic organisms.

The volume of eukaryotic cells is so large, relative to the cell membrane, that an additional role of the organelles is to expand the membrane area of the cell to allow controlled and rapid transport of waste products, newly synthesized proteins, and other products of cell metabolism into and out of the cell. Molecules move around rapidly and passively encounter each other with reasonable probability, but cells take a surprisingly active role in locating components where they are needed on a microscale, or in actively moving them around. They also *chaperone* molecules that protect them from degradation or interaction by other molecules in the cell on their way to the right location. Such mechanisms help fold proteins properly, prevent them from agglomerating with other proteins, and so on. Together, these organelles comprise the *endomembrane system*. The system includes the nuclear envelope, the endoplasmic reticulum, the Golgi apparatus, lysozymes, endosomes and peroxisomes, the plasma membrane, and various microbodies and vacuoles. The membranes can be in direct physical contact, such as that of the endoplasmic reticulum with the nuclear envelope, or connected by the transfer of cellular constituents via tiny vesicles. There is no equivalent system in prokaryotes, as the cells have no membrane-bounded organelles; all of the communication with the environment takes place directly across the plasma membrane.

The endomembrane system of eukaryotes allows these cells to take up macromolecules from the environment by *endocytosis* and deliver them to lysosomes for digestion. The resulting metabolites are delivered directly into the cytosol. The membrane system also regulates the delivery of newly synthesized proteins and the like to the exterior via *exocytosis*. Each molecule follows an elaborate pathway on its

way to the outside of the cell, and it can be modified each step of the way and stored for delivery to the cell surface at the appropriate time. The exocytic pathway extends from the endoplasmic reticulum to the Golgi apparatus out to the cell surface, whereas the endocytic route is from the plasma membrane inward to the endosomes and lysosomes. If the membranes of the various organelles are not contiguous, and thus able to pass molecules from one to the next directly, *transport vesicles* engulf the molecule and bud off from the membrane of the first compartment and fuse with the membrane of the next to deliver the molecule directly into the acceptor organelle, meanwhile managing to maintain the separate identity of each of the organelles involved.

An additional organelle in plant cells is the *vacuole*, which stores enzymes and waste products; it also provides the turgor (fluid) pressure that allows plants to stay upright. Vacuoles can take up much of the volume of a cell, sometimes as much as 90 percent. Higher plant cells contain additional partially autonomous organelles, called *plastids*, which have various functions, such as synthesis and storage of food, as well as the production of, among other cellular constituents, most amino acids and all fatty acids. In animal cells, fatty acids are made in the cytosol. *Chloroplasts* are the most commonly found plastid; they contain chlorophyll, giving green plants their color, and produce energy from sunlight by photosynthesis. Another is the *amyloplast*, which transforms glucose into starch and stores it within its membrane.

Mitochondria and plastids are membrane-encapsulated structures that contain their own DNA (mammalian mitochondrial DNA uses a coding system somewhat different from that of nuclear DNA) and are thought to be the descendants of once free-living bacteria that were endocytosed into cells as parasites several billion years ago (Margulis 1970). In this regard, incorporation through horizontal transfer, they are analogous to plasmids in bacteria, other horizontally transferred elements referred to in earlier chapters, and probably more to be discovered.

After millions of years of cohabitation, a combination of their own and the cell's gene products has evolved to control their activity, and the cell and these organelles now completely depend on each other. Although these organelles are similar in that mitochondria perform respiration-driven ATP (energy) synthesis and the synthesis of ATP is driven by light in chloroplasts, phylogenetic analysis of mitochondrial DNA and bacterial genomes suggests that, of currently available bacterial genomes, mitochondria are closest to the present-day *Rickettsia prowazekii* bacterium (which causes typhus in humans) and perhaps descended from its ancestor (Andersson et al. 1998), whereas chloroplasts appear to be descendants of oxygen-producing photosynthetic cyanobacteria (Alberts 1994).

FUNCTIONAL LOCALIZATION IN EUKARYOTIC CELLS

The nucleus is a highly organized organelle, with a nuclear matrix that controls the location of the chromosomes themselves, of transcription factors, transcriptionally competent or silent genes, and so forth, allowing the cell to precisely regulate the timing of gene expression (Lemon and Tjian 2000; Stenoien et al. 2000). At the same time, proteins move through the nucleus by passive diffusion, so that the encounter of any given molecule with its binding site also has a random component (Misteli 2001; 2001).

At the subcellular level, gene transcription, DNA replication, recombination, or DNA repair all depend on proteins having access to the target DNA. As described

earlier, DNA wraps around a core of histones to form nucleosomes, and the entire genome packaged into a series of nucleosomes constitutes chromatin. To carry out their role in the cellular machinery, transcription factors, repair proteins, and so forth must gain access to the nucleosome-bound DNA—to do so, they must outcompete nucleosomes for binding sites. That is, the conformation of the packaged DNA must be changed, and it is *chromatin-remodeling complexes* that do this, harnessing energy from ATP hydrolysis to loosen DNA-nucleosome bonds or to reposition nucleosomes, thus allowing proteins access to the DNA (King and Kingston 2001; Lemon et al. 2001).

In addition, transcription seems to occur in specific locations, called *transcription factories*, within the nucleus (Hume 2000; Navarro and Gull 2001). These are sites of active RNA polymerase II-mediated transcription; about 2,500 such sites are in every cell. Whether a specific protein-encoding gene winds up in the right place in the nucleus at the right time seems to be a partly random process (Hume 2000; McAdams and Arkin 1999). Gene expression seems to be digital—it happens or it doesn't in a given cell. That is, in some experimental situations, stimulating the expression of a gene causes it to be expressed in more cells, not necessarily at a higher level in a given or in every cell (Hume 2000). The average level of transcription varies among genes, cells, and circumstances (there can be many more copies of mRNA for some genes than for others in a given cell). Whether transcription of a specific gene happens in a specific cell seems to be probabilistic; however, when it is “time” for a gene to be expressed, the essential thing is that it happens in enough cells in a tissue or organ or organism to matter. Similarly, the level of expression appears to be based on various aspects of the *cis*-acting regulatory elements (number of copies of a given RE, its specific recognition sequence, location relative to the gene, etc.), although the use of these elements, too, has a stochastic component. Gene expression is a population-level event—the product of transcription in many cells in a tissue, not a single-cell event. Therefore, although it is stochastic at the cell level, at the population level, gene expression is highly predictable. As long as the tissue overall does what is needed, the organism can suffer the failure of some of its cells to do so (though this doubtlessly is dependent on how many spare cells there are—not many in creatures like *Caenorhabditis elegans* with only a restricted number of cells in total).

Structures such as ion channels and receptors for neurotransmitters are on the cell surface, sometimes localized into subregions to control the level of possible cross-reactions that could confuse signals or other information transfer or undo local conditions from their intended state. Supramolecular islands of receptor complexes or ion channels are produced by aggregation via mechanisms such as shared protein-protein binding domains or protein carriers (Sheng 2001; Sheng and Lee 2001; Sheng and Sala 2001). Similarly, cells often have chemical as well as shape polarity, with location-specific and/or cell-surface concentrations of proteins, asymmetric secretion, and the like modulated by a variety of trafficking processes known as *constitutive cycling* (Royle and Murrell-Lagnado 2003).

ION CHANNELS AND OSMOREGULATORY UNITS

One of the most important aspects of sequestration brought about by the packaging of biochemical materials within protective membranes was that it allowed the cell to control internal chemical conditions. This meant keeping some basic

chemical properties under control relative to the surrounding (presumably ocean-like) environment. Much of this is achieved by shuttling elements into and out of the cell. Some adjustment can be made by *carrier proteins* that bind some kinds of molecules, carry them through the membrane, and release them on the other side. In this case, the surrounding environment is the womb of cells, with shared characteristics so that a more tightly controlled barrier is important. In particular, ions affect the pH of the fluid in which cellular reactions take place, and pH is an important aspect of how those reactions work. Cells can titrate (adjust) the relative concentrations of many aspects of their internal environment relative to their external concentrations—for example, adjusting the salt concentrations inside vs. outside the cell. However, cells need means to manage the passage of ions and small molecules through their uncharged cell membranes. Several ways to do this have evolved, including *facilitated diffusion* and *active transport*.

Facilitated diffusion of ions takes place through water-filled pores or *ion channels* in the cell membrane. These pores are formed by proteins or groups of proteins that connect the cytosol of eukaryotic cells to the exterior. They are narrow, highly selective, and *gated*, that is, they open or close in response to a signal from the environment. Some ion channels are *ligand gated*; others are *voltage gated*, *mechanically gated*, *light gated*, or *stretch activated*, but in all cases they allow only some ions to pass only under specific conditions.

More than 100 types of ion channels are known, used in different cellular contexts and formed from channel-specific proteins coded for by different sets of genes. Neural cells are a good illustration of how ion channels function. The charge in a cell is related to the relative concentration of positive and negative ions on each side of its cell membrane. When the relative charge difference across the cell membrane reaches a critical point, ion channels are activated and a pulse of electrical charge moves along the cell, or a burst of neurotransmitter molecules is released, to be picked up by receptors on adjoining cells and sent further along the neural line.

Voltage and ligand gating are the most frequently found gating mechanisms. Voltage-gated channels open and close to ion transport depending on the membrane potential of the cell, that is, the relative internal charge. When the membrane potential is not in equilibrium or the concentration of ions inside the cell is too high or too low (the cell is *polarized*), Na^+ , K^+ , and/or Ca^{2+} ion channels open, admitting or releasing ions and depolarizing the cell. Each type of channel preferentially allows transmission of one particular ion type (for example, Na^+ , K^+ , Ca^{2+} , or Cl^-) and responds to the relative concentration of that ion inside the cell. Transport is usually passive, with ions flowing along the charge difference.

Ligand-gated channels respond to small signaling molecules received by receptors in the cell surface that protrude either into the extracellular matrix or into the cytosol. There are a few main classes, each activated by a particular neurotransmitter molecule or class of molecules (e.g., glutamate, GABA, acetylcholine). Each class has its own structural characteristics and behavior, but always receipt of the ligand toggles the channel to open or close and thus regulate the transmission of ions.

Ion channels serve many important functions within the cell, including maintenance of osmotic balance, neurotransmission, and muscle cell function. Voltage-gated channels are more passive respondents to the external environment (and are present on single-celled organisms) compared with ligand-gated channels, which require a ligand typically produced elsewhere in the organism as an intercellular

signaling phenomenon. In this sense, a multicellular organism actively controls the environment to which its cells respond, by evolving its own environmental constituents—the ligands—to provide organism-specific cues. This is how a chain of nerve cells can carry a message or how a nerve cell can activate some other type of cell.

Ion channels of different types can be present in multiple copies on the surface of a given cell. However, they can work by activating similar intracellular molecules. This raises the question as to how specificity can be achieved and why the cell is not awash in chaotic cross-reactions among different channels with different function. One way seems to be that given channel types are clustered together in floating “islands” in the cell membrane, which are actively located by the cell into specific regions of its surface. This sequestration mechanism, which can even localize and compartmentalize functional aspects of a given cell, is produced by evolution.

Small hydrophilic molecules, such as some sugars, pass readily through cell membranes by facilitated diffusion through transmembrane proteins that create pores filled with water through which the molecules diffuse following their concentration gradient.

Active transport is the pumping of ions or molecules through the cell membrane, *against* their concentration gradient, using a protein transporter and an energy source, usually ATP. For example, because the H^+ concentration of gastric acid is about 3 million times that of the blood parietal cells in the stomach, very low pH gastric acid is secreted into the stomach via active transport. Very different gradients of K^+ and Na^+ concentrations in and outside of animal cells are maintained by a specific kind of active transport called the Na^+/K^+ pump, and in mitochondria (the energy factories of the cell), the H^+ pump actively pumps hydrogen atoms into the intermembrane space of the organelle for the process of making ATP.

OSMOSIS

Water passes through lipid membranes by *osmosis*, from the level of higher concentration to the level of lower, and thus maintains the fluid balance of a cell with its environment, preventing the cell from shrinking or bursting. Osmosis is a specialized form of passive transport, although water concentration can be affected by the active transport of solutes across a membrane.

The nature of osmosis can be illustrated by its use in the production of urine in the waste-disposal process in vertebrate kidneys, where much of the animal's waste is removed as solutes, highly concentrated by the kidneys. One quarter of the blood volume pumped by every heartbeat passes through the kidneys where it is filtered through a very complex network of blood vessels; both waste and useful molecules are passed into the urine. This “early” urine is then refiltered through the nephrons and sent into the tubules where much of its initial water volume is released, by osmosis, into the interstitial spaces around the tubules, again following a concentration gradient in the interstitial spaces that goes from low to high. The urine gets more concentrated as it passes through the tubules; osmotic flow increases along the concentration gradient to pull more water from the urine on its way to the center of the kidneys. The concentrated urine is then passed to the ureters for transport to the bladder; the final volume of urine going to the bladder is approximately one percent of its initial volume with the remainder of the fluid returning to the blood.

Most of the solutes are recaptured as well, for reabsorption and reuse. This concentrating of the urine saves an animal the size of an adult human 2–3 liters of water a day.

CELL REPLICATION

REPLICATION BY FISSION

Perhaps the single most important characteristic of life is its ability to replicate. This will be the topic of Chapter 8, but a few comments can be made here. Today, biomolecules replicate mainly via the elaborate DNA-RNA-protein coding system, which is an intracellular mechanism. This can be direct, as in the repeated production of the same protein, or indirect, as in the production of a biomolecule through the action of enzymes (proteins coded for by genes).

An important stage in evolution was, of course, the acquisition of the ability of the whole cell as an organized structure to replicate. In bacteria, this is done by *binary fission*, that is, by the division of a cell into two “daughter” cells. Prokaryotes replicate quickly, usually asexually, dividing into two identical cells that each contain a copy of the entire original genome and hence are genetically identical (except for any mutations that might occur during replication).

In binary fission, the bacterial cells first replicate their ribosomes, enzymes, and other cell components and duplicate their chromosome. When the replication process is complete, the two chromosomes anchor to different parts of the cell wall, and new cell wall material lines up down the cell midline, one chromosome on each side, and the cell splits in two. Under most normal conditions for prokaryotes, the new daughter cells contain approximately the same nongenomic material that was in the parent cell; in some special instances, however, cell division is not completely symmetric, and the two daughter cells may differ.

Replication is much more complex in eukaryotic cells because they are larger and because they must replicate multiple chromosomes, including the chromosome within the mitochondria (and within plastids in plants), as well as all the membrane-bounded organelles. Eukaryotic replication is a multistage process, from *interphase* to *mitosis* to *cytokinesis*. Interphase is the longest stage of the cycle, during which the cell prepares for replication and the DNA in the nucleus is replicated. Mitosis includes the process of nuclear division up to the point of cell division, cytokinesis, when the cell splits into two genetically identical *daughter* cells—cells that each contain the same chromosome complement. There are variations on these themes, as is always to be expected. One, for example, is the syncytium of some muscle, insect eggs, or other single large cells with many nuclei. Syncytia can form by incomplete cell division, or by the fusion of cells, and so forth.

REPLICATION IS NOT ALWAYS SYMMETRIC

Contrary to what is typically thought, simple binary fission may be singular in nature. In particular, the two daughter products of eukaryotic cell division are often, perhaps usually, different from each other. Indeed, this asymmetry is at the foundation of the hierarchical nature of development and of functional specialization

among tissues in complex organisms. The daughter cells may look similar physically, and they do typically both inherit the complete genetic complement. Often, however, the daughter cells do not look alike, and genes can be used differently in each.

A ready example of physical asymmetry is when a mammalian egg cell is produced (three smaller cells, known as “polar bodies” are also produced in two rounds of cell division). Less widely appreciated, however, is that a differentiated organism depends fundamentally on genetic differentiation among cells that often takes place at or as a part of cell division. For example, many differentiated animal tissues arise from a layer of *stem cells*, partially differentiated cells that divide and produce one daughter stem cell, and a second cell that begins a sojourn of differentiation to become, for example, a sperm cell, an intestinal lining cell, or a skin cell. In mammals, hematopoietic cells divide into, among other things, red blood cells that eventually get rid of their nucleus and thus no longer have nuclear genes (and concomitantly limited lifespan and physiological capabilities).

Asymmetry of gene expression in daughter cells occurs quite regularly during embryogenesis and in some senses is the basis of organogenesis and morphogenesis. Cells sharing a common ancestral cell turn on, or off, genes that were repressed (or expressed) in the former. Some of this, at least in vertebrates, is regulated by gene-specific chemical modification (e.g., methylation of C's in CpG pairs), which sometimes affects only one of the two homologous chromosomes.

An important concept in embryogenesis is that of the morphogenetic “field,” in which some cells produce, and others receive, molecular signals. Although signals can pass in various ways among cells, so that a cell can change its genetic program in the resting phase of the cell cycle (and hence, develop gene expression differences unrelated to cell division), clearly a cascade of events differentiates cells from stage to stage that, at least in some instances, requires asymmetric behavior of cell division products.

Prokaryotes, too, have mechanisms for the equivalent of sexual reproduction and the transfer of genetic material from one cell to another, thereby altering the symmetry of simple binary fission. *Conjugation* is the one-way transfer of genetic information, usually plasmid but occasionally chromosomal, from a donor bacterium to a recipient via the pili. *Transformation* is the process by which bacteria take up unpackaged DNA from the environment. Only cells that are “competent” can receive DNA in this fashion. Cells can be made competent, however, via a routine procedure in molecular labs where introduction of foreign DNA into bacteria is fundamental to recombinant DNA technology. In *transduction*, viruses move DNA from one cell to another.

A somewhat different type of asymmetric cell division, used across the eukaryote realms, results in the production of germ cells (sperm and egg). It is the two-stage process known as *meiosis*. Beginning with a diploid precursor, the final functional product is haploid, containing only one copy of each chromosome (spermatogenesis yields four haploid sperm cells, and oogenesis yields one haploid functional egg and three polar bodies destined to die). There are variations on this theme. Some plants, for example, are tetraploid rather than diploid, but the essentials of the logic and net result of their meiosis are the same. Although traditionally thought of as unique compared with the common notion of evenly dividing cells in mitosis, from the foregoing discussion it is clear that meiosis, although relatively rare, is but one of many asymmetric aspects of cell division.

RELATIONSHIP OF CELL STRUCTURE TO THE CONCEPT OF A TREE OF LIFE

Exactly what counts as the oldest fossil evidence of life is somewhat controversial. However, many accept that the very earliest evidence includes structures known as *stromatolites*, which are made of layers of bacteria-like organisms of various types (Schopf 2000). Details are controversial with such old and often ambiguous material, but the evidence suggests that at least some stromatolites radiometrically date to about 3.85 billion years ago. Because the Earth is only around 5 billion years old, these fossils suggest that cells were already quite elaborate relatively early in the history of life. Furthermore, they attest to the robust nature of the early cellular “designs” because some stromatolites are indistinguishable from microbes alive today. This fact is sometimes used to treat time as if it were compressed, and the cell as we know it as an inevitable or even automatic consequence of biochemistry. Nonetheless, we need to keep things in perspective and not assume that reaching this stage was very early in the history of life on earth: it has been guesstimated that perhaps up to 0.5 billion years were required from the time of the first self-replicating molecules that we might call “life” to the appearance of the cells fossilized in these stromatolites, suggesting that life began on Earth around 4–4.3 billion years ago.

The most striking thing (from the perspective of biologists) that happened in the subsequent 3.85 billion years was the evolution of the much larger and more complex eukaryotic cell and the proliferation of multicellular plant and animal life. Of course, it is difficult to judge by just how much unicellular (or more simple viral) life has changed since its external shapes were first fossilized. As will be seen throughout this book, the elaboration of multicellular life was extensive, but it again and again employed basic mechanisms, and very similar if not identical biochemistry, that presumably were present in early eukaryotic cells, in whatever form, or kind of organism, that first arose.

The classical idea of a universal Tree of Life is based on a single origin of cells and generally divides all the forms of life found today into three main domains: Eucarya, Archaea, and Bacteria. The latter two branches comprise unicellular prokaryotes, and the former, the Eucarya, includes organisms from tiny protists (unicellular nucleated eukaryotes) to the most complex organisms on Earth. Because prokaryotes are less complex than eukaryotes, it had long been thought that the universal ancestor of all life must have been a simple prokaryotic cell, from which eukaryotes descended by adding structures over time.

As discussed in Chapters 1 and 2, current phylogenetic evidence built on DNA sequences from a diversity of genomes suggests that this unitary view of life is at least partially wrong. Primordial life may have consisted of a kind of primitive bacterial *biofilm*, not a single cell, with genomic material encased in permeable membranes that were not very efficient at sequestering the cell’s contents from the external world. Lateral exchange of genetic information between cells and the acquisition of characteristics from anywhere in the biotic mass probably occurred much more readily than is possible between cells with the enclosed genetic systems that subsequently evolved. The primordial cells probably lacked sophisticated translation mechanisms, and it is likely that proteins were simple and short and that replication was inaccurate. This type of simple beginning could account for the evolution of the codon system, of genes by exon shuffling, and of the transition from RNA to

DNA discussed in earlier chapters. Woese calls these primitive cells *progenotes* (e.g., Woese 1998). According to some views, as we have noted, there was so much horizontal transfer of information among progenotes that the idea of a single ancestral life form cannot be correct. See Figure 2-1 for a tree supporting this view (Doolittle 1999).

Stromatolites are prokaryotic organisms that most closely resemble modern cyanobacteria (Schopf 2000), single-celled, photosynthetic organisms that contain a blue pigment in addition to chlorophyll and that live singly or in colonies. Their descendants or closely related organisms still thrive; cyanobacteria are commonly found as the green scum on a stagnant pond. Interestingly, if indeed these earliest life forms were cyanobacteria, this would mean that photosynthesis had already evolved more than 3.5 billion years ago. Green plants use photosynthesis to synthesize carbohydrates from carbon dioxide and water, using light as the source of energy, thus suggesting that water and carbon dioxide were present very early in Earth's history (Schopf 1992, 1994, 2000).

Oxygen is the byproduct of most forms of photosynthesis; various geological formations from around 3.75 billion years ago show evidence that oxygen was already in the atmosphere, probably being produced by cells capable of photosynthesis (Schopf 1992, 2000). This implies that life already existed long before 3.75 billion years ago, perhaps at least another 0.5 billion years before (Schopf 1992). If the Earth was formed roughly 5 billion years ago, this would mean that life began relatively soon after the environment cooled from its extremely hostile early state. Fossil evidence shows that eukaryotic cells had arisen around 1 billion years ago. This is a minimum time estimate, based on the size of the cells in microfossils; the organelles that comprise eukaryotes, which would be definitively diagnostic, generally do not preserve as fossils. This represents a huge gap in time between the appearance of prokaryotes and eukaryotes, but the evidence shows that this evolution happened before, rather than as part of, the evolution of complex differentiated organisms: the first eukaryotic fossils are of apparently single-celled organisms. Given the successful persistence of prokaryotes to the present day, it is curious both that single-celled eukaryotes would evolve and that they would take so long in doing so.

Modern genomic techniques show that eukaryotes are not simply prokaryotes with more complex cellular machinery added on. Indeed, as we have seen in Chapter 2, comparisons of the DNA sequences of different components of eukaryotic organisms, compared with bacteria and archaea, suggest that eukaryotes are chimeric, the result of a fusion between (or at least sharing the intrinsic characteristics of) archaeobacteria, eubacteria, and a cytoskeleton-bearing prokaryote (Doolittle and Logsdon 1998; Katz 1999; Vellai and Vida 1999). Similarities between genes involved in transcription, DNA replication, and translation in eukaryotes and archaea suggest that the genetic mechanisms for these functions in eukaryotes came from archaea; other genes, however, are closer to those of bacteria, and still others are too divergent from either lineage for their ancestry to be determined. Phylogenetic analysis of mitochondria and chloroplasts suggests that these organelles were once free bacteria that entered the cell as the result of an endosymbiosis event, the phagocytosis of a bacterium by perhaps an archaeobacterium. An endosymbiotic origin for other eukaryotic organelles has also been suggested (Margulis 1970). There is also recent evidence of other levels of gene transfer, including between eukaryotes and

prokaryotes and even between eukaryotes (Andersson et al. 2003; Archibald et al. 2003).

EVOLUTION OF COMPLEXITY

Multicellular organisms appear to have arisen first in the Vendian period of the late Precambrian, about 600 million years ago (Gerhart and Kirschner 1997), and 3–3.5 billion years after the development of the first single-celled organism. The reason for the immense time lag between the development of single and multicellular organisms is not specifically known, but we can make educated guesses, keeping in mind that (like Darwin) we are somewhat blinded by, among other things, the rather minimal fossil record. The standard default assumption is that multicellularity evolved because increased complexity was favored by natural selection for some reason(s). However, any such selective advantage did not mean that single-celled life became obsolete. The extent to which characteristics are shared between contemporary single and multicellular organisms suggests that single celled life was probably pretty much as sophisticated then as it is today. Single-celled organisms predominate on Earth today and are at least as diverse as any kind of multicellular organism.

However, if *no inherent* advantage is associated with multicellular life, at some time(s) and at some place(s), opportunity(ies) must have arisen at which cells that aggregated gained some sort of local advantage—and we should keep open the possibility that the first instance(s) were due to chance and became assimilated or committed over time, subsequently ramified by selection and drift. Because this did not put single-celled life out of business, a default presumption is that this represented a new kind of niche rather than obsolescence of the old one.

Initially, multicellular life probably occurred by little if anything more than the failure of dividing cells to fully separate or by the adhesion of similar cells. Some primitive multicellular life today is little more than this; bacterial biofilms are an example: they consist of sheets of bacteria that can survive as single cells but that under some natural circumstances can mass together to form a film with different characteristics and different genes expressed compared with their single-celled planktonic counterparts (Prigent-Combaret et al. 1999).

COMMON CELLULAR FEATURES OF MULTICELLULAR LIFE

Once cellular differentiation became possible (whether it happened many times or only once), an advantage or some resource was eventually gained by the use of *both* intercellular signaling and signaling between *different kinds of cells*. As a general guide to cellular diversity as an index of complexity, prokaryotes manifest a very small number of different cell types. Plants have about 30–40 cell types, and higher animals like ourselves have about 150–200. To achieve whatever advantage there is to becoming complex in this way, cells had to become able to differentiate and, at the same time, to communicate with different cell types. This may be why multicellular life took so long after the origins of single-celled organisms to arise. Today, several classes of activity are important across the spectrum of multicellular life. Most of these clearly have their origin in unicellular times.

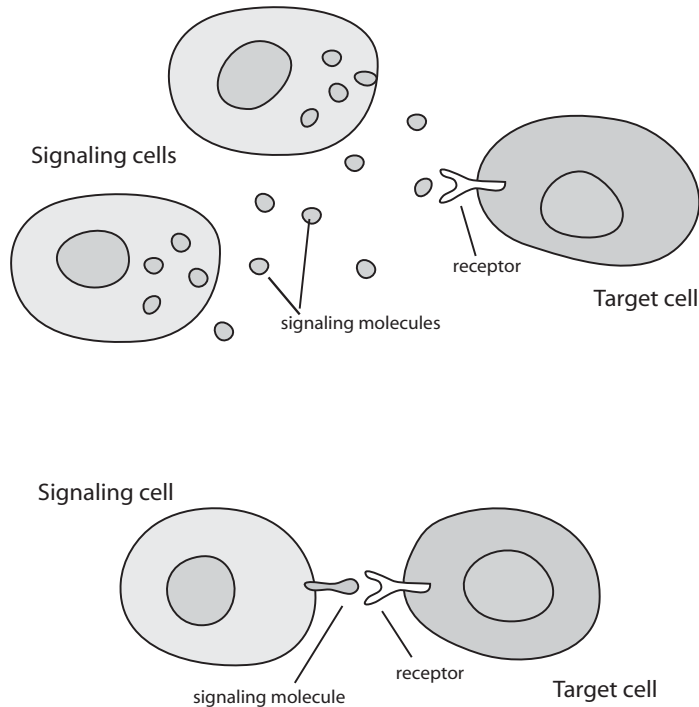


Figure 6-4. Schematic of cell signaling and information transfer mechanisms.

CELL-CELL SIGNALING

Cells in multicellular organisms communicate via hundreds of kinds of signaling molecules, using many pathways (not all of which would necessarily be found in the same cell type). Figure 6-5 schematically shows some of these pathways. Most of these molecules are ligands secreted by a signaling cell and received by a specific receptor protein on a target cell; the receptor-ligand binding then activates a response inside the cell. Signal molecules can be secreted by active transport mechanisms through the cell membrane into the extracellular space to be picked up at either short or long range by target cells, or the signal can be tightly bound to the surface of the producing cell and transmitted only to target cells that come into contact with the signal.

Some small signaling molecules, like retinoic acid, thyroid hormone, vitamin D₃, and steroid hormones, are hydrophobic and diffuse through cell membranes but then are bound by specific *binding proteins*, at which time their information is used to activate or repress specific genes. However, most *signaling factors* (SFs) cannot directly enter a cell because the lipid membrane, being uncharged, is largely impermeable to water-soluble molecules or because the molecules are too large; therefore, the information they carry is transmitted by the binding of the SF to a receptor on the cell surface, which communicates the occurrence of the event to the cell.

Signaling can occur at any distance from a cell, from intracellularly, to a few to thousands of cell diameters distant, or indeed even among distant organisms. A variety of terms are used to indicate the distance relationships, although as usual

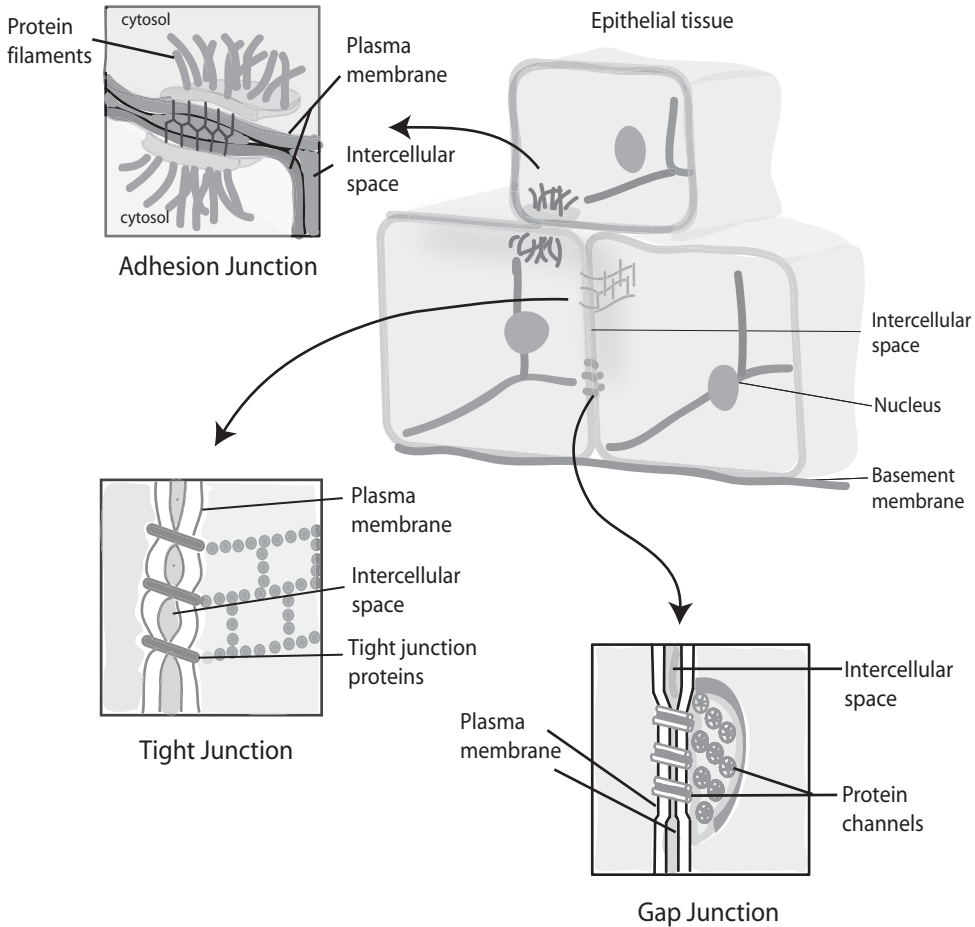


Figure 6-5. Some types of cell junctions.

they are not unambiguous and can overlap. Short-range signaling between adjacent cells is called *paracrine signaling*. The transmission of signals between specialized sites, called synapses, in the cell for long-range signaling, such as along nerve cells, is *synaptic signaling* (the role of ion channels in this process was referred to briefly above). *Endocrine signaling* in the form of a *hormone* molecule secreted from endocrine cells into an animal's bloodstream or the sap of a plant or from one cell to its neighboring cells or even for intracellular use targets cells anywhere in the organism. Hormonal signaling from one organism to others works through *pheromones*, specific molecules released into the air or water for which conspecifics have receptors.

These kinds of signaling mechanisms pass messages between different types of cells, but cells can also send signals that are bound by their own receptors or those of the same type of cell. This is called *autocrine signaling* and is important in development, as it allows all cells of a single type to respond identically to the same differentiation signals. Other signaling is transmitted through shared "pores" among adjacent cells (see below). In this case, whatever the signaling molecule is, and

whether it simply diffuses through the pore to the adjacent cell or is actively transported, when inside, it works as it would within the cell in which it was produced. Signaling to neighbors directly via their surface constituents is known as *juxtacrine* signaling.

CELL JUNCTIONS

A number of different cell-cell and cell-extracellular matrix pores form the foundation of the network of interactions between cells or between a cell and the extracellular matrix. They are of three main classes: *occluding junctions*, which seal together cells in an epithelial sheet to prevent leakage through the sheet; *anchoring junctions*; which attach adjacent cells to one another or attach cells to the extracellular matrix; and *communication junctions*, which regulate the passage of chemicals or electrical impulses between the cytoplasm of two cells. These are illustrated schematically in Figure 6-5.

Perhaps the most important tissue to develop in the evolution of complex animals was the *epithelial* sheet of cells. Tight sheets of epithelial cells form the skin and line the digestive system, body cavities, organs and glands, and one of their important functions is to serve as a selective sequestering barrier, preventing the leakage of fluids from one side to the other. Occluding or *tight junctions* help create that barrier by sealing together the cells in the sheet.

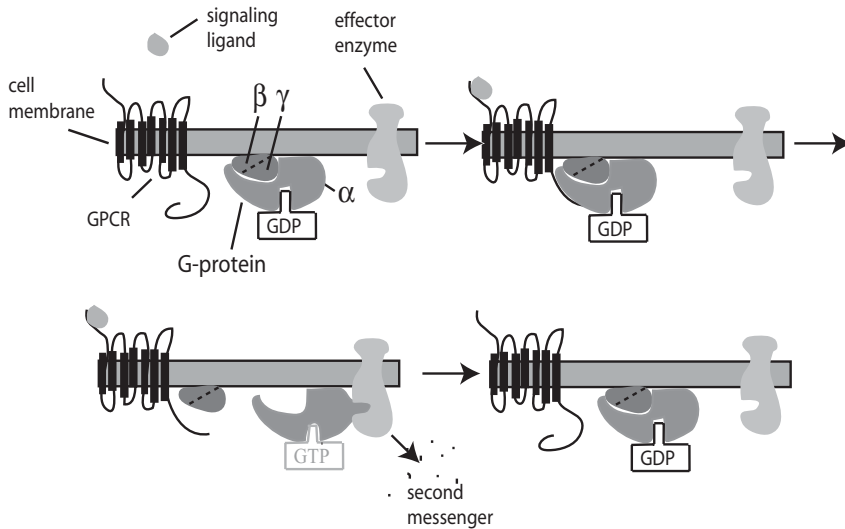
Anchoring junctions connect filaments of the cytoskeleton of one cell to that of another, allowing a large number of cells to function as a structural group. They are especially abundant in tissues that undergo stress, such as skin and muscle. Three forms of anchoring junctions have been described: *adherens* junctions, *desmosomes*, and *hemidesmosomes*. A number of genes that code for these structures are known, in part because they are associated with skin diseases, such as skin fragility and carcinomas. Adherens junctions and desmosomes both play roles in cell-to-cell connections, whereas hemidesmosomes help to connect the basal surface of a cell to the adjacent connective tissue.

Gap junctions, *chemical synapses*, and *plasmodesmata* are *communicating junctions*. Gap junctions are found in most cells in most tissues and in almost all animals, both vertebrates and invertebrates. Gap junctions are channels in cell membranes formed by two neighboring cells that allow the two cells to communicate, to share cytoplasmic ions, small regulatory molecules, and macromolecular substrates, selected principally by size. Cells can readily share small molecules, passing them from cytoplasm to cytoplasm via gap junctions, but they cannot share larger molecules, nucleic acids, or proteins in this way.

Gap junctions are *gated*, that is, sometimes open and sometimes closed (like ion channels), and transient; they do not necessarily exist for the life of the cell. They are important for the normal functioning of organs such as the heart or the intestine, which requires constant calibration of ion concentrations. Gap junctions seem to be important in development as well; the existence of these communication channels allows a group of cells to function as a whole (Kumar and Gilula 1996; Wilson et al. 2000). In very early embryos, cells are electrically coupled to each other, and this is maintained by gap junctions. Figure 6-6 illustrates several signaling mechanisms.

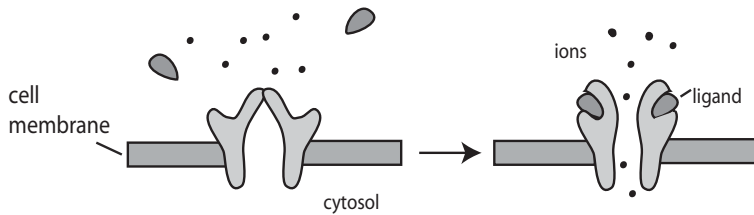
Higher plant cells have a communication mechanism similar to gap junctions called *plasmodesmata*. These, in fact, are the only form of intercellular communica-

G-protein-linked receptor

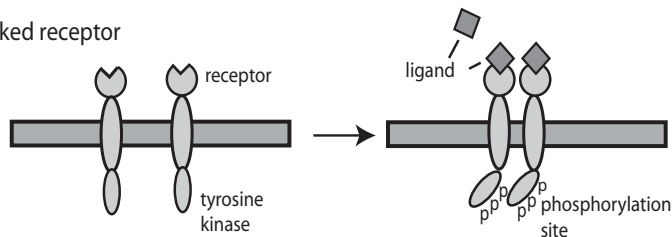


A

Ion-channel linked receptor



Enzyme-linked receptor



B

Figure 6-6. Schematic of some communicating pores in cells. (A) G protein-linked receptors; ion-channel linked receptor; enzyme-linked receptor; (B) plasmodesmata in plants.

tion plants have because, with their rigid cell wall, they have no need of anchoring junctions. Plasmodesmata connect every cell in a plant with its neighboring cells, allowing the passage of cytoplasm from cell to cell. In a sense, then, plant cells connected in this way form one mega-cell containing many nuclei.

The outer membranes of bacteria, mitochondria, and chloroplasts are permeated by pore proteins called *porins* that function similarly to gap junctions, although they are structurally very different. They allow the passage of small molecules through the membrane by passive transfer.

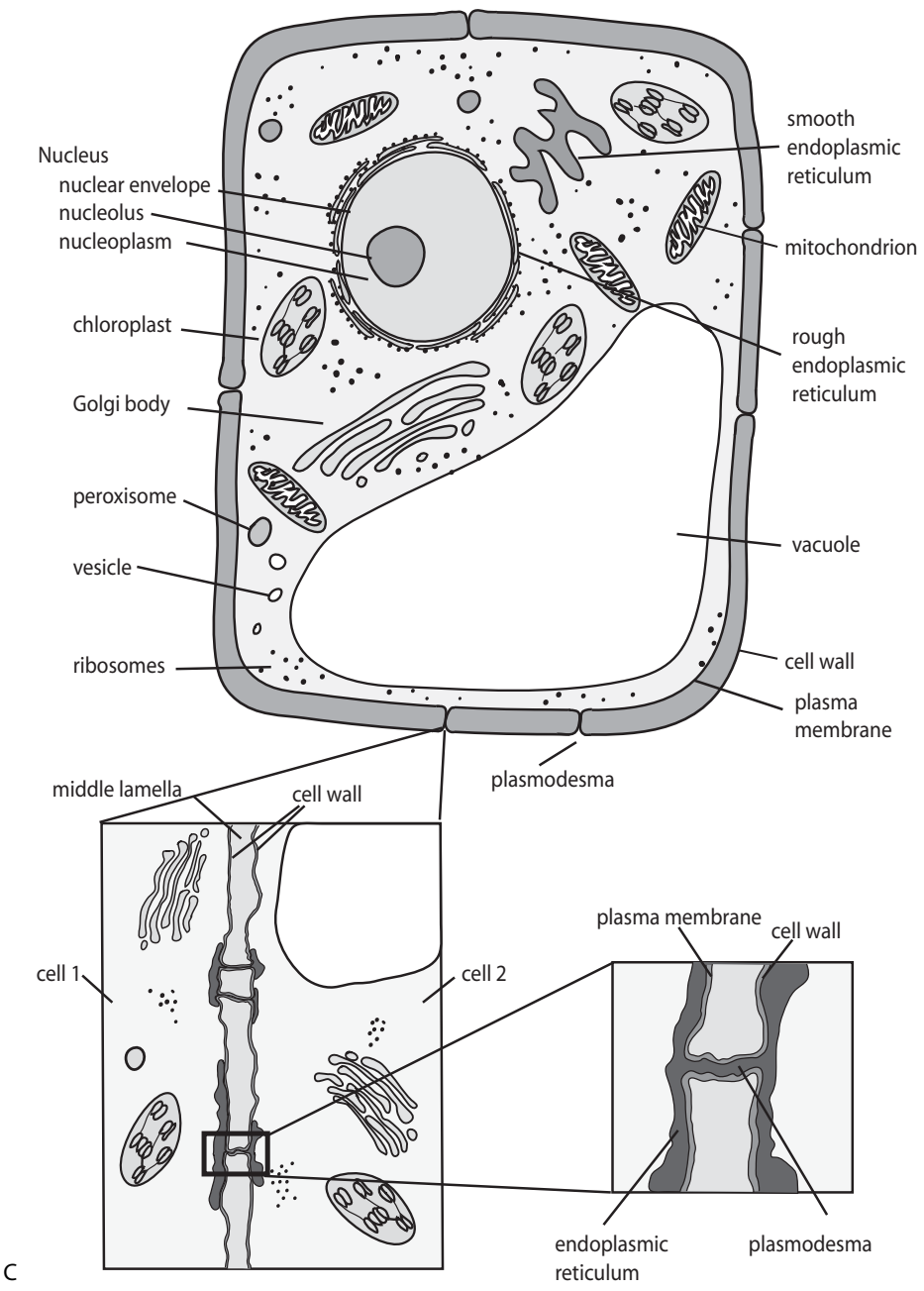


Figure 6-6. Continued

But a hole in a cell is a dangerous thing! In fact, one ancient part of our immune system, the *complement* system, functions mainly to poke holes in invading bacterial cells. The cell leaks its guts out, so to speak. In general, cells don't want drafty open doors, yet they must communicate. This explains the diverse and highly elaborated means animal and plant have for getting things into and out of themselves and of communicating with other cells without becoming too vulnerable.

SOME CONCLUDING GENERALIZATIONS ABOUT MULTICELLULAR LIFE

The organizational advances necessary for the cell as we know it to exist and predominate in the biosphere were many and included a cell membrane and internal structures of some kind in which controlled reactions could take place. Such evolutionary steps alone resulted in single-celled organisms such as bacteria and protozoa, which still comprise more than half of the total biomass on Earth. Of course, we do not know how much more “advanced” current unicellular organisms are versus those that look so similar in the 3.85-billion-year-old mud; however, in terms of external appearance they are very similar, which is how we can even recognize the earliest fossils. The extent and durability of their success show that evolution does not automatically favor complexity or overwhelming change. Nonetheless, as with conserved fossils like horseshoe crabs, there *is* evolution at the DNA level and this is why deep phylogenies—despite their horizontal connections—can be constructed. Appearances can in some sense be deceiving.

Despite this conservation of form and basic physiology, some circumstance somewhere was favorable to or tolerated the evolution of multicellular life in some way, and it proliferated into diverse forms, doing things far more complex than what unicellular forms do (although the latter are found in essentially all the same niches, and more).

Perhaps for a long time, adhesion and interaction among individual cells were rather fleeting or incidental and not necessary for survival, or for survival only in a simple physical ganging-together sense. Eventually, size and coordinated organization provided advantages over simple free-floating biomolecules or single cells with ephemeral interactions. It is interesting to speculate about what the advantage of cooperation between cells was, which extended from occasional interaction to complete mutual dependency.

Indeed, many cells currently undergo programmed self-destruction (*apoptosis*) during development and modeling of tissues. Generally, in this process, a cell activates a series of internal biochemical reaction steps that lead to its death: its nucleus shrinks, the cell condenses, and it fragments without spilling any of its cytosolic contents; the fragments then are quickly phagocytosed by macrophages or other nearby cells. Apoptosis is usually actively “ordered” (signaled) by other cells, and the ill-fated cell cooperates by suicide, the sincerest form of cooperation.

Apoptosis is in some ways logically like many other kinds of apparently altruistic behavior in life. Sterile ants, self-sacrifice of individuals to protect their group, and the abscission layers that allow leaves and fruit to separate from plants are other well-known examples. These are signal-driven or in other ways “programmed” into the behavioral or genetic repertoire of the organisms concerned; the bases for abscission or apoptosis and some aspects of sterility in social insects are at least partly known. Coordinated communication is often involved. As noted earlier, there

are debates about whether every such action must have evolved through selfish competition, but we understand the choreographed interactions in life at least as well by examining what is achieved by cooperation between the cells, organisms, or within the community in the process.

Unicellular organisms may be more pervasive and more ominous for us, but much of what we as humans find most interesting in the world today has to do with the function and evolution of multicellular organisms. The evolution of these inter- and intra-organismal dependencies has led to the complex traits that are the main subject of this book. From this point of view, cells are reasonably stable and uniform entities, whose common general properties are used in a diversity of contexts that make complex organisms what they are and made it possible for complex organisms to exist.

However, we do not want to leave the impression that the workings of cellular machinery are elegant and precise. It must be noted that not all cells of the same kind even in the same individual are identical. Many of the mechanisms of the cell cycle are error-prone and can introduce errors of many kinds at any stage. DNA or RNA polymerases are inaccurate to varying degrees, gene transcription and translation are imprecise, and cell division can go awry. A number of enzymes that repair replication errors, mitotic error, or errors caused by environmental degradation (for example, ultraviolet UV exposure) are ubiquitous in cells. Endonucleases, exonucleases, proteases, tumor suppressors, which we will discuss further in Chapter 7, perhaps even the immune system itself, because it recognizes and destroys foreign proteins, all keep the very imprecise cellular functions in line. Still, error (that is, deviation from the norm) happens.

In bacteria, perhaps 1 in every 10,000 amino acids or more is misincorporated into proteins as they are being synthesized (Bouadloun et al. 1983; Kurland and Gallant 1996; Kurland 1992). Perhaps, about 20 percent of newly synthesized proteins are degraded because they are misfolded or prematurely terminated or otherwise mis-synthesized. Another 10–20 percent of new proteins are found associated with molecular chaperones, which suggests a role for these molecules in the folding of proteins during synthesis (Wickner et al. 1999). These quality-control mechanisms recognize exposed hydrophobic regions in the polypeptides; proteases degrade them, whereas chaperones fold them properly.

In a very real sense, because cellular mechanisms can be so imprecise, the evolution of elaborate repair mechanisms to clean up the nontrivial errors of cellular functions is an essential facet of life as we know it. In fact, perhaps two percent of the energy of a cell goes into error repair (Scriver 2002). However, errors are not always repaired, or are not repairable by the tools of the trade, and thus fall outside the acceptable range of noise in the system. They may fall far enough outside the range of acceptability that they cause disease, including cancers and neurodegenerative diseases such as Alzheimer's, Parkinson's, Huntington's, and Creutzfeldt-Jakob, that develop when cells do not rid themselves of misfolded proteins and they aggregate into amyloid fibrils (Wickner, Maurizi et al. 1999). As with many other biological systems, there is some tolerance of errors in cellular mechanics, but too much error left unchecked can be momentous, if not disastrous.

In Milton's *Paradise Lost*, Satan is accompanied by hell hounds of Rumor, Chance, Tumult, Confusion, and Discord, dedicated to frustrating the good intentions of God with respect to humans at every possible turn. This is supposed to lead

to the human struggle to be good, to be vigilant and to overcome. Similarly, error and variation are central to evolution, were at the heart of Darwin's central notions, and have been a fundamental force that has kept the gods of selection ever on the alert during the long evolution of all of the pathways we have discussed in this book.

Chapter 7

A Repertoire of Basic Genetic Mechanisms

Observing the very similar chemical constituents of diverse life forms, biologists long ago realized that life (on Earth, at least) is a single subject. This edifying realization did not require knowledge of anything about genes and was basically understood even by Aristotle and others in ancient times. However, we now have genetic knowledge, and it confirms the unity of life in exquisite detail and adds whole new sets of phylogenetic relationships that go beyond and in some ways are independent of those of species.

Genes play a variety of roles. The differentiation of complex organisms into a variety of tissues depends on the fact that cells typically only express a fraction of their genes in any given context. This requires mechanisms to regulate context-specific gene expression, which as we have seen largely resides in the genome. Despite the diversity of genes and mechanisms, general statements can be made that apply broadly, from single-celled organisms to plants and animals. Chapter 4 described generalities concerning the ways DNA carries various kinds of “information,” Chapter 5 described ways in which the modular nature of the genome is used to produce complex function, and Chapter 6 described many of the diverse characteristics of cells and their behavior. Table 7-1 provides a few reminders, summary pointers, and general principles. In this chapter, we will discuss some of the types of genes that have arisen in terms of mechanisms of gene action.

Our objective is to identify different kinds of gene action and to describe a few salient features that can help identify generalizations about phenogenetic relationships. In each case, there is a diversity of genes, always involving at least one gene family (usually several). It would not be possible (for us) to enumerate all of these, but there would also be no point to that. They share the general characteristics of gene families: divergent sequence, divergent but related function, and so forth. Online references are available and easily located by keyword searching; they include sequences, functional and evolutionary aspects, protein structure and

TABLE 7-1.

Some basic pointers, reminders, and generalizations about gene functions.

1. Proteins are modular in structure, characterized by discrete, identifiable *functional domains*, which often correspond to their exon-intron structure.
2. Genes arise by duplication, or function domains may be introduced to a gene by translocation of exons from other genes. These events are individually rare but over evolutionary time have been very important.
3. Many gene products are characterized by internal repeated protein motifs that probably arose by tandem duplication.
4. Gene family members diverge in function over time, but these divergent functions may retain some similarities or redundancies, buffering the organism against mutation and enabling us to reconstruct their descent from a common ancestral gene.
5. Similar functions may be achieved by genes with no detectable evolutionary relationship.
6. Phylogenetic relationships among genes in a gene family reflect their history of gene duplication. There can be substantial differences in the numbers of members of gene families, especially among distant taxa, but even among more closely related ones.
7. It is not easy to define exactly what a “gene” is because genetic activity is so diverse, and genes can be variably spliced.
8. Only a small fraction of genes code for classic structural proteins or metabolic enzymes.
9. A large number of genes code for products to manage the processing of other genes, including transcription, translation, and chaperones that help fold, protect, or transport proteins in the cell to their appropriate destination transport.
10. Many genes code for RNA molecules that have direct biological rather than coding function, including regulation of the production of other genes.
11. A large number of genes are involved in communication among cells or between cells and the outside world, enabling cell specialization, homeostasis, and development to occur.
12. Gene family relationships are sometimes related to the arrangement of genes on chromosomes because duplicates sometimes stay in tandem linkage relationships with each other; but gene family members can also be translocated to other chromosomes.
13. Syntenic linkage arrangements of genes (that is, location together on the same chromosome) may, but need not, be related to their coordinated expression.
14. Many if not most genes are expressed in many cell types and have multiple, often unrelated functions. In many instances, the functions can be shown to descend from some common ancestral function (like cell adhesion).
15. To the extent that differentiation is the vital key element of complex organisms—or even of the existence of single cells—gene regulation is as important as gene coding.
16. Genomes are filled with regulatory or response elements (REs), sequences whose chemical qualities “attract” transcription factors that bind to those sequences and help express or repress the gene. These are functional units, though not usually called “genes” and that can be more variable in sequence, number, and location than the coding genes that the REs regulate.
17. Regulatory sequences, like most other functional (and indeed many nonfunctional) elements of DNA structure, can be grouped into related sequences that have some internal variability. Except when found in tandem arrays, these short (approximately 5–15 bp) sequences appear to arise repeatedly by de novo accretion and sequence evolution.
18. There are many genes and mechanisms for policing or preventing errors in a cell. These protect proteins when the cell is under stress, degrade misformed proteins, degrade or repair DNA or RNA, or perform other similar error-moderating functions.

TABLE 7-1.
Continued.

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19. Almost everything that goes on in a cell is affected by chance or environmental events. As a result, no two organisms are complete twins or completely symmetric, cells even of the same type don't express exactly the same genes at the same time (see Hume 2000), and a cell never divides into identical daughter cells. Stochasticity, including the often unpredictable effects of gene-environment interaction, has driven the evolution of various systems but can also drive the pathways themselves.
 20. Many processes of importance to organisms are not genetic, and nongenetic inheritance can be important in the short as well as the long term.
-

gene-evolution diagrams, and tools for analyzing the details of any example. Interested readers can use these as they go along through this and subsequent chapters.

SOME GENERAL TYPES OF GENES

The Central Dogma of biology, that genes code for protein, led to the general notion that genes specify individual structural proteins and enzymes that control basic physiology. The idea (not always explicitly stated) was that an organism is built up of separate identifiable functions and that one gene coded for one function. We know this is an overstatement, and one that can be quite misleading, but it is a view that has persistent effects on biology and biological research. The traditional kinds of genes, perhaps those most easily understood, code for extracellular structural proteins like collagen, intracellular structural proteins like actins and spectrins, carrier molecules like albumin or hemoglobin, or enzymes that trigger reactions like insulin or hormones, as well as for catalysts of basic energy storage and release or catalysts of reactions in the production and manipulation of basic biomolecules such as nucleic acids, amino acids, carbohydrates such as sugars and starches, cellulose, lipids, steroids and vitamin D derivatives, and so on. Other genes have fundamental roles in controlling DNA packaging and replication and have many general cellular housekeeping functions. Large families of genes exist for these purposes. Such "end-product" genes have each traditionally been associated with a biological function itself and, hence, from an evolutionary point of view, the targets of natural selection. Of course, not one of them acts alone, nor without a chain of cofactors related to their use, protein maturation, expression, and the like.

Such functions are vital to life, and many must have been present in the first cells. But essential as they are, these generic functions are not the focus of this book. We are concerned with the intricate specialized functions of differentiated organisms and the many ways the sequence in DNA works to serve these biological functions. Their discovery has broadened our understanding of how integrated organismal complexity is achieved and evolved and deepens the unifying view that genes are the core functional elements of life. But while the phenogenetic relationships being discovered display logical regularities, they are generally anything but simple.

Genes are discovered in many ways, sometimes by identifying coding sequence and sometimes via effects of mutation or studies of function. One reason that genes can be challenging to classify functionally is that as soon as we think we know what a gene does, other uses and expression may be discovered. Genes originating with one function can evolve others that initially may be related but eventually become

at best distantly so. One way this is a problem in scientific practice is that genes are often named for the function for which they were first discovered, which can be very misleading when, as so often happens, diverse additional functions are identified; often, the “original” function is minor or misperceived (e.g., the gene called “eyeless” is not a gene evolved to make—much less remove—eyes but is part of a class of general developmental genes).

Keeping in mind that genes are not necessarily “for” any particular thing (even within a given species), we can at least generally classify them into various functional types, as shown in Figure 7-1. As the figure makes clear, a rather small fraction of genes appear to have “function” in the classical sense. Although not shown in the figure, for each functional group there is a common theme: one or more families of genes descended by gene duplication from a common ancestor, usually with widespread sharing of members of the family among plants, or among animals, or both, and typically also with single-celled organisms. What are some of these types of genes?

PROCESSING GENES

A large class of genes codes for proteins involved in the processing of mRNA or of polypeptides after they have been translated. The processing functions include transport, packaging, error repair, scavenging, activating, inactivating, and modifying proteins or RNA. One interesting example are the heat-shock proteins, often called *chaperonins*. These genes were discovered because they are expressed under

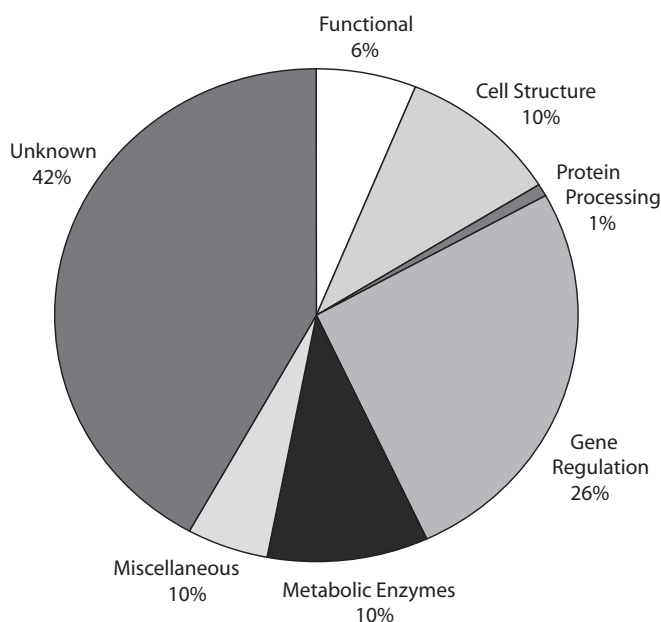


Figure 7-1. Distribution of human gene types based on annotation of human genome sequence. Definitions are rather imprecise and some genes overlap in function. For up to date details see the Gene Ontology website (Gene Ontology Consortium 2003).

conditions of stress. Their intracellular functions include the folding and protection of proteins (normally and/or to resist stress of a high-temperature environment, hence “heat shock”) and detecting unfolded protein or infection. These genes are found in all major life forms. Perhaps, chaperonins are (or were) also evolutionarily primitive intercellular signaling molecules, as they can be found outside of cells and adhere to various cell-surface receptors.

CELL ADHESION AND RELATED OR DESCENDANT FUNCTION

Multicellular life involves cell-to-cell contact in many ways, as described in Chapter 6. Some of this is achieved with members of a large and ancient family of *cell adhesion molecule* (CAM) genes. These typically involve a single transmembrane domain and an extracellular domain that binds to a molecule on an adjacent cell, holding the cells together. Some function as homo- or heterodimers (that is, two copies of the same gene product or one or more copies of each of more than one gene product forming a single functional unit). They may also require a cofactor element such as calcium (e.g., cadherins). Desmosomes and adherens junctions that stitch cells together are formed with CAMs.

Most cells in multicellular organisms have CAMs on their surface, so that they can be properly arranged in relation to other cells. This gives the organism its physical architecture. However, not all CAM binding is between cells; *integrins* can bind to extracellular matrix, for example. *Selectins* and integrins in some situations facilitate cell shape or movement along a substrate (which can be another cell or a blood vessel, for example).

This ancient class of genes has been involved in the evolution of many functions that would not now perhaps be thought of as simply involving architectural adhesion, but their properties are recognizable and the DNA/protein structure shows the common ancestry. Developing and migrating neurons form bundles that depend in part on a combinatorial cadherin “code,” in which cells adhere if they express the same specific set of cadherin gene products. This helps organize neural function (Redies and Puelles 2001). There are also other neural-specific CAMs (called N-CAMs).

These latter genes are members of a large family of immunoglobulin-like CAMs. As will be seen in Chapter 11, the immune system deals with internal damage and foreign molecules by recognition and binding processes that seem to be descendants of simpler cell adhesion phenomena. The immune system recognizes and handles infected or inflamed cells using the CAMs selectins as well as integrins. As things have evolved today, antigen-antibody binding goes beyond cell-cell binding to involve molecular fragments derived from invading cells or molecules. In vertebrates, immunoglobulin-like CAMs include antibody and histocompatibility genes involved in self-non-self recognition.

CAM genes are scattered over the genome, sometimes in large coordinately regulated clusters as in the immunoglobulin and T-cell receptor clusters in mammals.

CELL ENVIRONMENT CONTROL

As described in Chapter 6, cells have pores that enable them to adjust their internal conditions relative to that of adjacent cells and the extracellular space around them. This allows them to keep their internal conditions under control for their

particular needs by regulating what goes into and out of the cell. These pores are modular in that they are controlled by different families of genes with different evolutionary histories, although they share the type of functional role they play.

Ion Channels

Ligand-gated and voltage-gated ion channels are separate types of structure that were described in Chapter 6. Both are formed by complexes of proteins that make controlled passages into the cell. The ion channel may be composed of a single protein with multiple domains that pass through the cell membrane or of homo- or heteromultimers of various channel proteins. Voltage-gated channels typically involve four subunits, with the main subunit having multiple (e.g., five or six), similar transmembrane complexes, arranged in a circular fashion around the pore. Changes in the molecular configuration of the channel complex open or close the channel to ion flow. Whole families of genes code for the ion channels that preferentially transmit specific ions. That is, there are related genes that specify channels for Na^+ or K^+ or Ca^{2+} .

Ligand-gated channels also have multiple copies of components (often 5), including multiple membrane-spanning elements around the pore, along with receptor and response domains. There are three known superfamilies of ligand-gated channel genes (cys-loop receptors responsive to GABA and other signaling molecules, ATP responsive, and glutamate activated). Each class is coded by its own gene family, whose members serve different physiological functions but via shared mechanisms and channel-structure characteristics.

Gap Junctions

Like ion channels that face intercellular spaces, gap junctions are pores that connect cells directly to each other. The gap junction is formed by a circular complex, a *connexon*, composed of four to six protein subunits called *connexins*, which can be repeats of the same subcomplex. What can pass through a gap junction is determined by which gene family members code for the proteins of which it is comprised. In invertebrates, the proteins are coded for by members of the *innexin* gene family; interestingly, these are not homologous to connexins at the DNA sequence level, and they also appear to be unrelated to the CAM cell-contact genes.

Genes to Keep Things in Balance

If genes are generally thought of as making something, an obvious issue that arises is what happens when enough is enough. Getting rid of something can be as important as making more of it. Excess levels can be toxic, and genes related to development or homeostasis may be very damaging if expressed at the wrong times. At some point, a structure has to *stop* developing. Not surprisingly, a large number of mechanisms have evolved whose function is to adjust the balance of constituents in a cell, by removing or modifying things that are already there. Some of these involve repressor REs near genes, but others are genes specific to the purpose. As a generalization, almost anything produced in a cell is regulated in one or more of these ways.

Complex genetic mechanisms exist to destroy incoming bacterial parasites, and internal wounds can be repaired in part by first destroying or removing damaged

cells. This can be achieved by genes that code for products that degrade DNA or RNA or protein or that break down cell membranes or cell surface components.

One large class of such genes codes for *proteases* (sometimes called *peptidases*). Proteases typically cut proteins at the ends or internally or at particular amino acids. They are expressed in particular contexts and may serve special functions, such as the breakdown of extracellular matrix proteins. Plants, animals, and even bacteria have proteases, and there are several families of protease genes, each sharing a specific cutting mechanism (for example, that cut proteins using serine residues in their enzyme structure), which presumably characterized their common ancestor (Proteases 2003). There are four basic, ubiquitous classes (serine, cysteine, aspartic, and the ancient metalloproteases found even in bacteria, which use a metal ion, usually zinc, as a cofactor).

Proteases can degrade proteins no longer needed within a cell, or they may be secreted to work outside the cell, for example, to destroy extracellular matrix as tissue remodeling occurs during development or cell movement. Proteases can be used in defense against incoming proteins, or in the attack as in the breakdown of a host's proteins by toxins introduced by a parasite or predator (e.g., mosquitoes or snakes).

Corresponding to these and with analogous kinds of function are various RNases and DNases that break down RNA and DNA, including nucleases to break down nucleotides, *endonucleases*, and *exonucleases*, which cut DNA internally or at its ends, and a variety of gene products that help control cell growth, detect and repair mutations, and other functions related to DNA/RNA integrity and use. A well-known class of genes is one that codes for *restriction endonucleases*, enzymes that are used by bacteria to destroy incoming DNA (e.g., from viruses). This is a large class; each endonuclease specializes in cutting DNA at a particular short sequence. There is a large repertoire of restriction sites recognized by the products of this group of genes, which humans have appropriated for our own daily use, in molecular genetics laboratories around the world, to cut DNA at specific places of our choosing.

We have mentioned apoptosis, or programmed cell death, which occurs in numerous situations such as tissue remodeling during the complex spatiotemporal processes of development. A variety of genetic means are used for this purpose, including the degradation of mitochondria, cellular protein, or DNA or signals released by a cell that are recognized by cytotoxic mechanism in the organism's immune system. The mechanisms can destroy or disable the cell or prevent it from going through the mitotic cell cycle.

Another diverse set of genes is the *cytochrome P450 (CYP)* genes (the name has to do with their means of discovery and isolation in the laboratory). The products of these genes use iron to oxidize and inactivate classes of toxic compounds; the human liver uses these gene products to breakdown environmental toxins, for example. There is speculation that this huge class of hundreds of genes evolved in the competition between animals and plants—plants producing toxins to avoid being eaten and animals evolving CYP genes to enable them to keep nibbling.

Not surprisingly, in the checks and balances of life, every destructive action can also be overdone, and the amount, timing, and location of degradation activity is controlled by cells; defenses such as protease inhibitors protect against exuberant degradation. This is another manifestation of the widespread homeostatic mechanisms in complex organisms.

GENES FOR REGULATING OTHER GENES

RNA "GENES" AND THE REGULATION OF RNA

mRNA is an indirect code bearer for the translation of protein, but there are more direct RNA-based coding mechanisms. One mechanism for posttranscriptional gene silencing (known in these acronym laden days as PTGS) is *RNA interference* (RNAi). Stretches of DNA that include *antisense* sequences that are complementary to parts of specific mRNA, are transcribed into RNA that folds into a double-stranded (dsRNA) form. As currently understood, the latter is recognized and cleaved by an RNase (RNA-degrading nuclease) enzyme in the *Dicer* gene family, releasing small interfering RNA (siRNA) fragments of 20–25 bp. The siRNA has antisense sequence to part of the target mRNA and together with a protein complex called RISC (RNA-induced silencing complex) binds the target, making it thus partly double-stranded so that it is degraded by RNases (Banerjee and Slack 2002; Cerutti 2003; Grosshans and Slack 2002; Hannon 2002; Sharp 1999; Tijsterman et al. 2002; Zamore 2002) (see Figure 7-2). There may be an intermediate step in which the dsRNA is amplified to many copies so the inhibition can be more effective in the cell. These chains of events are diverse within and between species and involve carrier and facilitator proteins coded by a gene family called *Argonaute*.

Another and related mechanism involves similar processing machinery to activate small temporal RNA (stRNA). The roughly 70-bp stRNA transcript self-hybridizes to form a stem-loop structure that is inactive until a short segment

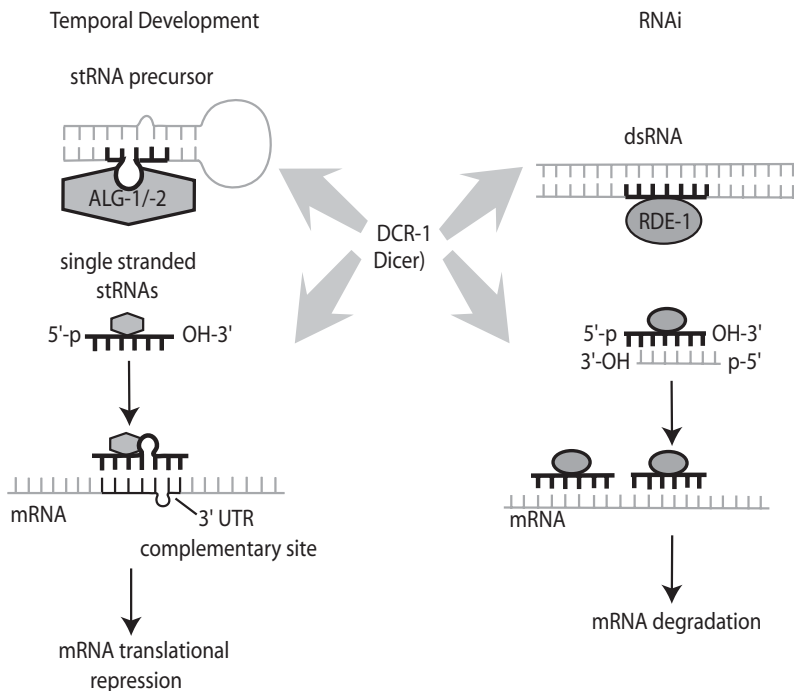


Figure 7-2. Basic antisense regulating mechanisms as currently understood. RDE and ALG are proteins that participate in the process. Redrawn after Banerjee and Slack, 2002.

(about 20bp) is excised. This piece is diced to single-stranded antisense fragments that bind to 3'-UTR sequences in target mRNA. The binding does not appear to cause the RNA to be destroyed, but instead prevents translation, which effectively down-regulates the target gene. Some stRNAs discovered early were the *Let7* and *Lin4* genes in the nematode *Caenorhabditis elegans*, but there are probably many more. In the nematode, these genes are used to regulate heterochronic aspects of development, that is, highly timed expression of each of these genes is required for the transitions through the four stages of larval development, ensuring that patterns of developmental-gene expression and tissue differentiation occur at the right times and places. A *Let7* gene may block translation of specific mRNAs, that have *Let7*-complementary sites in their 3'-UTRs. In one proposed scenario, a temporal chain of interactions appears to occur: a target of *Lin4* inhibits *Let7*, which had been blocking translation of *its* target, *Lin41* mRNA; the latter is an inhibitor of the final adult developmental moult. These genes are widely conserved among animals and seem to have similar developmental function in *Drosophila melanogaster* (Banerjee and Slack 2002; Grosshans and Slack 2002; Sempere et al. 2002), in which *Let7* is temporally regulated by the steroid molting-related hormone *ecdysone*. However, the two stRNA genes and their critical sequences appear to be evolutionarily unrelated. There are numerous other genes of similar function already known, and likely many more to be found.

The inactivation of mRNA may be involved in quick changes of gene expression that can be important in some stages of development or cell differentiation. RNAi also may clean up dysfunctional RNA in the cell and may even affect chromatin or DNA modification (Cerutti 2003). RNAi involving homologous mechanisms is found in animals, plants, fungi, and protozoa, with the usual diversity of related gene family members, reflecting the ancient age of this system, which means that what we see today in terms of function (or the function in which a system was first discovered by us) may not be its original function(s). RNAi may, for example, have originally served an immune function because plants and other species activate RNAi when invaded by infectious agents like viruses; at least some plant viruses appear to have counteracting genes (Tijsterman, Ketting et al. 2002). This would be a mechanism functionally analogous to the use of restriction enzymes by bacteria to degrade incoming DNA, although RNAi works by complementary base-pairing, whereas restriction enzymes bind and cut double-stranded DNA at specific short sequences. Like restriction enzymes that molecular biologists use to cut and manipulate DNA in the laboratory, RNAi is proving to be useful for experimentally interfering with gene expression in the laboratory, by introducing selected antisense fragments into cells at particular times. It is likely that many more genes will be found related to the large RNAi system.

Searches of genome sequences and cDNA libraries have also revealed numerous natural antisense RNA transcripts (NATs) that are complementary to mRNA sequences (see www.hgmp.mrc.ac.uk, and Cerutti 2003; Lehner et al. 2002). Many imprinted genes—their DNA chemically modified to inhibit transcription factor (TF) binding and hence expression—appear to have associated antisense transcripts as another way to inactivate them. A considerable fraction of *cis* antisense RNA sequences are transcribed from the antisense strand of regions of genomic DNA coding for the gene (that is, so the result is complementary to the mRNA being transcribed from the template strand); the antisense sequence is thus automatically available for use so long as the appropriate transcription start signals are available

on this complementary strand. There are also a substantial number of possible *trans* antisense transcripts located elsewhere in the genome (Lehner, Williams et al. 2002). The importance of these, if any, is not known but may indicate elaborate *trans* activation/regulation mechanisms.

Many types of cells have biochemical asymmetry, or polarity, which can be produced by differential distribution of mRNA. For example, asymmetric polarity in the egg cell is fundamental to fly development. In *Drosophila*, the protein coded by the *Bicoid* gene binds to a short 3'-untranslated recognition sequence in the mRNA of the *Caudal* gene. This inactivates the latter. Initially, caudal mRNA is uniformly distributed in the egg, and *Bicoid* is more concentrated at the anterior end (determined by the mother when the egg is produced). The result is an asymmetric concentration gradient of functional Caudal protein, which establishes locally different environments for the expression during the first stages of differentiation of the egg (Rivera-Pomar et al. 1996; Sauer et al. 1996).

ADAR

Other *RNA editing* phenomena affect the translation of a gene's amino acid code. Besides mRNA splice variation, there is at least one other known pre-mRNA editing phenomenon, adenosine deaminase acting on RNA (ADAR), which has been observed in both invertebrates and vertebrates. What ADAR does is change the sequence of particular codons so that the mRNA is translated to contain a different amino acid from that nominally coded in the gene itself (e.g., Reenan 2001). In the target regions, the DNA **A** is changed to a nucleotide inosine (**I**), which for chemical reasons the ribosomes read as **G**, because **I** chemically pairs like **G**. This change is an active enzymatic process; although the system is only recently discovered, an understanding of the targets in immature mRNA that will be converted is developing. No simple reason, however, is yet known regarding why such a mechanism would have evolved.

Most known examples of interference with translation were stumbled upon, rather than deliberately sought, because this is a phenomenon strange to the usual theory of the evolution and use of DNA. Among the known instances are interesting effects on neural function, suggesting that ADAR evolved in regard to some aspects of behavior. ADAR can alter the properties of ion channels in neural tissue and have major behavioral effects on flies (Reenan 2001). Because the degree of ADAR activity may vary, or may even be context-dependent, an element of behavioral flexibility may be introduced via ADAR. However, different uses may yet be discovered, and ADAR activity has been found in other tissues.

SIGNALING

It has been said that only a small percentage of the soldiers in an army actually do the fighting, whereas the rest are there to make the fighting possible by taking care of supplies, maintenance, food, administration, medical care, and the like. Biology is similar, as reflected in Figure 7-1. A majority of genes are used primarily in support of the functional end-products that, as we noted at the beginning of the chapter, comprise the usual image of what genes are. Among the most important support services, as in an army, is to communicate "instructions" by signaling among the many components of an organism.

Examples are legion and include all aspects of life. As will be detailed in Chapter 11, the detection and destruction of an infectious pathogen involves a chain of interactions in which one type of cell is signaled by another to go into action: cell-surface immunological receptors on mammalian white blood cells bind to fragments of pathogens presented on the surface of other immune-system cells. Antibodies secreted by B cells bind these, and once bound, the cell is marked for destruction by the complement system or neutralization by phagocytes. Macrophages and neutrophils have surface receptors that bind to circulating bacteria to target them for phagocytosis.

Metabolism involves signaling, as for example the use of insulin to stimulate cells to bring glucose in from the blood system, or apolipoproteins to stimulate the liver to take in cholesterol (and not make so much new cholesterol). Hunger and satiety responses are triggered by circulating signals. Chromophore molecules bind to photoreceptors, changing them when subsequently hit by photons of a particular energy to which they can respond. Steroid hormones and retinoic acid diffuse into cells where they become bound to specific binding proteins. Airborne odorant molecules are captured by binding proteins and transported to cell surface olfactory receptors. Small molecules diffuse between cells through gap junctions and through cell membrane ion channels.

In plants, diffusing auxins cause cell proliferation to promote upward or lightward growth or establish a dominant meristem to control plant shape. Abscissic acid causes the closure of leaf stomata and promotes seed dormancy by inhibiting cell growth, both reactions to dry conditions, for example. The gas ethylene causes fruit ripening (and is produced by ripe fruit) and is a ligand for transmembrane kinase receptors for the plant or neighboring plants (see below).

Similarly, development and differentiation depend heavily on the production, distribution, and detection of signals. Such signals are responsible for the establishment of polarity and segmentation and induce major cascades of differentiation.

A variety of gene-coded signaling factors (SFs) diffuse from cells to trigger these processes. Table 7-2 provides a description of some of the mechanisms. Signals can travel in various ways or for various distances. Cytokines or growth factors enter the circulation and travel around the body to stimulate growth, but some such signals are carried by circulating cells and only released locally, for example, to stimulate wound healing or immune responses. At the other extreme are pheromones that carry information between organisms, with similar kinds of effects and using very similar mechanisms. In a sense, this makes the members of a population of organisms a kind of large single super-organism, the connecting medium being air rather than blood or sap. We think of organisms as being different entities rather than serial homologs of each other the way vertebrae are but the communication processes are essentially the same. This is but one way in which the nature of what constitutes a “being” is more open than usually considered.

Passive Signaling

We mentioned in Chapter 6 that some small signaling molecules can diffuse through cell membranes directly into the cell. Steroid hormones are examples. These are nonprotein molecules. Once inside the cell, the signal molecule usually complexes with some binding factor, and this then affects gene expression or other aspects of cell biology. The action of various kinds of nuclear receptors is described in Table 7-3.

TABLE 7-2.

A sampler of receptor-mediated signal transduction mechanisms.

The logic of receptor-mediated signal transduction is similar among many classes of signaling factors and receptors. Here are some of the variations that exist on this theme. Note that gene names are historical; they have diverse, sometimes opposite function from those for which they were originally named. Also, we list only representative stereotypes; other examples are known and probably others have not yet been discovered. For example, there are several classes of growth factor that work in different ways; we only enumerate a few of them, because the basic ideas are similar.

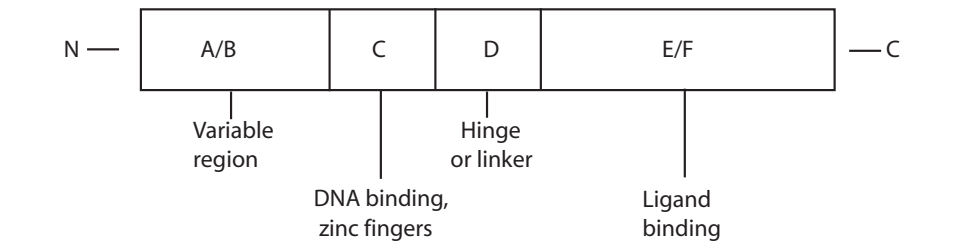
1. *FGFs* (fibroblast growth factors) are a class of around 20 diffusible mitogens (stimulants of cell division) that have numerous splicing isoforms (mRNA equivalents, that for example, make proteins that contain different complements of exon units). FGFs are ligands for a corresponding set of *FGF receptors* (*FGFRs*). Ligand binding is aided by cofactors (heparin sulfate proteoglycans, or *HSPGs*) that complex with the FGF and the FGFR. The intracellular FGFR domain is a tyrosine kinase. The extracellular part of an FGFR has several immunoglobulin-like domain loops, because of their origin by gene duplication from the cell-adhesion gene family that includes antibody and many other types of genes. However, the homology does not imply functional similarity. These receptors often are activated when two adjacent copies jointly bind ligand molecules. Other cell adhesion molecules may serve as ligands. FGFs stimulate growth and differentiation in many contexts and tissue types, mainly in embryos.
 2. *Wnt* signaling factors (called *Wingless*, in flies) are ligands for serpentine receptors in the *Frizzled* class (see Table 7-4). The receptor is ligated to an intracellular *Disheveled* gene product, which transduces the received signal through a pathway that releases a β -catenin molecule to bind with and activate a TF (e.g., *Lef/Tcf*). Different pathways respond to different members of the *Wnt* family, through different second messenger systems, and there are specific *Wnt* inhibitors (Niehrs 2001).
 3. *TGF β* (transforming growth factor) signals are diffusible signals that include the BMP (bone morphogenetic protein, called *Dpp* in flies) signaling factors. *TGF β* dimers bind between copies of two variant types of *TGF β* receptors. The type II receptor phosphorylates serine or threonine residues on the adjacent type I receptor molecule. The latter then phosphorylates and activates intracellular Smad proteins, intracellular signal mediators that dimerize to form active TFs in a complex set of pathways (von Bubnoff and Cho 2001).
 4. *Hedgehog* gene products are diffusible signaling molecules that travel only a few cell diameters. Hh proteins are ligands for cell-surface receptors of the *Patched* gene class bound to another protein called *Smoothened*. Receipt of the Hh ligand by this *Patched-Smoothened* complex leads to changes in *Smoothened* whose intracellular domain activates an intracellular transcription activator *Ci*. When *Smoothened* is quiescent, a modified *Ci* protein acts as an inhibitor of the same genes. *Hedgehog* signaling may work in relay fashion, with cells receiving signal and secreting *Hedgehog* protein of their own to travel the next few cell diameters.
 5. *Notch* class proteins reside in the cell surface and can receive several ligands, including proteins from the *Delta* (*Jagged*, *Serrate*, or other) class, which are bound to the membranes of adjacent cells. The binding event enables protease cleavage of an intracellular domain of the Notch protein. The cleaved fragment migrates to the nucleus where it complexes with and activates a TF (e.g., of the *CSL* class). Another consequence is that the Notch-presenting cell may be inhibited from producing its own *Delta*. *Notch-Delta* signaling generates zones of expression (cells presenting Notch) and inhibition zones in surrounding cells. This mechanism is used in neural structures, broadly defined, in vertebrates and invertebrates, often involving repetitive patterning.
-

TABLE 7-3.
Nuclear receptors and how they work.

Nuclear receptors (NRs) are ligand-dependent TFs that direct targeted gene expression when bound with steroid and thyroid hormones, retinoids, vitamin D, and other ligands. Ligand-bound NRs by and large interact with transcriptional coactivators, whereas ligand-free NRs interact with transcriptional corepressors. Ligand-binding significantly affects the distribution of receptors within the cell, both within the cytoplasm and the translocation of receptors from cytoplasm into the nucleus because ligand-binding releases chaperones and exposes nuclear localization signals (NLSs) (Black et al. 2001).

NRs activate gene expression by binding to a short DNA sequence, often (A/G)GGTCA, called a *hormone response element* (HRE). HREs are typically located in the promoter region flanking the coding region of the responding gene. This HRE is often present in two copies, either head to head as palindromes, as direct repeats, or tail to tail as inverted palindromes. The NR binds either as a homodimer or as a heterodimer, and the orientation of the HRE pairs determines to which HRE the receptor dimer binds.

The typical NR includes A/B, C, D, E, and sometimes F domains, each with a separate role in ligand binding and gene activation. C and E/F are conserved in basically all family members, whereas the A/B and D domains vary (Aranda and Pascual 2001). The structure of a typical nuclear receptor is diagrammed in Scheme 7-1. The A/B domain contains the ligand-independent transactivation domain. Region C recognizes specific DNA sequences. Linker region D connects the DNA-binding domain to the ligand-binding domain, E/F.



Scheme 7-1. Typical nuclear receptor domains.

Receptor-mediated Signaling

Larger signaling molecules cannot diffuse through the membranes of receiving cells and instead serve as ligands for specific receptors protruding from the surface of cells sensitized thereby to receive the message. This is true of a variety of growth and signaling factors as well as of protein hormones like insulin and adrenaline. In fact, message exchange among cells is so important that cell membranes are often littered with receptors of many kinds.

Cell surface receptors have one or more ionically neutral (uncharged) domains of the right length to reside stably within the thickness of the cell membrane. Trans-membrane domains typically evolve by duplication events and often coincide with exon-intron boundaries in the receptor gene. The extracellular domain of the receptor is usually the ligand-binding site, and an intracellular domain of the receptor protein is altered by the physical event of a ligand binding to its extracellular

domain. The changes in the intracellular part trigger second-messenger events, described in Chapter 6.

There are three general classes of cell surface receptor proteins, defined by the manner in which they transduce signals into the cell. We have already discussed some of them and their attributes. *Ion-channel-linked receptors* transmit inorganic ions between electrically excitable cells. *G protein-coupled receptors* (GPCRs) are a large family of transmembrane proteins (more than 2,000 have been identified) found in all eukaryotes, although fewer have been found in plants than in animals; aspects of this family are given in Table 7-4. The table shows the diversity of known function fulfilled by the various classes of these receptors, with class subdivision based on sequence phylogeny, indicating some, but not total, relationship between function and evolutionary origin. These are seven-transmembrane spanning helical proteins (e.g., see below). Binding of a ligand (signal molecule) to the extracellular domain causes a conformational change in the intracellular domain, which activates the G protein bound there. The G protein dissociates from the receptor and carries the signal to an intracellular target, which can be an enzyme or ion channel. Many transmembrane signaling mechanisms have evolved and will be discussed more fully below. *Enzyme-linked receptors* act either as enzymes directly or in association with enzymes.

A particularly important class of receptors, including GPCRs, that we will see more of in later chapters is described in Table 7-5. These are called seven-transmembrane membrane receptors (7TMRs) or “serpentine” receptors because their seven domains “snake” through the cell membrane. Many of these are receptor tyrosine kinase (RTK) genes, meaning that the signal receipt triggers the intracellular phosphorylation of tyrosine residues on responding proteins. Other kinase-based receptors phosphorylate serine or threonine amino acid residues on their target proteins. The phosphates are transferred from ATP molecules by kinases and removed by *phosphatases* to deactivate the protein. Phosphorylation produces structural change in a protein, causing it to bind or release another molecule. Protein phosphorylation is a predominant means by which activity is controlled inside the cell. It has been estimated that perhaps one-third of the proteins in a mammalian cell are phosphorylated at any given time, and that 2–3 percent of a eukaryotic genome encodes kinases (Hunter 1995). The initial response to ligand binding in turn activates *second messenger* molecules inside the cell. Through cascades of subsequent interactions, activated messengers are transported to the nucleus where they serve to activate (or repress) the expression of specific genes.

As would have to be the case, signaling genes have corresponding receptor genes. The two families coevolve in the face of mutation, so the signal and its receptor mechanism still work, and there can be cross-reactivity among them; thus, a given SF can bind to the product of more than one receptor of the same class. The binding may not be as efficient, but this cross-reactivity provides some functional redundancy, perhaps at the loss of some specificity. This is an example of the partial nature of functional modularity or sequestration that is so prevalent in life. The presence of cross-reactivity also suggests that during evolution the SF and receptor duplication events do not need to be tightly coordinated: mutational divergence of duplicate SF genes can lead each eventually to bind preferentially to one of the evolving, duplicating family of receptors.

TABLE 7-4.
G Protein-Coupled Receptor Superfamily.

Family	Group					
	I	II	III	IV	V	VI
Receptors related to rhodopsin/ β-adrenergic receptor	Olfactory, adenosine, melanocortin, cannabinoid, several orphan receptors	Serotonin, α- and β-adrenergic receptors, dopamine, histamine, muscarinic, octapamine, and orphan receptors	Bombesin/neuromedin, cholecystokinin, endothelin, growth hormone secretagogue, neuropeptide Y, neurotensin, opsins, tachykinin, thyrotropin-releasing hormone, and orphan receptors	Bradykinin invertebrate opsins, and orphan receptors	Angiotensin, C3a and C5a, chemokine, conopressin, eicosanoid, fMLP, FSH LH and TSH, galanin, GnRH, leukotriene, P2 (nucleotide), opioid, oxytocin, platelet activating factor, thrombin and protease activated, somatostatin, vasopressin, vasotocin, and orphan receptors	Melatonin and orphan receptors
Receptors related to the calcitonin and parathyroid hormone receptors	Calcitonin, calcitonin gene-related peptide, corticotropin-releasing factor, insect diuretic hormone, and orphan receptors	Parathyroid hormone/parathyroid-related peptide receptors, orphan receptors	Glucagon-like peptide, glucagons, gastric inhibitory peptide growth hormone releasing hormone, pituitary adenyl cyclase activating peptide, vasoactive intestinal peptide, secretin, and orphan receptors	Latrotoxin and orphan receptors		

TABLE 7-4.
Continued.

Family	Group					
	I	II	III	IV	V	VI
Receptors related to the metabotropic glutamate receptors	Metabotropic glutamate receptors	Extracellular calcium ion sensor receptor	GABA-B receptor	Putative pheromone receptors		
Receptors related to the STE2 pheromone receptor	STE2- α -factor pheromone receptor					
Receptors related to the STE3 pheromone receptor	STE3- α -pheromone receptor					
Receptors related to the cAMP receptor	<i>Dictyostelium discoideum</i> CAR2 to CAR4 receptors					

From: (Conn and Means 2000). These groupings are provisional because the phylogeny of the *FTMR* genes is still tentative.

TABLE 7-5.

The special case of the Serpentine receptors.

A particularly interesting and important class of cell-surface signal receptors are those known as the seven-transmembrane G protein-coupled receptors (7TM-GPCRs, or 7TMRs). There are hundreds of genes in this class, involved in diverse aspects of information transfer to cells. As shown in Scheme 7-2, these receptor molecules contain seven helical transmembrane domains, with additional extra- and intracellular domains.

Transmembrane information transfer that relies on secondary messengers often involves G protein cascades. G protein-coupled cell surface receptors (GPCRs) are a large and ancient class of seven transmembrane receptors, cellular signaling mechanisms found in many different kinds of receptor-ligand interactions in a wide range of cell types and organisms, including single-celled yeast (the receptors for yeast mating factors are an example) (Alberts 1994). In vertebrates, close to 2,000 GPCRs have been identified, including more than 1,000 for odorant and pheromone detection alone. About 1,100 GPCRs are known in *C. elegans* (Bockaert and Pin 1999).

When GPCRs are compared at the amino acid level, by phylogenetic relationship and the type of ligand to which they can bind, they cluster in at least six main families; GPCRs from the different families show little sequence homology at the DNA level (GPCR 2003; Bockaert and Pin 1999). These tentative groups are itemized in Table 7-4. Class I GPCRs constitute the largest and includes many small peptide or weakly hydrophobic organic molecules. This class is basically for chemoreception and chemotaxis and includes olfactory and taste receptors, for which the ligands are weakly hydrophobic, organic molecules, or small peptides. Class II GPCRs antedate the protostome-deuterostome divergence, and this class comprises receptors for biogenic amines (acetylcholine, catecholamine, and indoleamine) largely related to neurological function. Class III genes include small neuropeptide receptors and receptors for chromophores (vertebrate visual pigments); these are the opsins used in light sensing. Class IV genes are insect and molluscan opsins, which bind to small odorant molecules of various types. Class V genes are receptors for small hormone molecules of various kinds that are involved in immune chemotaxis and blood-clotting; these genes are also involved in the secretion of steroid hormones. In vertebrates, some similar neurological and endocrine peptides are mediated by single-transmembrane receptors. Class VI GPCRs are members of the *smoothened/frizzled* family of vomeronasal receptors.

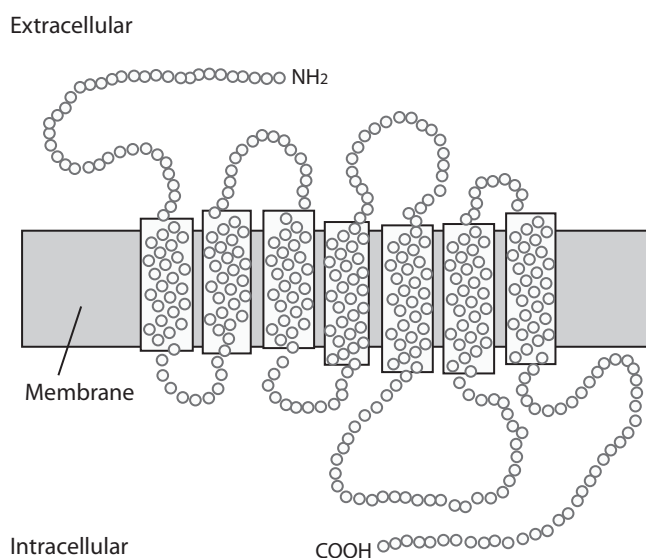
Additional GPCRs continue to be found and classified. No amino acids are conserved among all the classes of these genes, and few are conserved even among many. Within a given class, a relatively small number of sites determine the receptors' specificity to ligands.

GPCRs set into motion a chain of events that alter the concentration of intracellular signaling molecules, which in turn affect the behavior of other cellular components. Most G protein cascades involve alterations in the cellular concentrations of cAMP, cGMP, or Ca^{2+} , which induces activation of an effector molecule that produces a response specific to the receptor and cell type. This system is common in but not restricted to sensory perception; regulation of heart rate, contraction of smooth muscle, and secretion of hormones or growth factors are examples of other systems that use this mechanism. The effects can be achieved in various ways, including the regulation of membrane ion transport channels, transmission of signal pulses to adjacent cells as in the transmission of nerve impulses from retinal neurons, or the regulation of transcription factors. Similar mechanisms are fundamental or even ubiquitously used for the transfer of cell-differentiating signals in both plant and animal development (Chapter 8).

TABLE 7-5.

Continued.

Extracellular workings: A G protein cascade is activated when a ligand binds to a receptor (see Scheme 7-2). As 7-spanning transmembrane proteins, the free NH_2 terminus of the protein remains outside of the cell, and the COOH terminus floats in the cytoplasm of the cell. The receptor-ligand binding causes a conformational change (phosphorylation) in the three-dimensional structure of the receptor, and this results in the transfer of signal from the receptor to a G protein, which is in the plasma membrane of the cell.



Scheme 7-2. Stereotype of 7TMR protein structure. Seven regions are hydrophobic and insert stably in the cell membrane. The intracellular domain is involved in signal transduction. Signal may be received by the extracellular domain(s) or in some instances occurs in the intracellular or membrane domains. Specific amino acid variants in the appropriate domains affect the nature of the ligand (or light energy frequency) that triggers a response, and hence is how divergence and specialization in function is brought about.

Intracellular workings: G proteins are *heterotrimers*, proteins with three subunits: α , β , and γ . The structure of the α -subunit usually defines the type of G protein. The α -subunit binds a guanine nucleotide (thus giving this class of proteins their name). When the guanine is in the form of GDP (guanine nucleotide *di*-phosphate, rather than *tri*-), the α -subunit binds with the $\beta\gamma$ -subunits. When the nucleotide becomes a GTP (guanine nucleotide *tri*-phosphate), which is induced by the receptor-G protein interaction, the G protein is activated, and the α -subunit is freed from the complex (Lewin 2000).

What the α -subunit does next depends on the class of G protein-coupled receptor; despite the large number of GPCRs, there are only a few basic ways in which they affect signal transduction. Some GPCRs regulate the activity of enzymes, and some affect the concentration of cyclic nucleotides or calcium inside the cell, with the effect of activating or inactivating ion channels. Others directly activate or inactivate ion channels, but the effect of G protein activity involved in sensory transduction is to alter the ion permeability, and thus the excitability, of the membrane of the target cell.

TABLE 7-5.
Continued.

Of the GPCRs that modulate cyclic nucleotides, some regulate *adenylate cyclase* activity. Activation of adenylyl cyclase leads to the production of cAMP as a second messenger, and cAMP in turn leads to the activation of ion channels. Odorant receptors are an example of this type of GPCR. Photoreceptors are a class of GPCR coupled to the G protein *transducin*, which activates phosphodiesterase, an enzyme that causes a drop in the level of cGMP, which leads to the closing of sodium and calcium channels and *hyperpolarization* (an increase in the negative charge) of the cell.

These are general, stereotypical descriptions. Not all aspects of all of these many genes have been carefully checked. Some things are reasonably well-studied, however. The ligand binding characteristics differ among genes and among classes and will be discussed in several places in this book. The point is that receipt of an odorant by an olfactory receptor GPCR is “wired” to the olfactory part of the brain, whereas the receipt of a photon by a chromophore GPCR ligand is wired to the visual system. We have given this brief summary of this class of genes because its members are used in so many diverse ways relevant to this book and because it is central to communication between a cell and its outside world. But it is the logic and modular nature of the various uses of these common signal-related molecules that is of importance here. For biochemical details, seek appropriate sources.

REGULATION OF GENE EXPRESSION

Signaling between cells may affect a variety of functions. Usually this means changes in gene expression in the receiving cell, which implies that the signal causes the activation or inactivation of one or more TFs. There are several TF gene families (Table 7-6), characterized by their DNA binding domains of their coded protein. The most famous of these is the homeobox after which a major TF class is named. The homeobox is a region coding for a 60-amino acid homeodomain (so called because of the effect discovered in the 1980s when mutations in these genes produced famous replicated or altered segmental structures in flies). Each of the many subclasses of homeobox genes shares one or more variants in the homeodomain, and they bind to somewhat different response or regulatory element (RE) sequences; however, as a rule, this region of the class of genes is much less variable than other active domains, and there is a lot of sharing of the binding sequence (e.g., **TAAT**) among homeobox genes; the other domains are more varied and have different functions, not all of which are known (Alberts 1994; Transfac 2003).

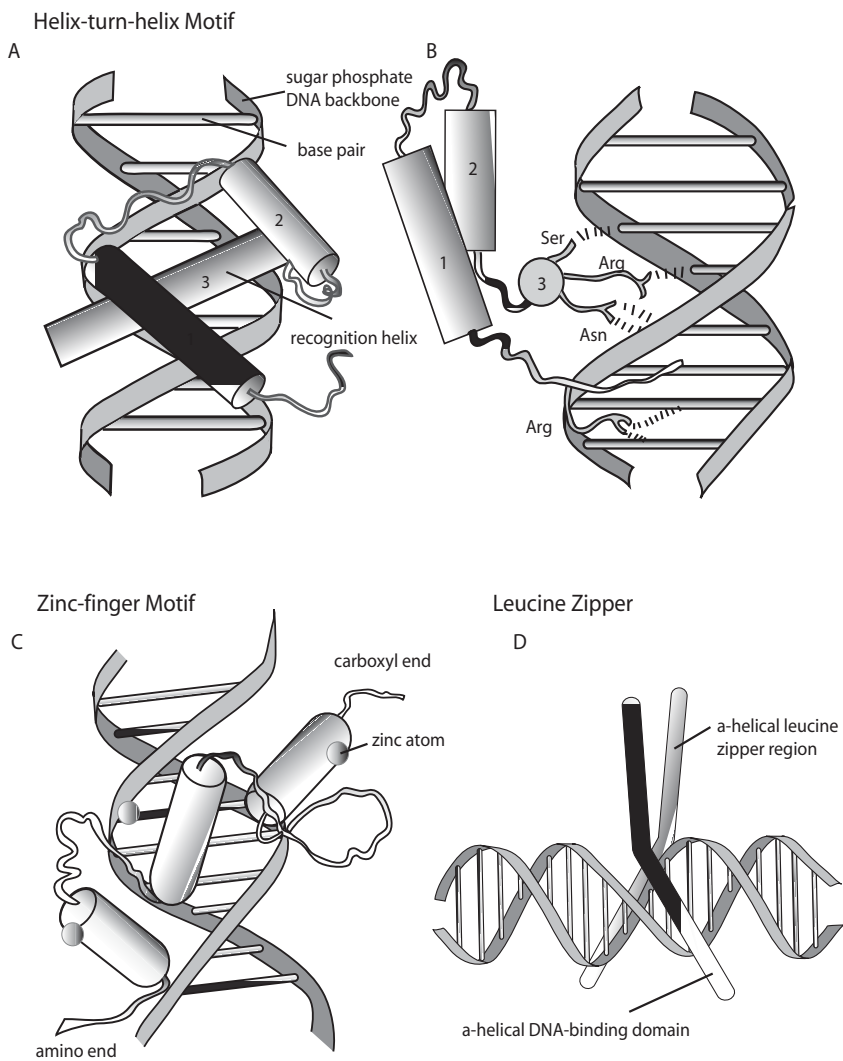
CHAPTERS 4–5–6–7: COMPLEX TRAITS REVISITED

Signaling and gene regulation are so fundamental to complex organisms that we can expect there to be an elaborate set of mechanisms, and there are. We can now look at them in light of concepts covered in this and previous chapters. The importance of the modular organization of the genome and the various information encoded in DNA sequence is clear.

The major classes of genes for signaling appear to have evolved from multiple essentially independent beginnings. However, the *logic* of signaling mechanisms

TABLE 7-6.
Characteristics of the major transcription factor classes.

Transcription factors (TFs) share the generic property of binding to specific DNA sequences (which are often palindromic so the binding of both strands can occur in anti-parallel ways). But each family of TFs varies in detail in how its DNA binding is accomplished. Scheme 7-3 shows some of the major mechanisms (this section is not exhaustive but again is intended to suggest the modular nature and logic of the process and the diversity of agents that bring it about).



Scheme 7-3. DNA binding motifs. Helix-turn-helix motif; (A) homeodomain bound to its specific DNA sequence; (B) specific amino acids of helix 3 (recognition helix) make contact with the major groove of the target DNA at specific sites; (C) zinc-finger motif: this motif uses zinc fingers of specific amino acids to recognize DNA binding sites; (D) leucine zipper motif; two α -helical DNA-binding domains join to form a leucine zipper dimer that binds to a specific DNA sequence.

TABLE 7-6.
Continued.

Homeodomain TFs (coded for by homeobox genes) have a basic 60 or 61-amino acid helix-turn-helix (bHTH) protein domain that, depending on its amino acid sequence, binds specific DNA sequences; the third helix extends into the major groove of the DNA that it “recognizes.” Amino acids in the NH₂-terminal portion of the homeodomain also contact the bases in the minor groove of the DNA double helix. A class called *POU* TFs also has a second DNA-binding domain. *Lim* and *Pax* genes comprise two other homeodomain gene classes. The *Hox* class is the most well-known; they are involved in neural differentiation, anatomic axis specification (e.g., main anterior-posterior, or proximal-distal appendage axes), and head and eye development.

Zinc finger TFs form domains that include loops made by the binding of spaced pairs of cysteine and histidine around a zinc molecule, leaving a projection (“finger”). Between such fingers is a sequence-specific DNA-binding region. *Zf* genes include the *Sox* family that bend DNA to assist in the assembly of regulatory protein complexes, sex differentiation, cartilage development, and hormone and retinoic acid signal reception. They are part of a large number of chromatin-associated genes with a High Mobility Group (HMG) box, a DNA-binding domain that recognizes a consensus sequence **(A/T)AACAAT**. There are many *Sox* genes, and several classes of *HMG* genes that are not closely related to this family, which may or may not include a zf domain.

Leucine zipper TFs are formed of two polypeptides joined by regularly spaced leucine residues (the “zipper”) adjoining a DNA-binding region of basic (positively charged) amino acids.

Basic helix-loop-helix (*bHLH*) TFs form dimers in each of two helical domains, that are separated by a loop; DNA binding is by a region of 10–13 basic amino acids at the end of the first helix in each of the polypeptides. The two molecules may be of the same or from a related bHLH gene. bHLH genes are important in a variety of neural and sensory developmental functions, and include the *MyoD* genes used in muscle development.

MADS box genes regulate many functions but are particularly prominent in flower morphogenesis (the acronym is the initial letter of the first four family members identified). Along with a variety of other functional domains, these genes have a characteristic 57 amino acid DNA binding motif, and a domain for forming homo- or heterodimers.

Other classes of TFs, such as the “paired” group, have one or two DNA binding domains that are variants of these classes. The TF families are old and have accumulated family members through duplication events in long phylogenetic lineages; there are therefore numerous members overall, that vary among species. For example, there are tens of genes in the *Hox* homeobox superfamily (39 in humans). Typically, representative homologs in the animal TF families are found across the animal world.

(e.g., receptor-ligand binding leading to TF activation and *cis*-regulation) is generic and development and its evolution are now frequently viewed in terms of regulatory pathways (Carroll et al. 2001; Davidson 2001; Wilkins 2002). A regulatory pathway or circuit goes from inducing signal to TF activation and hence an activated (or repressed) developmentally *downstream* target gene. Development is hierarchical in that one circuit can activate subsequent circuits—for example, when the first

TF activates a second TF that then activates subsequent genes. A set of genes regulated by the same circuit has been called a *battery* of genes. Interactions among contemporary circuits or batteries comprise a regulatory *network*. It might be worth noting that *meshwork* could be even more evocative, because two-dimensional networks of regulation are connected to each other through time, with some circuits active and others missing at different times, or with a pathway having different uses at different times, and effects feeding back upon themselves, depending (presumably) on what else is active at each time.

An indicator of the basic importance of this kind of genetic logic is that SFs, their receptors, and TF classes are widely represented in vertebrates and invertebrates, sometimes even plants and bacteria. It is possible in many instances to identify specific orthologs among gene family members in very distantly related groups, such as flies and mammals. Homologous interaction pathways are often deeply conserved, such as between a given ligand, its receptor, and/or the TFs it activates. In addition, such conservation may effect similar function (such as activation in neural or gut tissue), revealing surprisingly deep homologies in basic animal or plant functions.

Members of TF families may be linked in chromosomal clusters that have been conserved for long evolutionary time periods, and the conserved linkage arrangement is related to the control of the expression of the genes in the cluster. However, this is not an automatic need for TF regulation, because multiple TFs, including members of the same families as those in clusters, are chromosomally off by themselves.

REGULATORY CONTROL OCCURS IN DIFFERENT WAYS

Some TFs are autoregulatory; that is, once expressed they enhance their own continued transcription—copies of their own RE sequences are located 5' to the TF genes themselves. Just as apoptosis and proteases remove cells or substances in a variety of remodeling or repair contexts, gene expression can also be down-regulated or inhibited.

Inhibition can be achieved in various ways. One way is pretranscriptional, when other regulatory repressor proteins bind areas around a gene to prevent activation by TFs. Chemical modification of the regulatory region has the same effect, preventing access or binding by TFs. The DNA cannot be opened (e.g., histones removed) to permit access by an activating TF. Earlier, we described posttranscriptional regulation by RNAi. There is also posttranslational inhibition. For example, bHTH TFs (Table 7-6) bind REs as heterodimers. The TF protein has separate dimerization and DNA binding domains. There are genes that code for truncated proteins that lack the DNA-binding domain; they form normal dimers, but the result cannot bind DNA properly, which prevents or inhibits the expression of the gene a normal dimer would activate. Similarly, there are receptors that lack one of their domains, and hence reside in the membrane where they receive signal, but the receipt is not transmitted internally to the cell (e.g., Kroiher et al. 2001).

Expression of a particular gene or cell-specific set of genes depends on the required set of TFs and/or inhibiting factors being present at the same time in the cell, meaning that the appropriate set of receptors, second messengers, and the like must also be expressed in the cell, along with the TFs themselves. Cells may require more than one signal for a given action to take place and thus all the requisite receptors may need to be triggered; this means that the extracellular as well as the intra-

cellular environments must be appropriately prepared. Thus, itemizing a pathway by itself is somewhat of an illusion of simplicity.

AN EMBARRASSMENT OF RICHES, OR NEO-VICTORIAN BEETLE COLLECTIONS?

A description of the basic logic of signaling and gene-regulation mechanisms, along with a catalog of their major classes, does not do justice to what actually is going on within a cell. The cascades of regulation that bring about even relatively simple aspects of complex traits are themselves complex (e.g., Carroll, Grenier et al. 2001; Davidson 2001) and often involve parallel shunts or redundancy. Regulatory and signaling pathways can be investigated by a host of experimental *in vivo* and *in vitro* methods, including cell culture, animal models, and the direct manipulation of individual genes or regulatory sequences. Experiments dissect pathways component by component in a way to reveal their individual effects (*under the particular laboratory conditions being used*). As shown in Figure 7-3, regulatory pathways can be intricate, and also, the apparent complexity of a pathway can depend on the method used to study it. For example, in part A, the pathway seen from normal development of relatively simple model organisms can seem quite straightforward. In part B of the figure, the results of different experiments, in different cell types under different conditions, in more complex organisms, can make the “same” pathway appear more complex, because the latter approach reveals more of the phenogenetic equivalence (redundancy, interconnectedness) of “the” pathway.

Experiments with individual known pathways can be supplemented with genomic approaches that identify which of the entire set of an organism’s genes are used in a given cell type. With the use of various technologies, it is becoming possible to identify and quantify all the genes that are being expressed in a given cell type. This *expression profiling* can identify, for example, genes whose expression changes quantitatively or qualitatively before and after cells are subjected to nutritional stress or those that are expressed differently between tumor and normal cells from the same tissue (e.g., Eisen et al. 1998; Tamayo et al. 1999). Expression profiles are useful when the situation is relatively simple or well-controlled experimentally, but they can also reveal the quantitative and complex nature of cell behavior. Redundancies and other interactions may be difficult to discern from an expression list (even with quantitative data) alone.

Expression profiles can reveal whether genes with coordinated expression under the tested circumstances are (1) in coregulated chromosomal clusters, (2) coregulated members of gene families, or (3) scattered genes coregulated because they have related function (e.g., networks of TFs and downstream expression cascades). Co-regulation within clusters can be achieved in various ways, and genes on different chromosomes that are coordinately expressed can use the same or different regulatory mechanisms. That is, this can be due to similar REs, or different REs whose coordinated usage evolved; in some instances, phenogenetic drift is probably responsible, if different genes have had a similar common expression but have evolved different regulatory mechanisms. The alpha- and beta-globin genes, and different means of coordinate regulation of *Hox* genes in vertebrates probably represent instances of the former. But the function could have been achieved by recruiting different, otherwise unrelated genes to some new function.

In some ways, we have in recent years been presented with a sudden embarrassment of riches. There is a danger in taking the details too seriously, mistaking

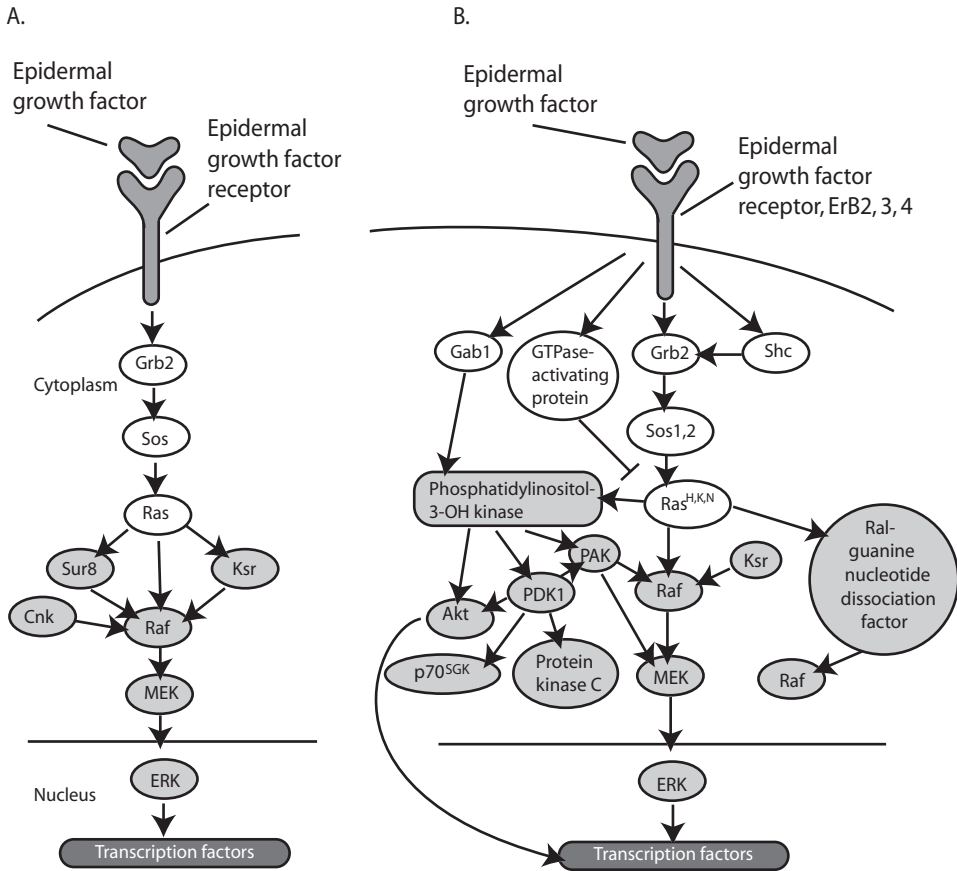


Figure 7-3. Two views of the *Ras* regulatory pathway that responds to growth-factor signal. (A) Genetic view; (B) view from molecular cell biology. Ovals indicate individual factors whose specific names are unimportant here. Grayed ovals indicate factors classified as “effectors;” white ovals as “intracellular mediators” of the signal. Redrawn from (Downward 2001) with permission; see original for details and terms, text for explanation. Original figure copyright Nature Publishing Group 2001.

enumeration for understanding, underestimating phenogenetic equivalence and its importance or prevalence, and viewing genotype-phenotype relationships as being more prescriptive and unitary than they are in nature. In this sense, this new work resembles the stage of biology in the 19th century, when naturalists traveled around the world collecting the beetle specimens that were then dispersed throughout the museums of Europe. This led to an accumulation of species lists that we tend today to view as merely descriptive natural history. Until we have a better ability to synthesize and digest the information from gene expression studies, the difference between then and now may be less than we think.

One problem is that manipulation of gene expression in the laboratory is almost always artificial and can even make systems seem misleadingly complex (or blind investigators to alternative pathways) (Downward 2001; Niehrs and Meinhardt 2002). This is illustrated by comparing Figures 7-3A and 7-3B, and that does not

even consider variation within species. Sequence variation in all the elements, from TF protein sequences and regulation to the REs of regulated genes and their surrounding sequence and chromatin structure, can lead to quantitative variation in traits, while experimental studies in development generally focus on the invariant aspects of mechanisms (at least, as seen in model animal strains). Experiments cannot in a practical sense test all combinations of variants, and strain-by-strain variation has raised caution flags that should not be ignored.

Most pathway diagrams seen in the literature are partial, even when they appear as 7-3B, in part because they do not include all the other pathways required to activate the components that are shown. More thorough diagrams can be intimidating in complexity (Davidson et al. 2002; Davidson et al. 2002) even if the pathways shown are an exhaustive list and even if they are all correct and always operative. Individual studies report experiments on pieces of networks, often those known by experience with natural or randomly engineered mutation, but then specifically followed up experimentally. This can sometimes give the impression that the tested genes are the only genes of importance, but this can be highly misleading (expression profiling shows this as well).

Reports of only subsets of genes are understandable for practical reason, but often they consist of a mixed bag of genes, including receptors, TFs, some second messengers and the like. One can view these as akin to throwing darts randomly at a complex diagram, and then studying the result as if those were the only important genes. These facts should be kept in mind in reading this book. Many examples will follow in which we present these kinds of partial sets of genes, and in discussion of other phenomena we simply omit pathway details that do not apply to the logic of the system we are discussing.

Considering the complex nature of such cascades one can see the many points where mutation can introduce variation. Nonetheless, despite all these caveats, there is considerable and tractable order. Some broad generalizations have already emerged, and we can consider some of them and what they mean.

A BASIC FUNCTIONAL TOOLKIT

The logic of *cis*-regulation, messenger-transduction, modular genome organization, and combinatorial expression provides a powerful new way to understand how new function arises and complex organisms are assembled: a small number of components can generate a huge number of combinations, and a new combination can in principle add novelty without destroying existing traits in an organism, for example, by only affecting specific enhancer locations, not a TF itself.

Pleiotropy Aplenty

The same TFs, SFs, receptors, and REs are typically used in numerous developmental or histological contexts in an organism or even at different times and different ways during the ontogeny of a single tissue or organ. This in a sense is what is implied by the term “toolkit”, but there is an important implication: if expression control is combinatorial and each gene has many uses, we can’t expect to understand the development of a given structure until we know most or all of the regulatory factors involved.

Pleiotropy can appear to involve changing sets of functions, and phenogenetic drift can bring that about by substituting a new regulatory mechanism for the orig-

inal one. A well-studied example is the Stripe 2 enhancer region responsible for expressing the developmental gene *Eve* in the second of the many stripes in the very early fruit fly embryo. *Eve* activity is similar between closely related species of fly, but the REs and TFs involved bear only limited homology (Carroll, Grenier et al. 2001; Ludwig et al. 2000; Ludwig and Kreitman 1995; Ludwig et al. 1998). However, the multiple uses of a gene or pathway in organisms today can be an indicator of phylogenetic relationships among those uses: they may have diverged from some common ancestral function.

Development is Functionally “Arbitrary”

Just as the codon system for specifying amino acids is completely arbitrary relative to the chemical properties of the proteins being coded for, regulatory DNA sequences are arbitrary relative to the function they enable (Weiss 2002d; 2002e). There is nothing about regulatory genes or RE sequences that has anything directly to do with the function of a regulated protein. There is nothing about the chemical nature of SFs or their receptors that is related to the physical or functional properties of the limbs, guts, feathers, teeth, collagen, or chitin, whose production they induce. Regulatory sequence information is highly specific because ligands only fit particular receptors or TFs only recognize particular REs, but the same signaling machinery can operate with any downstream function. As shown in Figure 7-1, a large fraction of the genome is indeed of this logically necessary but functionally arbitrary type. This again is implied by, or a consequence of, the modular, repetitive, combinatorial use of the same elements for sequestered differentiation that is deeply built into the nature of life.

Hierarchies of Regulatory Circuits, Batteries, and Networks

Some TFs are known as “selectors”, meaning that they activate a hierarchical cascade of change in the expression of other, developmentally “downstream” genes. This is comparable to the notion of an embryological “organizer” that we will see in later chapters. Another term for the idea is “master control” genes, but it can be misleading to think of genes so metaphorically in terms of our own culture’s social structure; rarely if ever is one gene the master of a whole organ in a very meaningful sense. Selectors cause a major differentiation commitment or branch point. They activate a series of other TFs that in turn activate still more TFs that ultimately alter the expression of structural genes, enzymes, and the like. *Hox*, *Pax*, and *MyoD* genes are examples. Different selectors work at different levels or stages of embryological development (Carroll, Grenier et al. 2001). But they don’t act alone, and, typically, other factors (perhaps controlling their own downstream effects) are also involved.

We will elaborate further on this in Chapter 9, but here we can indicate the uses of linkage relationships among selector genes. The members of the *Hox* class of homeobox TF genes have become an archetype of high-level selectors in which linkage arrangements are important. The *Hox* genes are linked in chromosomal clusters (shown in Figure 4-4), whose individual genes are expressed in a colinear, sequential, cumulative fashion in cells from anterior to posterior regions of the early embryo, respectively. That is, the gene at the 3’ end of a *Hox* cluster is expressed earlier in development, in cells in the most anterior parts of the embryo. A bit later, as the next most posterior region of the embryo begins to develop, the next most 5’ *Hox* gene in the cluster also becomes expressed (that is, along with the *Hox* gene

that is already “on”). This continues down the line of genes in the cluster to the most 5' gene, when the most posterior anatomic segments develop, which is when most or all of the genes are being expressed. Because the latest gene to be activated as more posterior segments differentiate has its effect on those segments, even though earlier (more “anterior”) genes may still be expressed, the effect is known as “posterior prevalence.”

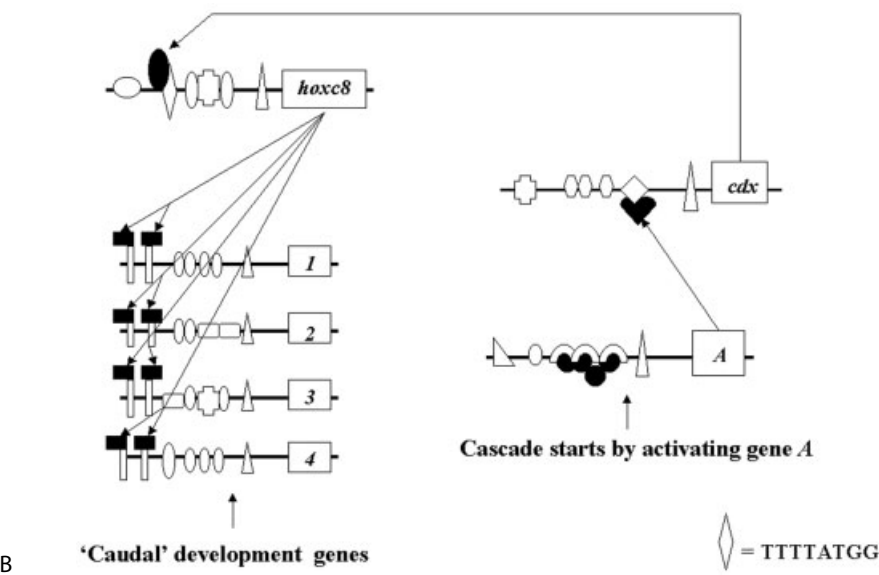
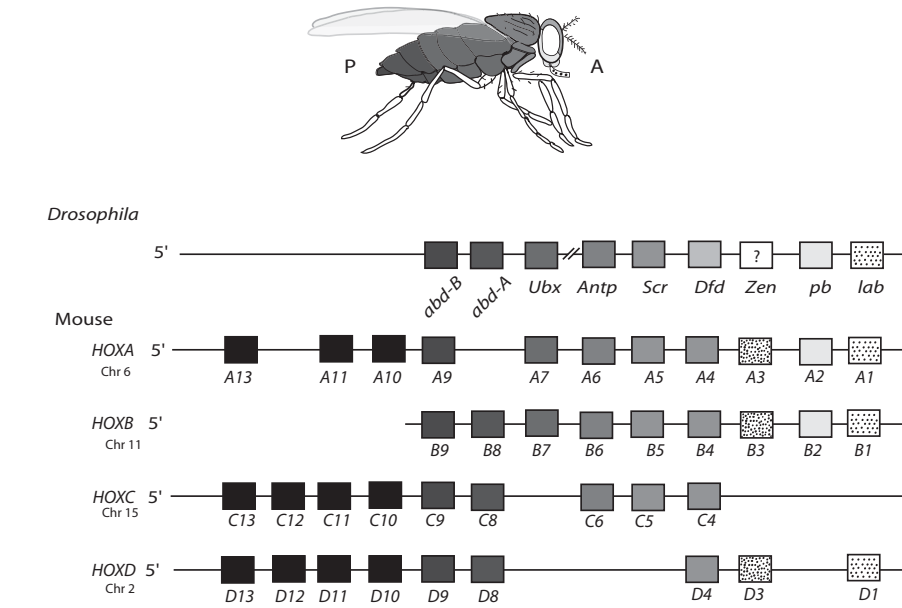
Figure 7-4A repeats part of Figure 4-4 to show the gene clusters in *Drosophila* and in mammals, and is another oft-used figure because of its great importance in the recent history of developmental biology and genetics. The colinear summed expression pattern sets up combinatorial sets of *Hox* transcripts that effect gene expression cascades that differ locally along an anatomic axis. The figure shows the similarity in this aspect of the expression pattern that is shared between vertebrates and invertebrates, first documented in a famous paper by Lewis in flies (Lewis 1978) and later shown in vertebrates.

Essentially, the same summed colinear expression system is used to specify the primary AP axis in species across much of the animal world, although the details of what the various combinations achieve vary. However, the *Hox* cluster genes in vertebrates are also used in the development of many other structures, including the segmentation of the gut and limbs. Thus, a *Hox* gene is not a “vertebra” gene, but can be viewed as a selector gene, or in some contexts a patterning selector gene in several structures. The use is functionally arbitrary as noted above and not inherent in the genes themselves, and can only work because of the contingent, hierarchical, sequestered nature of development (e.g., limb buds are not the same kinds of cells as are early vertebral precursors).

Not all pathways are as deeply conserved as the *Hox* system, nor rely on conserved linkage relationships. Even this classic system is subject to all the sources of variation that one might expect; in fact, the more intensely it is studied, the less completely conserved it seems to be (e.g., Levine et al. 2002), including the variety of ways the genes in the cluster may be regulated to coordinate their spatial and temporal domains of expression (Kmita and Duboule 2003). Nonetheless, much is conserved, and similar situations apply to a number of other differentiation hierarchies, such as the role of *Pax6* in eye development (Chapter 14), or *MyoD* in muscle development (Gerhart and Kirschner 1997). Figure 7-4B shows the general logic of a selector gene, and the way that REs are used to start and proliferate a cascade of downstream effects on gene expression.

A regulatory gene that acts early in helping trigger differentiation cascades may be widely expressed in an organism; some TFs seem to help regulate a high fraction of all the genes in the genome, as seems to be the case with the *Ftz* and *Eve* genes in *Drosophila* (Liang and Biggin 1998). However, the spatiotemporal distribution of most regulatory genes is restricted at least somewhat, and like a Venn diagram of the embryo, the set of regulatory signals activated in a given cell is assumed to be correlated with the functional genes activated in that cell. It is assumed that there is a unique combination for each function, or at least for each context in an organism; although this is far from proven, it is central to the present view of differentiation by sequestered combinations of *cis*-regulating factors.

A critical aspect of this that enables so much combinatorial pleiotropy to work is that gene activation states can be mitotically heritable. This is a most important sequestration factor. Once a differentiation cascade has been initiated, a cell and its descendants in the organism can become strikingly autonomous, either self-



sustaining or changing in a contingent way. The state of differentiation is not washed out by being blended with that of neighboring cells. One cell or few cells can be the progenitor of a whole complex structure even if isolated from the rest of the organism (e.g., grown *in vitro*). As cancer shows, and may be important to many normal traits as well, a mutant somatic cell can generate a clone of cells with altered function. Cellular commitments are achieved in various ways; for example, the differentiation state can affect chromatin structure that leaves some genes open for TF binding, whereas others are closed to regulatory machinery (Davidson 2001; Davidson, Rast et al. 2002; Davidson, Rast et al. 2002; Lieb et al. 2001).

In the chicken-and-egg discussion raised in Chapter 4 about RNA and DNA, the regulation of an organism is in this view necessarily hierarchical, all kick-started by parental mRNA and proteins produced in the zygote. The increasingly committed nature of cells that occurs during development is the result. The new zygote is but another form of single-cell precursor to an autonomously developing tissue cascade. Of course, this is hierarchical with respect to the zygote but not the process itself. The process is circular, since it was the same genome that made the maternal mRNA and will eventually make that of the germ cells of the next generation.

Because sequestration is necessarily incomplete, cells are able to continue differentiating in contingent, context-dependent ways, but in some cases can re- or de-differentiate. The degree of reversibility that occurs in nature varies. In a mature animal embryo, most cells are unable to differentiate into more than a limited set of tissue or cell types; artificial cloning from somatic (body, as opposed to germ) cells in mammals, for example, requires special treatment of the cells to dedifferentiate them. However, the nature of commitment depends on species and even context, and not all cells are fully committed to some end state. Reptiles can regenerate entire tails or limbs, and humans can regenerate blood vessels and skin, from cells that are partially differentiated even in adults. Cells in a developing brain may in some instances be entrained to behave like cells amongst which they intercalate (see Chapter 15). Plants are rather different in this respect, with much more flexibility built into the basic way they do business (for example, in some, the tip of any branch able to form flowers).

Implications of Pleiotropy in Complex Regulatory Circuits

If the pleiotropic use of a limited set of regulatory genes makes complex evolution easier to explain, it entails the potentially serious problem that mutation in a TF could disrupt *all* of its different functions simultaneously, which could be a disaster



Figure 7-4. Aspects of regulation by and of genes. (A) *Hox* combinatorial expression figure; similar anterior-posterior colinear expression affects both invertebrates and vertebrates; gray shade is used to indicate the gene class and the most anterior location of its expression along the axis where the gene has its predominant effect, as shown by corresponding shading in the two types of animal; for example, darker-shaded genes are expressed posteriorly and affect caudal regional development; (B) schematic of regulatory circuits, hierarchies, and batteries, illustrated by a schematic of the use of the TF *Cdx* to regulate *Hoxc8* that in turn is a selector for many other genes, in patterning early caudal vertebrate development; the diamond symbol represents an enhancer sequence that flanks *Hoxc8* (e.g., see Shashikant et al. 1998).

for the organism (e.g., Losick and Sonenshein 2001; Struhl 2001). However, because transcription factors work via regulatory elements, there is a ready escape: mutations that add, delete, or modify a particular RE will only affect the expression of its nearby gene and hence only that particular use of the TF that recognizes the RE.

This provides room for tremendous evolutionary flexibility, but there is also a potential flip side. If the same TF is used in many contexts, an organism is vulnerable to a high degree of multiple jeopardy. A given TF may use hundreds of REs scattered across the genome in its different functional contexts. Relying on one sequence in too many ways raises the likelihood that one or more of these REs will be mutated, and that could be harmful. There are at least two ways to protect a gene against this kind of mutational regulatory “noise” (Sengupta et al. 2002). One is to flank it with multiple copies of the relevant RE; Figure 4-7 provided an example of enhancers upstream of lens protein genes (this shows between-species variation, but the principle is the same). As was shown in Table 4-5, another mutational buffer is for a TF to evolve tolerance for variation in its recognition sequence. Not all the sequences are equally efficient at binding the TF, but they are at least recognized.

However, this same robustness that buffers a gene’s enhancers from being erased by mutation makes it more difficult for evolution to remove that gene from the TF’s set of regulated genes—it could take many mutations to delete all the relevant REs for a given gene to completely destroy the binding site (as opposed just to making it less efficient and, for example, lowering the expression level of the gene). This could tie functions together over evolutionary time once the use of REs for a common TF is established, yielding suites of traits that evolve together, at least for a period until mutational variation accumulates. During that time, traits might appear to share adaptive constraint, and could possibly be wrongly interpreted as of apparent canalization of traits, of selectively constrained variation, evolutionary stability, or range of response to environmental changes (e.g., Siegal and Bergman 2002). This is speculative, but a balance between robustness to mutation and stabil-

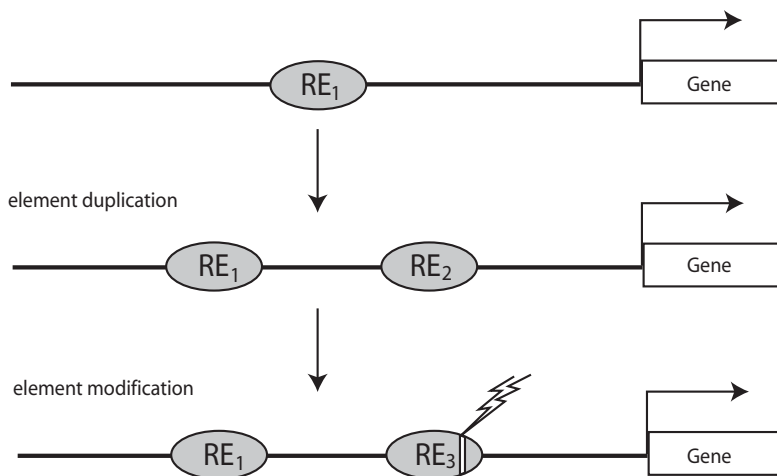


Figure 7-5. Schematic of regulatory element evolution by duplication and modification by mutation (lightning). Short elements can also be created or destroyed by mutation in otherwise random sequence. Bent arrow indicates start and direction of gene transcription.